# Gauge-Covariant Variational Free Energy
# Supplement B

Base Transport, Infinite Networks, and the Lattice Cluster

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# Part B-I

# Base Transport and Multi-Point Designs

# Chapter B0

# Introduction

The main manuscript develops the theory at one fixed base point $c_*$: the design is $\{c_*\}$, every agent carries one common frame, and the only transports are the independent interaction links. This supplement collects what the theory requires once contexts vary, once networks become infinite, and once agents are placed on a lattice. It answers one question for each extension: which statements of the single-point theory survive, and exactly which additional data (a connection, a cover, a design, a base measure, a scale identification, or a lattice realization) each extension consumes. Base transport is deferred until the single-point theory is complete, and nothing in the main manuscript depends on this supplement. The relative frame $h_i$ between belief and model frames, which the main manuscript removes by a gauge choice because no energy, kernel, or evaluator at $c_*$ reads it, returns to the main manuscript only if frames become dynamical agent states.

Part B-I treats base transport and multi-point designs. Chapter B1 introduces base curves, relative frames, chosen connections and covariant defects, the Čech picture, finite designs with their shared-link stabilizer (Proposition B1.26), and scale maps over base maps. Chapter B2 pulls the fiber Fisher and Amari–Chentsov tensors back to the base through a chosen connection and proves the exact signed comparison of fine and coarse pullbacks (Theorem B2.25) together with the isotropy criterion for a vanishing horizontal defect (Theorem B2.31). Chapter B3 lifts inference histories to spaces of sections, and Chapter B4 constructs parents over covers and states the connection-descent criterion. Part B-II, beginning with Chapter B5 and continuing in Chapter B6, holds the asymptotic apparatus that the main manuscript does not use: mode exponents, tempered comparisons, interaction beta functions, fixed points, spectra, and two-index limits. Part B-III holds the lattice cluster, Chapter B7, Chapter B8, and Chapter B9, whose typing, as agents at $c_*$ or as base points, is Open Problem B7.1. The appendices hold the proofs of the base-direction classifications (Appendix BA), of the constructive spatial family (Appendix BB), and of the thermodynamic Fisher theorems (Appendix BC), followed by the notation table, the open problems, and the numerical index of this supplement (Appendix BD).

The supplement cites the main manuscript and Supplement G; its chapters are numbered B0, B1, and so on, so that references into the other two documents are unambiguous. It uses the symbols of the main manuscript with their meanings there and adds its own. The horizontal lift $H_e^x$, the relative frame $h_i$, the design index $d$ with incidence set $I_D$, the scale bundle morphism $\Psi$, the covariant defect $\mathcal{D}\Phi$ of a cross map, and the base-cocycle residual $\mathcal{R}$ belong to this supplement. Here $f$ denotes a base map and $s$ a section; the fast and slow blocks of the main manuscript do not

occur. Frame re-expressions follow the convention of the main manuscript, $u' = u \cdot g^{-1}$, separately in each channel when two atlases are used, and the common-frame case is $g_i^b = g_i^m = g_i$. The lattice part declares its local symbols in one table at its head.

The contextual base is not identified with physical space or spacetime; graph links are not identified with base transports except under the declared realization Hypothesis B1.13; no probability law on a continuum space of sections is constructed; and the asymptotic Wilsonian vocabulary is not applied to a finite network at $c_*$, where finite depth has only finite-step rates (Proposition B5.15). [NOT-CLAIMED] The continuum law theory is Open Problem B1.22 and the lattice typing is Open Problem B7.1.

# Chapter B1

# Base Geometry and Multi-Point Designs

At the fixed base point of the main manuscript every agent's frame is a point of one principal fiber, the independent links are the only transports, and the design is $\{c_*\}$. This chapter asks what additional structure appears when contexts vary, and which of it is forced by the one principal bundle rather than declared. Two local sections of that bundle determine a unique relative frame, and the active gauge image at a context remains one copy of $G$ inside the passive product of independent belief and model re-expressions (Propositions B1.1 and B1.2). Chosen connections define covariant defects of the cross maps, whose vanishing is exactly parallelness, and bundle triviality, graph flatness, connection flatness, and trivial loop holonomy are four logically independent conditions (Propositions B1.9 and B1.15). On a finite design, constraining one link to be shared across design points leaves exactly the context-independent rechoices multiplied by a central twist common to both endpoints (Proposition B1.26). The connections and defects are used by the base-direction pullbacks of Chapter B2, the designs and section responses by the section-space histories of Chapter B3, and the scale maps over base maps by the covers of Chapter B4.

The connections of Definition B1.6 are chosen data, and no curvature or transport is inferred from the agent frames; the matrix defect laws are not asserted for nonlinear law fibers; graph links are not identified with base transports except under Hypothesis B1.13; and the finite constructions of this chapter supply no probability law on a continuum space of sections. [NOT-CLAIMED] The continuum theory is Open Problem B1.22, the realization of independently specified links by base transport and the recovery of a connection from records are Open Problems B4.14 and B4.15, and the regular quotient of the passive coordinate action is Open Problem B1.30.

## B1.1 Base curves and contexts

Away from $c_*$ the contextual base $\mathcal{C}$ of Definition 2.1 is a smooth manifold whose curves order contexts. A connection on a bundle over $\mathcal{C}$ defines transport in that bundle along base curves; unless a connection on $T\mathcal{C}$ is separately chosen, it does not transport tangent vectors of the base. A finite design (Definition B1.18) is a finite subset of $\mathcal{C}$, not a continuum limit or a discretization of it. A base curve does not by itself represent the evolution of an agent. Change of a belief or model at one context is a curve inside one associated fiber, simultaneous change over all contexts is a curve in a declared space of sections, and a trajectory whose projection moves in $\mathcal{C}$ needs the horizontal and vertical splitting of a chosen connection; the three curve types are defined in Definition B3.1.

## B1.2 Relative frames and the two-atlas apparatus

The main manuscript uses one frame per agent, a gauge choice made because no energy, kernel, or evaluator at $c_*$ reads a relative frame, and it records at one fiber how two atlases reduce to one (Proposition 2.7). Belief and model frames may instead be chosen as two local sections of the same principal bundle over the supports $\mathcal{C}_i$. In this supplement a passive re-expression of either frame is written with the convention of the main manuscript, $u' = u \cdot g^{-1}$, separately in each channel.

**Proposition B1.1 (Relative principal frame over a support).** For each agent $i$ let $u_i^b, u_i^m : \mathcal{C}_i \to \mathscr{P}_G$ be smooth local sections of the principal bundle. (i) The relative frame $h_i$ of (2.13), taken at every point of $\mathcal{C}_i$, is a smooth map $h_i : \mathcal{C}_i \to G$, and its existence uses only $\mathscr{P}_G$, not the representations $\rho_b, \rho_m$. (ii) Under independent smooth passive re-expressions

$$u_i^{b\prime} = u_i^b \cdot (g_i^b)^{-1}, \qquad u_i^{m\prime} = u_i^m \cdot (g_i^m)^{-1}, \qquad g_i^b, g_i^m : \mathcal{C}_i \longrightarrow G, \tag{B1.1}$$

associated coordinates change by $\beta'_x = \widehat{\rho}_x(g_i^x)\beta_x$ and $h_i$ obeys (2.14) pointwise. (iii) At $c \in \mathcal{C}_i \cap \mathcal{C}_j$ the comparisons $T_{ij}^x \in G$ defined by $u_j^x(c) = u_i^x(c) \cdot T_{ij}^x$ transform by $T_{ij}^{x\prime} = g_i^x T_{ij}^x (g_j^x)^{-1}$ and satisfy (2.15). (iv) The represented pair $(\rho_b(h_i), \rho_m(h_i))$ determines $h_i$ only modulo $\ker \rho_b \cap \ker \rho_m$. A basis of an associated vector fiber that is not induced by a section of $\mathscr{P}_G$ need not be related to an induced basis by any element of $G$: for $G = \mathrm{SO}(2)$ acting on $\mathbb{R}^2$ by its defining representation, the basis obtained from an induced one by $\mathrm{diag}(2, 1)$ is not. [ESTABLISHED]

*Proof.* (i) Pointwise existence and uniqueness is Proposition 2.7; in a local trivialization in which $u_i^x(c) = (c, U^x(c))$ with smooth $U^x$, $h_i = (U^b)^{-1}U^m$ is smooth. (ii) The quotient convention (2.7) gives $[u, \beta] = [u' \cdot g, \beta] = [u', \widehat{\rho}_x(g)\beta]$, and $u_i^{m\prime} = u_i^b h_i (g_i^m)^{-1} = u_i^{b\prime} \cdot g_i^b h_i (g_i^m)^{-1}$ with uniqueness in the torsor gives the law. (iii) From $u_j^{x\prime} = u_i^x T_{ij}^x (g_j^x)^{-1} = u_i^{x\prime} \cdot g_i^x T_{ij}^x (g_j^x)^{-1}$ follows the transformation law, and the two expressions $u_j^m = u_j^b h_j = u_i^b T_{ij}^b h_j$ and $u_j^m = u_i^m T_{ij}^m = u_i^b h_i T_{ij}^m$ give the relation. (iv) $\rho_x(h) = \rho_x(h')$ exactly when $h^{-1}h' \in \ker \rho_x$. An induced basis is carried to another induced basis by $\rho_b(k) \in \mathrm{SO}(2)$, and $\mathrm{diag}(2, 1) \notin \mathrm{SO}(2)$. □

The relative frame compares two principal sections even when $\rho_b$ and $\rho_m$ are inequivalent or act on fibers of different dimensions. It produces two represented operators, $\widehat{\rho}_b(h_i)$ on $\mathcal{B}_b$ and $\widehat{\rho}_m(h_i)$ on $\mathcal{B}_m$, and no operator from $\mathcal{B}_b$ to $\mathcal{B}_m$; a comparison of channel points still needs a cross map of Definition 2.9. At every level $\ell$ of a scale diagram the same construction applies: on a common trivializing neighborhood $\mathcal{U}_{\ell,i}$ of the level base, two sections determine

$$u_{\ell,i}^m = u_{\ell,i}^b \cdot h_{\ell,i}, \qquad h_{\ell,i} : \mathcal{U}_{\ell,i} \longrightarrow G_\ell, \tag{B1.2}$$

with the law (2.14) under independent rechoices, by Proposition B1.1 at level $\ell$. The common-frame convention of the main manuscript is $h_i = e_G$ with $g_i^b = g_i^m$.

The active gauge group is not enlarged by the second atlas. At one context the argument of Proposition 2.6 applies with an $h_i$-twist in the model channel.

**Proposition B1.2 (Active gauge image under independent atlases).** Fix $c \in \mathcal{C}$ with $V_c = \{i \in V : c \in \mathcal{C}_i\}$ nonempty, and evaluate $h_i$ and $T^x_{ij}$ at $c$. For a right-$G$-equivariant bijection $F_c$ of the fiber $(\mathscr{P}_G)_c$ define $\varphi^x_i \in G$ by $F_c(u^x_i(c)) = u^x_i(c) \cdot \varphi^x_i$. Then

$$\varphi^x_j = (T^x_{ij})^{-1} \varphi^x_i T^x_{ij}, \qquad \varphi^m_i = h_i^{-1} \varphi^b_i h_i, \qquad i, j \in V_c, \tag{B1.3}$$

the second being (2.16), and $F_c \mapsto (\varphi^b_i, \varphi^m_i)_{i \in V_c}$ is a group isomorphism from $\mathrm{Aut}_G((\mathscr{P}_G)_c)$ onto the subgroup $\mathcal{A}_c \subseteq \prod_{i \in V_c} (G \times G)$ cut out by (B1.3). For each $i_0 \in V_c$ the projection onto $\varphi^b_{i_0}$ is an isomorphism $\mathcal{A}_c \cong G$. For nontrivial $G$, $\mathcal{A}_c$ is a proper subgroup of the passive product $\prod_{i \in V_c} (G \times G)$, and, as soon as $|V_c| \geq 2$, a proper subgroup of the product $\prod_{i \in V_c} \{(k, h_i^{-1} k h_i) : k \in G\}$ of pointwise twisted diagonals. The local functions at $c$ of every gauge automorphism of $\mathscr{P}_G$ form one element of $\mathcal{A}_c$. [ESTABLISHED]

*Proof.* Equivariance gives $F_c(u^x_j) = F_c(u^x_i T^x_{ij}) = u^x_i \varphi^x_i T^x_{ij} = u^x_j (T^x_{ij})^{-1} \varphi^x_i T^x_{ij}$, and uniqueness in the torsor gives the first relation; the second follows from $F_c(u^m_i) = F_c(u^b_i h_i) = u^b_i \varphi^b_i h_i = u^m_i h_i^{-1} \varphi^b_i h_i$. Since $F_c(F'_c(u)) = u \cdot \varphi\varphi'$, the map is a homomorphism, and it is injective because $\varphi^b_{i_0}$ determines $F_c$ on the whole torsor. Given an element of $\mathcal{A}_c$, the map $u^b_{i_0}(c) \cdot g \mapsto u^b_{i_0}(c) \cdot \varphi^b_{i_0} g$ is an equivariant bijection whose local functions are the given ones, and every $\varphi^b_{i_0} \in G$ occurs, which proves $\mathcal{A}_c \cong G$. For $k \neq e_G$, the element with $\varphi^b_i = k$, $\varphi^m_i = h_i^{-1} k h_i$ and $\varphi^b_j = e_G = \varphi^m_j$ for $j \neq i$ lies in the product of twisted diagonals but violates the first relation of (B1.3); the passive product is larger still. A gauge automorphism restricts to an equivariant bijection of each fiber. □

Whether a given element of $\mathcal{A}_c$ is the value of a global gauge automorphism is a separate question about the base.

**Proposition B1.3 (Realization of fiber automorphisms by gauge automorphisms).** Let $c_1, \dots, c_M \in \mathcal{C}$ be distinct and let $F_d \in \mathrm{Aut}_G((\mathscr{P}_G)_{c_d})$ for $d = 1, \dots, M$. If $G$ is connected, there is a smooth gauge automorphism $F$ of $\mathscr{P}_G$ over $\mathrm{id}_{\mathcal{C}}$ that restricts to $F_d$ at every $c_d$. For disconnected $G$ this can fail: gluing $[0,1] \times S_3$ by $(1, g) \sim (0, \mathring{g} g)$ for a 3-cycle $\mathring{g}$ gives a principal $S_3$-bundle over the circle, and the fiber automorphism $(0, g) \mapsto (0, \lambda g)$ with $\lambda$ a transposition is the restriction of no gauge automorphism. [ESTABLISHED]

*Proof.* Choose pairwise disjoint neighborhoods of the $c_d$ carrying local sections $\sigma_d$, define $\kappa_d \in G$ by $F_d(\sigma_d(c_d)) = \sigma_d(c_d) \cdot \kappa_d$, and write $\kappa_d = \exp(\xi_1) \cdots \exp(\xi_r)$ with $\xi_1, \dots, \xi_r \in \mathfrak{g}$, which is possible because a connected Lie group is generated by the image of its exponential map. Put $\gamma_d(t) = \exp(t\xi_1) \cdots \exp(t\xi_r)$. With smooth bump functions $\chi_d$ of compact support in those neighborhoods and $\chi_d(c_d) = 1$, the map $F(\sigma_d(y) \cdot g) = \sigma_d(y) \cdot \gamma_d(\chi_d(y)) g$ there and $F = \mathrm{id}$ elsewhere is a smooth equivariant automorphism restricting to $F_d$ at $c_d$. For the $S_3$ bundle, a gauge automorphism acts on the strip by $(t, g) \mapsto (t, \kappa g)$ with $\kappa$ constant because $S_3$ is discrete, and compatibility with the gluing forces $\kappa \mathring{g} = \mathring{g} \kappa$. The centralizer of a 3-cycle in $S_3$ is the cyclic group it generates, which contains no transposition. □

Graph links between fibers over different contexts are compared through link maps. Anchor every vertex at a context $c_i \in \mathcal{C}_i$, evaluate the frames and $h_i$ there, and for an edge copy $e : j \to i$ of the interaction multigraph of Definition 2.11 define $\mathfrak{t}_e^x : (\mathscr{P}_G)_{c_j} \to (\mathscr{P}_G)_{c_i}$ by $\mathfrak{t}_e^x(u_j^x \cdot g) = u_i^x \cdot \Theta_e^x g$.

**Proposition B1.4 (Identified links under independent atlases).** Each $\mathfrak{t}_e^x$ is a right-$G$-equivariant bijection that does not depend on the vertex frames of its channel when the links transform by $\Theta_e^{x\prime} = g_i^x \Theta_e^x (g_j^x)^{-1}$. The two channels carry one link on $e$, meaning $\mathfrak{t}_e^b = \mathfrak{t}_e^m$, exactly when

$$\Theta_e^m = h_i^{-1} \Theta_e^b h_j, \qquad e : j \to i, \tag{B1.4}$$

and this relation is invariant under independent re-expressions of both atlases. The literal equality $\Theta_e^b = \Theta_e^m$ is not invariant under them; under the common-frame convention $h_i = h_j = e_G$, it expresses $\mathfrak{t}_e^b = \mathfrak{t}_e^m$, which is the form (2.31) of the main manuscript. More generally, equal link coordinates $\Theta_e^b = \Theta_e^m = \Theta_e$ represent equal link maps exactly when $h_i^{-1} \Theta_e h_j = \Theta_e$. [ESTABLISHED]

*Proof.* Every point of $(\mathscr{P}_G)_{c_j}$ is $u_j^x \cdot g$ for a unique $g$, so $\mathfrak{t}_e^x$ is well defined, equivariant, and invertible. After a rechoice, $\mathfrak{t}_e^{x\prime}(u_j^{x\prime} \cdot g) = u_i^x (g_i^x)^{-1} g_i^x \Theta_e^x (g_j^x)^{-1} g = u_i^x \Theta_e^x (g_j^x)^{-1} g = \mathfrak{t}_e^x (u_j^x (g_j^x)^{-1} g) = \mathfrak{t}_e^x (u_j^{x\prime} \cdot g)$. Next, $\mathfrak{t}_e^b(u_j^m \cdot g) = \mathfrak{t}_e^b(u_j^b \cdot h_j g) = u_i^b \Theta_e^b h_j g = u_i^m \cdot h_i^{-1} \Theta_e^b h_j g$, which equals $\mathfrak{t}_e^m(u_j^m \cdot g) = u_i^m \Theta_e^m g$ for every $g$ exactly when (B1.4) holds; invariance follows because both link maps are frame independent. A rechoice of the model atlas alone with $g_j^m = e_G$ and $g_i^m = k \neq e_G$ replaces $\Theta_e^m$ by $k\Theta_e^m$ and leaves $\Theta_e^b$ fixed, so it breaks the literal equality. □

Cross maps are the remaining place where two atlases matter, because their local representatives must be converted to one atlas before they can be read as sections of a map bundle.

**Proposition B1.5 (Cross maps under independent atlases).** Let $\Phi, \widetilde{\Phi}$ be cross morphisms (Definition 2.9) with local representatives $\phi_i = \Phi_i^{bm}$ and $\widetilde{\phi}_i$ in the frames $u_i^b, u_i^m$. (i) Under (B1.1), $\phi_i' = \widehat{\rho}_m(g_i^m) \circ \phi_i \circ \widehat{\rho}_b(g_i^b)^{-1}$ and $\widetilde{\phi}_i' = \widehat{\rho}_b(g_i^b) \circ \widetilde{\phi}_i \circ \widehat{\rho}_m(g_i^m)^{-1}$. (ii) On overlaps $\phi_i \circ \widehat{\rho}_b(T_{ij}^b) = \widehat{\rho}_m(T_{ij}^m) \circ \phi_j$ and $\widetilde{\phi}_i \circ \widehat{\rho}_m(T_{ij}^m) = \widehat{\rho}_b(T_{ij}^b) \circ \widetilde{\phi}_j$. (iii) In the belief atlas the converted representatives

$$\check{\phi}_i := \widehat{\rho}_m(h_i) \circ \phi_i, \qquad \check{\widetilde{\phi}}_i := \widetilde{\phi}_i \circ \widehat{\rho}_m(h_i^{-1}), \tag{B1.5}$$

the first being $\Phi_i^{\mathrm{conv}}$ of (2.17), satisfy $\check{\phi}_i \circ \widehat{\rho}_b(T_{ij}^b) = \widehat{\rho}_m(T_{ij}^b) \circ \check{\phi}_j$ and $\check{\widetilde{\phi}}_i \circ \widehat{\rho}_m(T_{ij}^b) = \widehat{\rho}_b(T_{ij}^b) \circ \check{\widetilde{\phi}}_j$, so they are local representatives of sections of the associated map bundles with actions $\phi \mapsto \widehat{\rho}_m(g) \circ \phi \circ \widehat{\rho}_b(g)^{-1}$ and its interchange. (iv) If $\Phi$ is induced by an intertwiner $\phi$ as in (2.26), then $\phi_i = \widehat{\rho}_m(h_i^{-1}) \circ \phi$ and $\check{\phi}_i = \phi$. [ESTABLISHED]

*Proof.* A point $[u_i^b, \beta]$ is sent to $[u_i^m, \phi_i \beta]$. In new coordinates its input is $\widehat{\rho}_b(g_i^b)\beta$ and its output $\widehat{\rho}_m(g_i^m)\phi_i\beta$, which gives (i). Over $c \in \mathcal{C}_i \cap \mathcal{C}_j$, $[u_j^b, \beta] = [u_i^b, \widehat{\rho}_b(T_{ij}^b)\beta]$ has image $[u_j^m, \phi_j\beta] = [u_i^m, \widehat{\rho}_m(T_{ij}^m)\phi_j\beta]$ computed in $j$ and $[u_i^m, \phi_i \widehat{\rho}_b(T_{ij}^b)\beta]$ computed in $i$, which gives (ii); the second map is the same computation. For (iii), $\check{\phi}_i \widehat{\rho}_b(T_{ij}^b) = \widehat{\rho}_m(h_i T_{ij}^m)\phi_j = \widehat{\rho}_m(T_{ij}^b h_j)\phi_j = \widehat{\rho}_m(T_{ij}^b)\check{\phi}_j$ by (2.15), and $\check{\widetilde{\phi}}_i \widehat{\rho}_m(T_{ij}^b) = \widetilde{\phi}_i \widehat{\rho}_m(T_{ij}^m h_j^{-1}) = \widehat{\rho}_b(T_{ij}^b)\check{\widetilde{\phi}}_j$. For (iv), $[u_i^b, \beta] \mapsto [u_i^b, \phi\beta] = [u_i^m h_i^{-1}, \phi\beta] = [u_i^m, \widehat{\rho}_m(h_i^{-1})\phi\beta]$. □

The compatible family $(\phi_i)$ is therefore a map-bundle section only after the conversion (B1.5), and $h_i$ does not replace the cross-fiber map.

## B1.3 Connections, parallel transport, and covariant defects

**Definition B1.6 (Two connections on the common principal bundle).** Under Hypothesis 2.3, choose principal connections

$$\omega_b, \omega_m \in \Omega^1(\mathscr{P}_G, \mathfrak{g}). \tag{B1.6}$$

They induce horizontal distributions $H^b, H^m$ on $\mathcal{E}_b, \mathcal{E}_m$, with horizontal lift $H^x_e X$ of $X \in T_c\mathcal{C}$ at $e \in (\mathcal{E}_x)_c$, and, for every piecewise-smooth $\gamma : [0,1] \to \mathcal{C}$, parallel transports

$$\Omega_\gamma : (\mathcal{E}_b)_{\gamma(0)} \longrightarrow (\mathcal{E}_b)_{\gamma(1)}, \qquad \widetilde{\Omega}_\gamma : (\mathcal{E}_m)_{\gamma(0)} \longrightarrow (\mathcal{E}_m)_{\gamma(1)}, \tag{B1.7}$$

obtained by carrying the associated-law coordinate along the $\omega_x$-horizontal lift of $\gamma$ through a point of $(\mathscr{P}_G)_{\gamma(0)}$.

This is transport over the base curve, not transport of vectors in $T\mathcal{C}$; a tangent-vector transport needs a separately chosen affine connection on $T\mathcal{C}$, unless $\mathscr{P}_G$ and its representation have been identified with the frame bundle of $\mathcal{C}$. The two connections need not agree. Their difference $\omega_m - \omega_b$ is horizontal and $\mathrm{Ad}$-equivariant, hence descends to an $\mathrm{Ad}(\mathscr{P}_G)$-valued one-form on $\mathcal{C}$. In an associated vector-bundle realization the horizontal distributions define covariant derivatives $\nabla^b, \nabla^m$; no linear covariant derivative is assumed on a law fiber. The equal-connection specialization declares $\omega_b = \omega_m$; the principal holonomy is then common, while its actions on belief and model laws remain $\widehat{\rho}_b$ and $\widehat{\rho}_m$. It constrains the connections only, so it is separate from the common frame and from identified links, which constrain frames and link data.

**Proposition B1.7 (Frame laws of connections and transports).** Let $A^x_i = (u^x_i)^* \omega_x$. In matrix notation for the group, the re-expressions (B1.1) give

$$A^{b\prime}_i = \mathrm{Ad}_{g^b_i} A^b_i - dg^b_i (g^b_i)^{-1}, \tag{B1.8}$$

$$A^{m\prime}_i = \mathrm{Ad}_{g^m_i} A^m_i - dg^m_i (g^m_i)^{-1}. \tag{B1.9}$$

If the endpoint coordinates of $\gamma$ are re-expressed by $g_0, g_1$ in the belief frames and by $\widetilde{g}_0, \widetilde{g}_1$ in the model frames, the represented transports obey

$$\Omega'_\gamma = \widehat{\rho}_b(g_1) \circ \Omega_\gamma \circ \widehat{\rho}_b(g_0)^{-1}, \tag{B1.10}$$

$$\widetilde{\Omega}'_\gamma = \widehat{\rho}_m(\widetilde{g}_1) \circ \widetilde{\Omega}_\gamma \circ \widehat{\rho}_m(\widetilde{g}_0)^{-1}. \tag{B1.11}$$

[ESTABLISHED]

*Proof.* In a trivialization with $u(x) = (x, U(x))$, a principal connection has $u^*\omega = U^{-1}A_0U + U^{-1}dU$ for the local form $A_0$ of the identity section. For $U' = Ug^{-1}$ this gives $gU^{-1}A_0Ug^{-1} + gU^{-1}dUg^{-1} + g\mathrm{d}(g^{-1}) = \mathrm{Ad}_g A - (\mathrm{d}g)g^{-1}$, because $\mathrm{d}(g^{-1}) = -g^{-1}(\mathrm{d}g)g^{-1}$. The transport is a coordinate-free

map between fibers, and the coordinates at each endpoint change by $\beta' = \widehat{\rho}_x(g)\beta$. These are the standard laws for the quotient convention (2.7) [17, 22]. □

The endpoint laws also show why a same-point frame comparison and transport along a nonconstant curve are different objects: the first involves one fiber, the second two.

**Definition B1.8 (Covariant defects).** Let $e \in \mathcal{E}_b$ and $e' \in \mathcal{E}_m$ lie over $c$, and let $X \in T_c\mathcal{C}$. For smooth associated fibers and smooth cross morphisms define the vertical-valued tensors

$$(\mathcal{D}\Phi)_e(X) = T_e\Phi\big(H^b_e X\big) - H^m_{\Phi(e)}X, \tag{B1.12}$$

$$(\mathcal{D}\widetilde{\Phi})_{e'}(X) = T_{e'}\widetilde{\Phi}\big(H^m_{e'} X\big) - H^b_{\widetilde{\Phi}(e')}X. \tag{B1.13}$$

The cross map is parallel when its defect vanishes identically.

**Proposition B1.9 (Covariant defects and parallelness).** (i) Both defects take vertical values. (ii) $\mathcal{D}\Phi = 0$ and $\mathcal{D}\widetilde{\Phi} = 0$ hold exactly when every piecewise-smooth base curve satisfies

$$\widetilde{\Omega}_\gamma \circ \Phi_{\gamma(0)} = \Phi_{\gamma(1)} \circ \Omega_\gamma, \tag{B1.14}$$

$$\Omega_\gamma \circ \widetilde{\Phi}_{\gamma(0)} = \widetilde{\Phi}_{\gamma(1)} \circ \widetilde{\Omega}_\gamma, \tag{B1.15}$$

respectively. (iii) In a fiberwise-linear associated-vector-bundle specialization with representation spaces $\mathbb{V}_b, \mathbb{V}_m$, the defects reduce to

$$D\Phi = \nabla^m \circ \Phi - \Phi \circ \nabla^b, \qquad D\widetilde{\Phi} = \nabla^b \circ \widetilde{\Phi} - \widetilde{\Phi} \circ \nabla^m, \tag{B1.16}$$

the old-to-new coordinate maps induced by (B1.1) are

$$R^b_i = \rho_b(g^b_i) : \mathbb{V}_b \longrightarrow \mathbb{V}_b, \qquad R^m_i = \rho_m(g^m_i) : \mathbb{V}_m \longrightarrow \mathbb{V}_m, \tag{B1.17}$$

and the defect matrices obey

$$(D\Phi)'_i = R^m_i (D\Phi)_i (R^b_i)^{-1}, \qquad (D\widetilde{\Phi})'_i = R^b_i (D\widetilde{\Phi})_i (R^m_i)^{-1}. \tag{B1.18}$$

[ESTABLISHED]

*Proof.* (i) The two terms of each defect project to the same $X$. (ii) Vanishing defect says that the image of every horizontal lift is horizontal; integration along $\gamma$ gives the two commutative squares, and differentiation at the initial point gives the converse. (iii) In the linear specialization $T\Phi$ of a horizontal lift minus the horizontal lift of the image is the covariant derivative of the Hom-bundle section; the coordinate maps are $R = \rho_x(g)$ by $\beta' = \widehat{\rho}_x(g)\beta$; and with $\nabla^{x\prime} = R^x \nabla^x (R^x)^{-1}$ and $\Phi' = R^m \Phi (R^b)^{-1}$ one gets $\nabla^{m\prime}\Phi' - \Phi'\nabla^{b\prime} = R^m(\nabla^m\Phi - \Phi\nabla^b)(R^b)^{-1}$. □

The coordinate maps $R^x_i$ are invertible re-expressions within one channel, not cross-channel morphisms or coarse restriction operators.

## B1.4 The Čech picture and bundle topology

Let $\mathcal{C}_0 = \bigcup_i \mathcal{C}_i$, and let $\mathcal{N}$ be the nerve of this open cover,

$$\{i_0, \dots, i_r\} \in \mathcal{N} \quad \Longleftrightarrow \quad \mathcal{C}_{i_0 \cdots i_r} = \bigcap_{k=0}^{r} \mathcal{C}_{i_k} \neq \varnothing. \tag{B1.19}$$

Write $\underline{G}$ for the sheaf of smooth $G$-valued maps.

**Proposition B1.10 (Transition cocycles and the principal class).** (i) For either atlas $x \in \{b, m\}$, the comparisons $T^x_{ij} : \mathcal{C}_{ij} \to G$ obey

$$T^x_{ii} = e_G, \qquad T^x_{ji} = (T^x_{ij})^{-1}, \qquad T^x_{ij} T^x_{jl} = T^x_{il}. \tag{B1.20}$$

(ii) The two cocycles are cohomologous and define one nonabelian Čech class,

$$[T^b] = [T^m] = [\mathscr{P}_G|_{\mathcal{C}_0}] \in \check{H}^1(\{\mathcal{C}_i\}; \underline{G}). \tag{B1.21}$$

(iii) The class is distinguished exactly when $\mathscr{P}_G|_{\mathcal{C}_0}$ is trivial, equivalently when, for either atlas, there are smooth $U^x_i : \mathcal{C}_i \to G$ with

$$T^x_{ij} = U^x_i (U^x_j)^{-1}. \tag{B1.22}$$

Over a single context every principal bundle is trivial and every class is distinguished. (iv) The single principal class does not make the associated bundles isomorphic: for the nontrivial principal $\{\pm 1\}$-bundle over the circle, the trivial and the sign representations on $\mathbb{R}$ give the trivial line bundle and the Möbius line bundle, which are not isomorphic. (v) A global section of an associated bundle need not trivialize $\mathscr{P}_G$: if a fiber contains a $G$-fixed point $\beta_0$, then $c \mapsto [p, \beta_0]$ is a global section for every $\mathscr{P}_G$. [ESTABLISHED]

*Proof.* (i) On a triple overlap, composing the unique change from $l$ to $j$ with the unique change from $j$ to $i$ gives the unique change from $l$ to $i$. (ii) Relation (2.15) is a coboundary equivalence of the two cocycles. (iii) If (B1.22) holds, the rechoice with $g^x_i = (U^x_i)^{-1}$ in (B1.1) makes every transition the identity, so the sections glue to a global section, which trivializes a principal bundle; conversely a global trivialization supplies the $U^x_i$. This is the global-section characterization of trivial principal bundles [12, 17, 29]. A single context is a point, over which every bundle is trivial. (iv) A nowhere-vanishing continuous section of the Möbius bundle would be a continuous $\sigma : [0, 1] \to \mathbb{R} \setminus \{0\}$ with $\sigma(1) = -\sigma(0)$, which the intermediate value theorem excludes, while the trivial line bundle has the section $1$. (v) $[p \cdot g, \beta_0] = [p, g \cdot \beta_0] = [p, \beta_0]$, so the value does not depend on the representative. □

**Hypothesis B1.11 (Common principal trivialization).** Whenever global principal-frame coordinates are used, $\mathscr{P}_G|_{\mathcal{C}_0}$ is trivial and one reference section $\sigma_0$ is chosen, so that

$$u^b_i = \sigma_0 \cdot (U^b_i)^{-1}, \qquad u^m_i = \sigma_0 \cdot (U^m_i)^{-1}, \qquad h_i = U^b_i (U^m_i)^{-1}. \tag{B1.23}$$

The hypothesis excludes nontrivial restrictions $\mathscr{P}_G|_{\mathcal{C}_0}$. It is one topological hypothesis, not two channelwise ones. At $c_*$ the main manuscript needs no such hypothesis, because a reference point of the single fiber supplies the potentials (Definition 2.2).

### B1.4.1 Independent channel frames in the Gaussian realization

Supplement G realizes the directed law with Gaussian kernels in the common frame. With independent atlases, invoke Hypothesis B1.11 on a declared region $\mathcal{C}_0$ containing the contexts under study, restrict each frame to $\mathcal{C}_i \cap \mathcal{C}_0$, and write the independent frame coordinates as

$$U_i^b, U_i^m : \mathcal{C}_i \cap \mathcal{C}_0 \longrightarrow G, \qquad u_i^x = \sigma_0 \cdot (U_i^x)^{-1}, \tag{B1.24}$$

so that $\sigma_0$ is common only on the trivialized region and the relative frame is

$$h_i = U_i^b (U_i^m)^{-1}, \qquad u_i^m = u_i^b h_i. \tag{B1.25}$$

Its represented actions $\rho_b(h_i)$ and $\rho_m(h_i)$ live in different endomorphism spaces and define neither the bridge matrix of the Gaussian model nor either cross morphism. An independent pair $(g_i^b, g_i^m) \in G \times G$ has represented matrices $R_i^b = \rho_b(g_i^b)$ and $R_i^m = \rho_m(g_i^m)$; the pair records two trivializations of bundles associated to one $\mathscr{P}_G$ and does not replace the structure group by $G \times G$.

**Remark B1.12 (Gaussian kernel covariance under independent frames).** The transformation laws of the Gaussian directed parameters in Supplement G, eqs. (G2.40) to (G2.46), hold verbatim with $R_i^b = \rho_b(g_i^b)$ and $R_i^m = \rho_m(g_i^m)$ for independent $g_i^b, g_i^m$ and links transforming by $\Theta_e^{x\prime} = g_i^x \Theta_e^x (g_j^x)^{-1}$, and the kernel covariance of Proposition G2.17 holds under every independent channel-frame re-expression. [ESTABLISHED]

*Proof.* Each law and each step of the proof of Proposition G2.17 uses the belief matrix only on belief coordinates and the model matrix only on model coordinates: with $F' = R_i^b F (R_i^b)^{-1}$, $\rho_b(\Theta_e^{b\prime}) = R_i^b \rho_b(\Theta_e^b)(R_j^b)^{-1}$, $B' = R_i^b B (R_i^m)^{-1}$, and $\eta' = R_i^b \eta$, the transformed belief mean at transformed inputs is $R_i^b$ times the original mean, and the covariance congruence makes each transformed kernel the receiving-frame pushforward. No step uses a relation between $R_i^b$ and $R_i^m$. □

## B1.5 Graph links and base transport

The links of Definition 2.11 are declared independently of $\mathscr{P}_G$ and of any connection. Relating them to base transport is a separate declaration.

**Hypothesis B1.13 (Graph links realized by base transport).** Contexts $c_i \in \mathcal{C}_i$ are chosen, and for every edge copy $e : j \to i$ a piecewise-smooth curve $\gamma_e$ from $c_j$ to $c_i$ is assigned, with $\gamma_{\bar{e}}$ the

reverse of $\gamma_e$. In the chosen endpoint frames,

$$\widehat{\rho}_b(\Theta_e^b) = \Omega_{\gamma_e}, \qquad \widehat{\rho}_m(\Theta_e^m) = \widetilde{\Omega}_{\gamma_e}. \tag{B1.26}$$

The hypothesis excludes links that are not represented transports of the chosen connections along the assigned curves.

**Proposition B1.14 (Loop products under base-transport realization).** Under Hypothesis B1.13, for every closed edge-labeled walk $p = (e_0, \dots, e_{r-1})$ with $e_k : i_{k+1} \to i_k$ and $i_r = i_0$,

$$\widehat{\rho}_b\big(\mathrm{Hol}^b(p)\big) = \Omega_{\gamma_p}, \qquad \widehat{\rho}_m\big(\mathrm{Hol}^m(p)\big) = \widetilde{\Omega}_{\gamma_p},$$

where $\gamma_p$ is the base loop at $c_{i_0}$ that traverses $\gamma_{e_{r-1}}, \gamma_{e_{r-2}}, \dots, \gamma_{e_0}$ in this order. [ESTABLISHED]

*Proof.* The representation is a homomorphism, so $\widehat{\rho}_x(\Theta^x_{e_0} \cdots \Theta^x_{e_{r-1}})$ is the composite of the represented links with $\widehat{\rho}_x(\Theta^x_{e_{r-1}})$ applied first; consecutive factors are expressed in the same frame at their common endpoint, and parallel transport along a concatenation is the composite of the transports. □

Without the curve assignment and the equalities (B1.26), graph holonomy and base-connection holonomy are unrelated, and even an exact assignment concerns connection holonomy, not topological nontriviality of $\mathscr{P}_G$. The hypothesis is satisfiable by declaration, by setting each link equal to a chosen base transport, and no result of this supplement realizes it for independently specified links; the cover construction of Chapter B4 declines that identification, and the operational record of a declared base holonomy is Theorem B4.13.

Whether independently specified links $\Theta_e^b, \Theta_e^m$ are realized by principal connections and base curves satisfying (B1.26), and whether a connection can be recovered from the laws of a declared class of population records, are Open Problems B4.14 and B4.15; what holds for declared realizations is Proposition B1.14.

The single-point theory cannot separate the following conditions, because over one context every bundle is trivial and every connection vanishes. Over a base they separate completely.

**Proposition B1.15 (Four independent notions of flatness).** Consider (a) triviality of $\mathscr{P}_G$ over a region, equivalently a distinguished class in Proposition B1.10; (b) graph flatness of the links of one channel in the sense of Proposition 2.13; (c) flatness of a principal connection, meaning vanishing curvature; and (d) triviality of the holonomy of that connection around one specified base loop. Then (a) implies neither (c) nor (d), (c) implies neither (a) nor (d), and (d) does not imply (c); and (b) is implied by none of (a), (c), (d) and implies none of them. Explicitly, for the trivial $\mathrm{U}(1)$-bundle with connection form $\mathrm{i}\alpha$ for a real one-form $\alpha$, whose holonomy around a loop $\gamma$ is $e^{\pm \mathrm{i} \oint_\gamma \alpha}$ with the sign fixed by the orientation convention: $\alpha = x\mathrm{d}y$ on $\mathbb{R}^2$ is not flat and has trivial holonomy around the circle of radius $\sqrt{2}$; $\alpha = \lambda(x\mathrm{d}y - y\mathrm{d}x)/(x^2+y^2)$ on $\mathbb{R}^2 \setminus \{0\}$ with $\lambda \notin \mathbb{Z}$ is flat with nontrivial holonomy around the unit circle; and the principal $S_3$-bundle of Proposition B1.3 is nontrivial while its unique connection is flat, with holonomy $\mathring{g}$ around the base circle. [ESTABLISHED]

*Proof.* For $\alpha = x\mathrm{d}y$, $\mathrm{d}\alpha = \mathrm{d}x \wedge \mathrm{d}y \neq 0$, while $\oint x\mathrm{d}y$ over the positively oriented circle of radius $\sqrt{2}$ is its enclosed area $2\pi$, so the holonomy is trivial; this proves that (d) does not imply (c) and, since the bundle is trivial, that (a) does not imply (c). For $\alpha = \lambda(x\mathrm{d}y - y\mathrm{d}x)/(x^2+y^2)$, $\mathrm{d}\alpha = 0$ on the punctured plane and $\oint \alpha = 2\pi\lambda \notin 2\pi\mathbb{Z}$ around the unit circle, so the connection is flat on a trivial bundle with nontrivial holonomy; hence (a) does not imply (d) and (c) does not imply (d). For the $S_3$-bundle, a continuous section of the strip is a constant $(t, s)$ because $S_3$ is discrete, and the gluing requires $\mathring{g}s = s$, which fails, so the bundle is nontrivial; the Lie algebra of $S_3$ is zero, so its only connection has horizontal space equal to the whole tangent space and zero curvature, and its holonomy around the circle is $\mathring{g} \neq e$. Hence (c) does not imply (a). Finally the links are declared independently of the bundle and the connection: the all-identity links are graph flat over every bundle and connection, and links on a triangle whose loop product differs from $e_G$ are not graph flat over the trivial bundle with the zero connection, which satisfies (a), (c), and (d). □

## B1.6 Optional product-gauge extension

**Definition B1.16 (Product-gauge extension).** An application that requires independently noniso-morphic principal bundles or two independent gauge symmetries replaces $\mathscr{P}_G$ by

$$\pi_b : P_b \to \mathcal{C}, \qquad \pi_m : P_m \to \mathcal{C}, \qquad Q = P_b \times_{\mathcal{C}} P_m, \tag{B1.27}$$

where $Q$ is a principal $G_b \times G_m$-bundle and each associated bundle is formed with its own factor. In a category admitting the mapping object, a belief-to-model morphism is a section of

$$Q \times_{G_b \times G_m} \mathrm{Map}(\mathcal{B}_b, \mathcal{B}_m), \qquad (g_b, g_m) \cdot \phi = \widehat{\rho}_m(g_m) \circ \phi \circ \widehat{\rho}_b(g_b)^{-1}. \tag{B1.28}$$

At every level $\ell$ of a scale diagram the same construction uses $P_{\ell,b}, P_{\ell,m} \to \mathcal{C}_\ell$ and $Q_\ell = P_{\ell,b} \times_{\mathcal{C}_\ell} P_{\ell,m}$.

**Proposition B1.17 (Comparison data for the product extension).** (i) If $G_b = G_m = G$ and $B : P_b \to P_m$ is a principal $G$-bundle isomorphism over $\mathrm{id}_{\mathcal{C}}$, local sections $u_i^b$ of $P_b$ and $u_i^m$ of $P_m$ determine a unique relative frame $h_i$ with $u_i^m = B(u_i^b) \cdot h_i$. (ii) If $\kappa : G_b \to G_m$ is a homomorphism and $B : P_b \to P_m$ a bundle map over $\mathrm{id}_{\mathcal{C}}$ with $B(p \cdot g_b) = B(p) \cdot \kappa(g_b)$, then every fiber map $\phi : \mathcal{B}_b \to \mathcal{B}_m$ with $\phi \circ \widehat{\rho}_b(g_b) = \widehat{\rho}_m(\kappa(g_b)) \circ \phi$ induces the cross morphism $[p, \beta] \mapsto [B(p), \phi(\beta)]$. (iii) A constant fiber map $\phi$ defines a section of (B1.28) in any chosen trivialization of $Q$. This construction is independent of the choice of trivialization exactly when $(g_b, g_m) \cdot \phi = \phi$ for every $(g_b, g_m) \in G_b \times G_m$. [ESTABLISHED]

*Proof.* (i) $B(u_i^b)$ and $u_i^m$ are local sections of $P_m$, so the torsor argument of Proposition B1.1 applies. (ii) The representatives $(p \cdot g_b, \beta)$ and $(p, \widehat{\rho}_b(g_b)\beta)$ have images $[B(p)\kappa(g_b), \phi\beta]$ and $[B(p), \phi\widehat{\rho}_b(g_b)\beta] = [B(p), \widehat{\rho}_m(\kappa(g_b))\phi\beta]$, which agree. (iii) The class $[q, \phi]$ is independent of the chosen principal frame over its base point exactly when $[q \cdot (g_b, g_m), \phi] = [q, (g_b, g_m) \cdot \phi]$ equals $[q, \phi]$. □

Without such comparison data the product bundle supplies neither a relative frame valued in one group nor either cross morphism, and the invariance required in (iii) under independently varying

$(g_b, g_m)$ is restrictive. The extension carries additional topology and an independent product gauge freedom, so it is invoked only when those degrees of freedom are intended, not because belief and model use different representations of one group.

## B1.7 Finite designs, context-indexed sections, and sampling compatibility

**Definition B1.18 (Finite design and incidence).** A *design* is a finite set of distinct contexts

$$D = \{c_d\}_{d=1}^{M} \subseteq \mathcal{C}, \tag{B1.29}$$

indexed by $d$. The agent–design incidence set is $I_D = \{(i,d) : c_d \in \mathcal{C}_i\}$, with $D_i = \{d : (i,d) \in I_D\}$ and $V_d = \{i : (i,d) \in I_D\}$; design points with no participating agent are omitted, so $V_d \neq \varnothing$. At each $(i,d) \in I_D$ measurable state and model fibers $(\mathsf{K}_{i,d}, \mathscr{K}_{i,d})$ and $(\mathsf{M}_{i,d}, \mathscr{M}_{i,d})$ are declared, the latent space is $\mathsf{Y}_D = \prod_{(i,d)\in I_D}(\mathsf{K}_{i,d} \times \mathsf{M}_{i,d})$ with its product sigma-algebra, and $\mathsf{O}_D$ is the observation space of the records declared on the design. When $D \subseteq \bigcap_{i\in V} \mathcal{C}_i$ the incidence is rectangular, $I_D = V \times \{1, \dots, M\}$. The construction of the main manuscript is the case $D = \{c_*\}$.

A sparse design is handled by indexing over declared incidences, never by writing a product over all $(i,d)$ and deleting undefined coordinates, which would silently change the sample space. Missing or aggregated observations are likewise represented by a declared observation map or kernel, which changes $\mathsf{O}_D$ and not $\mathsf{Y}_D$. The recognition marginals at an incidence, $q_{i,d}^{b;o,X}$ and $q_{i,d}^{m;o,X}$, are the marginals of $\mathbb{Q}_{V,o,X}$ on $\mathsf{K}_{i,d}$ and $\mathsf{M}_{i,d}$, as in Definition 3.13.

**Hypothesis B1.19 (Measurable context-indexed section response).** For every agent $i$, and for the structural space $\mathsf{X}$ of Definition 3.2, measurable spaces $(\mathfrak{S}_i^b, \mathscr{S}_i^b)$ and $(\mathfrak{S}_i^m, \mathscr{S}_i^m)$ are declared whose underlying sets satisfy $\mathfrak{S}_i^x \subseteq \Gamma(\mathcal{C}_i, \mathcal{E}_x|_{\mathcal{C}_i})$, such that each evaluation map $\mathrm{eval}_{i,c}^x : \mathfrak{S}_i^x \to (\mathcal{E}_x)_c$, $s \mapsto s(c)$, is measurable, together with a measurable response map

$$\mathrm{Resp}_i : (\mathsf{O}_D \times \mathsf{X}, \mathscr{O}_D \otimes \mathscr{X}) \longrightarrow (\mathfrak{S}_i^b \times \mathfrak{S}_i^m, \mathscr{S}_i^b \otimes \mathscr{S}_i^m), \qquad (o, X) \longmapsto (q_i^{b;o,X}, q_i^{m;o,X}). \tag{B1.30}$$

At every $(i,d) \in I_D$ the fixed fiber identifications send the evaluated values into $\mathcal{P}(\mathsf{K}_{i,d}, \mathscr{K}_{i,d})$ and $\mathcal{P}(\mathsf{M}_{i,d}, \mathscr{M}_{i,d})$, so design-point evaluations are measurable functions of $(o, X)$. For fixed $(o, X)$ these sections occupy the section slots of (2.20). The hypothesis excludes responses whose section values are not measurable in the observation and the structural data.

Evaluation and section evaluation have different codomains. For a sample-field realization $y = ((k_i, m_i))_{i\in V}$ on the admitted incidences, the evaluation $\big((k_i(c_d), m_i(c_d))_{i\in V_d}\big)_{d=1}^{M}$ is a point of $\mathsf{Y}_D$, whereas $q_i^{b;o,X}(c_d)$ is a probability measure on $\mathsf{K}_{i,d}$. The continuum sections and the finite recognition law are separately declared objects, related only by the hypothesis below.

### B1.7.1 Continuum sections versus finite evaluation

Each state or model-coordinate law at $(i, d)$ can be obtained in two ways: by sampling the section at $c_d$,

$$\mathrm{Samp}_{D,i}\big(q_i^{x;o,X}\big) = \big(q_i^{x;o,X}(c_d)\big)_{d\in D_i}, \qquad x \in \{b, m\}, \tag{B1.31}$$

or by marginalizing the finite law. The sampling map acts on the response spaces $\mathfrak{S}_i^x$, the marginal maps act on $\mathcal{P}(\mathsf{Y}_D, \mathscr{Y}_D)$, and neither is defined through the other.

**Proposition B1.20 (Compatibility determines neither side).** Suppose the compatibility relation

$$q_i^{b;o,X}(c_d) = q_{i,d}^{b;o,X}, \qquad q_i^{m;o,X}(c_d) = q_{i,d}^{m;o,X} \qquad \text{for every } (i,d) \in I_D \tag{B1.32}$$

holds. (i) When at least two local agent-block laws are non-Dirac, (B1.32) does not determine $\mathbb{Q}_{V,o,X}$. (ii) If some support satisfies $\mathcal{C}_i \not\subseteq D$ and the selected fiber of channel $x$ carries a global chart onto $\mathbb{R}^n$ with $n \geq 1$, and the admitted section class is closed under the smooth local chart perturbations supported off $D$ used below, then (B1.32) does not determine the sections $q_i^{x;o,X}$. This closure holds for the unrestricted smooth section class; it need not hold for a declared response subfamily. (iii) If $\mathcal{C}_i \subseteq D$ for every $i$, which forces every nonempty support to be zero dimensional, the compatible values determine the sections pointwise. [ESTABLISHED]

*Proof.* (i) By Proposition 3.17 there are distinct laws on $\mathsf{Y}_D$ with the same coordinate marginals, and (B1.32) constrains the joint only through those marginals. (ii) Fix $c^\star \in \mathcal{C}_i \setminus D$, a neighborhood $W$ of $c^\star$ disjoint from the finite set $D$ over which $\mathcal{E}_x$ is trivialized, the chart $\psi$ of the fiber, and a smooth $\chi : \mathcal{C}_i \to [0, 1]$ supported in $W$ with $\chi(c^\star) = 1$; when $c^\star$ is isolated, $W = \{c^\star\}$. For $v \neq 0$ replace the local coordinate $\beta(c)$ of the section by $\psi^{-1}(\psi(\beta(c)) + \chi(c)v)$ on $W$. The result belongs to the admitted class by the local-perturbation hypothesis, differs at $c^\star$, and agrees on $D$, so it satisfies (B1.32) against the same marginals. (iii) Equality at every point of the support is equality of sections. A nonempty open set contained in a finite set consists of isolated points. □

**Hypothesis B1.21 (Sampling and marginalization are compatible).** Where a result transfers a statement between the continuum sections and the finite recognition law, relation (B1.32) is adopted. The hypothesis excludes configurations in which an agent's section and the finite law disagree at a design point, which is possible because the two objects are declared independently. A result stated entirely about the finite construction does not use it. Under the respective hypotheses of Proposition B1.20, adopting it reconstructs neither the joint from its marginals nor a section from finitely many values; a restricted response family can be identifiable from finitely many evaluations.

## B1.8 What the finite construction does not supply

**Open Problem B1.22 (Continuum law theory).** The finite construction supplies probability spaces and kernels for a declared design $D$. It supplies no probability measure on a space of sections over $\mathcal{C}$, no unique extension of the finite marginals to such a space, and no continuum limit,

and finite-dimensional consistency alone does not produce them. Construct a refining family of designs; a projective system of normalized finite laws indexed by them, so that a Kolmogorov-type extension theorem applies [16]; a declared topology and sigma-algebra on a space of sections, with tightness that places the extension measure on that space; convergence of the evidence bound, the connections, and the observables that the finite theory computes; control of the gauge choices and of the reference measures under refinement, neither of which is canonical; and a normalizable continuum interaction energy.

The last item is where the obstruction is concrete. The reference measures of Section 3.3 are finite products, and their sigma-finiteness is inherited from finitely many sigma-finite factors. The same construction does not extend to infinitely many non-probability factors, so a continuum theory must declare a measure on the section space directly. Until that is done a continuum energy has no density to be the exponent of, and every evidence-bound statement of the main manuscript and of this supplement is a statement about a finite design.

## B1.9 The multi-point generative law

On a design the structural data contain a finite directed acyclic graph $\Gamma = (V, E)$ whose incidence is parent closed: if $(i, d) \in I_D$ and $j \in \mathrm{pa}(i)$, then $(j, d) \in I_D$. The induced graph $\Gamma_d$ on $V_d$ therefore contains every parent of each participating vertex and is acyclic. At each design point the one-point latent law $\mathbb{P}^Y_{\theta,V,d}(dy_d \mid X)$ is the directed law of the main manuscript on the agents $V_d$, composed from the incidence mechanisms $G_{\theta,i,d}$ of (4.4) (Definition 4.2).

**Hypothesis B1.23 (Design-point product).** Distinct design points are conditionally independent given $X$:

$$\mathbb{P}^Y_{\theta,V}(dy \mid X) = \bigotimes_{d=1}^{M} \mathbb{P}^Y_{\theta,V,d}(dy_d \mid X). \tag{B1.33}$$

Equivalently, each agent contributes the product mechanism

$$G_{\theta,i,D} := \bigotimes_{d \in D_i} G_{\theta,i,d}. \tag{B1.34}$$

The hypothesis excludes residual dependence across design points.

When the component kernels have densities against the declared component reference measures, the latent density is the product over design points of the one-point densities of (4.8), $p^Y_{\theta,V}(y \mid X) = \prod_{d=1}^{M} \prod_{i \in V_d} g_{\theta,i,d}$.

**Corollary B1.24 (Factorwise decomposition over a design).** Assume Hypotheses B1.23 and 5.3, and assume the factorwise log-integrability of proposition 5.15 for every design-point factor and every record factor. Write $E_{a,o}$ for the negative log density of the record kernel of $a \in \mathfrak{A}$ and $\partial a$

for its scope. Then

$$\begin{aligned}\mathcal{F}_V[\mathbb{Q}_{V,o,X}; X, o] = &\sum_{d=1}^{M}\sum_{i\in V_d}\mathbb{E}_{\mathbb{Q}_{V,o,X}}\big[-\log p^{m}_{\theta,i,d}\big(m_{i,d} \mid m_{\mathrm{pa}(i),d}, X\big)\big]\\ &+\sum_{d=1}^{M}\sum_{i\in V_d}\mathbb{E}_{\mathbb{Q}_{V,o,X}}\big[-\log p^{k}_{\theta,i,d}\big(k_{i,d} \mid k_{\mathrm{pa}(i),d}, m_{i,d}, X\big)\big]\\ &+\sum_{a\in\mathfrak{A}}\mathbb{E}_{\mathbb{Q}_{V,o,X}}\big[E_{a,o}(Y_{\partial a})\big]+\mathbb{E}_{\mathbb{Q}_{V,o,X}}\big[\log q_{V,o,X}(Y)\big],\end{aligned}$$

where at a root the parent tuples are empty and the state factor is the model-to-belief bridge. The generative sectors contain $2|I_D|+|\mathfrak{A}|$ factors, which equals $2MN+|\mathfrak{A}|$ on a rectangular design, and the last term is one joint recognition log density, not a sum of marginal entropies. [ESTABLISHED]

*Proof.* Under Hypothesis B1.23 the complete log density is the sum over design points of the one-point latent log densities plus the record log densities, and each one-point term is the sum of its model and state factors. The integrability assumption allows each expectation to be taken separately, so the argument of Proposition 5.15 applies term by term, with the one joint recognition term appearing once on each side. □

The design-point product carries one copy of each link per design point. For an edge copy $e : j \to i$ put $D_e := D_i \cap D_j$; the structural data carry copies $\Theta^x_{d,e} \in G$ exactly for $d \in D_e$. Independent frame re-expressions at the incidences, $g^x_{d,i} := g^x_i(c_d)$, act on each copy by the vertex law of the main manuscript, $\Theta^{x\prime}_{d,e} = g^x_{d,i}\Theta^x_{d,e}(g^x_{d,j})^{-1}$ (Equation (4.21)). At one base point a constraint tying copies to one link is preserved automatically, because all copies transform by the same pair (Equation (4.22)); across design points the copies of one edge transform by different pairs, and preservation becomes a condition.

**Definition B1.25 (Shared-link admissibility and the setwise stabilizer).** A submodel may constrain the link of a constrained edge copy $e$ to be one value $\Theta^x_e$ at every point of $D_e$. At a fixed shared-link datum, a rechoice is admissible when for every constrained $e$ and all $d, d' \in D_e$

$$g^x_{d,i}\Theta^x_e(g^x_{d,j})^{-1} = g^x_{d',i}\Theta^x_e(g^x_{d',j})^{-1}. \tag{B1.35}$$

The admissible family of the datum is the solution set of (B1.35). The setwise stabilizer is the set of rechoices that map the constraint set, the set of all link assignments whose constrained links are independent of $d \in D_e$, into itself.

The admissible family depends on the datum, so admissible rechoices compose as arrows of the action groupoid of rechoices on link data: $g$ may follow $\widetilde{g}$ when $g$ is admissible at the transformed datum $\widetilde{g} \cdot \Theta$, not at $\Theta$. The group that acts on the whole constrained submodel is the setwise stabilizer, and a larger transformation law requires an explicitly declared parameter compensation.

**Proposition B1.26 (Setwise stabilizer of shared links).** Fix a channel $x$, let $Z(G)$ be the center of $G$, and suppose every constrained link ranges over all of $G$. A rechoice $(g^x_{d,i})$ lies in the setwise

stabilizer exactly when, for every constrained edge copy $e : j \to i$ and all $d, d' \in D_e$,

$$(g^x_{d',i})^{-1} g^x_{d,i} = (g^x_{d',j})^{-1} g^x_{d,j} \in Z(G). \tag{B1.36}$$

Equivalently, on the design points of each constrained edge the rechoice is a context-independent pair $(\bar{g}^x_i, \bar{g}^x_j)$ multiplied by a context-dependent central twist common to both endpoints, $g^x_{d,i} = \bar{g}^x_i z_d$ and $g^x_{d,j} = \bar{g}^x_j z_d$ with $z_d \in Z(G)$. The context-independent rechoices are the subgroup with trivial twists, and when $Z(G)$ is trivial the stabilizer consists exactly of the rechoices that are constant over each $D_e$ of a constrained edge. [ESTABLISHED]

*Proof.* The rechoice maps the constraint set into itself exactly when $g^x_{d,i}\Theta(g^x_{d,j})^{-1} = g^x_{d',i}\Theta(g^x_{d',j})^{-1}$ for every constrained edge, all $d, d' \in D_e$, and every $\Theta \in G$, that is, when $\lambda_i\Theta = \Theta\lambda_j$ for all $\Theta$, with $\lambda_i = (g^x_{d',i})^{-1} g^x_{d,i}$ and $\lambda_j = (g^x_{d',j})^{-1} g^x_{d,j}$. Taking $\Theta = e_G$ gives $\lambda_i = \lambda_j$, and then $\lambda_i$ commutes with every element of $G$. Conversely $\lambda_i = \lambda_j \in Z(G)$ gives $\lambda_i\Theta = \Theta\lambda_j$. Fixing $d' \in D_e$ and setting $\bar{g}^x_v = g^x_{d',v}$ and $z_d = \lambda_i$ gives the product form. The set is a group, since for two rechoices with twists $z_d$ and $\widetilde{z}_d$ the product has twist $z_d\widetilde{z}_d$ at both endpoints because central elements commute past $\widetilde{g}^x_{d',v}$, and the inverse has twist $z_d^{-1}$; so mapping the constraint set into itself already gives setwise invariance. □

**Proposition B1.27 (Shared-link admissibility need not be closed under composition).** For $G = \mathrm{GL}(2, \mathbb{R})$, two design points at which both endpoints participate, one constrained edge, and $\Theta_e = I$, the solution set of (B1.35) is not closed under pointwise multiplication. [ESTABLISHED]

*Proof.* Let $S = \mathrm{diag}(2, 1)$ and $C = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. The rechoice $g_{1,i} = g_{1,j} = I$, $g_{2,i} = g_{2,j} = S$ sends the shared link to $I$ at both design points, and $\widetilde{g}_{d,i} = C$, $\widetilde{g}_{d,j} = I$ sends it to $C$ at both. Their pointwise product sends it to $C$ and to $SCS^{-1} = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$. □

The nonclosure is the composition law of the action groupoid: the product acts as $g \cdot (\widetilde{g} \cdot \Theta)$, and $g$ is admissible at $I$ but not at $C$. The first rechoice is not in the setwise stabilizer because $S$ is not central. In the Gaussian realization of Supplement G the links enter the kernels only through their represented copies, so the same stabilizer acts on a Gaussian submodel with shared links.

**Definition B1.28 (Passive coordinate product).** At the unconstrained finite-design level, independently declared coordinate re-expressions form the passive product $\prod_{(i,d)\in I_D}(G \times G)$ of pairs $(g^b_{d,i}, g^m_{d,i})$. It records two independently chosen trivializations of bundles associated to one $\mathscr{P}_G$ and is not a second principal structure group. A context-independent re-expression is an element for which $g^x_{d,i}$ is constant as $d$ ranges over $D_i$.

**Proposition B1.29 (Images in the passive product).** (i) If the re-expressions must arise as values $g^x_{d,i} = g^x_i(c_d)$ of smooth maps $g^x_i : \mathcal{C}_i \to G$, the realized group is the image of this evaluation, and it is the whole passive product when $G$ is connected. (ii) The local functions at the design points of the gauge automorphisms of $\mathscr{P}_G$ form a subgroup of $\prod_{d=1}^{M} \mathcal{A}_{c_d}$, with $\mathcal{A}_{c_d}$ as in Proposition B1.2, equal to it when $G$ is connected and possibly strictly smaller when $G$ is disconnected. (iii) For nontrivial $G$ this active image is a proper subgroup of the passive product, and a proper subgroup

of the product of pointwise $h_i$-twisted diagonals as soon as one design point carries two agents. [ESTABLISHED]

*Proof.* (i) Given values at the distinct points $c_d$, the bump-function construction in the proof of Proposition B1.3, applied to $G$-valued maps in disjoint coordinate neighborhoods, interpolates them, and at isolated points any values are smooth. (ii) The containment is Proposition B1.2 at each design point; equality for connected $G$ is Proposition B1.3, and the $S_3$-bundle there with its base point in $D$ gives strictness. (iii) Each factor $\mathcal{A}_{c_d} \cong G$ is proper in $\prod_{i \in V_d}(G \times G)$, and in the product of twisted diagonals when $|V_d| \geq 2$, by Proposition B1.2. □

**Open Problem B1.30 (Regular frame-coordinate quotient).** Determine the generic stabilizers of the passive $G \times G$ action generated by independent belief and model trivializations on the full parameter space, and identify strata on which its orbit space is a regular quotient manifold. The coordinate action is not the principal structure group, and it can have nontrivial stabilizers, so subtracting the dimension of the coordinate group from a parameter count is unjustified without freeness and properness. Nothing in the main manuscript depends on this quotient.

## B1.10 Scale maps over base maps

The scale maps of the main manuscript at $c_*$ are torsor maps fixed by the image of one point (Equation (11.2)). Over a base map their existence is a topological condition. Let $\pi_\ell : \mathscr{P}_\ell \to \mathcal{C}_\ell$ be principal $G_\ell$-bundles over smooth level bases, $f_\ell : \mathcal{C}_\ell \to \mathcal{C}_{\ell+1}$ smooth, and $\kappa_\ell : G_\ell \to G_{\ell+1}$ a Lie-group homomorphism.

**Proposition B1.31 (Existence of a principal scale map over a base map).** A map $\mathcal{P}_\ell : \mathscr{P}_\ell \to \mathscr{P}_{\ell+1}$ covering $f_\ell$ with $\mathcal{P}_\ell(p \cdot g) = \mathcal{P}_\ell(p) \cdot \kappa_\ell(g)$ exists if and only if the extended bundle $\mathscr{P}_\ell \times_{\kappa_\ell} G_{\ell+1}$ is isomorphic to the pullback $f_\ell^* \mathscr{P}_{\ell+1}$ as a principal $G_{\ell+1}$-bundle over $\mathcal{C}_\ell$. The condition can fail: with the Hopf $\mathrm{U}(1)$-bundle over $S^2$ as source, the trivial bundle as target, $f_\ell = \mathrm{id}$, and $\kappa_\ell(z) = z^n$ with $n \neq 0$, no such map exists, while for $n = 0$ the map $p \mapsto (\pi_\ell(p), 1)$ exists for every source bundle. [ESTABLISHED]

*Proof.* Given $\mathcal{P}_\ell$, the map $[p, g'] \mapsto (\pi_\ell(p), \mathcal{P}_\ell(p)g')$ is well defined, since $[p \cdot g, \kappa_\ell(g)^{-1}g']$ has the same image, and it is a $G_{\ell+1}$-equivariant map of principal bundles over the identity, hence an isomorphism. Conversely, an isomorphism $\Xi$ gives $\mathcal{P}_\ell(p) = \mathrm{pr}_2\, \Xi([p, e])$, and $[p \cdot g, e] = [p, e] \cdot \kappa_\ell(g)$ gives the equivariance. For the Hopf bundle, the extension by $z \mapsto z^n$ is the frame bundle of the $n$-th tensor power of the associated line bundle, whose first Chern number is $n$ times that of the Hopf bundle, which is nonzero; the trivial bundle has Chern number zero. For $n = 0$ the extension is trivialized by $[p, g'] \mapsto (\pi_\ell(p), g')$. □

Given $\mathcal{P}_\ell$, a fiber map $q_{\ell,x}$ descends to a bundle map $C_{\ell,x} : \mathcal{E}_{\ell,x} \to \mathcal{E}_{\ell+1,x}$ covering $f_\ell$ exactly under the intertwining condition (11.4), by the representative computation that proves (11.3), which is pointwise in the base. For a multi-step scale functor the base maps compose as well, $f_{\ell+1} \circ f_\ell$, and the composition laws of the main manuscript hold with them.

**Proposition B1.32 (Relative frames across scales).** Let $h_\ell$ and $h_{\ell+1}$ be the relative frames (B1.2) of an agent at levels $\ell$ and $\ell+1$, and let comparison functions $\varsigma_\ell^x$ satisfy $\mathcal{P}_\ell(u_\ell^x(c)) = u_{\ell+1}^x(f_\ell c)\cdot\varsigma_\ell^x(c)$. Then

$$\varsigma_\ell^b \kappa_\ell(h_\ell) = (h_{\ell+1}\circ f_\ell)\varsigma_\ell^m. \tag{B1.37}$$

[ESTABLISHED]

*Proof.* $\mathcal{P}_\ell(u_\ell^m) = \mathcal{P}_\ell(u_\ell^b h_\ell) = u_{\ell+1}^b \varsigma_\ell^b \kappa_\ell(h_\ell) = u_{\ell+1}^m \cdot (h_{\ell+1}\circ f_\ell)^{-1}\varsigma_\ell^b\kappa_\ell(h_\ell)$, and uniqueness in the torsor against $\mathcal{P}_\ell(u_\ell^m) = u_{\ell+1}^m\varsigma_\ell^m$ gives the identity. □

The two represented frame channels therefore remain compatible across scales even when their fiber dimensions or actions differ.

**Definition B1.33 (Connection defects along base curves).** Let each level carry principal connections $\omega_{\ell,b}, \omega_{\ell,m}$ with associated transports $\Omega_{\ell,\gamma}, \widetilde{\Omega}_{\ell,\gamma}$, and cross morphisms $\Phi_\ell, \widetilde{\Phi}_\ell$. For a base curve $\gamma : c \to c'$, the cross-scale connection defects are the ordered pairs

$$\begin{aligned}
\mathbf{D}_{\ell,\gamma}^b &= \left(C_{\ell,b,c'}\circ\Omega_{\ell,\gamma}, \Omega_{\ell+1,f_\ell\gamma}\circ C_{\ell,b,c}\right), \\
\mathbf{D}_{\ell,\gamma}^m &= \left(C_{\ell,m,c'}\circ\widetilde{\Omega}_{\ell,\gamma}, \widetilde{\Omega}_{\ell+1,f_\ell\gamma}\circ C_{\ell,m,c}\right),
\end{aligned} \tag{B1.38}$$

and the within-scale nonparallelness pairs are

$$\begin{aligned}
\mathbf{A}_{\ell,\gamma}^{\Phi} &= \left(\widetilde{\Omega}_{\ell,\gamma}\circ\Phi_{\ell,c}, \Phi_{\ell,c'}\circ\Omega_{\ell,\gamma}\right), \\
\mathbf{A}_{\ell,\gamma}^{\widetilde{\Phi}} &= \left(\Omega_{\ell,\gamma}\circ\widetilde{\Phi}_{\ell,c}, \widetilde{\Phi}_{\ell,c'}\circ\widetilde{\Omega}_{\ell,\gamma}\right).
\end{aligned} \tag{B1.39}$$

A defect vanishes when the two components of its pair are equal. Subtraction of the two components is reserved for a declared vector or affine realization.

**Corollary B1.34 (Covariance of the connection defects).** Under admissible local-frame changes of the belief and model bundles at every level, both components of every pair in (B1.38) and (B1.39) acquire the same endpoint coordinate changes, so the equality locus of each pair is gauge invariant, and in a fiberwise-linear realization their difference obeys the left and right law of a Hom-bundle section. [ESTABLISHED]

*Proof.* The transports change by (B1.10) and (B1.11), the cross-scale maps by the endpoint coordinate changes at $c$ and $f_\ell(c)$, and the cross morphisms by Proposition B1.5; substitution is the computation of Proposition 11.3, applied to the connection pairs. □

**Proposition B1.35 (A principal sufficient condition).** If

$$\mathcal{P}_\ell^*\omega_{\ell+1,x} = \mathrm{d}\kappa_\ell\circ\omega_{\ell,x}, \tag{B1.40}$$

then $\mathbf{D}_{\ell,\gamma}^x$ vanishes for every base curve. If $\rho_x$ is not faithful, vanishing of the represented defect does not imply (B1.40). [ESTABLISHED]

*Proof.* Under (B1.40), $\omega_{\ell+1,x}(T\mathcal{P}_\ell Y) = \mathrm{d}\kappa_\ell(\omega_{\ell,x}(Y)) = 0$ for horizontal $Y$, so $\mathcal{P}_\ell$ carries horizontal lifts of $\gamma$ to horizontal lifts of $f_\ell\gamma$, and the associated maps intertwine the transports. For the trivial representation every represented transport is the identity, so the represented defect vanishes for all connections, while (B1.40) can fail. □

The infinitesimal failure of (B1.40) is the scale-connection defect form (B2.61); the represented horizontal defect vanishes along a coarse section exactly under the isotropy criterion of Theorem B2.31, and the defects of composable steps obey the ordered law (B2.65).

An associated-bundle isomorphism $\Xi_\ell : \mathcal{E}_{\ell,b} \to \mathcal{E}_{\ell,m}$ may also be declared. Its existence is additional global data: equal fiber type is necessary but not sufficient, as the trivial and Möbius line bundles of Proposition B1.10 show, and $\Xi_\ell$ is unavailable when the associated bundles are not isomorphic. Its parallel and scale-preservation conditions are $\widetilde{\Omega}_{\ell,\gamma}\Xi_{\ell,c} = \Xi_{\ell,c'}\Omega_{\ell,\gamma}$ and $C_{\ell,m}\Xi_\ell = \Xi_{\ell+1}C_{\ell,b}$. No such isomorphism is needed to define $\Phi_\ell$ and $\widetilde{\Phi}_\ell$.

## B1.11 Variable contexts and ordered Kron frames

Supplement G classifies the Gaussian coefficient families that pass the fixed-anchor pair test at one anchor (Theorem G7.12). When the anchor varies over contexts, the classification holds locally and its ordering can fail globally.

**Theorem B1.36 (Local frames and a global ordering obstruction).** On a smooth context domain $\mathcal{U} \subseteq \mathcal{C}$, let $Q_0(c) \succ 0$ vary smoothly, let $\mathscr{W}_c$ be an inclusion-maximal family for the fixed-anchor pair test at $Q_0(c)$, and suppose that the whitened spans

$$\mathscr{D}_c = \mathrm{span}_{\mathbb{R}}\{Q_0(c)^{-1/2} A Q_0(c)^{-1/2} : A \in \mathscr{W}_c\} \tag{B1.41}$$

form a smooth rank-$K$ vector subbundle of $\mathcal{U} \times \mathbb{S}^K$; no pair using coefficients from different contexts is admitted. Then every $c_0 \in \mathcal{U}$ has a neighborhood $\mathcal{U}_0$ and a smooth $O : \mathcal{U}_0 \to \mathrm{O}(K)$ with

$$\begin{aligned} \mathscr{D}_c &= O(c)\,\mathrm{Diag}(K)O(c)^\top, \\ \mathscr{W}_c &= \mathsf{Q}(c)\,\mathrm{Diag}_+(K)\mathsf{Q}(c)^\top, \\ \mathsf{Q}(c) &= Q_0(c)^{1/2}O(c). \end{aligned} \tag{B1.42}$$

A global continuous ordered diagonal frame need not exist, even for a smooth maximal family over the circle with $Q_0 = I_2$. Smooth global coefficients nevertheless give smooth Schur complements and log determinants wherever the precision blocks remain positive definite, and if every cross-context pair were tested against one common fixed anchor, the fixed-anchor classification would apply to the union and force one common frame. [ESTABLISHED]

*Proof.* The implicit-function theorem applied on symmetric matrices to $S^2 - Q_0 = 0$ at the positive square root has derivative $X \mapsto SX + XS$, which multiplies entry $ij$ by $s_i + s_j > 0$ in an eigenbasis of $S$; so $Q_0^{1/2}$ and its inverse are smooth, and uniqueness of the positive square root identifies the local solutions. By Theorem G7.12, $\mathscr{D}_{c_0}$ contains a matrix with $K$ distinct eigenvalues, and a smooth

local frame of the subbundle extends it to a smooth section $\mathsf{E}(c) \in \mathscr{D}_c$. For $\det(\lambda I - \mathsf{E}(c)) = 0$, the derivative in $\lambda$ at each initial eigenvalue is the product of its nonzero spectral gaps, so the implicit-function theorem supplies $K$ distinct smooth eigenvalues $\lambda_k(c)$ on a smaller neighborhood, with spectral projectors

$$\mathsf{P}_k(c) = \prod_{l \neq k} \frac{\mathsf{E}(c) - \lambda_l(c) I}{\lambda_k(c) - \lambda_l(c)}. \tag{B1.43}$$

In an eigenbasis this polynomial is one on the $k$-th eigenspace and zero on the others, so these are smooth orthogonal rank-one projectors. For a unit $v_k \in \operatorname{im} \mathsf{P}_k(c_0)$, $e_k(c) = \mathsf{P}_k(c) v_k / \|\mathsf{P}_k(c) v_k\|$ is smooth after shrinking the neighborhood, and its columns form $O(c)$. Every member of $\mathscr{D}_c$ commutes with $\mathsf{E}(c)$ and preserves its eigenlines, and both spaces have dimension $K$, so the diagonal containment is equality; undoing the whitening gives (B1.42).

For the obstruction write $t \in S^1 = \mathbb{R}/(2\pi\mathbb{Z})$, let $\operatorname{Rot}(\phi)$ be the planar rotation by $\phi$, and set

$$\mathscr{W}_t = \operatorname{Rot}(t/4) \operatorname{Diag}_+(2) \operatorname{Rot}(t/4)^\top, \qquad \mathsf{E}_t = \begin{pmatrix} \cos(t/2) & \sin(t/2) \\ \sin(t/2) & -\cos(t/2) \end{pmatrix}. \tag{B1.44}$$

Increasing $t$ by $2\pi$ rotates by another $\pi/2$, which permutes the coordinate axes and preserves the cone. The algebra is $\operatorname{span}\{I_2, \mathsf{E}_t\}$, with trace-inner-product projection

$$X \longmapsto \frac{1}{2} \operatorname{Tr}(X) I_2 + \frac{1}{2} \operatorname{Tr}(\mathsf{E}_t X) \mathsf{E}_t, \tag{B1.45}$$

which is smooth and periodic because $\mathsf{E}_{t+2\pi} = -\mathsf{E}_t$, and has constant rank two, so it defines the required subbundle. The two trace-one extreme projectors $(I_2 \pm \mathsf{E}_t)/2$ interchange after a full circuit. A continuous ordered first projector $\mathsf{P}(t)$ has $\operatorname{Tr}(\mathsf{E}_t \mathsf{P}(t)) \in \{1, -1\}$, a continuous function on $[0, 2\pi]$, so its sign is constant, and its endpoint is the opposite projector from its initial value, contradicting periodicity; an ordered orthogonal diagonal frame would supply such a projector. Matrix inversion is the adjugate divided by a nonzero determinant and $\log\det$ is smooth on positive-definite matrices, so no diagonal frame is needed for Schur complements or normalizers. The last statement is Theorem G7.12 applied to the union of the families. □

# Chapter B2

# Base-Direction Information Geometry

The fiber geometry of the main manuscript lives on one associated fiber: the Fisher metric $g_x^F$ and the Amari–Chentsov tensor $T_x^F$ of the regular channel models (Hypothesis 7.1), their descent to the associated bundles (Proposition 7.3), and the vertical Fisher defect of a Markov coarse morphism with its cocycle (Corollary 7.12 and theorem 7.13). Over $\{c_*\}$ the base tangent space is zero, so the covariant first jet $D^\omega s$ of a section, its pullback tensors $h_s^\omega$ and $c_s^\omega$, the horizontal defect $A_\Psi$ of a bundle morphism, and the isotropy criterion below all vanish identically. This chapter asks what the fiber geometry induces on the base once contexts vary and connections are chosen. The pullbacks are gauge invariant but connection relative, with exact change formulas (Theorem B2.2 and proposition B2.4). Across a bundle morphism over a base map, fine and coarse pullbacks differ by an exact signed identity with two retained anomaly tensors (Theorem B2.25); the horizontal defect of an induced morphism is the fundamental field of a scale-connection defect form and vanishes exactly under an isotropy criterion (Theorem B2.31); and the base Fisher defect composes by a cocycle up to an explicit residual (Theorem B2.35), whose relaxed pointwise defect problems have complete finite-dimensional parameterizations (Theorems B2.39, B2.41 and B2.42). These results are used by the section-space histories of Chapter B3, the covers of Chapter B4, and the perceived-geometry beta of Chapter B5. At $c_*$ the counterpart of a scale-connection defect is the defect of the dressed interaction links under agent coarse-graining, treated in Chapter 11 of the main manuscript. With additional common-target comparison data, a collective Fisher minimizer also defines an aggregate-moment jet; its rank can be four even when every law is a centered Gaussian (propositions B2.11 and B2.13). This supplies a local example for the conditional determinant geometry of proposition B2.14.

No connection-independent tensor on the base is obtained from the bundle and the statistical models alone, no canonical connection is derived from the change formulas or from passive covariance, and the pullback tensors need not be nondegenerate without rank hypotheses. A scalar gauged sigma energy would further require a base cometric, a base density, channel weights, boundary conditions, and a decision whether the connection is fixed or dynamical, none of which is selected here. The classifications of Section B2.11 assert no global solution with prescribed boundary data across changing ranks, no infinite-dimensional closed-range result, and no compatible connection for every scale map. The collective construction selects neither its message sector nor its comparison maps, and its local example establishes no persistence under inference, uniform large-network bound, or physical dynamics; preservation through a specified inference and coarse family is open problem B2.15. [NOT-CLAIMED]

## B2.1 Covariant first jets and pullback tensors

Fix one channel $x$ and write $E = \mathcal{E}_x$, $\mathcal{B} = \mathcal{B}_x$, $\varpi = \varpi_x$, $g^F = g^F_x$, $T^F = T^F_x$, and $\omega = \omega_x$, with the fiber tensors of eqs. (7.1) and (7.2) descended as in Proposition 7.3. The connection induces a horizontal subbundle $H^\omega E \subset TE$ and a vertical projection $\mathrm{ver}^\omega : TE \to VE$. For $e \in E$ write $\mathsf{J}_e : \mathrm{Ad}(\mathscr{P}_G)_{\varpi(e)} \to V_e E$ for the fundamental vertical map of the represented action, and $\mathsf{f}_\xi$ for the fundamental vertical field of $\xi \in \mathfrak{g}$ in a local frame.

**Definition B2.1 (Covariant first jets and informational pullbacks).** For a smooth section $s : \mathcal{U} \to E$ over an open set $\mathcal{U} \subseteq \mathcal{C}$, the covariant first jet is

$$D^\omega s := \mathrm{ver}^\omega \circ Ts : T\mathcal{U} \longrightarrow s^* VE, \tag{B2.1}$$

and the connection-relative Fisher and Amari–Chentsov pullbacks of $s$ are

$$h^\omega_s(X, Y) = g^F_{s(c)}(D^\omega s(X), D^\omega s(Y)), \tag{B2.2}$$

$$c^\omega_s(X, Y, Z) = T^F_{s(c)}(D^\omega s(X), D^\omega s(Y), D^\omega s(Z)), \tag{B2.3}$$

for $X, Y, Z \in T_c\mathcal{C}$. Thus $h^\omega_s \in \Gamma(\mathrm{Sym}^2 T^*\mathcal{U})$ and $c^\omega_s \in \Gamma(\mathrm{Sym}^3 T^*\mathcal{U})$; the first is called a semimetric until its nondegeneracy is proved.

For nonlinear statistical fibers, $D^\omega s$ is a vertical-valued one-form along $s$, not a linear covariant derivative on a vector space of sections. In the Gaussian realization of Supplement G the fiber metric is the moment-chart Fisher metric of Proposition G5.1, and Definition B2.1 pulls it back to a base tensor that depends on the section and the connection; by Theorem B2.7 its rank is the rank of the first jet. The Gaussian component formulas do not identify the base with the sample or parameter spaces.

**Theorem B2.2 (Passive gauge covariance of the covariant pullbacks).** Under a passive local change of frame $u' = u \cdot g^{-1}$, with the connection and the section coordinates transformed by their associated laws, $h^\omega_s$ and $c^\omega_s$ are unchanged as tensors on $\mathcal{U}$. They are not invariant under active gauge transformations at a fixed connection: on the trivial principal $\mathbb{R}$-bundle over $\mathbb{R}$ with the normal location fiber $\{\mathcal{N}(\mu, 1)\}$, the zero connection, and the zero section, $h = 0$, while the bundle automorphism $F(u(x)g) = u(x)(x + g)$ carries that section to $s'(x) = \mathcal{N}(x, 1)$ and $h$ to $dx^2$. [ESTABLISHED]

*Proof.* In a local frame the vertical jet transforms by the tangent representation,

$$D^{\omega'} s'(X) = T\widehat{\rho}_x(g) D^\omega s(X), \tag{B2.4}$$

because coordinates change by $\beta' = \widehat{\rho}_x(g)\beta$ under the quotient convention (2.7), and the connection changes by (B1.8) or (B1.9). By Proposition 7.3 the tangent representation is an isometry of $g^F$ and preserves $T^F$, so the frame change cancels in every argument of (B2.2) and (B2.3). For the active

example, with zero connection the vertical jet of a section of means $\mu(x)$ is $d\mu$, and the location Fisher metric is $1$. □

Passive covariance says that two coordinate descriptions of one pair $(\omega, s)$ give the same tensor. It does not say that two connections give the same pullback, which is the subject of the next section.

## B2.2 Exact dependence on the chosen connection

**Definition B2.3 (Connection difference and its vertical image).** For a second principal connection $\omega'$ on $\mathscr{P}_G$, the difference $\omega' - \omega$ is horizontal and $\mathrm{Ad}$-equivariant; its descended form is

$$\mathfrak{d} \in \Omega^1(\mathcal{C}, \mathrm{Ad}(\mathscr{P}_G)). \tag{B2.5}$$

Along a section $s$, its vertical image is

$$\mathsf{J}^s_{\mathfrak{d}}(X) = \mathsf{J}_{s(c)}(\mathfrak{d}_c(X)). \tag{B2.6}$$

The sign convention is that in a local trivialization the horizontal lift of $X$ at $(c, \beta)$ is $(X, -\mathsf{f}_{A(X)}\beta)$, compatibly with (2.7). For a law fiber that is not a vector space the linear expression $\widehat{\rho}_{x*}(A)$ is undefined; it is available only in a fiberwise-linear realization, where it agrees with $\mathsf{f}_{A(X)}$.

**Proposition B2.4 (Connection-change formulas).** For $\omega' = \omega + \mathfrak{d}$, the vertical jets and Fisher pullbacks obey

$$D^{\omega'} s = D^{\omega} s + \mathsf{J}^s_{\mathfrak{d}}, \tag{B2.7}$$

$$h^{\omega'}_s(X, Y) = h^{\omega}_s(X, Y) + g^F(\mathsf{J}^s_{\mathfrak{d}} X, D^{\omega} sY) + g^F(D^{\omega} sX, \mathsf{J}^s_{\mathfrak{d}} Y) + g^F(\mathsf{J}^s_{\mathfrak{d}} X, \mathsf{J}^s_{\mathfrak{d}} Y), \tag{B2.8}$$

and the Amari–Chentsov pullback satisfies the polarized identity

$$c^{\omega'}_s(X, Y, Z) = T^F\big((D^{\omega} s + \mathsf{J}^s_{\mathfrak{d}})X, (D^{\omega} s + \mathsf{J}^s_{\mathfrak{d}})Y, (D^{\omega} s + \mathsf{J}^s_{\mathfrak{d}})Z\big), \tag{B2.9}$$

which contains the original term, three terms with one occurrence of $\mathsf{J}^s_{\mathfrak{d}}$, three with two, and one with three. The dependence occurs already for the trivial translation bundle over $\mathbb{R}$ with fiber $\{\mathcal{N}(\mu, 1)\}$ and the constant section $\mu = 0$: the zero connection gives $h = 0$, and the local connection $A' = \mathrm{c}_1 dx$ gives

$$h^{A'} = \mathrm{c}_1^2 dx^2. \tag{B2.10}$$

[ESTABLISHED]

*Proof.* In one local frame, replacing the horizontal lift $(X, -\mathsf{f}_{A(X)}\beta)$ by $(X, -\mathsf{f}_{A(X)+\mathfrak{d}(X)}\beta)$ shifts the vertical part of $Ts$ by $\mathsf{J}^s_{\mathfrak{d}}(X)$, which proves (B2.7). Bilinearity of $g^F$ and trilinearity of $T^F$ give the other two formulas, which are frame independent because $\mathfrak{d}$ is $\mathrm{Ad}(\mathscr{P}_G)$-valued and every contraction is gauge invariant. In the example the vertical jet of the constant section is $A'(X)\partial_\mu$ and the Fisher metric is $1$. □

The example refutes a connection-independent reading of $h^{\omega}_{s}$; it does not refute the gauge invariance of a fixed connection and section.

## B2.3 Two channel geometries

Restoring the channel labels gives the ordered pair

$$(h^{\omega_b}_{i,b}, h^{\omega_m}_{i,m}) \in \Gamma(\mathrm{Sym}^2 T^*\mathcal{U}) \times \Gamma(\mathrm{Sym}^2 T^*\mathcal{U}) \tag{B2.11}$$

for the sections $q^b_i$ and $q^m_i$, and an analogous pair of cubic tensors. The pair is determined by the two statistical models, sections, and connections; a single scalar geometry is not part of those data.

**Definition B2.5 (Weighted product geometry).** If one joint geometry is required, positive constants $w_b, w_m$ are declared and

$$h^{\mathrm{prod}} = w_b h^{\omega_b}_{i,b} + w_m h^{\omega_m}_{i,m}. \tag{B2.12}$$

The weights fix the relative normalization of the two channels; unit weights are a specialization, not a consequence of the common principal bundle.

**Proposition B2.6 (Radical of the weighted product).** Under Hypothesis 7.1 in both channels,

$$\operatorname{rad} h^{\mathrm{prod}} = \ker D^{\omega_b} q^b_i \cap \ker D^{\omega_m} q^m_i, \tag{B2.13}$$

so two degenerate channel tensors are jointly nondegenerate exactly when their null distributions intersect in zero. [ESTABLISHED]

*Proof.* $h^{\mathrm{prod}}(X,X) = w_b g^F_b(D^{\omega_b} q^b_i X, D^{\omega_b} q^b_i X) + w_m g^F_m(D^{\omega_m} q^m_i X, D^{\omega_m} q^m_i X)$ is a sum of nonnegative terms with positive weights, which vanishes exactly when both vertical jets vanish; for a positive-semidefinite form the diagonal null set is the radical. □

The common group acts on the two channels separately and supplies no bilinear cross tensor on $V\mathcal{E}_b \times V\mathcal{E}_m$; such a term needs a separately declared invariant bilinear form, and a comparison of channel points needs $\Phi$ or $\widetilde{\Phi}$ of Definition 2.9. Neither the relative frame nor equal channel dimensions supplies either object.

## B2.4 Rank, radicals, and quotient geometry

**Theorem B2.7 (Constant-rank quotient theorem).** Let $s : \mathcal{U} \to E$ be a smooth section under Hypothesis 7.1. Pointwise,

$$\operatorname{rad} h^{\omega}_{s} = \ker D^{\omega} s, \qquad \operatorname{rank} h^{\omega}_{s} = \operatorname{rank} D^{\omega} s. \tag{B2.14}$$

If $D^\omega s$ has constant rank on $\mathcal{U}$, then $K_s^\omega = \ker D^\omega s$ is a smooth subbundle, and

$$Q_s^\omega = T\mathcal{U}/K_s^\omega \tag{B2.15}$$

carries a positive-definite metric and a symmetric cubic tensor defined by

$$\bar{h}_s^\omega([X],[Y]) = h_s^\omega(X,Y), \tag{B2.16}$$
$$\bar{c}_s^\omega([X],[Y],[Z]) = c_s^\omega(X,Y,Z), \tag{B2.17}$$

and the induced map $Q_s^\omega \to \operatorname{im} D^\omega s$ is an isometric vector-bundle isomorphism. [ESTABLISHED]

The proof is in Appendix BA. The quotient is a vector bundle, not automatically the tangent bundle of a quotient manifold, and the following records what the manifold reading requires.

**Proposition B2.8 (Requirements for a quotient manifold).** Assume the constant-rank case of Theorem B2.7. (i) If there is a smooth Hausdorff manifold $\overline{\mathcal{U}}$ and a surjective submersion $q_K : \mathcal{U} \to \overline{\mathcal{U}}$ whose fibers are the null leaves and whose tangent kernel is $K_s^\omega$, then

$$D^\omega s([X,Y]) = 0 \quad \text{for all } X, Y \in \Gamma(K_s^\omega); \tag{B2.18}$$

involutivity does not supply such a quotient, even at constant rank. (ii) Tensors on $\overline{\mathcal{U}}$ whose pullbacks are $h_s^\omega$ and $c_s^\omega$ exist only if

$$\mathcal{L}_Z h_s^\omega = 0, \qquad \mathcal{L}_Z c_s^\omega = 0 \quad \text{for every } Z \in \Gamma(K_s^\omega), \tag{B2.19}$$

and this basicness can fail although a smooth Hausdorff quotient exists. (iii) For any $g^F$-compatible connection $\nabla$ on $s^*VE$, with the covariant Lie derivative

$$(\mathcal{L}_Z^\nabla \alpha)(X) = \nabla_Z(\alpha(X)) - \alpha([Z,X]) \tag{B2.20}$$

of an $s^*VE$-valued one-form $\alpha$, every $Z \in \Gamma(K_s^\omega)$ satisfies

$$(\mathcal{L}_Z h_s^\omega)(X,Y) = g^F\big((\mathcal{L}_Z^\nabla D^\omega s)(X), D^\omega s Y\big) + g^F\big(D^\omega s X, (\mathcal{L}_Z^\nabla D^\omega s)(Y)\big), \tag{B2.21}$$

and the right side does not depend on the choice of $\nabla$. [ESTABLISHED]

The proof, with a dense-leaf witness for (i) and a non-basic witness for (ii), is in Appendix BA. Constant rank is itself needed, and integrability of the radical is a further condition.

**Proposition B2.9 (A constant-rank nonintegrable radical).** For the normal location family, the trivial principal translation bundle over $\mathbb{R}^3$ with coordinates $(x,y,z)$, the constant section $\mu = 0$, and the local connection form

$$A = \mathrm{d}z - x\mathrm{d}y, \tag{B2.22}$$

one has $D^\omega s = A$,

$$h_s^\omega = (\mathrm{d}z - x\mathrm{d}y)^2, \tag{B2.23}$$

of constant rank one, and

$$A \wedge \mathrm{d}A = -\mathrm{d}z \wedge \mathrm{d}x \wedge \mathrm{d}y \neq 0, \tag{B2.24}$$

so $\ker D^\omega s$ is a contact distribution and not a foliation. Without constant rank the quotient bundle fails: for the same family over $\mathcal{C} = \mathbb{R}$ with zero connection and $s(x) = \mathcal{N}(x^2, 1)$,

$$h_s = 4x^2 dx^2 \tag{B2.25}$$

has rank one away from the origin and rank zero at it. [ESTABLISHED]

*Proof.* For a section of means $\mu$, the vertical jet is $(d\mu + A)\partial_\mu$ by the sign convention of Definition B2.3, and the location Fisher metric is $1$. Then $\mathrm{d}A = -\mathrm{d}x \wedge \mathrm{d}y$ and $x\mathrm{d}y \wedge \mathrm{d}x \wedge \mathrm{d}y = 0$, and the Frobenius criterion identifies $\ker A$ as nonintegrable. In the second example the vertical jet is $d\mu = 2x\mathrm{d}x$, whose square is (B2.25). □

## B2.5 Collective Fisher minimization and aggregate soldering

The rank question can be posed after choosing a connection from several law sections. The construction below uses the orbit Fisher form of corollary 7.7, with explicit comparison data before any laws or their derivatives are combined.

**Definition B2.10 (Common-target collective data).** Let $\mathscr{W}^{\mathrm{msg}} \to \mathcal{U}$ be a complex rank-two message bundle. Choose complex rank-two source bundles $\mathscr{W}_i^{\mathrm{msg}}$, smooth fiber isomorphisms $\mathsf{S}_i : \mathscr{W}_i^{\mathrm{msg}} \to \mathscr{W}^{\mathrm{msg}}$, and source-law sections $q_i$. Their target laws $p_i = (\mathsf{S}_i)_\# q_i$ lie in finite-dimensional regular statistical families closed under the $\mathrm{GL}(2,\mathbb{C})$ pushforward action, with invariant positive Fisher metrics. Their second moments exist and depend smoothly on the law coordinates. Independent passive source changes $g_i$ and a target change $g$ act by

$$q_i' = (g_i)_\# q_i, \qquad \mathsf{S}_i' = g\mathsf{S}_i g_i^{-1}. \tag{B2.26}$$

For positive constants $w_i$, write $\boldsymbol{p} = (p_i)_i$ and set

$$\begin{aligned} \mathsf{G}_{\boldsymbol{p}}^{F,\mathrm{col}}(r,t) &= \sum_i w_i g_{p_i}^F(r_i, t_i), \\ \mathsf{J}_{\boldsymbol{p}}\mathsf{a} &= (\mathsf{a}_{\mathcal{B}}(p_i))_i, \qquad \mathfrak{g}_{\boldsymbol{p}} = \ker \mathsf{J}_{\boldsymbol{p}}, \qquad \mathsf{H}_{\boldsymbol{p}} = (\operatorname{im} \mathsf{J}_{\boldsymbol{p}})^{\perp_{\mathsf{G}^{F,\mathrm{col}}}}, \\ \mathfrak{k}_{\boldsymbol{p}}(\mathsf{a},\mathsf{b}) &= \mathsf{G}_{\boldsymbol{p}}^{F,\mathrm{col}}(\mathsf{J}_{\boldsymbol{p}}\mathsf{a}, \mathsf{J}_{\boldsymbol{p}}\mathsf{b}), \\ \mathcal{M}(\boldsymbol{p}) &= \sum_i w_i \int z z^\dagger p_i(dz). \end{aligned} \tag{B2.27}$$

Here $\mathsf{a}, \mathsf{b} \in \mathfrak{gl}(2,\mathbb{C})$ are real Lie-algebra variables and $\mathcal{M}$ takes values in the real rank-four Hermitian-tensor bundle $\mathrm{Herm}(\mathscr{W}^{\mathrm{msg}})$. The weights fix both the product metric and the aggregate; dividing $\mathcal{M}$

by $\sum_i w_i$ gives the moment of the corresponding mixture law. These comparisons and weights are additional data.

**Proposition B2.11 (Collective minimizing connection class).** Under definition B2.10, assume $\mathsf{J}_{\boldsymbol{p}(c)}$ has constant rank on $\mathcal{U}$. For every $v \in T_c\mathcal{U}$, the minimizers of

$$\mathsf{a} \longmapsto \|d\boldsymbol{p}(v) + \mathsf{J}_{\boldsymbol{p}}\mathsf{a}\|^2_{\mathsf{G}^{F,\mathrm{col}}} \tag{B2.28}$$

form one affine class $\mathsf{A}^F(v) + \mathfrak{g}_{\boldsymbol{p}}$, characterized by

$$\begin{aligned} \mathfrak{k}_{\boldsymbol{p}}(\mathsf{A}^F(v), \mathsf{b}) &= -\mathsf{G}^{F,\mathrm{col}}_{\boldsymbol{p}}(d\boldsymbol{p}(v), \mathsf{J}_{\boldsymbol{p}}\mathsf{b}), \\ \mathsf{r}(v) &:= d\boldsymbol{p}(v) + \mathsf{J}_{\boldsymbol{p}}\mathsf{A}^F(v) \\ &= P_{\mathsf{H}_{\boldsymbol{p}}} d\boldsymbol{p}(v). \end{aligned} \tag{B2.29}$$

The class has smooth local representatives, linear in $v$. Under eq. (B2.26), the target laws and minimizing classes obey

$$\begin{aligned} p'_i &= g_{\#}p_i, \\ [\mathsf{A}^{F\prime}] &= [g\mathsf{A}^F g^{-1} - (dg)g^{-1}], \\ \mathsf{r}' &= T\widehat{\rho}(g)\mathsf{r}. \end{aligned} \tag{B2.30}$$

Thus the residual is independent of the source coordinates and of the stabilizer-valued choice of representative. The class alone need not specify a full principal connection. [ESTABLISHED]

*Proof.* Orthogonally project $-d\boldsymbol{p}(v)$ onto $\operatorname{im}\mathsf{J}_{\boldsymbol{p}}$. Every preimage minimizes the square, and its other preimages differ exactly by $\ker\mathsf{J}_{\boldsymbol{p}}$. Expanding the square gives the displayed normal equation, including its minus sign. Constant rank makes the kernel and image smooth subbundles; on a local complement of the kernel, $\mathfrak{k}_{\boldsymbol{p}}$ is positive definite, so its inverse gives a smooth linear solution. Composition of pushforwards cancels $g_i$ in $p'_i$. Differentiating $\boldsymbol{p}' = \widehat{\rho}(g)\boldsymbol{p}$ gives the extra fundamental field of $(dg)g^{-1}$; it cancels the inhomogeneous term in eq. (B2.30). Fisher invariance then carries every minimizer and its orthogonal residual to the stated ones. No averaging of coordinate-induced source connection potentials enters this construction. □

**Corollary B2.12 (Aggregate horizontal rank criterion).** Under proposition B2.11, let $M^{\mathrm{col}} = \mathcal{M}(\boldsymbol{p})$. Its minimizing covariant derivative is the well-defined Hermitian-valued one-form

$$\begin{aligned} \mathsf{e}^{\mathrm{col}}(v) &= dM^{\mathrm{col}}(v) + \mathsf{A}^F(v)M^{\mathrm{col}} + M^{\mathrm{col}}\mathsf{A}^F(v)^\dagger \\ &= d\mathcal{M}_{\boldsymbol{p}}\big(P_{\mathsf{H}_{\boldsymbol{p}}} d\boldsymbol{p}(v)\big). \end{aligned} \tag{B2.31}$$

It transforms as $\mathsf{e}^{\mathrm{col}\prime} = g\mathsf{e}^{\mathrm{col}}g^\dagger$. On a four-dimensional base it is a soldering exactly when $d\mathcal{M}_{\boldsymbol{p}} \circ \mathsf{r} : T_c\mathcal{U} \to \mathrm{Herm}(\mathscr{W}_c^{\mathrm{msg}})$ has rank four. Surjectivity of $d\mathcal{M}_{\boldsymbol{p}}|_{\mathsf{H}_{\boldsymbol{p}}}$ is necessary for this rank and is sufficient for the existence of a first jet with that rank at the given law tuple; it need not hold for a specified section's first jet. [ESTABLISHED]

*Proof.* The moment map is equivariant, so $d\mathcal{M}_{\boldsymbol{p}}(\mathsf{J}_{\boldsymbol{p}}\mathsf{a}) = \mathsf{a}M^{\mathrm{col}} + M^{\mathrm{col}}\mathsf{a}^{\dagger}$. The chain rule proves eq. (B2.31) and shows that isotropy changes contribute zero. Covariance follows from eq. (B2.30). The rank assertion is linear algebra. If the restricted differential is onto, choose four horizontal preimages of a Hermitian basis and realize them as the first derivatives of a local chart-valued section. □

**Proposition B2.13 (A local Gaussian collective soldering).** There are four smooth centered proper complex Gaussian law sections on a neighborhood of $0 \in \mathbb{R}^4$, with identity comparison maps and $w_i = 1$, whose collective isotropy is exactly $i\mathbb{R}I_2$ and whose $\mathsf{e}^{\mathrm{col}}$ has rank four throughout that neighborhood. Their covariances at the origin are

$$M_0^0 = I_2, \quad M_1^0 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad M_2^0 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad M_3^0 = \begin{pmatrix} 2 & i \\ -i & 2 \end{pmatrix}. \tag{B2.32}$$

At the origin a minimizing representative is zero and $\mathsf{e}^{\mathrm{col}}(\partial_{c^\mu}) = \boldsymbol{\sigma}_\mu$, where $\boldsymbol{\sigma}_0 = I_2$ and

$$\boldsymbol{\sigma}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \boldsymbol{\sigma}_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \boldsymbol{\sigma}_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{B2.33}$$

By contrast, minimizing from one centered Gaussian covariance alone always gives zero covariant derivative of that covariance. [ESTABLISHED]

*Proof.* For the density $(\pi^2 \det M_i)^{-1} \exp(-z^\dagger M_i^{-1} z)$, the score in a Hermitian direction $r_i$ is $-\operatorname{Tr}(M_i^{-1} r_i) + z^\dagger M_i^{-1} r_i M_i^{-1} z$. Whitening and the identity $\mathbb{E}[\bar{z}_a z_b \bar{z}_d z_f] = \delta_{ab}\delta_{df} + \delta_{af}\delta_{db}$ for independent standard proper complex Gaussian coordinates give

$$\begin{aligned} g^F_{M_i}(r_i, t_i) &= \operatorname{Tr}(M_i^{-1} r_i M_i^{-1} t_i), \\ \sum_i g^F_{M_i}(r_i, \mathsf{a}M_i + M_i\mathsf{a}^\dagger) &= 2 \operatorname{Re} \operatorname{Tr}\left(\mathsf{a}^\dagger \sum_i M_i^{-1} r_i\right). \end{aligned} \tag{B2.34}$$

The fourth-moment identity follows by expanding into independent real centered Gaussian coordinates of variance $1/2$. The real pairing with arbitrary $\mathsf{a}$ is nondegenerate, so a jet is horizontal exactly when $\sum_i M_i^{-1} r_i = 0$.

Choose $r_{i\mu} = \sum_{\nu=0}^{3} \mathsf{R}^{\mathrm{jet}}_{4i+\nu,\mu} \boldsymbol{\sigma}_\nu$, where rows are ordered by $(i, \nu)$, each from $0$ to $3$, and

$$\mathsf{R}^{\mathrm{jet}} = \frac{1}{2} \begin{pmatrix} -5 & 1 & -1 & -1 \\ 4 & -2 & 4 & -4 \\ 0 & 0 & 2 & 0 \\ 3 & -3 & 3 & -1 \\ 4 & -4 & 4 & 4 \\ -4 & 4 & -4 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 3 & 3 & 3 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -3 & 3 & -3 & 3 \\ 0 & 0 & -6 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{B2.35}$$

Using $(M_1^0)^{-1} = \mathrm{diag}(1/2, 1)$, $(M_2^0)^{-1} = \frac{1}{3}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$, and $(M_3^0)^{-1} = \frac{1}{3}\left(\begin{smallmatrix} 2 & -i \\ i & 2 \end{smallmatrix}\right)$, multiplication of each of the four columns gives the two matrix identities

$$\sum_{i=0}^{3} (M_i^0)^{-1} r_{i\mu} = 0, \qquad \sum_{i=0}^{3} r_{i\mu} = \boldsymbol{\sigma}_\mu \quad (\mu = 0, 1, 2, 3). \tag{B2.36}$$

These identities prove both minimization at zero and the identity matrix of aggregate Pauli coefficients. For the stabilizer, $\mathsf{a} M_0^0 + M_0^0 \mathsf{a}^\dagger = 0$ first makes $\mathsf{a}$ anti-Hermitian. Commutation with $M_1^0$ makes it diagonal, and commutation with $M_2^0$ makes its diagonal entries equal. Thus the joint kernel is $i\mathbb{R} I_2$.

Set $M_i(c) = M_i^0 + \sum_{\mu=0}^{3} c^\mu r_{i\mu}$. Since every initial covariance has minimum eigenvalue one, all covariances remain positive on a ball where

$$\max_i \left\| \sum_{\mu=0}^{3} c^\mu r_{i\mu} \right\|_{\mathrm{op}} < \frac{1}{2}.$$

The action map has real rank seven at zero and everywhere has rank at most seven, since the central phase acts trivially. A nonzero rank-seven minor remains nonzero nearby, so the rank is constant there. Solving on a fixed complement of $i\mathbb{R} I_2$ gives a smooth local $\mathsf{A}^F$. Its aggregate jet equals the Pauli basis at zero by eq. (B2.36); continuity keeps its determinant nonzero on a smaller neighborhood. For one covariance and any Hermitian jet $r$, the choice $\mathsf{a} = -rM^{-1}/2$ gives $r + \mathsf{a}M + M\mathsf{a}^\dagger = 0$, attaining the zero minimum. This proves the single-law assertion without applying it to the collective cost. □

**Proposition B2.14 (Conditional determinant metric and circle completion).** Suppose $\mathsf{e}^{\mathrm{col}}$ is an isomorphism on a four-dimensional region, and choose a positive section $\kappa_{\det}$ of the determinant-scale line, with local transformation $\kappa'_{\det} = |\det g|^2 \kappa_{\det}$. Then

$$\eta_{\det}(H,K) = \tfrac{1}{2}[\mathrm{Tr}(HK) - \mathrm{Tr}(H)\,\mathrm{Tr}(K)], \qquad \mathsf{G}^{\det}(v,w) = \kappa_{\det}^{-1}\eta_{\det}(\mathsf{e}^{\mathrm{col}}v, \mathsf{e}^{\mathrm{col}}w) \tag{B2.37}$$

defines a gauge-invariant metric with one negative and three positive directions. Its positive Hermitian cone supplies a time orientation. On a base of dimension greater than four, a surjective $\mathsf{e}^{\mathrm{col}}$ instead gives this metric on the rank-four quotient bundle by its kernel; a quotient manifold still requires an integrable kernel, a regular leaf quotient, and basicness of $\mathsf{G}^{\det}$, as in proposition B2.8.

On the four-dimensional region, let $\mathscr{Q}_{\det}$ be the unit circle bundle of $\det \mathscr{W}^{\mathrm{msg}}$ for the line metric with coefficient $\kappa_{\det}^{-1}$, and $\pi_{\det} : \mathscr{Q}_{\det} \to \mathcal{U}$ its projection. If a global real principal circle connection $\omega_{\det}$, normalized to one on the generator of period $2\pi$, and a smooth positive scalar radius $r_{\det}$ are supplied, then

$$\mathsf{G}^{\mathrm{KK}} = \pi_{\det}^{*}\mathsf{G}^{\det} + (r_{\det} \circ \pi_{\det})^2 \omega_{\det} \otimes \omega_{\det} \tag{B2.38}$$

is a circle-invariant metric with one negative and four positive directions. A chosen global $\mathrm{GL}(2,\mathbb{C})$ connection lifting the minimizing class gives such a circle connection locally as $d\theta_{\det} + \mathrm{Im}\,\mathrm{Tr}\,\mathsf{A}$, where $\theta'_{\det} = \theta_{\det} + \arg\det g$. For the centered Gaussian example the minimizing class fixes neither this phase component nor the positive circle radius. The determinant form is an additional moment-space form; the Fisher tensor remains positive. [ESTABLISHED]

*Proof.* In the basis of eq. (B2.33), $\eta_{\det}(H,H) = -\det H = -(h^0)^2 + \sum_{a=1}^{3}(h^a)^2$. Multiplicativity of the determinant and polarization give $\eta_{\det}(gHg^\dagger, gKg^\dagger) = |\det g|^2 \eta_{\det}(H,K)$. The compensator cancels this factor, and an invertible soldering preserves the inertia. Congruence preserves the positive cone. In the surjective case the radical is exactly the kernel because the target form is nondegenerate. The circle connection splits its total tangent bundle into horizontal and vertical summands; the displayed metric is the base metric on the first and the positive form $r_{\det}^2 d\theta_{\det}^2$ on the second. Its invariance follows from the principal connection law. In local coordinates, $\mathrm{Im}\,\mathrm{Tr}\,\mathsf{A}' = \mathrm{Im}\,\mathrm{Tr}\,\mathsf{A} - d\arg\det g$, which cancels the angle change. A central $g = e^{i\beta}I_2$ changes that angle by $2\beta$; its action on all centered proper Gaussian laws and on their moments is trivial, so its connection component and a positive circle norm are absent from the collective normal equations. □

**Open Problem B2.15 (Preserving the collective interface through inference and coarse-graining).** Consider connected sparse Gaussian quadratic-record models with held invertible links, positive local and edge record precisions, centered product-Gaussian recognition, and joint Fisher natural gradient descent of the collective free energy. Supply smooth context-dependent records and common-target comparisons on a four-dimensional region. At each prescribed coarse step, use the exact pushforward through a recognition-independent, covariant linear block map with disjoint blocks of at least two agents, surjective onto a complex rank-two output per block, leaving the observations unchanged. Require the fine and coarse initial interfaces to have rank four and their collective orbit maps to have rank seven. The problem is to give structural, quantitatively

checkable conditions on this family that preserve the message-bundle comparisons and yield a uniform positive lower bound for the smallest singular value of $\mathrm{e}^{\mathrm{col}}$, measured in specified reference metrics, on a prescribed positive horizon and through a prescribed nonzero coarse depth, along a family of connected graphs of unbounded size. The estimate must follow from the actual coupled equations and retained context dependence, with constants controlled as network size grows, rather than assuming the desired rank along the trajectory. An admissible class must contain nonstationary inference, nonzero edge couplings, and genuinely reducing channels; channels that already destroy initial rank and repetitions of the identity channel are excluded. Proposition B2.13 establishes local realizability but does not provide such a dynamical and coarse-preservation criterion.

## B2.6 Transported divergence along base curves

Let $\mathscr{D}$ be a contrast satisfying Definition 7.5, with induced metric $g_{\mathscr{D}}$ of (7.5), mixed third jets $\Gamma_{\mathscr{D}}$ and $\Gamma^*_{\mathscr{D}}$ of eqs. (7.6) and (7.7), and cubic tensor $T^F_{\mathscr{D}} = \Gamma_{\mathscr{D}} - \Gamma^*_{\mathscr{D}}$ of (7.8). For a smooth base curve $\gamma$ with $\gamma(0) = c$, let $\Omega^{\omega}_{0\to\epsilon}$ be the channel transport of Definition B1.6 along $\gamma|_{[0,\epsilon]}$, and bring the section value back to the initial fiber:

$$\widehat{s}_\gamma(\epsilon) = (\Omega^{\omega}_{0\to\epsilon})^{-1} s(\gamma(\epsilon)). \tag{B2.39}$$

**Corollary B2.16 (Quadratic transported divergence).** The transported section curve has initial velocity

$$\frac{d}{d\epsilon}\widehat{s}_\gamma(\epsilon)\Big|_{\epsilon=0} = D^{\omega} s(\dot\gamma(0)), \tag{B2.40}$$

and with $h^{\omega}_{s,\mathscr{D}} = (D^{\omega}s)^* g_{\mathscr{D}}$,

$$\mathscr{D}(s(c), \widehat{s}_\gamma(\epsilon)) = \frac{\epsilon^2}{2} h^{\omega}_{s,\mathscr{D}}(\dot\gamma(0), \dot\gamma(0)) + O(\epsilon^3). \tag{B2.41}$$

If $g_{\mathscr{D}} = g^F$, as for the regular KL contrast of Proposition 7.6, the two pullbacks coincide, $h^{\omega}_{s,\mathscr{D}} = h^{\omega}_s$. [ESTABLISHED]

*Proof.* A horizontal curve over $\gamma$ is carried by $(\Omega^{\omega}_{0\to\epsilon})^{-1}$ to a constant, so the derivative of $\widehat{s}_\gamma$ at zero is the vertical part of $Ts(\dot\gamma(0))$, which is (B2.40). A contrast and its first derivative vanish on the diagonal and its quadratic Taylor term in the second argument is the induced metric, as in the proof of Corollary 7.7; substitution gives the expansion. □

The cubic coefficient of such a one-sided expansion is not determined by $c^{\omega}_{s,\mathscr{D}} := (D^{\omega}s)^* T^F_{\mathscr{D}}$ alone, and it can be computed exactly. In a chart write $\Gamma_{\mathscr{D},ij,k} = \Gamma_{\mathscr{D}}(\partial_i, \partial_j, \partial_k)$ and $\Gamma^*_{\mathscr{D},ij,k} = \Gamma^*_{\mathscr{D}}(\partial_i, \partial_j, \partial_k)$, and let $\nabla^*_{\mathscr{D}}$ be the torsion-free connection on $\mathcal{B}$ with $g_{\mathscr{D}}(\nabla^*_{\mathscr{D},\partial_i}\partial_j, \partial_k) = \Gamma^*_{\mathscr{D},ij,k}$.

**Proposition B2.17 (Cubic coefficient of a one-sided contrast expansion).** The connection $\nabla^*_{\mathscr{D}}$ is invariant under the represented action and descends to each fiber of $E$. For every $C^3$ curve

$c$ in $\mathcal{B}$ with $c(0) = p$, $\dot{c}(0) = v$, and covariant acceleration $a^* = (\nabla^*_{\mathscr{D},\dot{c}}\dot{c})(0)$,

$$\mathscr{D}(p, c(\epsilon)) = \frac{\epsilon^2}{2} g_{\mathscr{D}}(v, v) + \epsilon^3 \left[\frac{1}{2} g_{\mathscr{D}}(v, a^*) + \frac{1}{6} T^F_{\mathscr{D}}(v, v, v)\right] + o(\epsilon^3). \tag{B2.42}$$

For the transported section curve, $v = D^\omega s(\dot{\gamma}(0))$ and the cubic coefficient in (B2.41) is $\frac{1}{2} g_{\mathscr{D}}(v, a^*) + \frac{1}{6} c^\omega_{s,\mathscr{D}}(\dot{\gamma}(0), \dot{\gamma}(0), \dot{\gamma}(0))$ with $a^*$ the $\nabla^*_{\mathscr{D}}$-acceleration of $\widehat{s}_\gamma$; it equals $\frac{1}{6} c^\omega_{s,\mathscr{D}}$ exactly when $a^*$ is $g_{\mathscr{D}}$-orthogonal to $v$, for instance along a $\nabla^*_{\mathscr{D}}$-geodesic. For regular KL, $\Gamma^*_{\mathscr{D},ij,k} = \mathbb{E}_\theta[\mathbf{s}_k \partial_i \mathbf{s}_j]$ with $\mathbf{s}_i = \partial_i \log p_\theta$, the lowered coefficients of the exponential connection, which vanish in the natural chart of an exponential family, and $T^F_{\mathscr{D}}$ is the Amari–Chentsov tensor. [ESTABLISHED]

*Proof.* Invariance of $\mathscr{D}$ under the simultaneous represented action makes its mixed jets, hence $\nabla^*_{\mathscr{D}}$, invariant. Differentiating the diagonal identity $\partial'_i \partial'_j \mathscr{D}(\theta, \theta) = (g_{\mathscr{D}})_{ij}(\theta)$, obtained in the proof of Corollary 7.7, gives $\partial'_i \partial'_j \partial'_k \mathscr{D} = \partial_k (g_{\mathscr{D}})_{ij} + \Gamma^*_{\mathscr{D},ij,k}$ on the diagonal, and differentiating $(g_{\mathscr{D}})_{ij}(\theta) = -\partial_i \partial'_j \mathscr{D}(\theta, \theta)$ gives $\partial_k (g_{\mathscr{D}})_{ij} = \Gamma_{\mathscr{D},ki,j} + \Gamma^*_{\mathscr{D},kj,i}$. The third derivative of $\epsilon \mapsto \mathscr{D}(p, c(\epsilon))$ at zero is $(\partial'_i \partial'_j \partial'_k \mathscr{D}) v^i v^j v^k + 3 (g_{\mathscr{D}})_{ij} v^i \ddot{c}^j(0)$, which by these identities equals $T^F_{\mathscr{D}}(v, v, v) + 3\Gamma^*_{\mathscr{D}}(v, v, v) + 3 g_{\mathscr{D}}(v, \ddot{c}(0)) = T^F_{\mathscr{D}}(v, v, v) + 3 g_{\mathscr{D}}(v, a^*)$, because $a^{*k} = \ddot{c}^k(0) + \Gamma^{*k}_{\mathscr{D},ij} v^i v^j$ with the index raised by $g^{-1}_{\mathscr{D}}$. The lower-order coefficients are those of Corollary B2.16, and Taylor's theorem gives (B2.42). For regular KL the last line of (B.1) gives $\Gamma^*_{\mathscr{D}}$ [1]. □

The divergence-induced cubic tensor is therefore the difference of the two mixed third jets, not an arbitrary one-sided cubic Taylor coefficient.

## B2.7 Morphisms over base maps

Consider smooth associated bundles $E \to \mathcal{C}$ and $\bar{E} \to \bar{\mathcal{C}}$ with connections $\omega$ and $\bar{\omega}$, a smooth map $f : \mathcal{C} \to \bar{\mathcal{C}}$, and a smooth bundle morphism $\Psi : E \to \bar{E}$ over $f$, with vertical differential $T^V \Psi : VE \to V\bar{E}$. For a section $s$, the composite $\Psi \circ s$ is only a section of the pullback bundle $f^* \bar{E}$.

**Definition B2.18 (Related sections).** Sections $s$ of $E$ and $\bar{s}$ of $\bar{E}$ are related by $\Psi$ when

$$\Psi \circ s = \bar{s} \circ f. \tag{B2.43}$$

For a surjective submersion $f$, this excludes fine sections whose $\Psi$-images vary along a fiber of $f$.

**Definition B2.19 (Horizontal defect of a bundle morphism).** Following the covariant defect (B1.12), the horizontal defect of $\Psi$ at $e \in E_c$ is

$$(\mathcal{D}\Psi)_e(X) = A_\Psi(e; X) := T_e\Psi(H^\omega_e X) - H^{\bar{\omega}}_{\Psi(e)}(T_c f X), \qquad X \in T_c\mathcal{C}, \tag{B2.44}$$

with the transported fine horizontal lift first and the coarse horizontal lift subtracted. Its type is $A_\Psi \in \Gamma(E; \varpi^* T^*\mathcal{C} \otimes \Psi^* V\bar{E})$.

Both terms of (B2.44) project to $T_c f X$, so the defect is vertical and equals $\mathrm{ver}^{\bar{\omega}}(T_e\Psi(H^\omega_e X))$. Along a section, $A_\Psi(s; \cdot) \in \Gamma(\mathcal{U}; T^*\mathcal{U} \otimes f^* \bar{s}^* V\bar{E})$.

**Theorem B2.20 (Sharp section descent).** Let $f$ be a surjective smooth submersion, $\Psi$ a smooth bundle morphism over $f$, and $Q \in \Gamma(\mathcal{C}, E)$. Consider (P1) there is $\bar{Q} \in \Gamma(\bar{\mathcal{C}}, \bar{E})$ with $\Psi \circ Q = \bar{Q} \circ f$; (P2) $\Psi \circ Q$ is constant on each fiber of $f$; and

$$\textbf{(P3)} \qquad T^V\Psi\left(D^\omega Q(X)\right) + A_\Psi\left(Q(c); X\right) = 0 \quad \text{for every } c \text{ and every } X \in \ker T_c f. \tag{B2.45}$$

Then (P1) and (P2) are equivalent, both imply (P3), and (P3) implies (P2) when the fibers of $f$ are connected. Under (P2) the descended $\bar{Q}$ is unique, smooth, and a section. The submersion hypothesis cannot be dropped: for $f(x) = x^3$ on $\mathbb{R}$, the trivial bundle with fiber $\{\mathcal{N}(\mu, 1)\}$, $\Psi$ the identity on fibers, and $Q(x) = \mathcal{N}(x, 1)$, the unique descent $\bar{Q}(y) = \mathcal{N}(y^{1/3}, 1)$ is not differentiable at $y = 0$. [ESTABLISHED]

*Proof.* That (P1) implies (P2) is immediate. A surjective smooth submersion is a smooth quotient map, so a smooth map constant on its fibers descends uniquely to a smooth map; surjectivity gives uniqueness, and $\bar{\varpi}\bar{Q}f = \bar{\varpi}\Psi Q = f\varpi Q = f$ gives $\bar{\varpi}\bar{Q} = \mathrm{id}$. For (P2) implying (P3), take $X \in \ker T_c f$, split $T_c Q X = H^\omega_{Q(c)} X + D^\omega Q(X)$, apply $T\Psi$, and use $T_c f X = 0$: what remains is $T(\Psi Q)(X) = A_\Psi(Q(c); X) + T^V\Psi(D^\omega Q(X))$, which fiber constancy forces to vanish. The same computation shows that (P3) makes $T(\Psi Q)$ annihilate $\ker Tf$, and a smooth map with vanishing derivative along a connected embedded submanifold is constant on it. In the example $f$ is a bijection that fails to be a submersion at $0$. □

**Proposition B2.21 (A pointwise bundle morphism is not a map on configurations).** Suppose $f$ is a surjective submersion, $\ker T_{c_0} f \neq 0$ at some $c_0$, there are $e_0 \in E_{c_0}$ and $w \in V_{e_0}E$ with $T^V\Psi(w) \neq 0$, and a global smooth section $Q_0$ satisfies $Q_0(c_0) = e_0$. Then the projectable set $\Gamma_{\mathrm{proj}}(\Psi)$ of sections satisfying (P2) is a proper subset of $\Gamma(\mathcal{C}, E)$. For the total collapse $S^1 \to \{*\}$ with trivial bundles carrying $\{\mathcal{N}(\mu, 1)\}$, $\Psi$ the identity on fibers, and $Q(x) = \mathcal{N}(\sin x, 1)$, no coarse section exists, and in the linear realization on $L^2(S^1)$ the descendable set is the constants, a closed subspace of infinite codimension. If $f$ is a diffeomorphism every section descends. [ESTABLISHED]

*Proof.* Choose a chart in which $\ker Tf$ is spanned by $\partial_1, \ldots, \partial_k$ with $k \geq 1$. If $Q_0$ fails (P2) there is nothing to prove. Otherwise let $W$ be a smooth vertical field in a local trivialization with $W(Q_0(c_0)) = w$ and local flow $\Phi^W$, take $\chi \in C^\infty_c$ supported in the chart with $\chi(c_0) = 0$ and $\partial_1\chi(c_0) \neq 0$, and set $Q_\epsilon(c) = \Phi^W_{\epsilon\chi(c)}(Q_0(c))$ on the support and $Q_\epsilon = Q_0$ off it. Since $\chi(c_0) = 0$, $A_\Psi(Q_\epsilon(c_0); \partial_1)$ and $T^V\Psi$ at that point do not depend on $\epsilon$, and differentiating the left side of (B2.45) at $\epsilon = 0$ gives $\partial_1\chi(c_0)T^V\Psi(w) \neq 0$, so (P3) fails for small $\epsilon \neq 0$. In the collapse example $A_\Psi = 0$, $\ker Tf = TS^1$, and $T^V\Psi(D^\omega Q\partial_x) = \cos(x)\partial_\mu \neq 0$; a descendable section is constant on $S^1$. A diffeomorphism has trivial kernel and every fiber is a point. □

**Theorem B2.22 (Covariant first-jet chain rule).** For sections related by (B2.43),

$$D^{\bar{\omega}}\bar{s}(Tf X) = T^V\Psi(D^\omega s X) + (\mathcal{D}\Psi)_{s(c)}(X). \tag{B2.46}$$

Hence the square of covariant first jets commutes exactly when $\mathcal{D}\Psi$ vanishes along $s$, and then every vertical covariant tensor $\bar{\tau}$ on $\bar{E}$ satisfies

$$f^*[(D^{\bar{\omega}}\bar{s})^*\bar{\tau}] = (D^{\omega}s)^*[(T^V\Psi)^*\bar{\tau}]. \tag{B2.47}$$

[ESTABLISHED]

*Proof.* Differentiate (B2.43) and decompose $TsX = H^{\omega}_s X + D^{\omega}sX$. Applying $T\Psi$ to the horizontal term gives the coarse horizontal lift plus $(\mathcal{D}\Psi)_{s(c)}(X)$ by (B2.44); applying it to the vertical term gives $T^V\Psi(D^{\omega}sX)$. The vertical part of the other side is $D^{\bar{\omega}}\bar{s}(Tf X)$. When the defect vanishes, applying $\bar{\tau}$ to the intertwined jets gives (B2.47). □

## B2.8 The base Fisher comparison

A bundle morphism $\Psi$ over $f$ is *fiberwise Markov* when Hypothesis 7.1 holds at both scales and each fiber map $E_c \to \bar{E}_{f(c)}$, read in frames at $c$ and $f(c)$, satisfies Definition 7.11. The coarse metric is invariant under the represented action (Proposition 7.3), so $(T^V\Psi)^*\bar{g}^F$ does not depend on the frames, and the vertical Fisher defect $\Delta^{\Psi}_F = g^F - (T^V\Psi)^*\bar{g}^F$ of (7.15) is defined along $\Psi$ for every smooth $\Psi$.

**Definition B2.23 (Base comparison tensors).** Along related sections write $\mathsf{u}_X = D^{\omega}sX$ for the fine vertical jet, $\Psi_* = T^V\Psi$, $\mathsf{a}_X = A_{\Psi}(s(c); X)$ for the horizontal defect, and $\bar{\mathsf{u}}_X = D^{\bar{\omega}}\bar{s}(T_c f X) = \Psi_*\mathsf{u}_X + \mathsf{a}_X$ for the coarse vertical jet, by (B2.46). Define

$$\delta_{\Psi}(X,Y) = \Delta^{\Psi}_F(\mathsf{u}_X, \mathsf{u}_Y), \tag{B2.48}$$

$$\mathcal{X}_{\Psi}(X,Y) = \bar{g}^F(\Psi_*\mathsf{u}_X, \mathsf{a}_Y) + \bar{g}^F(\mathsf{a}_X, \Psi_*\mathsf{u}_Y), \tag{B2.49}$$

$$\mathcal{Q}_{\Psi}(X,Y) = \bar{g}^F(\mathsf{a}_X, \mathsf{a}_Y). \tag{B2.50}$$

**Corollary B2.24 (Base Fisher contraction).** Let $\Psi$ be fiberwise Markov, let $s, \bar{s}$ be related, and suppose $\mathcal{D}\Psi = 0$ along $s$. Then

$$h^{\omega}_s - f^*h^{\bar{\omega}}_{\bar{s}} = (D^{\omega}s)^*\Delta^{\Psi}_F \succeq 0, \tag{B2.51}$$

with equality on $X$ exactly when the fine score in the direction $D^{\omega}sX$ is measurable with respect to the coarse output. [ESTABLISHED]

*Proof.* In a frame pair $\Delta^{\Psi}_F$ is the vertical Fisher defect of the fiber map, so Corollary 7.12 gives $\Delta^{\Psi}_F \succeq 0$ with the stated equality case (7.18). With vanishing defect, (B2.47) for $\bar{\tau} = \bar{g}^F$ gives $f^*h^{\bar{\omega}}_{\bar{s}} = (D^{\omega}s)^*(T^V\Psi)^*\bar{g}^F$, and subtracting this from (B2.2) proves (B2.51). □

**Theorem B2.25 (Exact signed base Fisher comparison).** For sections related by (B2.43), with no hypothesis on the defect,

$$h_s^\omega - f^*\bar{h}_{\bar{s}}^{\bar{\omega}} = \delta_\Psi - \mathcal{X}_\Psi - \mathcal{Q}_\Psi \tag{B2.52}$$

as symmetric two-tensors on the fine domain. Here $\mathcal{X}_\Psi$ is symmetric and sign indefinite and $\mathcal{Q}_\Psi \succeq 0$, and both enter with a minus sign. [ESTABLISHED]

*Proof.* By (B2.46), $f^*\bar{h}(X,Y) = \bar{g}^F(\Psi_*\mathsf{u}_X + \mathsf{a}_X, \Psi_*\mathsf{u}_Y + \mathsf{a}_Y) = \bar{g}^F(\Psi_*\mathsf{u}_X, \Psi_*\mathsf{u}_Y) + \mathcal{X}_\Psi(X,Y) + \mathcal{Q}_\Psi(X,Y)$, and $\bar{g}^F(\Psi_*\mathsf{u}_X, \Psi_*\mathsf{u}_Y) = g^F(\mathsf{u}_X, \mathsf{u}_Y) - \delta_\Psi(X,Y)$. Subtracting from $h_s^\omega(X,Y) = g^F(\mathsf{u}_X, \mathsf{u}_Y)$ proves the identity. □

**Theorem B2.26 (Exact signed positivity criterion).** Let $\Psi$ be fiberwise Markov, so that $\delta_\Psi \succeq 0$. At a fixed $c$ the following are equivalent: $h_s^\omega - f^*\bar{h}_{\bar{s}}^{\bar{\omega}} \succeq 0$; the coarse jet is no longer than the fine jet, $\|\bar{\mathsf{u}}_X\|_{\bar{g}^F} \le \|\mathsf{u}_X\|_{g^F}$ for every $X$; and

$$2\bar{g}^F(\Psi_*\mathsf{u}_X, \mathsf{a}_X) + \|\mathsf{a}_X\|_{\bar{g}^F}^2 \le \delta_\Psi(X,X) \quad \text{for every } X. \tag{B2.53}$$

Vanishing of the horizontal defect is therefore sufficient for base positivity and not necessary, and the triangle-inequality margin $\|\mathsf{a}_X\|_{\bar{g}^F} \le \sqrt{h(X,X)} - \sqrt{h(X,X) - \delta_\Psi(X,X)}$ is sufficient and not necessary. Both failure directions occur: over $\mathcal{C} = \bar{\mathcal{C}} = \mathbb{R}$ with $f = \mathrm{id}$, fine fiber $\{\mathcal{N}(\mu,1)\}$, the channel adding independent $\mathcal{N}(0,1)$ noise, the section $s(x) = \mathcal{N}(x,1)$, zero source connection, and a target connection contributing $\mathsf{a}_X = \mathrm{c}\partial_\mu$, the comparison equals $1 - \frac{1}{2}(1+\mathrm{c})^2$, which is $\frac{79}{200}$ at $\mathrm{c} = \frac{1}{10}$ and $\frac{23}{25}$ at $\mathrm{c} = -\frac{3}{5}$, with nonzero defect in both cases and the margin violated in the second, and $-\frac{1}{8}$ at $\mathrm{c} = \frac{1}{2}$; with the identity kernel, constant related sections, source horizontal field $\partial_x$, and target horizontal field $\partial_x + \mathrm{c}\partial_\mu$ with $\mathrm{c} \neq 0$, the comparison is $-\mathrm{c}^2 dx^2 \prec 0$ at zero information loss. [ESTABLISHED]

*Proof.* The first two conditions are (B2.52) on the diagonal, with $h(X,X) = \|\mathsf{u}_X\|^2$ and $f^*\bar{h}(X,X) = \|\Psi_*\mathsf{u}_X + \mathsf{a}_X\|^2$. Expanding the square and substituting $\|\Psi_*\mathsf{u}_X\|^2 = \|\mathsf{u}_X\|^2 - \delta_\Psi(X,X)$ gives (B2.53). In the first example $g^F = 1$, $\bar{g}^F = \frac{1}{2}$, $\Delta_F^\Psi = \frac{1}{2}$, $\mathsf{u} = 1$, and $\bar{\mathsf{u}} = 1 + \mathrm{c}$; at $\mathrm{c} = -\frac{3}{5}$, $\|\mathsf{a}\| = \frac{3}{5\sqrt{2}}$ exceeds $1 - \frac{1}{\sqrt{2}}$. In the second, $\mathsf{u} = 0$ and $\mathsf{a} = -\mathrm{c}\partial_\mu$. □

Without the guard $\mathcal{D}\Psi = 0$, the vertical conclusion of Corollary 7.12 is unchanged, while (B2.51) is replaced by (B2.52) and its criterion (B2.53).

When weighted products (Definition B2.5) at two scales are compared, two further declarations are consumed. Writing $\mu_\mathcal{C}$ and $\bar{\mu}_{\bar{\mathcal{C}}}$ for finite positive base measures used to integrate the fine and coarse tensors and $w_x, \bar{w}_x$ for the two sets of channel weights, they are

$$\textbf{(X1)} \quad f_\# \mu_\mathcal{C} = \bar{\mu}_{\bar{\mathcal{C}}}, \qquad \textbf{(X2)} \quad \bar{w}_x \circ f \le w_x \quad \text{for } x \in \{b, m\}. \tag{B2.54}$$

**Proposition B2.27 (Cross-scale comparison of weighted products).** Under a channelwise comparison, with $h_x$ and $\bar{h}_x$ the channel pullbacks,

$$h^{\mathrm{prod}} - f^*\bar{h}^{\mathrm{prod}} = \sum_{x\in\{b,m\}} \left[w_x \left(h_x - f^*\bar{h}_x\right) + \left(w_x - \bar{w}_x \circ f\right) f^*\bar{h}_x\right]. \tag{B2.55}$$

If each channel is fiberwise Markov with vanishing horizontal defect, so that $\delta_x := h_x - f^*\bar{h}_x \succeq 0$ by Corollary B2.24, then (X2) makes the difference positive semidefinite, and without (X2) positivity fails on $X$ exactly when

$$\sum_{x\in\{b,m\}} (\bar{w}_x \circ f - w_x) f^*\bar{h}_x(X,X) > \sum_{x\in\{b,m\}} w_x \delta_x(X,X); \tag{B2.56}$$

for one channel with $f^*\bar{h}_x(X,X) = \delta_x(X,X) = \frac{1}{2}$, as for the unit-variance normal location section followed by additive unit-variance normal noise, unit fine weight gives $1 - \frac{1}{2}\bar{w}_x$, positive at $\bar{w}_x = \frac{3}{2}$ and negative at $\bar{w}_x = 3$. For $f$-related vector fields $X$ and $\bar{X}$ with finite integrals, (X1) gives

$$\int h^{\mathrm{prod}}(X,X)\mathrm{d}\mu_{\mathcal{C}} - \int \bar{h}^{\mathrm{prod}}(\bar{X},\bar{X})\mathrm{d}\bar{\mu}_{\bar{\mathcal{C}}} = \int (h^{\mathrm{prod}} - f^*\bar{h}^{\mathrm{prod}})(X,X)\mathrm{d}\mu_{\mathcal{C}},$$

which is nonnegative under the preceding hypotheses and (X2); without (X1) the left side can be negative with vanishing fiberwise defect: for $\mathcal{C} = \bar{\mathcal{C}}$ the disjoint union of two copies of $\mathbb{R}$, $f = \mathrm{id}$, $h$ zero on the first copy and $dx^2$ on the second, and unit point masses $\mu_{\mathcal{C}}$ and $\bar{\mu}_{\bar{\mathcal{C}}}$ on the first and second copy, the fine integral is $0$ and the coarse one $1$. [ESTABLISHED]

*Proof.* Expand $w_x h_x - (\bar{w}_x \circ f) f^*\bar{h}_x$. Under the hypotheses both summands in (B2.55) are positive semidefinite when (X2) holds, and in general the diagonal value is negative exactly under (B2.56). In the one-channel witness the value is $\frac{1}{2} + (1 - \bar{w}_x)\frac{1}{2}$. For $f$-related fields $\bar{h}(\bar{X},\bar{X}) \circ f = f^*\bar{h}(X,X)$, and the change-of-variables formula for $f_{\#}\mu_{\mathcal{C}} = \bar{\mu}_{\bar{\mathcal{C}}}$ moves the coarse integral to the fine domain. □

**Corollary B2.28 (Coarse or meta-agent perceived geometry).** The coarse section $\bar{s}$ carries its own connection-relative semimetric

$$\bar{h}^{\bar{\omega}}_{\bar{s}} = (D^{\bar{\omega}}\bar{s})^*\bar{g}^F. \tag{B2.57}$$

For related sections with $\mathcal{D}\Psi = 0$ along $s$,

$$f^*\bar{h}^{\bar{\omega}}_{\bar{s}} = (D^{\omega}s)^*(T^V\Psi)^*\bar{g}^F, \tag{B2.58}$$

and if $\Psi$ is fiberwise Markov the fine-minus-coarse perceived-geometry defect is

$$h^{\omega}_s - f^*\bar{h}^{\bar{\omega}}_{\bar{s}} = (D^{\omega}s)^*\Delta^{\Psi}_F \succeq 0. \tag{B2.59}$$

[ESTABLISHED]

*Proof.* The first display is Definition B2.1 at the coarse scale, the second is (B2.47) with $\bar{\tau} = \bar{g}^F$, and the third is Corollary B2.24. □

The comparison is contravariant: the coarse tensor is pulled back to the fine tangent space, and no pushforward of a Riemannian metric is used. If the coarse operation is a restriction along an embedding, an energy precomposition, or a parameter-dependent kernel, the Markov hypothesis fails and the corollary does not apply. A Galerkin aggregation of a Gaussian law is the pushforward through a precision-dependent map, which is a fixed kernel on each fixed-precision fiber and on no larger domain (Proposition G7.5); the corollary applies along such a fiber and not across precisions.

When levelwise agent or meta-agent sections and regular vertical Fisher tensors are declared at every level of a scale diagram, the geometric state may include their connection-relative perceived semimetrics, which the associated bundles alone do not supply. A meta-agent thus has its own pullback geometry, and its cross-scale naturality is Corollary B2.28. No tensor on one level base is subtracted from a tensor on another before both are transported to a common reference space.

**Definition B2.29 (Perceived-geometry beta).** If the reference-space declarations of Chapter B5 include base diffeomorphisms $i_\ell : \mathcal{C}_\ell \to \mathcal{C}_\star$ and a block factor $b_\ell > 1$, the discrete perceived-geometry beta on $\mathcal{C}_\star$ is

$$\widetilde{h}_\ell = (i_\ell^{-1})^* h_\ell, \qquad \mathfrak{B}_{h,\ell} = \frac{\widetilde{h}_{\ell+1} - \widetilde{h}_\ell}{\log b_\ell}, \tag{B2.60}$$

and on a differentiable reference-space curve $t \mapsto \widetilde{h}_t$ its continuous counterpart is $\partial_t \widetilde{h}_t$. Both are conditional on the declared identifications and scale normalization; without them $h_{\ell+1} - h_\ell$ is not a well-typed difference.

## B2.9 The scale-connection defect and the isotropy criterion

The horizontal defect of an induced morphism is not an arbitrary vertical field. Suppose $\Psi$ is induced by an equivariant principal scale map $\mathcal{P} : \mathscr{P} \to \bar{\mathscr{P}}$ over $f : \mathcal{C} \to \bar{\mathcal{C}}$ with Lie-group homomorphism $\kappa : G \to \bar{G}$ and law-fiber map $q$, in the sense of (11.3) and Proposition B1.31.

**Definition B2.30 (Scale-connection defect form).** The scale-connection defect form of $\mathcal{P}$ is

$$\mathfrak{A}_{\mathcal{P}} = \mathcal{P}^* \bar{\omega} - \mathrm{d}\kappa \circ \omega \in \Omega^1(\mathscr{P}, \bar{\mathfrak{g}}). \tag{B2.61}$$

**Theorem B2.31 (Isotropy criterion for a vanishing horizontal defect).** The form $\mathfrak{A}_{\mathcal{P}}$ is horizontal and $\mathrm{Ad} \circ \kappa$-equivariant, hence descends to an element of $\Omega^1(\mathcal{C}; f^* \operatorname{Ad} \bar{\mathscr{P}})$, and

$$A_\Psi(e; X) = \mathsf{J}_{\Psi(e)} \left( \mathfrak{A}_{\mathcal{P}}(X) \right), \tag{B2.62}$$

so the horizontal defect is the fundamental vertical field of a $\bar{\mathfrak{g}}$-valued one-form on the base. For related sections, $A_\Psi(s(c); X) = 0$ for every $c$ and every $X \in T_c\mathcal{C}$ if and only if

$$\mathfrak{A}_{\mathcal{P}}(X) \in \bar{\mathfrak{g}}_{\bar{s}(f(c))} \quad \text{for every } c \text{ and every } X, \tag{B2.63}$$

the isotropy subalgebra of the coarse section value. The principal identity $\mathfrak{A}_{\mathcal{P}} = 0$ is sufficient; where the isotropy subalgebra is trivial, meaning the action is locally free at $\bar{s}(f(c))$, the criterion forces $\mathfrak{A}_{\mathcal{P}}$ to vanish on $T_c\mathcal{C}$, and where it is nontrivial the criterion permits nonzero values in it. Moreover $\mathcal{Q}_\Psi(X,Y) = \mathfrak{k}_{\bar{s}(f(c))}(\mathfrak{A}_{\mathcal{P}}(X), \mathfrak{A}_{\mathcal{P}}(Y))$, where $\mathfrak{k}_{\bar{e}}(\xi,\eta) = \bar{g}^F(\mathsf{J}_{\bar{e}}\xi, \mathsf{J}_{\bar{e}}\eta)$ is the orbit Fisher form (7.9) of the coarse fiber, positive semidefinite with radical the isotropy subalgebra (Corollary 7.7). [ESTABLISHED]

*Proof.* Evaluating $\mathcal{P}^*\bar{\omega}$ on the fundamental field of $\xi$ gives $\bar{\omega}(\bar{\mathsf{f}}_{\mathrm{d}\kappa\xi}) = \mathrm{d}\kappa\xi$, so the difference annihilates vertical vectors; equivariance follows from $R_g^*\mathcal{P}^*\bar{\omega} = \mathrm{Ad}_{\kappa(g)^{-1}}\mathcal{P}^*\bar{\omega}$ and $\mathrm{d}\kappa \circ \mathrm{Ad}_{g^{-1}} = \mathrm{Ad}_{\kappa(g)^{-1}} \circ \mathrm{d}\kappa$. In local trivializations write $\Psi(c,\beta) = (f(c), \psi_c\beta)$ with $\psi_c = \widehat{\bar{\rho}}(\varsigma(c)) \circ q$. Using $H^{\omega}_{(c,\beta)}X = (X, -\mathsf{f}_{A(X)}\beta)$ and the differentiated intertwining $Tq \circ \mathsf{f}_\xi = \bar{\mathsf{f}}_{\mathrm{d}\kappa(\xi)} \circ q$, which is (11.4) at $g = \exp(t\xi)$, gives $A_\Psi(X) = \bar{\mathsf{f}}_{\mathfrak{a}(X)}(\psi_c\beta)$ with $\mathfrak{a} = \mathrm{Ad}_\varsigma(u^*\mathfrak{A}_{\mathcal{P}})$, which is (B2.62). A fundamental field vanishes at $\bar{\beta}$ exactly when its generator lies in the isotropy subalgebra, and conjugation by $\mathrm{Ad}_\varsigma$ moves both together, which gives (B2.63). The last statement is (B2.62) substituted into (B2.50). □

**Definition B2.32 (Connection compatibility).** A principal scale map is compatible with the connections along related sections when (B2.63) holds. No weaker or informal reading of compatibility is used in this supplement.

**Theorem B2.33 (Ordered composition of horizontal defects).** For composable morphisms $E_0 \xrightarrow{\Psi_{01}} E_1 \xrightarrow{\Psi_{12}} E_2$ over $f_{01}, f_{12}$, with $\Psi_{02} = \Psi_{12} \circ \Psi_{01}$ and $f_{02} = f_{12} \circ f_{01}$,

$$A_{\Psi_{02}}(e;X) = T^V\Psi_{12}\big|_{\Psi_{01}(e)}\left(A_{\Psi_{01}}(e;X)\right) + A_{\Psi_{12}}\left(\Psi_{01}(e); T_c f_{01} X\right), \tag{B2.64}$$

and for induced morphisms, at the connection level,

$$\mathfrak{A}_{\mathcal{P}_{02}} = \mathcal{P}_{01}^*\mathfrak{A}_{\mathcal{P}_{12}} + \mathrm{d}\kappa_{12} \circ \mathfrak{A}_{\mathcal{P}_{01}}. \tag{B2.65}$$

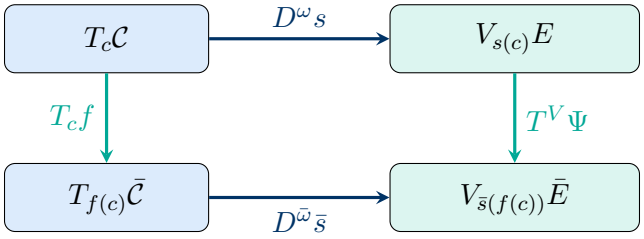
Consequently $A_{\Psi_{02}}(e;X) = 0$ holds if and only if

$$T^V\Psi_{12}\big|_{\Psi_{01}(e)} A_{\Psi_{01}}(e;X) = -A_{\Psi_{12}}\left(\Psi_{01}(e); T_c f_{01} X\right). \tag{B2.66}$$

Factorwise vanishing is sufficient and not necessary: on the abelian three-level instance with $f_{01}(x) = 2x$, $f_{12}(y) = 3y$, local connection forms $0$, $\mathrm{c}_1 dy$, $\mathrm{c}_2 dz$, and identity fiber maps, the composite defect is $2\mathrm{c}_1 + 2(3\mathrm{c}_2 - \mathrm{c}_1) = 6\mathrm{c}_2$, so $\mathrm{c}_2 = 0$ and $\mathrm{c}_1 \neq 0$ give a vanishing composite defect with both factor defects nonzero. [ESTABLISHED]

*Proof.* Expand $T\Psi_{01}(H^{\omega_0}_e X) = H^{\omega_1}_{\Psi_{01}e}(Tf_{01}X) + A_{\Psi_{01}}(e;X)$ and apply $T\Psi_{12}$. The horizontal summand contributes $H^{\omega_2}_{\Psi_{02}e}(Tf_{02}X) + A_{\Psi_{12}}(\Psi_{01}e; Tf_{01}X)$ by (B2.44), and the vertical summand contributes $T^V\Psi_{12}(A_{\Psi_{01}}(e;X))$ because a bundle morphism preserves verticality; subtracting $H^{\omega_2}(Tf_{02}X)$ gives (B2.64). The connection-level form follows from $T^V\Psi_{12} \circ \mathsf{J} = \mathsf{J} \circ \mathrm{d}\kappa_{12}$, or directly from $(\mathcal{P}_{12}\mathcal{P}_{01})^*\omega_2 - \mathrm{d}\kappa_{12}\mathrm{d}\kappa_{01}\omega_0 = \mathcal{P}_{01}^*(\mathcal{P}_{12}^*\omega_2 - \mathrm{d}\kappa_{12}\omega_1) + \mathrm{d}\kappa_{12}(\mathcal{P}_{01}^*\omega_1 - \mathrm{d}\kappa_{01}\omega_0)$. In the instance, the first defect is $2\mathrm{c}_1 \mathrm{d}x$ and the second is $(3\mathrm{c}_2 - \mathrm{c}_1)dy$, evaluated on the pushed vector $2\partial_y$. □

The order in (B2.64) is not cosmetic. The earlier defect takes values in $VE_1$ and the later one in $VE_2$, so the unweighted sum of the two is a type error before it is an error of fact: the earlier defect must be pushed by the later vertical differential, and the later one evaluated at the image point on the pushed base vector. In the instance, dropping the base pushforward gives $\mathrm{c}_1 + 3\mathrm{c}_2$.




$\mathcal{D}\Psi = 0$: the first-jet square commutes

**Figure B2.1:** Naturality of the connection-split first jet. The square commutes exactly under the isotropy criterion (B2.63). When the horizontal defect is nonzero, (B2.46) records the missing vertical term and (B2.52) records the two retained base tensors $-\mathcal{X}_\Psi$ and $-\mathcal{Q}_\Psi$.

**Corollary B2.34 (Null directions under a parallel coarse morphism).** If $\mathcal{D}\Psi = 0$ along $s$, then

$$Tf(\ker D^\omega s) \subseteq \ker D^{\bar\omega}\bar s, \tag{B2.67}$$

so a coarse morphism can create null directions, and first-jet naturality does not imply rank preservation. [ESTABLISHED]

*Proof.* Set $D^\omega sX = 0$ in (B2.46) and use $\mathcal{D}\Psi = 0$. □

## B2.10 The base Fisher defect cocycle

For composable bundle morphisms the vertical cocycle (7.19) holds at every point, since its proof is the contravariance of tensor pullback. Pulled back to the base it is not additive in general, and the residual is available in closed form. Fix sections $s_0, s_1, s_2$ related at each stage, write $\delta_{jk} = (D^{\omega_j} s_j)^* \Delta_F^{\Psi_{jk}}$ for the three base defects, and along $s_0$ write $\mathsf{p}_X = T^V\Psi_{01}(D^{\omega_0}s_0X)$, $\mathsf{a}_X = A_{\Psi_{01}}(s_0; X)$, and $\bar{\mathsf{u}}_X = D^{\omega_1}s_1(Tf_{01}X) = \mathsf{p}_X + \mathsf{a}_X$.

**Theorem B2.35 (Base Fisher defect cocycle: exact residual and sharp criterion).** The residual $\mathcal{R} := \delta_{02} - \delta_{01} - f_{01}^*\delta_{12}$ has the three equivalent exact forms

$$\mathcal{R}(X,Y) = \Delta_F^{\Psi_{12}}(\mathsf{p}_X, \mathsf{p}_Y) - \Delta_F^{\Psi_{12}}(\bar{\mathsf{u}}_X, \bar{\mathsf{u}}_Y), \tag{B2.68}$$

$$= -\Big[\Delta_F^{\Psi_{12}}(\bar{\mathsf{u}}_X, \mathsf{a}_Y) + \Delta_F^{\Psi_{12}}(\mathsf{a}_X, \bar{\mathsf{u}}_Y)\Big] + \Delta_F^{\Psi_{12}}(\mathsf{a}_X, \mathsf{a}_Y), \tag{B2.69}$$

$$= -\Big[\Delta_F^{\Psi_{12}}(\mathsf{p}_X, \mathsf{a}_Y) + \Delta_F^{\Psi_{12}}(\mathsf{a}_X, \mathsf{p}_Y)\Big] - \Delta_F^{\Psi_{12}}(\mathsf{a}_X, \mathsf{a}_Y). \tag{B2.70}$$

The sign of the quadratic term is coupled to the argument of the cross terms, and mixing the two conventions inflates the residual by exactly $2\Delta_F^{\Psi_{12}}(\mathsf{a}_X, \mathsf{a}_Y)$. The sharp base cocycle $\delta_{02} = \delta_{01} + f_{01}^*\delta_{12}$

holds if and only if

$$\Delta_F^{\Psi_{12}}(\mathsf{p}_X, \mathsf{p}_X) = \Delta_F^{\Psi_{12}}(\bar{\mathsf{u}}_X, \bar{\mathsf{u}}_X) \quad \text{for every } X, \tag{B2.71}$$

equivalently $\Delta_F^{\Psi_{12}}(\mathsf{a}_X, 2\mathsf{p}_X + \mathsf{a}_X) = 0$ for every $X$; if the second arrow is fiberwise Markov this is equality of two $\Delta_F^{\Psi_{12}}$-seminorms. [ESTABLISHED]

*Proof.* Applying $(D^{\omega_0} s_0)^*$ to (7.19) gives $\delta_{02} = \delta_{01} + \Delta_F^{\Psi_{12}}(\mathsf{p}_X, \mathsf{p}_Y)$, and by definition $f_{01}^* \delta_{12}(X, Y) = \Delta_F^{\Psi_{12}}(\bar{\mathsf{u}}_X, \bar{\mathsf{u}}_Y)$, which gives (B2.68). Substituting $\bar{\mathsf{u}} = \mathsf{p} + \mathsf{a}$ in the second term gives (B2.70), and $\mathsf{p} = \bar{\mathsf{u}} - \mathsf{a}$ in the first gives (B2.69). A symmetric form vanishes exactly when it vanishes on the diagonal. □

**Proposition B2.36 (Sufficient conditions for the sharp base cocycle).** Each of the following implies (B2.71): the stage-one horizontal defect vanishes, which by Theorem B2.31 is (B2.63) at the first arrow; the second arrow is Fisher lossless, $\Delta_F^{\Psi_{12}} = 0$; or $\mathsf{a}_X \in \operatorname{rad} \Delta_F^{\Psi_{12}}$ for every $X$. The first two each imply the third and are not ordered by strength, and none of the three is necessary. On the three-level instance with normal location fibers of variance $1, 2, 3$, sections $x \mapsto x$, $y \mapsto y/2$, $z \mapsto z/6$ of the means, base maps $f_{01}(x) = 2x$ and $f_{12}(y) = 3y$, additive normal noise channels, and local connection forms $0$, $\mathrm{c}_1 dy$, $\mathrm{c}_2 dz$, one has $\Delta_F^{\Psi_{12}} = \frac{1}{6}$, stage-one defect $2\mathrm{c}_1$, and $\mathcal{R} = -\frac{2}{3}\mathrm{c}_1(\mathrm{c}_1 + 1)$, which vanishes at $\mathrm{c}_1 = -1$ where the stage-one defect is $-2$; the convention-mixed expression is $\frac{2}{3}\mathrm{c}_1(\mathrm{c}_1 - 1)$, and at $\mathrm{c}_1 = \frac{1}{10}$ the two values are $-\frac{11}{150}$ and $-\frac{3}{50}$. [ESTABLISHED]

*Proof.* Every term of (B2.70) contains $\Delta_F^{\Psi_{12}}\mathsf{a}$, and each of the first two conditions makes it zero. At $\mathrm{c}_1 = 0$ of the instance the stage-one defect vanishes while $\Delta_F^{\Psi_{12}} = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$, and replacing its second arrow by the identity gives $\Delta_F^{\Psi_{12}} = 0$ with stage-one defect $2\mathrm{c}_1 \neq 0$ for $\mathrm{c}_1 \neq 0$. With the sign convention of Definition B2.3, the jets are $\mathsf{p} = 1$ and $\bar{\mathsf{u}} = 2(\frac{1}{2} + \mathrm{c}_1) = 1 + 2\mathrm{c}_1$ per $dx$, so $\mathcal{R} = \frac{1}{6}(1 - (1 + 2\mathrm{c}_1)^2)$, and the mixed expression $-2\Delta_F^{\Psi_{12}}(\mathsf{p}, \mathsf{a}) + \Delta_F^{\Psi_{12}}(\mathsf{a}, \mathsf{a})$ equals $\frac{1}{6}(4\mathrm{c}_1^2 - 4\mathrm{c}_1)$. □

The sharp base cocycle requires only the stage-one condition (B2.71). Separately, for each arrow $jk$, the base defect $\delta_{jk}$ equals the information loss $h_j - f_{jk}^* h_k$ exactly when $\mathcal{X}_{\Psi_{jk}} + \mathcal{Q}_{\Psi_{jk}} = 0$, by Theorem B2.25. Vanishing of the horizontal defect is sufficient for this identification, not necessary: the signed cross term can cancel the quadratic term. When these identifications hold, the sharp base cocycle is an additive decomposition of information loss on the base.

## B2.11 Intrinsic classification of connection defects

The positivity criterion (B2.53) and the cocycle criterion (B2.71) are conditions on the horizontal defect, and by Theorem B2.31 the defect is the fundamental field of the connection defect. One can therefore ask for all target connections that satisfy them. Fix the sections, the principal scale map, the fiber morphism, and the source connection. At one base point write $\mathcal{V} = T_c\mathcal{C}$, $E_0 = V_{s(c)}E$, and $E_1 = V_{\bar{s}(f(c))}\bar{E}$ with Fisher inner products $g_0$ and $g_1$; $\mathsf{D} = D^\omega s$ at $c$, $\Psi_* = T^V\Psi$, and the pushed jet $\mathsf{p} = \Psi_*\mathsf{D}$; $\mathfrak{a}_c = (f^* \operatorname{Ad} \bar{\mathscr{P}})_c$ and the fundamental-action map $\mathsf{J} = \mathsf{J}_{\bar{s}(f(c))} : \mathfrak{a}_c \to E_1$. A descended connection defect $\eta : \mathcal{V} \to \mathfrak{a}_c$ produces the anomaly $\mathsf{J}\eta$, with image in the orbit tangent $\mathcal{O} = \operatorname{im} \mathsf{J}$.

**Definition B2.37 (Relaxed pointwise defect problems).** The positivity problem and the cocycle problem are

$$(\mathsf{p}+\mathsf{J}\eta)^*g_1(\mathsf{p}+\mathsf{J}\eta)\preceq \mathsf{D}^*g_0\mathsf{D}, \tag{B2.72}$$

$$(\mathsf{p}+\mathsf{J}\eta)^*\Delta(\mathsf{p}+\mathsf{J}\eta)=\mathsf{p}^*\Delta\mathsf{p}, \tag{B2.73}$$

where $\Delta$ in the second is the second-arrow vertical Fisher defect. They are separate predicates; imposing both intersects their solution sets for the same $\eta$. Varying the source connection changes $\mathsf{D}$ and is a different problem.

**Proposition B2.38 (Relaxation and the kernel constraint).** Every defect realized by a target connection under related sections satisfies

$$(\mathsf{p}+\mathsf{J}\eta)|_{\ker T_cf}=0, \tag{B2.74}$$

which is condition (P3) of Theorem B2.20 at $c$. The problems of Definition B2.37 do not impose it, so the parameterizations below classify the relaxed pointwise problems, and a solution violating (B2.74) is realized by no lift. [ESTABLISHED]

*Proof.* By Theorem B2.22, $D^{\bar\omega}\bar s(T_cfX)=(\mathsf{p}+\mathsf{J}\eta)X$, and the left side vanishes for $X\in\ker T_cf$. The reference defect $\eta_0$ of Theorem B2.44 satisfies the constraint, so for a violating $\eta$ the difference $\mathsf{J}(\eta-\eta_0)$ does not vanish on $\ker T_cf$, and neither does $\eta-\eta_0$, which the annihilation clause of that theorem excludes for every lift. □

**Theorem B2.39 (Complete positivity parameterization).** Let $P_{\mathcal O}$ be the $g_1$-orthogonal projection onto $\mathcal O$ and set

$$\mathsf{p}_\perp=(I-P_{\mathcal O})\mathsf{p},\qquad \mathsf{p}_{\mathcal O}=P_{\mathcal O}\mathsf{p},\qquad \mathsf{H}=\mathsf{D}^*g_0\mathsf{D}-\mathsf{p}_\perp^*g_1\mathsf{p}_\perp. \tag{B2.75}$$

The positivity problem (B2.72) has a pointwise solution if and only if $\mathsf{H}\succeq 0$. In that case let $\mathrm{pr}_{\mathsf{H}}:\mathcal V\to\mathcal V_{\mathsf{H}}=\mathcal V/\operatorname{rad}\mathsf{H}$ be the quotient with the inner product induced by $\mathsf{H}$. All solutions are exactly

$$\mathsf{J}\eta=\mathsf{k}\,\mathrm{pr}_{\mathsf{H}}-\mathsf{p}_{\mathcal O},\qquad \mathsf{k}:(\mathcal V_{\mathsf{H}},\mathsf{H})\longrightarrow(\mathcal O,g_1)\text{ a linear contraction}, \tag{B2.76}$$

every such right side has a lift through $\mathsf{J}$, and the lift ambiguity is $\mathrm{Hom}(\mathcal V,\ker\mathsf{J})$. Zero-dimensional spaces and noninjective contractions are included. [ESTABLISHED]

**Corollary B2.40 (Markov first arrow: a convex feasible set).** If the first arrow is fiberwise Markov, then $\delta=\mathsf{D}^*(g_0-\Psi_*^*g_1\Psi_*)\mathsf{D}\succeq 0$ and

$$\mathsf{H}=\delta+\mathsf{p}_{\mathcal O}^*g_1\mathsf{p}_{\mathcal O}\succeq 0, \tag{B2.77}$$

so the feasible set of (B2.72) is nonempty, contains the zero anomaly, and is convex, as is its set of lifts. These facts do not establish descent to a target connection. [ESTABLISHED]

**Theorem B2.41 (Complete positive-semidefinite cocycle parameterization).** Suppose $\Delta \succeq 0$. Let $\mathrm{pr}_\Delta : E_1 \to E_\Delta = E_1/\operatorname{rad}\Delta$ be the metric quotient and define

$$\mathsf{J}_\Delta = \mathrm{pr}_\Delta \mathsf{J}, \qquad \mathcal{O}_\Delta = \operatorname{im}\mathsf{J}_\Delta, \qquad \mathsf{p}_\Delta = \mathrm{pr}_\Delta \mathsf{p}, \qquad \mathsf{w}_\Delta = P_\Delta \mathsf{p}_\Delta, \tag{B2.78}$$

with $P_\Delta$ the orthogonal projection onto $\mathcal{O}_\Delta$. The solutions of (B2.73) are exactly

$$\mathsf{J}_\Delta \eta = (\mathsf{T} - \iota)\mathsf{w}_\Delta, \qquad \mathsf{T} : \operatorname{im}\mathsf{w}_\Delta \longrightarrow \mathcal{O}_\Delta \text{ a linear isometric embedding}, \tag{B2.79}$$

with $\iota$ the inclusion and lift ambiguity $\mathrm{Hom}(\mathcal{V}, \ker \mathsf{J}_\Delta)$, which contains both action isotropy and action directions invisible to $\Delta$. The radical and isotropy sufficient conditions correspond to $\mathsf{T} = \iota$; $\mathsf{T} = -\iota$ gives $\mathsf{J}_\Delta\eta = -2\mathsf{w}_\Delta$ without a vanishing anomaly or a lossless second arrow; if $\mathsf{w}_\Delta = 0$ the equation forces $\mathsf{J}_\Delta\eta = 0$, whereas $\Delta = 0$ makes every $\eta$ a solution. Solutions realized under related sections additionally satisfy (B2.74). [ESTABLISHED]

**Theorem B2.42 (Complete signed cocycle parameterization).** Let $\Delta$ be any symmetric form on $E_1$. Put $\mathcal{O}_0 = \operatorname{rad}(\Delta|_\mathcal{O})$, choose a complement $\mathcal{O} = \mathcal{O}_1 \oplus \mathcal{O}_0$, and write $\Delta_1$ for the nondegenerate restriction to $\mathcal{O}_1$, with positive and negative indices $(n_+, n_-)$. Define $\mathsf{w} : \mathcal{V} \to \mathcal{O}_1$ and $\mathsf{l}_x \in \mathcal{O}_0^*$ by

$$\Delta_1(\mathsf{w}x, o) = \Delta(\mathsf{p}x, o) \quad (o \in \mathcal{O}_1), \qquad \mathsf{l}_x(n) = \Delta(\mathsf{p}x, n) \quad (n \in \mathcal{O}_0). \tag{B2.80}$$

For a linear map $\mathsf{n} : \mathcal{V} \to \mathcal{O}_0$ set

$$\mathsf{G}_\mathsf{n}(x, y) = \Delta_1(\mathsf{w}x, \mathsf{w}y) - \mathsf{l}_x(\mathsf{n}y) - \mathsf{l}_y(\mathsf{n}x). \tag{B2.81}$$

Admit exactly those $\mathsf{n}$ for which the positive and negative indices $(m_+, m_-)$ of $\mathsf{G}_\mathsf{n}$ obey $m_+ \le n_+$ and $m_- \le n_-$. With $\mathcal{K}_\mathsf{n} = \operatorname{rad}\mathsf{G}_\mathsf{n}$, choose a complement $\mathcal{V} = \mathcal{V}_\mathsf{n} \oplus \mathcal{K}_\mathsf{n}$, an isometric embedding $\mathsf{I}_\mathsf{n} : (\mathcal{V}_\mathsf{n}, \mathsf{G}_\mathsf{n}|_{\mathcal{V}_\mathsf{n}}) \to (\mathcal{O}_1, \Delta_1)$, and a linear map $\mathsf{Z} : \mathcal{K}_\mathsf{n} \to \mathsf{I}_\mathsf{n}(\mathcal{V}_\mathsf{n})^{\perp_{\Delta_1}}$ with totally isotropic image, $\Delta_1(\mathsf{Z}x, \mathsf{Z}y) = 0$ for all $x, y$. Define

$$\mathsf{k}(v + k) = \mathsf{I}_\mathsf{n} v + \mathsf{Z}k, \qquad \mathsf{J}\eta = \mathsf{k} - \mathsf{w} + \mathsf{n} \quad (v \in \mathcal{V}_\mathsf{n},\ k \in \mathcal{K}_\mathsf{n}). \tag{B2.82}$$

These are all solutions of (B2.73), with arbitrary lifts through $\mathsf{J}$ and ambiguity $\mathrm{Hom}(\mathcal{V}, \ker \mathsf{J})$; zero spaces are allowed, and $\mathsf{Z}$ may vanish, with rank at most $\min(n_+ - m_+, n_- - m_-)$. No Markov hypothesis is used. The restricted radical cannot be replaced by the ambient one: for $\Delta = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\mathcal{O} = \operatorname{span}(e_1)$, and $\mathsf{p}(X) = Xe_2$, one has $\mathcal{O}_0 = \mathcal{O}$, and $\mathsf{n}(X) = \mathrm{c}Xe_1$ gives $\mathsf{G}_\mathsf{n}(X, X) = -2\mathrm{c}X^2$, so the cocycle forces $\mathrm{c} = 0$, while for $\mathsf{p} = 0$ every $\mathsf{n}$ is allowed. [ESTABLISHED]

**Proposition B2.43 (Smooth solutions on regular strata).** Let the preceding data be smooth finite-rank bundles and bundle maps on a paracompact region of the base. For positivity assume constant ranks of $\mathsf{J}$ and $\mathsf{H}$ with $\mathsf{H} \succeq 0$; for the positive-semidefinite cocycle, constant ranks of $\Delta$, $\mathsf{J}_\Delta$, and $\mathsf{w}_\Delta$. Then the smooth solutions are exactly the smooth contractions or isometric bundle embeddings of eqs. (B2.76) and (B2.79), with smooth lifts and arbitrary smooth kernel-valued maps.

For the signed case assume constant ranks of J and $\Delta|_{\mathcal{O}}$, choose a smooth complement $\mathcal{O}_1$, and restrict to smooth n for which $G_n$ has constant rank and signature; the smooth solutions are exactly the smooth $I_n, Z$ and lifts of (B2.82). Inertia bounds give local smooth $I_n$ but not a global isometric embedding into the given $\mathcal{O}_1$ bundle, and agreement of the resulting k on overlaps is part of the parameterization. The classified sets are intrinsic: passive frame changes carry $D, \Psi_*, J, p$, their kernels and images, and the Fisher or signed forms by their tensor laws, and preserve contraction, isometry, inertia, and total isotropy; auxiliary metrics and splittings change only the parameters of one solution set, no positive invariant norm on a noncompact adjoint representation is assumed, and nothing extends automatically across a rank change. [ESTABLISHED]

**Theorem B2.44 (Exact realization by a target connection).** Fix the source connection $\omega$, the principal scale map $\mathcal{P}$, and a target reference connection $\bar{\omega}_0$, and let $\eta_0$ be the descended defect $\mathcal{P}^*\bar{\omega}_0 - \mathrm{d}\kappa \circ \omega$. An adjoint-valued one-form $\eta$ is realized by a target principal connection if and only if

$$\eta = \eta_0 + f^*\bar{\eta}, \qquad \bar{\eta} \in \Omega^1(\bar{\mathcal{C}}; \mathrm{Ad}\, \bar{\mathscr{P}}). \tag{B2.83}$$

For a surjective smooth submersion $f$, this holds exactly when $\delta\eta = \eta - \eta_0$ annihilates $\ker Tf$ and satisfies

$$\delta\eta_c(X) = \delta\eta_{c'}(X') \quad \text{whenever} \quad f(c) = f(c'), \quad T_c f X = T_{c'} f X', \tag{B2.84}$$

both values lying in the same target adjoint fiber and all components of each fiber included; the descended $\bar{\eta}$ is unique and smooth. The principal-realizable solutions are exactly the algebraic parameterizations, with all lift freedom retained, intersected with (B2.83). When $f$ is a diffeomorphism every smooth adjoint lift is realizable; for general $f$ the failure of one lift does not exclude another corrected by a kernel-valued one-form. No flatness or curvature equation is imposed on the target connection. [ESTABLISHED]

The proofs of the statements of this section and four witnesses for their boundaries, including an action-rank obstruction to smooth lifts and signed isotropic solutions outside the Markov domain, are in Appendix BA (Proposition BA.1).

# Chapter B3

# Section-Space Histories and Base Averaging

This chapter asks what becomes of the inference histories and Fisher clocks of chapter 7 when the state of an agent is a whole section over a contextual domain of positive dimension, rather than a point of its fibers over $c_*$. Three results carry the answer. A free, proper, isometric gauge action on a strong Riemannian Hilbert manifold of configurations has a smooth principal quotient on which the gauge-quotient speed is attained by a unique orthogonal lift, so quotient durations are well defined (theorem B3.10). The same conclusion fails for a weak Fisher metric even when the smooth quotient and a topological splitting exist, and the trace quotient of that example is not a fixed observation channel (propositions B3.11 and B3.12). Averaging sections over the fibers of a base map loses integrated Fisher information by an exact channel, weight, and context term under a joint-convexity condition on the fiber chart, and without that condition the loss can be negative (theorem B3.23 and proposition B3.24). The Hilbert theorem is the section-space counterpart of proposition 7.40, and the averaging defect is the counterpart of corollary 7.52 for a base map that collapses contexts; the covariant first jets and base-direction pullbacks used here are those of chapter B2.

Over the single point $c_*$ every construction of the chapter reduces to a fiber statement of the main manuscript. A section is then a point of the fiber, every base curve is constant, so every nonstationary curve is vertical and horizontality is vacuous; the design and continuum clocks reduce to the pointwise clock (7.20); the Hilbert quotient becomes one orthogonal projection; and averaging over the fibers of the identity of $\{c_*\}$ is the pointwise map of corollary 7.52.

The chapter declines to claim that the set of fine sections on which a pointwise bundle morphism induces a configuration map (theorem B2.20) is a smooth submanifold carrying a smooth induced arrow, so the coarse configuration map of section B3.7 is declared rather than constructed by section descent; that the base, the principal bundle, or the free energy selects a base measure, channel weights, or a strong manifold topology for a continuum clock; that the Hilbert theorem supplies a singular quotient when the action is not free or not proper; and that the tiers exhibited here classify section-space metrics. [NOT-CLAIMED] Whether a declared recognition family can be brought to an admitted configuration tier is open problem B3.14; the smooth structure of the projectable set is open problem B3.19; a probabilistic theory on continuum section spaces is open problem B1.22.

## B3.1 Fixed-base kinematics

**Definition B3.1 (Vertical, horizontal, and mixed curves).** For agent $i$ with contextual domain $\mathcal{C}_i$, restrict the belief and model bundles to $\mathcal{C}_i$ and form

$$\mathcal{E}_i = \mathcal{E}_b|_{\mathcal{C}_i} \times_{\mathcal{C}_i} \mathcal{E}_m|_{\mathcal{C}_i}, \qquad \varpi_i : \mathcal{E}_i \longrightarrow \mathcal{C}_i. \tag{B3.1}$$

The pair $\boldsymbol{\omega}_i = (\omega_b, \omega_m)$ of principal connections supplies a product horizontal distribution with horizontal lift $H^{\boldsymbol{\omega}_i}$ and a vertical projection $\mathrm{ver}^{\boldsymbol{\omega}_i}$. The vertical Fisher form of $\mathcal{E}_i$ is fixed only after positive channel weights or a joint statistical metric are declared. Let $\mathscr{I}$ be a connected interval and $\Gamma : \mathscr{I} \to \mathcal{E}_i$ a $C^1$ curve with base curve $\gamma = \varpi_i \circ \Gamma$. Relative to $\boldsymbol{\omega}_i$ its velocity splits uniquely as

$$\dot\Gamma = H^{\boldsymbol{\omega}_i}_{\Gamma}\dot\gamma + v_{\boldsymbol{\omega}_i}(\Gamma), \qquad v_{\boldsymbol{\omega}_i}(\Gamma) = \mathrm{ver}^{\boldsymbol{\omega}_i}(\dot\Gamma), \tag{B3.2}$$

and applying $T\varpi_i$, which is the identity on horizontal lifts and zero on vertical vectors, recovers $\dot\gamma$; this is the connection splitting of Kobayashi and Nomizu [17, Ch. II]. At a parameter value $r$ the velocity is *stationary* when $\dot\Gamma(r) = 0$; *vertical* when $\dot\gamma(r) = 0$ and $\dot\Gamma(r) \neq 0$; $\boldsymbol{\omega}_i$*-horizontal* when $v_{\boldsymbol{\omega}_i}(\Gamma)(r) = 0$ and $\dot\gamma(r) \neq 0$; and *mixed* when $\dot\gamma(r)$ and $v_{\boldsymbol{\omega}_i}(\Gamma)(r)$ are both nonzero. These four cases partition the parameter values. A curve is vertical, horizontal, or mixed on $\mathscr{I}$ when it has that type at every parameter value, so the interval-level labels are not exhaustive. Verticality depends only on the bundle projection; horizontality and the mixed decomposition depend on the connections.

A base curve carries no verticality predicate, since the predicate concerns a total-space velocity, and a curve of section configurations (section B3.2) carries one only through its evaluation at a fixed context, (B3.6). A vertical curve stays in one fiber, whereas a horizontal lift compares points of different fibers by the chosen connection; neither moves the base.

**Proposition B3.2 (Horizontality is connection relative).** The same nonvertical total-space curve can be mixed for one connection and horizontal for another. [ESTABLISHED]

*Proof.* Take the trivial line bundle $\mathbb{R}_c \times \mathbb{R}_y \to \mathbb{R}_c$ and the curve $\Gamma(r) = (r, r)$. For the horizontal distribution spanned by $\partial_c$, its vertical velocity is $\partial_y$, so the curve is mixed. For the distribution spanned by $\partial_c + \partial_y$, the same velocity is horizontal. The base trace $c = r$ is unchanged. □

**Proposition B3.3 (Lengths of vertical, horizontal, and section curves).** Fix one channel $x \in \{b, m\}$ and write $\omega = \omega_x$, $g^F = g^F_x$, and $E = \mathcal{E}_x$. For a $C^1$ curve $\Gamma : \mathscr{I} \to E$ over $\gamma$, let $v^\omega(\Gamma) = \mathrm{ver}^\omega(\dot\Gamma)$ and let $L^\omega(\Gamma) = \int_{\mathscr{I}} g^F(v^\omega(\Gamma), v^\omega(\Gamma))^{1/2} dr$ be its vertical Fisher length; on a curve confined to one fiber this is the Fisher length of (7.50). Then $L^\omega$ is invariant under orientation-preserving $C^1$ reparameterization; if $\Gamma$ is vertical, $L^\omega(\Gamma)$ does not depend on the connection; if $\Gamma$ is $\omega$-horizontal, $L^\omega(\Gamma) = 0$; and for a section curve $\Gamma = s \circ \gamma$, with the covariant first jet (B2.1),

$$L^\omega(s \circ \gamma) = \int_{\mathscr{I}} g^F\big(D^\omega s(\dot\gamma), D^\omega s(\dot\gamma)\big)^{1/2} dr, \tag{B3.3}$$

which is the length of $\gamma$ in the Fisher pullback (B2.2). If $\omega' = \omega + \Delta\omega$, with $\Delta\omega$ the descended difference form of (B2.5) and $\vartheta$ the fundamental vertical map of (B2.6), then along every curve

$$v^{\omega'}(\Gamma) = v^{\omega}(\Gamma) + \vartheta_{\Gamma}\big(\Delta\omega_{\gamma}(\dot\gamma)\big), \tag{B3.4}$$

so the length of a mixed curve generally depends on the connection. [ESTABLISHED]

*Proof.* Under $r = \phi(\tilde r)$ the vertical velocity is multiplied by $\phi'(\tilde r) > 0$, and positive homogeneity of the square root turns that factor into the Jacobian of the change of variables. For a vertical curve $\dot\gamma = 0$, so the horizontal part of $\dot\Gamma$ vanishes for every connection. Horizontality is the equation $v^{\omega}(\Gamma) = 0$. For $\Gamma = s \circ \gamma$ the chain rule gives $\dot\Gamma = Ts(\dot\gamma)$, hence $v^{\omega}(\Gamma) = D^{\omega}s(\dot\gamma)$, which is (B3.3). Finally, replacing $\omega$ by $\omega'$ changes the horizontal lift of $\dot\gamma$ at $\Gamma$ by $-\vartheta_{\Gamma}(\Delta\omega_{\gamma}(\dot\gamma))$ in the sign convention of (B2.6), and the vertical part of the fixed vector $\dot\Gamma$ changes by the opposite amount; for a section curve this is (B2.7). □

## B3.2 Curves of sections and pointwise vertical change

**Hypothesis B3.4 (Regular section configuration).** In this chapter $\mathfrak{S}_i$ is a declared space of pairs $Q = (q^b, q^m)$ of smooth sections of $\mathcal{E}_i \to \mathcal{C}_i$. Wherever a tangent-space calculation is used, $\mathfrak{S}_i$ is either a finite-dimensional submanifold of configurations or carries a specified smooth locally convex manifold structure for which the evaluation $(Q, c) \mapsto Q(c)$ is differentiable. This excludes the space of all smooth sections taken without a declared manifold structure, since none follows from writing the sections down.

Oriented histories in $\mathfrak{S}_i$ are the equivalence classes of regular representatives under orientation-preserving $C^1$ reparameterization, (7.41), exactly as in definition 7.31 with $\mathfrak{S}_i$ in place of the configuration manifold at $c_*$.

**Proposition B3.5 (Section histories change pointwise vertically).** Let $Q_i = (q_i^b, q_i^m) : \mathscr{I} \to \mathfrak{S}_i$ be a $C^1$ representative and define its adjoint family

$$\widehat{\mathrm{ev}}_i : \mathscr{I} \times \mathcal{C}_i \longrightarrow \mathcal{E}_i, \qquad \widehat{\mathrm{ev}}_i(r, c) = Q_i(r)(c), \qquad \varpi_i \circ \widehat{\mathrm{ev}}_i(r, c) = c. \tag{B3.5}$$

Then at every fixed context

$$T\varpi_i\big(\partial_r \widehat{\mathrm{ev}}_i(r, c)\big) = 0, \qquad \partial_r \widehat{\mathrm{ev}}_i(r, c) \in V_{Q_i(r)(c)}\mathcal{E}_i. \tag{B3.6}$$

With the connections held fixed, the family of Fisher pullbacks (B2.2) of the instantaneous sections,

$$\mathbf{P}_i(r) = \Big(\big(D^{\omega_b} q_i^b(r)\big)^* g_b^F, \big(D^{\omega_m} q_i^m(r)\big)^* g_m^F\Big), \tag{B3.7}$$

is an ordered family of base semimetrics that transforms under $\widetilde{Q}_i = Q_i \circ \phi$ by $\widetilde{\mathbf{P}}_i = \mathbf{P}_i \circ \phi$. [ESTABLISHED]

*Proof.* Differentiating $\varpi_i(\widehat{\mathrm{ev}}_i(r,c)) = c$ in $r$ at fixed $c$ gives (B3.6). Each entry of $\mathbf{P}_i(r)$ depends on $r$ only through $Q_i(r)$, because the connections are fixed, so reparameterization composes with $\phi$. □

A curve of full sections therefore produces a vertical statistical update at every fixed context although its state is an entire section, and it orders a family of perceived base geometries without making the base evolve. A separately declared connection history would add the correction terms of proposition B2.4 to (B3.7).

**Proposition B3.6 (Diagonal evaluation separates history change from base probing).** Let $Q_i : \mathscr{I} \to \mathfrak{S}_i$ be a $C^1$ representative, let $\gamma : \mathscr{I} \to \mathcal{C}_i$ be a $C^1$ base probe, set $\Gamma(r) = Q_i(r)(\gamma(r))$, and write

$$D^{\boldsymbol{\omega}_i} Q_i(r) = \big(D^{\omega_b} q_i^b(r), D^{\omega_m} q_i^m(r)\big) \tag{B3.8}$$

for the pair of covariant first jets (B2.1). Then

$$v_{\boldsymbol{\omega}_i}(\Gamma) = \partial_r \widehat{\mathrm{ev}}_i(r, \gamma(r)) + D^{\boldsymbol{\omega}_i} Q_i(r)\dot{\gamma}(r). \tag{B3.9}$$

At a parameter value with $\dot\gamma = 0$ the diagonal curve is vertical when the pointwise history velocity is nonzero and stationary when it vanishes; on an open subinterval where $\dot\gamma \neq 0$ it is mixed exactly when the sum in (B3.9) is nonzero and horizontal when the sum vanishes. Moreover every smooth section $Q$ of $\mathcal{E}_i \to \mathcal{C}_i$ is an embedding; a total-space curve has the form $Q \circ \gamma$ exactly when it lies in the image of $Q$, and then $\gamma = \varpi_i \circ \Gamma$ is unique. In particular a nonconstant curve confined to one fiber is not the image of a base curve under a section. [ESTABLISHED]

*Proof.* The chain rule for the jointly $C^1$ map $(r, c) \mapsto Q_i(r)(c)$, which hypothesis B3.4 supplies, gives $\dot\Gamma = \partial_r \widehat{\mathrm{ev}}_i(r, \gamma(r)) + TQ_i(r)\dot\gamma$. The first term is vertical by (B3.6), and $\mathrm{ver}^{\boldsymbol{\omega}_i} \circ TQ_i(r) = D^{\boldsymbol{\omega}_i} Q_i(r)$ by (B2.1); applying the vertical projection gives (B3.9). The type statements follow from definition B3.1, since $\dot\gamma = 0$ makes $\dot\Gamma$ equal to its vertical part. Since $\varpi_i \circ Q = \mathrm{id}$, the section is injective, $T\varpi_i \circ TQ = \mathrm{id}$ makes it an immersion, and $\varpi_i$ restricted to its image is a continuous inverse, so $Q$ is a homeomorphism onto its image. If $\Gamma = Q \circ \gamma$, then $\varpi_i \circ \Gamma = \gamma$; conversely a curve in the image satisfies $\Gamma = Q \circ \varpi_i \circ \Gamma$. A curve in one fiber has constant $\varpi_i \circ \Gamma$, so its only representation is a constant curve. □

## B3.3 Finite-design and continuum clocks

The pointwise clock of (7.20) uses only the vertical evaluation velocities of (B3.6) at one context. Over a base, two further clocks are available once their data are declared.

**Definition B3.7 (Finite-design and continuum clocks).** Fix positive channel weights $w_b, w_m$ on $\mathcal{C}_i$ and a $C^1$ section history $Q_i = (q_i^b, q_i^m)$. For a finite design $D = \{c_d\}_{d=1}^M \subset \mathcal{C}_i$ with positive design

weights $\mathsf{p}_d$, the finite-design squared speed is

$$\nu^2_{i,D} = \sum_{d=1}^{M} \mathsf{p}_d \left[ w_b(c_d) g^F_b\big(\partial_r q^b_i(c_d), \partial_r q^b_i(c_d)\big) + w_m(c_d) g^F_m\big(\partial_r q^m_i(c_d), \partial_r q^m_i(c_d)\big) \right]. \tag{B3.10}$$

For a finite positive base measure $\varrho_i$, measurable positive channel weights, and square-integrable evaluation velocities, the continuum squared speed is

$$\nu^2_{i,\varrho} = \int_{\mathcal{C}_i} \left[ w_b(c) g^F_b\big(\partial_r q^b_i(c), \partial_r q^b_i(c)\big) + w_m(c) g^F_m\big(\partial_r q^m_i(c), \partial_r q^m_i(c)\big) \right] d\varrho_i(c). \tag{B3.11}$$

Both clocks are declared data, and without their declarations only the pointwise clock is defined.

**Proposition B3.8 (The design clock has a large radical).** The finite-design speed (B3.10) is the product Fisher speed of the declared design and weights, and every section variation that vanishes at all $c_d$ lies in its radical. In particular every nonzero vertical variation in $T_Q\mathfrak{S}_i$ supported in an open set disjoint from $D$, such as a bump variation when $\mathfrak{S}_i$ admits one, is a nonzero element of the radical, and the design speed is then not positive definite on $T_Q\mathfrak{S}_i$. [ESTABLISHED]

*Proof.* Each summand of (B3.10) evaluates the variation only at a design point. □

**Lemma B3.9 (Local continuum metrics see no cross-context pairing).** Let a configuration metric on $T_Q\mathfrak{S}_i$ have the local single-integral form obtained by polarizing (B3.11). Then two variations whose contextual supports are disjoint have zero pairing. Consequently, if the exact Fisher bilinear form of a declared joint law of section configurations pairs two such variations nontrivially, that form is not of the local single-integral form, and replacing it by (B3.11) defines a local approximation. [ESTABLISHED]

*Proof.* For variations $V, W$ with disjoint supports the integrand $\sum_x w_x(c) g^F_x(V_x(c), W_x(c))$ vanishes at every $c$. □

Contextual dependence of a law alone therefore does not settle whether the continuum clock is its exact Fisher speed; that identity must be derived from the score covariance of the declared joint law, and any cross-context term of the exact metric must be retained.

## B3.4 Strong Hilbert quotients and attained duration

At $c_*$ with finitely many agents the configuration space is finite-dimensional, the gauge group is a finite-dimensional Lie group, and the gauge-quotient speed is one orthogonal projection (proposition 7.40). For configurations of sections the same formula needs a Hilbert category. For a smooth proper action of a Banach–Lie group on a Banach manifold, the infinitesimal orbit maps have closed range [15, Theorem 2.1], and the orbit space of a free proper action on a Hilbert manifold carries a unique Hilbert manifold structure for which the quotient map is a submersion [15, Corollary 2.5]; both use properness in the sense of (B3.12). The theorem below adds what

a quotient duration needs and the citation does not supply: the metric-orthogonal splitting, its attainment, the principal bundle structure, and a strong quotient metric making the projection a Riemannian submersion. Its proof is given in section B3.8.

**Theorem B3.10 (Strong Hilbert quotient duration).** Let $\mathcal{M}$ be a nonempty Hausdorff $C^\infty$ real Hilbert manifold with a $C^\infty$ strong Riemannian metric $\mathrm{g}$: the metric norm gives the declared tangent topology and, in each Hilbert chart, its positive metric operator is boundedly invertible and varies smoothly in operator norm. Let a Hausdorff $C^\infty$ real Banach–Lie group $\mathbb{G}$, with Lie algebra $\mathfrak{g}$, act jointly smoothly, freely, and isometrically on $\mathcal{M}$, and assume that the action map

$$\mathfrak{a} : \mathbb{G} \times \mathcal{M} \longrightarrow \mathcal{M} \times \mathcal{M}, \qquad \mathfrak{a}(g, Q) = (g \cdot Q, Q), \tag{B3.12}$$

is closed with compact fibers. Then the infinitesimal action

$$A_Q \xi = \frac{d}{dt}\Big|_{t=0} \exp(t\xi) \cdot Q \tag{B3.13}$$

is injective with closed range $V_Q$; the spaces $V_Q$ and $H_Q = V_Q^{\perp_{\mathrm{g}}}$ are smooth complementary subbundles; the orbit space $\mathcal{M}/\mathbb{G}$ is, componentwise, a Hausdorff smooth Hilbert manifold; and its projection $\pi$ is a smooth principal bundle and a Riemannian submersion for a strong quotient metric $\bar{\mathrm{g}}$. For every $Q \in \mathcal{M}$ and $v \in T_Q\mathcal{M}$,

$$\|d\pi_Q v\|^2_{\bar{\mathrm{g}}} = \min_{\xi \in \mathfrak{g}} \mathrm{g}_Q(v - A_Q\xi, v - A_Q\xi), \tag{B3.14}$$

the minimizing $\xi$ is unique and smooth in $(Q, v)$, and the quotient speed and its integrated duration do not depend on a $C^1$ history-dependent choice of representative. Neither metric completeness nor finite dimensionality of $\mathbb{G}$ is assumed. [ESTABLISHED]

Applied to a section configuration $(\mathcal{M}, \mathrm{g}) = (\mathfrak{S}_i, \mathsf{G}_i^F)$, the theorem requires that the Hilbert topology, the strong metric, the Banach–Lie gauge group, and the smooth proper isometric action be established first. Passive re-trivialization changes coordinates and adds no segment to a history.

## B3.5 A proper action with a weak Fisher quotient of zero speed

Closed range and a topological complement do not suffice for a weak metric, even when the smooth quotient and a global topological splitting exist.

**Proposition B3.11 (Proper weak-Fisher quotient obstruction).** Let $\mathcal{C} = \mathbb{R}/\mathbb{Z}$ carry normalized arclength $dc$, and set

$$\mathcal{M} = H^1(\mathcal{C}; \mathbb{R}), \qquad \|q\|^2_{H^1} = \int_{\mathcal{C}} (|q|^2 + |q'|^2) dc, \qquad \mathcal{K} = \{h \in \mathcal{M} : h(0) = 0\}, \tag{B3.15}$$

using the continuous periodic representative of each Sobolev class. The additive Hilbert Lie group $\mathcal{K}$, with the inherited $H^1$ topology, acts by $h \cdot q = q + h$. Equip $\mathcal{M}$ with

$$\mathrm{g}_q(v, w) = \int_{\mathcal{C}} v(c)w(c)dc. \tag{B3.16}$$

The action is smooth, free, proper, and isometric; its smooth quotient is $\mathbb{R}$, via $q \mapsto q(0)$; and $\mathcal{K}$ is closed and complemented in $H^1$. Nevertheless

$$\mathcal{K}^{\perp_{\mathrm{g}}} = \{0\}, \qquad \inf_{h\in\mathcal{K}} \|v - h\|_{\mathrm{g}}^2 = 0 \quad \text{for every } v \in H^1, \tag{B3.17}$$

and the infimum is attained exactly when $v(0) = 0$. The form g is the Fisher metric of the identifiable normalized joint observation family

$$P_q(dc, do) = \mathcal{N}\big(o; q(c), 1\big)(dc \otimes do), \tag{B3.18}$$

which is differentiable in quadratic mean in the $H^1$ parameter topology. Thus every nonzero quotient tangent has zero induced speed and no minimizing lift, although the quotient is smooth and proper. [ESTABLISHED]

*Proof.* Absolute continuity and the Cauchy–Schwarz inequality give

$$\|q\|_\infty \le \|q\|_2 + \|q'\|_2 \le \sqrt{2}\|q\|_{H^1}; \tag{B3.19}$$

for the first inequality bound $|q(c)|$ by $|q(t)| + \int_{\mathcal{C}} |q'|$ and average over $t$. The trace $\mathrm{tr}_0(q) = q(0)$ is therefore bounded, and

$$q \longmapsto \big(q - q(0)\mathbf{1}, q(0)\big) \quad \text{identifies} \quad H^1 \cong \mathcal{K} \oplus \mathbb{R} \tag{B3.20}$$

by bounded inverse maps. An $H^1$-Cauchy sequence converges uniformly by (B3.19) and its derivatives converge in $L^2$, so passing to the limit in $q(c) - q(0) = \int_0^c q'(t)dt$ gives an absolutely continuous periodic limit with that derivative; this verifies the Hilbert structure used for the group and the manifold.

The action is translation in the $\mathcal{K}$ factor of (B3.20); it is smooth and free, and its quotient is the displayed line. The action map $(h, q) \mapsto (q + h, q)$ is a homeomorphism onto the closed relation $\{(u, q) : \mathrm{tr}_0(u - q) = 0\}$, with inverse $(u, q) \mapsto (u - q, q)$, so it is closed with fibers that are singletons or empty, which is properness in the sense of (B3.12). Every translation has identity derivative and preserves g.

Let $\mathsf{b}(t) = \exp(1 - (1 - t^2)^{-1})$ for $|t| < 1$ and $\mathsf{b}(t) = 0$ otherwise, and for $n \ge 3$ let $\mathsf{b}_n(c) = \mathsf{b}(nc)$ in the signed circle coordinate $[-1/2, 1/2]$, extended periodically. These are smooth with $\mathsf{b}_n(0) = 1$, and with $\mathrm{c}_1 = \int_{-1}^1 \mathsf{b}^2 > 0$ and $\mathrm{c}_2 = \int_{-1}^1 (\mathsf{b}')^2 > 0$,

$$\|\mathsf{b}_n\|_2^2 = \frac{\mathrm{c}_1}{n}, \qquad \|\mathsf{b}_n'\|_2^2 = n\mathrm{c}_2, \qquad v^{(n)} = v - v(0)\mathsf{b}_n \in \mathcal{K}. \tag{B3.21}$$

Hence $\|v - v^{(n)}\|_2 = |v(0)|\sqrt{\mathrm{c}_1/n} \to 0$. If $v$ is g-orthogonal to $\mathcal{K}$, then $0 = \lim_n \mathrm{g}(v, v^{(n)}) = \|v\|_2^2$, so $v = 0$. The same approximation gives the zero infimum in (B3.17); attainment at $h$ would force $v - h = 0$, because the $L^2$ form is positive definite on $H^1$, and this is equivalent to $v(0) = 0$.

The metric is weak in the declared topology: its musical map is injective but does not represent the continuous trace covector, since $\mathrm{tr}_0 = \mathrm{g}(v, \cdot)$ would give

$$1 = \mathrm{tr}_0(\mathsf{b}_n) = \int_{\mathcal{C}} v\mathsf{b}_n dc, \qquad \left|\int_{\mathcal{C}} v\mathsf{b}_n dc\right| \le \|v\|_2\sqrt{\mathrm{c}_1/n} \longrightarrow 0. \tag{B3.22}$$

The derivative norm in (B3.21) diverges, so the bumps do not converge in $H^1$, and properness in that topology is not contradicted.

It remains to show that (B3.16) is a statistical tensor. The law (B3.18) is normalized by integrating each conditional Gaussian and then the unit circle measure. Equality of two joint laws gives equality of their conditional means almost everywhere in $c$, and continuity gives $q = \widetilde{q}$ everywhere, which is identifiability. Write $\mathrm{r}_\mu(o) = \mathcal{N}(o; \mu, 1)^{1/2}$ and $\mathrm{S}(q)(c, o) = \mathrm{r}_{q(c)}(o)$. The translation-invariant constant $\mathrm{c}_3 = \|\partial_\mu^2 \mathrm{r}_\mu\|_{L^2(do)}$ is finite, since the derivative is a quadratic polynomial times a Gaussian square root. Taylor's formula in $L^2(do)$, followed by integration over $c$, gives

$$\begin{aligned} D\mathrm{S}(q)[v](c, o) &= \tfrac{1}{2}v(c)\big(o - q(c)\big)\mathrm{S}(q)(c, o), \\ \|\mathrm{S}(q + h) - \mathrm{S}(q) - D\mathrm{S}(q)[h]\|_2 &\le \tfrac{1}{2}\mathrm{c}_3\|h\|_4^2 \le \tfrac{1}{2}\mathrm{c}_3\|h\|_\infty\|h\|_2 = O(\|h\|_{H^1}^2). \end{aligned} \tag{B3.23}$$

This is differentiability in quadratic mean with square-integrable directional score $\mathbf{s}_q[v](c, o) = v(c)(o - q(c))$. The conditional mean and variance of $O - q(C)$ are zero and one, so

$$\mathbb{E}_q\big[\mathbf{s}_q[v]\big] = 0, \qquad \mathbb{E}_q\big[\mathbf{s}_q[v]\mathbf{s}_q[w]\big] = \int_{\mathcal{C}} v(c)w(c)dc, \tag{B3.24}$$

the Fisher metric of the observed pair $(C, O)$, with $C$ retained. For each fixed $h \in \mathcal{K}$, the bimeasurable observation change $(c, o) \mapsto (c, o + h(c))$ pushes $P_q$ to $P_{q+h}$, so orbit equivalence is implemented by a declared coordinate action; it is not equality of these laws in fixed recorded coordinates. □

**Proposition B3.12 (The trace quotient is not a fixed observation channel).** No parameter-independent Markov kernel $K : \mathcal{C} \times \mathbb{R} \rightsquigarrow \mathbb{R}$ satisfies

$$P_q K = \mathcal{N}\big(q(0), 1\big) \quad \text{for every } q \in H^1(\mathcal{C}; \mathbb{R}). \tag{B3.25}$$

Attaching this normalized output family to the smooth parameter quotient of proposition B3.11 therefore gives neither a statistical coarse channel nor its quotient Fisher metric. [ESTABLISHED]

*Proof.* Use $\mathrm{TV}(P, P') = \sup_B |P(B) - P'(B)|$, half the $L^1$ distance for densities. Integrating the derivative of the standard normal density gives

$$\int_{\mathbb{R}} \big|\mathcal{N}(o; t, 1) - \mathcal{N}(o; 0, 1)\big| do \le |t| \int_{\mathbb{R}} \big|\partial_o \mathcal{N}(o; 0, 1)\big| do = |t|\sqrt{2/\pi}, \qquad \mathrm{TV}(P_q, P_0) \le \frac{\|q\|_1}{\sqrt{2\pi}}. \tag{B3.26}$$

Fix $\delta > 0$ and take $q_n = \delta \mathsf{b}_n$, with the bumps of the preceding proof. Its trace is $\delta$, whereas $\|q_n\|_1 = \delta n^{-1} \int_{-1}^{1} \mathsf{b} \to 0$. Every Markov kernel contracts total variation: for an output event $B$ the function $z \mapsto K(z, B)$ takes values in $[0, 1]$, and writing $K(z, B) = \int_0^1 \mathbf{1}_{\{K(z,B)>t\}} dt$ bounds its expectation difference by the total variation of the inputs. If (B3.25) held, it would give $\mathrm{TV}(\mathcal{N}(\delta, 1), \mathcal{N}(0, 1)) \to 0$. But the fixed event $(\delta/2, \infty)$ has probability difference

$$\mathcal{N}(\delta, 1)\big((\delta/2, \infty)\big) - \mathcal{N}(0, 1)\big((\delta/2, \infty)\big) = \int_{-\delta/2}^{\delta/2} \mathcal{N}(o; 0, 1) do > 0, \tag{B3.27}$$

a contradiction. A continuous conditional version at $C = 0$ exists, but selecting it at this zero-probability context is not the action of a fixed kernel on one joint observation. □

**Lemma B3.13 (Minimizing lifts are orthogonal lifts).** Let $V$ be a subspace of an inner-product space $T$. For $v \in T$, the element $k_* \in V$ minimizes $\|v - k\|$ over $k \in V$ exactly when $v - k_* \in V^\perp$. Hence every class of $T/V$ has a minimizing lift exactly when $T = V \oplus V^\perp$. [ESTABLISHED]

*Proof.* Differentiating $t \mapsto \|v - k_* - tk\|^2$ at $t = 0$ for every $k \in V$ proves necessity; if $v - k_* \in V^\perp$, Pythagoras gives $\|v - k\|^2 = \|v - k_*\|^2 + \|k_* - k\|^2$, which proves sufficiency. A class $v + V$ has a minimizing lift exactly when $v \in V + V^\perp$. □

A topological complement is therefore insufficient, as proposition B3.11 shows, and a smooth quotient metric built from minimizing lifts additionally needs a smooth compatible horizontal structure, which theorem B3.10 supplies in the strong case.

**Open Problem B3.14 (Admission of declared recognition families to a configuration tier).** The history theorems of chapter 7 hold on the tiers admitted by hypothesis 7.17, and theorem B3.10 covers free, proper, isometric actions on strong Riemannian Hilbert manifolds. A weak metric, a Fréchet group or manifold, a singular action, or a changing support is covered by neither. What is not settled is whether a recognition family that an application declares, in particular an infinite-dimensional law fiber or a space of sections, admits at every scale where a natural-gradient field or a Fisher duration is asserted a manifold calculus, its actual statistical metric, a smooth quotient, and a metric-orthogonal splitting. The problem is settled for a declared family by exhibiting these structures, or by exhibiting a family for which one of them provably fails while the history theorems are still invoked.

## B3.6 An exhibited finite-dimensional configuration tier

The finite-dimensional kind of hypothesis 7.17 admits every regular statistical manifold with a declared positive-definite metric. The following tier is a base-integrated Gaussian instance over a positive-dimensional base, with a strong constant metric, a rank test for nondegeneracy, and a closed gauge-orbit tangent, built from bundle data rather than assumed.

**Definition B3.15 (An exhibited finite-dimensional configuration tier).** Declare at scale $\ell$ a compact smooth base $\mathcal{C}_\ell$ with a finite positive Borel measure $\varrho_\ell$; the trivial principal bundle $\mathcal{C}_\ell \times G$ with $G = (\mathbb{R}^K, +)$ acting on the fiber by translation of the mean and $\omega$ the flat connection; the fiber $\mathcal{B} = \{\mathcal{N}(\mu, \Sigma_0) : \mu \in \mathbb{R}^K\}$ with fixed $\Sigma_0 \succ 0$, whose Fisher form is the constant $g^F = \Sigma_0^{-1}$; a measurable weight $w : \mathcal{C}_\ell \to (0, \infty)$ bounded above and below; and smooth fields $\psi_1, \dots, \psi_n : \mathcal{C}_\ell \to \mathbb{R}^K$ that are linearly independent in $L^2(w\varrho_\ell; \Sigma_0^{-1})$ and whose span contains every constant field. Set

$$\mathcal{Q}_\ell = \{s_{\mathbf{z}} : \mathbf{z} \in \mathbb{R}^n\}, \qquad s_{\mathbf{z}}(c) = \mathcal{N}\Big(\sum_j z_j \psi_j(c), \Sigma_0\Big). \tag{B3.28}$$

Let $\jmath_\ell : \mathbb{R}^K \to \mathbb{R}^n$ be the unique coefficient map with $\sum_j (\jmath_\ell g)_j \psi_j(c) = g$ for every $c$. Constant mean translations preserve the span and act on coefficients by

$$\mathscr{A}_g^\ell(\mathbf{z}) = \mathbf{z} + \jmath_\ell g. \tag{B3.29}$$

Only constant translations are declared: a context-dependent translation $c \mapsto g(c)$ maps $\mathcal{Q}_\ell$ into itself exactly when it lies in the span of the $\psi_j$, so the tier is not closed under general context-dependent gauge transformations. The configuration metric is the weighted product of marginal fiber metrics

$$\mathsf{G}_\ell(V, V) = \int_{\mathcal{C}_\ell} w(c) g^F\big(\partial_V s_{\mathbf{z}}(c), \partial_V s_{\mathbf{z}}(c)\big) d\varrho_\ell(c), \tag{B3.30}$$

not the pullback of a joint-law Fisher metric.

**Theorem B3.16 (The exhibited tier satisfies the regular metric hypothesis).** Under definition B3.15, $\mathcal{Q}_\ell$ is a nonempty smooth manifold diffeomorphic to $\mathbb{R}^n$, with $\mathbf{z} = 0$ the constant configuration $c \mapsto \mathcal{N}(0, \Sigma_0)$; the metric (B3.30) is the constant Gram form

$$\mathsf{G}_\ell = \mathsf{M}_\ell, \qquad (\mathsf{M}_\ell)_{jk} = \int_{\mathcal{C}_\ell} \psi_j(c)^\top \Sigma_0^{-1} \psi_k(c) w(c) d\varrho_\ell(c), \tag{B3.31}$$

independent of $\mathbf{z}$, finite, and positive definite, hence strong; every $C^1$ functional has the unique gradient $\mathsf{M}_\ell^{-1}\nabla\mathcal{F}$, and for $\mathcal{F} \in C^2$ the natural-gradient field is locally Lipschitz with locally unique integral curves; and the gauge-orbit tangent of constant mean translations is the closed complemented subspace $\jmath_\ell(\mathbb{R}^K)$, so the minimum in (B3.14) is attained at the $\mathsf{M}_\ell$-orthogonal projection. The tier is therefore an instance of the finite-dimensional kind of hypothesis 7.17. [ESTABLISHED]

*Proof.* The bundle is trivial and $s_{\mathbf{z}}$ is smooth jointly in $(c, \mathbf{z})$. If $s_{\mathbf{z}} = s_{\mathbf{y}}$, then $\sum_j (z_j - y_j)\psi_j = 0$ in $L^2(w\varrho_\ell; \Sigma_0^{-1})$, and independence gives $\mathbf{z} = \mathbf{y}$, so the family is diffeomorphic to its coefficient space. Since $\partial_V s_{\mathbf{z}}(c) = \sum_j v_j \psi_j(c)$ does not depend on $\mathbf{z}$, the integrand of (B3.30) is $\mathbf{v}^\top \mathsf{M}_\ell \mathbf{v}$; finiteness follows from compactness of the base and boundedness of $w$. If $\mathbf{v}^\top \mathsf{M}_\ell \mathbf{v} = 0$, then $\sum_j v_j \psi_j = 0$ in $L^2(w\varrho_\ell; \Sigma_0^{-1})$, so $\mathbf{v} = 0$. An inner product on a finite-dimensional space induces its norm topology and has an invertible musical map, which gives strongness and the gradient. The span condition defines $\jmath_\ell$ and gives $s_{\mathscr{A}_g^\ell(\mathbf{z})} = g \cdot s_{\mathbf{z}}$; a finite-dimensional subspace is closed and complemented, and lemma B3.13 gives the attained minimum. □

**Remark B3.17 (Nondegeneracy is a rank condition).** Nondegeneracy can be checked before any dynamics is written. With an $M$-atom design $\varrho_\ell = \sum_{d=1}^{M} \mathsf{p}_d \delta_{c_d}$, the Gram matrix has rank at most $MK$, so for $n > MK$ the metric is degenerate and, by proposition 7.34, the natural-gradient equation has no solution or many. For $\mathcal{C}_\ell = \mathbb{R}/2\pi\mathbb{Z}$ with normalized arclength, $K = 1$, $\Sigma_0 = 1$, $w \equiv 1$, and $\{\psi_j\} = \{1, \cos c, \sin c\}$, the Gram matrix is $\mathrm{diag}(1, \frac{1}{2}, \frac{1}{2})$, with determinant $\frac{1}{4}$, so $\mathcal{Q}_\ell \cong \mathbb{R}^3$ carries a positive-definite constant metric. [ESTABLISHED]

*Proof.* Under an $M$-atom design each field enters $\mathsf{M}_\ell$ only through its values at the atoms, so $\mathsf{M}_\ell = \mathsf{V}^\top \mathsf{D} \mathsf{V}$, where $\mathsf{V} \in \mathbb{R}^{MK \times n}$ stacks the values $\psi_j(c_d)$ and $\mathsf{D}$ is block diagonal with blocks $\mathsf{p}_d w(c_d) \Sigma_0^{-1}$. On the circle, orthogonality of $1, \cos c, \sin c$ under normalized arclength gives the diagonal entries $1, \frac{1}{2}, \frac{1}{2}$. □

**Proposition B3.18 (A strong $L^2$ tier and a weak Sobolev tier).** For the integrated normal-mean metric $\mathsf{G}^{L^2}(V, W) = \int V^\top \Sigma_0^{-1} W w d\varrho$ on $\mathcal{Q}^{L^2} = L^2(\varrho; \mathbb{R}^K)$, the bounds $0 < w_- \le w \le w_+ < \infty$ and $\mathrm{c}_- I \preceq \Sigma_0^{-1} \preceq \mathrm{c}_+ I$ give $w_- \mathrm{c}_- \|V\|_{L^2}^2 \le \mathsf{G}^{L^2}(V, V) \le w_+ \mathrm{c}_+ \|V\|_{L^2}^2$, so the musical map is a topological isomorphism and the metric is strong; if the $\varrho$-essential infimum of $w$ is zero or its essential supremum is infinite, the form is not equivalent to the $L^2$ norm. On the circle $\mathcal{C} = \mathbb{R}/2\pi\mathbb{Z}$ with normalized arclength, $K = 1$, and $\Sigma_0 = w = 1$, the same metric on a Sobolev tier $H^s$, $s > 0$, induces a strictly coarser topology than the manifold topology, and its musical map is injective and bounded but not surjective, so the metric is weak. The failure is realized for $\mathcal{Q} = H^1$ and $\mathcal{F}(Q) = \frac{1}{2} \int |Q'|^2$: at $Q = \sum_{j \ge 1} j^{-2} \sin jc$, which lies in $H^1$, a gradient would be $-Q'' = \sum_j \sin jc$, which is not in $L^2$, so no natural-gradient vector field exists at $Q$. [ESTABLISHED]

*Proof.* The two-sided bounds are the pointwise bounds integrated. If the essential infimum of $w$ is zero, the sets $\{w < 1/k\}$ have positive measure, and the normalized indicator functions of these sets times a unit vector have $\mathsf{G}^{L^2}(V, V) \le \mathrm{c}_+ / k$; an infinite essential supremum gives $\mathsf{G}^{L^2}(V, V) \ge \mathrm{c}_- k$ in the same way. On the circle, in Fourier coefficients $\hat{v}_j$, the $H^s$ norm is $\sum_j (1 + j^2)^s |\hat{v}_j|^2$ and the metric is $\sum_j |\hat{v}_j|^2$, so the two are not equivalent. The functional $\ell(v) = \sum_{j \ge 1} j^{s - 1/2} \log(1 + j)^{-1} \hat{v}_j$ is $H^s$-continuous, since $\sum_j j^{2s-1} \log(1 + j)^{-2} (1 + j^2)^{-s} < \infty$, but it is represented by no $L^2$ function, since $\sum_j j^{2s-1} \log(1 + j)^{-2} = \infty$. In the example $\|Q\|_{H^1}^2$ is a multiple of $\sum_j (1 + j^2) j^{-4} < \infty$, while $\|Q''\|_2^2$ is a multiple of $\sum_j 1 = \infty$, and $d\mathcal{F}_Q[V] = \int Q' V' = -\int Q'' V$ is $\mathsf{G}^{L^2}$-representable only by $-Q''$. □

The missing ingredient on a weak Sobolev tier is representability of $d\mathcal{F}_Q$ by a vector in its declared tangent space. A bound $|d\mathcal{F}_Q[V]| \le \mathrm{c}_Q \|V\|_{L^2}$ on a dense domain gives an $L^2$ Riesz representative, but a gradient on the $H^s$ tier additionally requires that representative to belong to $H^s$. For example, on the circle with normalized arclength, $f(c) = \sum_{j \ge 1} j^{-1} \sin(jc)$ lies in $L^2 \setminus H^1$, and $\mathcal{F}(Q) = \int fQ$ is a continuous linear functional on $H^1$ with the displayed bound but no $H^1$ gradient for the $L^2$ metric. The price of the strong $L^2$ tier is that $\mathcal{F}$ must be $C^1$ on $L^2$, which excludes gradient-energy objectives.

## B3.7 Base averaging and the coarse configuration map

A pointwise bundle morphism induces a map on configurations only on the projectable set of theorem B2.20, and under the reachability hypotheses of proposition B2.21 that set is a proper subset whenever the base map collapses a direction on which the vertical differential does not vanish, and it can have empty interior in the linear tier. The coarse configuration map $\mathsf{F}_\ell$ of (7.65) is therefore declared separately, and averaging over the fibers of a base map is its standard instance.

**Open Problem B3.19 (Smooth structure of the projectable set).** Theorem B2.20 characterizes the fine sections that descend, and proposition B2.21 shows that the descendable set is proper and can have empty interior. It is not known under which hypotheses that set is a smooth submanifold of a declared configuration manifold carrying a smooth induced configuration map. The problem is settled by a transversality or elliptic-regularity hypothesis on the base map and the recognition family under which the projectable set is proved to be such a submanifold, or by a family in which it provably is not.

**Definition B3.20 (Averaging coarse configuration map).** Let $f : \mathcal{C}_\ell \to \mathcal{C}_{\ell+1}$ be measurable between standard Borel spaces, let $\varrho$ be a finite positive measure on $\mathcal{C}_\ell$ with $\bar{\varrho} = f_\# \varrho$, and let $\{\varrho_{\bar{c}}\}$ disintegrate $\varrho$ over $f$. Let the coarse fiber $\bar{\mathcal{B}}$ be convex in a locally convex space on which the structure group acts by restrictions of linear maps, and let $\Psi$ act fiberwise as the restriction of a continuous affine map with linear part $L$, with the integrand Bochner integrable and its barycenter in the fiber. The averaging map is

$$\left(\mathsf{F}_\ell s\right)(\bar{c}) = \int_{f^{-1}(\bar{c})} \Psi\big(s(c)\big)\, \varrho_{\bar{c}}(dc). \tag{B3.32}$$

The affine chart of $\bar{\mathcal{B}}$ is *jointly convex* when $(\bar{\beta}, u) \mapsto \bar{g}^F_{\bar{\beta}}(u,u)$ is jointly convex on $\bar{\mathcal{B}}$ times its model space; the special case in which $\bar{g}^F$ is constant in the chart is written (JC-const).

**Proposition B3.21 (Averaging reproduces descended sections).** In definition B3.20, if $\Psi \circ s$ is $f$-basic with unique descended section $\bar{s}$, then $(\mathsf{F}_\ell s)(\bar{c}) = \bar{s}(\bar{c})$ for $\bar{\varrho}$-almost every $\bar{c}$. [ESTABLISHED]

*Proof.* For $\bar{\varrho}$-almost every $\bar{c}$ the integrand equals $\bar{s}(\bar{c})$ for $\varrho_{\bar{c}}$-almost every point of the fiber, and the disintegration is normalized. □

**Hypothesis B3.22 (Chart joint convexity).** Where theorem B3.23 asserts a nonnegative averaging loss, the affine chart of the coarse fiber is jointly convex in the sense of definition B3.20. This is a restriction on the chart, not a property of Fisher metrics, and it excludes charts such as the covariance sector of the Gaussian moment chart (proposition B3.24).

**Theorem B3.23 (Exact averaging defect).** Assume definition B3.20, hypothesis B3.22, the fiberwise Fisher contraction $\Delta^\Psi_F = g^F - L^* \bar{g}^F L \succeq 0$, which holds when $\Psi$ is a Markov coarse morphism by corollary 7.12, the weight condition $\bar{w} \circ f \le w$ of (B2.54), and finiteness of every

integral. Then $\mathsf{F}_\ell s$ is a section, $\mathsf{F}_\ell$ is gauge equivariant because the transition maps act linearly and commute with barycenters, and for every $Z \in T_s\mathcal{Q}_\ell$ the defect

$$\Delta_{\mathrm{avg}}(Z) = \int_{\mathcal{C}_\ell} w g^F_{s(c)}(Z,Z)d\varrho - \int_{\mathcal{C}_{\ell+1}} \bar{w}\bar{g}^F_{(\mathsf{F}_\ell s)(\bar{c})}\big(T\mathsf{F}_\ell Z, T\mathsf{F}_\ell Z\big)d\bar{\varrho} \tag{B3.33}$$

is nonnegative. Under (JC-const) it is exactly the sum of three nonnegative terms,

$$\Delta_{\mathrm{avg}}(Z) = \underbrace{\int_{\mathcal{C}_\ell} w\Delta^\Psi_F(Z,Z)d\varrho}_{\text{channel loss}} + \underbrace{\int_{\mathcal{C}_\ell} (w - \bar{w}\circ f)\big(L^*\bar{g}^F\big)(Z,Z)d\varrho}_{\text{weight gap}} + \underbrace{\int_{\mathcal{C}_{\ell+1}} \bar{w}\,\mathrm{Var}^{\bar{g}^F}_{\varrho_{\bar{c}}}(LZ)d\bar{\varrho}}_{\text{context gap}}, \tag{B3.34}$$

where $\mathrm{Var}^{\bar{g}}_{\varrho_{\bar{c}}}(U) = \int \bar{g}(U,U)d\varrho_{\bar{c}} - \bar{g}\big(\int U d\varrho_{\bar{c}}, \int U d\varrho_{\bar{c}}\big)$. [ESTABLISHED]

*Proof.* Affineness gives $T\mathsf{F}_\ell Z(\bar{c}) = \int LZ(c)\varrho_{\bar{c}}(dc)$, so $(\mathsf{F}_\ell s(\bar{c}), T\mathsf{F}_\ell Z(\bar{c}))$ is the $\varrho_{\bar{c}}$-barycenter of $(\Psi s(c), LZ(c))$. Joint convexity and Jensen's inequality give

$$\bar{g}^F_{\mathsf{F}_\ell s(\bar{c})}(T\mathsf{F}_\ell Z, T\mathsf{F}_\ell Z) \le \int \bar{g}^F_{\Psi s(c)}(LZ, LZ)d\varrho_{\bar{c}}.$$

Multiplying by $\bar{w}$, integrating against $\bar{\varrho}$, and using the disintegration turns the right side into $\int(\bar{w}\circ f)(L^*\bar{g}^F)(Z,Z)d\varrho$; the weight condition and $\Delta^\Psi_F \succeq 0$ then give nonnegativity. When $\bar{g}^F$ is constant in the chart, the Jensen step is an equality minus the variance term, and expanding $g^F = \Delta^\Psi_F + L^*\bar{g}^F$ separates the channel and weight contributions, which gives (B3.34). □

Corollary 7.52 is the case of a shared base, where the disintegration is a point mass and no convexity is needed; the averaging defect admits a collapsing base map at the price of affine-fiber, convexity, and barycenter hypotheses, so no general comparison of logical strength between the two results is asserted.

**Proposition B3.24 (Joint convexity cannot be dropped).** The joint-convexity hypothesis holds in the location sector of a Gaussian fiber with fixed covariance, where the metric is constant, and in the law chart, where the integrand $(p,\dot{p}) \mapsto \dot{p}^2/p$ has Hessian determinant zero and positive trace. It fails in the covariance sector of the Gaussian moment chart, where $(\Sigma,\dot{\Sigma}) \mapsto \dot{\Sigma}^2/(2\Sigma^2)$ has Hessian determinant $-\dot{\Sigma}^2\Sigma^{-6} < 0$ for $\dot{\Sigma} \neq 0$, and there the averaging defect can be negative: for two contexts with the uniform disintegration, unit weights, $\Psi$ the identity, the centered normal fiber $\{\mathcal{N}(0,\Sigma)\}$, the configuration $(\Sigma_1,\Sigma_2) = (1,\delta)$, and the tangent $Z = (1,0)$, the fine integrated metric is $\frac{1}{4}$, the image metric is $1/(2(1+\delta)^2)$, their ratio is $2/(1+\delta)^2$, equal to $\frac{20000}{10201}$ at $\delta = 10^{-2}$, and the defect is negative exactly for $\delta < \sqrt{2}-1$, tending to $-\frac{1}{4}$ as $\delta \to 0^+$. Every other hypothesis of theorem B3.23 holds on this datum. [ESTABLISHED]

*Proof.* The Hessians are direct computations. With the uniform disintegration the fine metric is $\frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot 0 = \frac{1}{4}$; the barycenter configuration is $\bar{\Sigma} = (1+\delta)/2$ with tangent $\frac{1}{2}$, whose metric is $(\frac{1}{2})^2/(2\bar{\Sigma}^2) = 1/(2(1+\delta)^2)$. The defect $\frac{1}{4} - 1/(2(1+\delta)^2)$ is negative exactly when $(1+\delta)^2 < 2$. $\Delta^\Psi_F = 0$ and the weights are equal, so only joint convexity fails. □

No contraction statement may therefore be attached to a generic averaging or variational coarse map.

**Proposition B3.25 (Smoothness and gauge descent at the exhibited tier).** Take the averaging map (B3.32) with $\bar{\varrho} = f_{\#}\varrho$, its disintegration $\{\varrho_{\bar{c}}\}$, and $\Psi$ acting on means as the identity. Extend definition B3.15 to scale $\ell+1$ with base $\mathcal{C}_{\ell+1}$, measure $\bar{\varrho}$, weight $\bar{w}$, coarse fiber $\{\mathcal{N}(\mu, \bar{\Sigma}_0)\}$, and a coarse basis $\bar{\psi}_1, \dots, \bar{\psi}_{\bar{n}}$ linearly independent in $L^2(\bar{w}\bar{\varrho}; \bar{\Sigma}_0^{-1})$ whose span contains every constant field, with constant coefficient map $\jmath_{\ell+1}$ as in (B3.29). Suppose the fiberwise average of each fine basis field lies in the coarse span, $\int_{f^{-1}(\bar{c})} \psi_j d\varrho_{\bar{c}} = \sum_k \mathbf{E}_{kj}\bar{\psi}_k(\bar{c})$ for $\bar{\varrho}$-almost every $\bar{c}$, with a constant matrix $\mathbf{E} \in \mathbb{R}^{\bar{n}\times n}$. Then the averaging map reads $\mathsf{F}_\ell(\mathbf{z}) = \mathbf{E}\mathbf{z}$ in coordinates; it is linear, hence $C^\infty$, with tangent map $\mathbf{E}$; it intertwines the coefficient representations of the constants,

$$\mathbf{E}\jmath_\ell = \jmath_{\ell+1}, \tag{B3.35}$$

so it is gauge equivariant, $\mathbf{E}\mathscr{A}_g^\ell = \mathscr{A}_g^{\ell+1}\mathbf{E}$; and its metric compatibility $\mathsf{F}_\ell^*\mathsf{G}_{\ell+1} \preceq \mathsf{G}_\ell$ is the matrix inequality

$$\mathbf{E}^\top \overline{\mathsf{M}}\mathbf{E} \preceq \mathsf{M}_\ell, \qquad \overline{\mathsf{M}}_{kk'} = \int \bar{\psi}_k^\top \bar{\Sigma}_0^{-1}\bar{\psi}_{k'}\bar{w}d\bar{\varrho}, \tag{B3.36}$$

which holds whenever $\bar{w}\circ f \le w$ $\varrho$-almost everywhere and $\bar{\Sigma}_0 \succeq \Sigma_0$. Neither sufficient condition can be dropped: with $f$ the identity, $\bar{\psi} = \psi$, and $\mathbf{E} = I$, the choice $\bar{\Sigma}_0 = \Sigma_0$, $\bar{w} = 100w$ gives $\mathbf{E}^\top\overline{\mathsf{M}}\mathbf{E} - \mathsf{M}_\ell = 99\mathsf{M}_\ell \succ 0$, and the choice $\bar{w} = w$, $\bar{\Sigma}_0 = \Sigma_0/2$ gives $\mathbf{E}^\top\overline{\mathsf{M}}\mathbf{E} - \mathsf{M}_\ell = \mathsf{M}_\ell \succ 0$. [ESTABLISHED]

*Proof.* Averaging a configuration of the declared family produces the mean field

$$\bar{c} \longmapsto \sum_j z_j \int \psi_j d\varrho_{\bar{c}} = \sum_k (\mathbf{E}\mathbf{z})_k \bar{\psi}_k(\bar{c}),$$

which is the coarse configuration $s_{\mathbf{E}\mathbf{z}}$. Averaging the identity $\sum_j (\jmath_\ell g)_j \psi_j(c) = g$ against the probability measure $\varrho_{\bar{c}}$ gives

$$\sum_k (\mathbf{E}\jmath_\ell g)_k \bar{\psi}_k(\bar{c}) = g = \sum_k (\jmath_{\ell+1} g)_k \bar{\psi}_k(\bar{c})$$

almost everywhere, and linear independence of the $\bar{\psi}_k$ gives (B3.35); then $\mathbf{E}\mathscr{A}_g^\ell(\mathbf{z}) = \mathbf{E}\mathbf{z} + \mathbf{E}\jmath_\ell g = \mathscr{A}_g^{\ell+1}(\mathbf{E}\mathbf{z})$. Both metrics are constant Gram forms, so the pullback of $\mathsf{G}_{\ell+1}$ is $\mathbf{E}^\top\overline{\mathsf{M}}\mathbf{E}$. For $\mathbf{v} \in \mathbb{R}^n$ and $U = \sum_j v_j\psi_j$, the two conditions give $wU^\top\Sigma_0^{-1}U \ge (\bar{w}\circ f)U^\top\bar{\Sigma}_0^{-1}U$ pointwise, and the disintegration followed by Jensen's inequality for the convex quadratic form $\bar{\Sigma}_0^{-1}$ gives

$$\mathbf{v}^\top\mathsf{M}_\ell\mathbf{v} \ge \int \bar{w}\int U^\top\bar{\Sigma}_0^{-1}U d\varrho_{\bar{c}}d\bar{\varrho} \ge \int \bar{w}\Big(\int U d\varrho_{\bar{c}}\Big)^\top\bar{\Sigma}_0^{-1}\Big(\int U d\varrho_{\bar{c}}\Big)d\bar{\varrho} = \mathbf{v}^\top\mathbf{E}^\top\overline{\mathsf{M}}\mathbf{E}\mathbf{v}.$$

This is the (JC-const) case of theorem B3.23, with channel loss $\Sigma_0^{-1} - \bar{\Sigma}_0^{-1}$. In the two counterexamples $\overline{\mathsf{M}} = 100\mathsf{M}_\ell$ and $\overline{\mathsf{M}} = 2\mathsf{M}_\ell$; for the Gram matrix $\mathrm{diag}(1, \frac{1}{2}, \frac{1}{2})$ of remark B3.17 the first difference has smallest eigenvalue $49.5$. □

**Proposition B3.26 (Integrated horizontal conformality).** Let the two scales share the base $\mathcal{C}_i$, the finite base measure $\varrho_i$, and the positive channel weights $w_x$; let $\mathsf{F}_\ell$ act pointwise as $(\mathsf{F}_\ell Q)(c) = \Psi_c(Q(c))$, each $\Psi_c$ the pushforward of a parameter-independent Markov kernel; let both configuration metrics be the weighted integrals of the fiber Fisher metrics; and let every integral be finite. Then for $Z, W \in T_Q\mathcal{Q}_\ell$

$$\mathsf{G}_\ell(Z,W) - \mathsf{G}_{\ell+1}(T\mathsf{F}_\ell Z, T\mathsf{F}_\ell W) = \int_{\mathcal{C}_i} \sum_x w_x \Delta_F^{\Psi_c}\big(Z_x(c), W_x(c)\big) d\varrho_i(c),$$

the integrated form of (7.73). If $\mathcal{H} = (\ker T\mathsf{F}_\ell)^{\perp_{\mathsf{G}_\ell}}$ and $T\mathsf{F}_\ell|_{\mathcal{H}}$ is onto, as for the submersions of proposition 7.54, then $\mathsf{F}_\ell$ is horizontally conformal with dilation $\varphi_\ell$ exactly when

$$\int_{\mathcal{C}_i} \sum_x w_x \Delta_F^{\Psi_c}\big(Z_x(c), W_x(c)\big) d\varrho_i(c) = (1-\varphi_\ell^2)\mathsf{G}_\ell(Z,W) \quad \text{for every } Z, W \in \mathcal{H}. \tag{B3.37}$$

The integrated identity does not imply the pointwise one: on a base consisting of two contexts of weight $\frac{1}{2}$, with fine fibers $\mathcal{N}(\mu, 1)$, the constant-mean tangent, and independent Gaussian noises of variances $3$ and $\frac{1}{3}$ added at the two contexts, the pulled-back location Fisher forms are $\frac{1}{4}$ and $\frac{3}{4}$, the integrated ratio is $\varphi_\ell^2 = \frac{1}{2}$, and the pointwise ratios differ. If $\mathcal{H}$ is closed under multiplication by nonnegative compactly supported smooth functions on the base, the integrated identity implies the pointwise identity $\varrho_i$-almost everywhere. [ESTABLISHED]

*Proof.* Differentiating the pointwise action and inserting it into the coarse metric gives the first display, whose integrand is the polarized vertical Fisher defect (7.15) of $\Psi_c$. Horizontal conformality means $\mathsf{G}_{\ell+1}(T\mathsf{F}_\ell Z, T\mathsf{F}_\ell W) = \varphi_\ell^2 \mathsf{G}_\ell(Z,W)$ on $\mathcal{H}$, with $T\mathsf{F}_\ell|_{\mathcal{H}}$ onto; subtracting from $\mathsf{G}_\ell(Z,W)$ and using the first display gives the equivalence. In the example the pushforward of $\mathcal{N}(\mu,1)$ by noise of variance $\sigma^2$ is $\mathcal{N}(\mu, 1+\sigma^2)$, whose location Fisher information is $1/(1+\sigma^2)$, namely $\frac{1}{4}$ and $\frac{3}{4}$; the fine metric of the constant-mean tangent is $1$ and the coarse metric is $\frac{1}{2}(\frac{1}{4}+\frac{3}{4}) = \frac{1}{2}$. For the last claim, apply the integrated identity to $\chi Z \in \mathcal{H}$ for smooth compactly supported $\chi \geq 0$: the integrable function $c \mapsto \sum_x w_x[\Delta_F^{\Psi_c} - (1-\varphi_\ell^2) g_x^F](Z_x(c), Z_x(c))$ integrates to zero against every $\chi^2$; monotone limits of such $\chi^2$ give the indicator functions of compact coordinate balls, so its integral over every such ball vanishes and the function vanishes almost everywhere. Polarization gives the bilinear identity. □

## B3.8 Proof of the strong Hilbert quotient theorem

*Proof of theorem B3.10.* Closedness of $\mathfrak{a}$ with compact fibers implies compact inverse images of compact sets: an open cover of such an inverse image has a finite subcover on each fiber, closedness gives a neighborhood of each target point whose inverse image lies in the union of that subcover, and finitely many such neighborhoods cover the compact target set. Conversely, compact-preimage properness gives closedness, since a convergent sequence of images of a closed set lifts into the compact inverse image of the sequence and its limit, and first countability gives a convergent subsequence of lifts.

The length distance of a strong metric induces the manifold topology on each connected component. In a Hilbert chart at $Q$, strongness and operator-norm continuity give constants $0 < \mathrm{c}_1 < \mathrm{c}_2 < \infty$ and a coordinate ball on which

$$\mathrm{c}_1\|u\|^2 \le \mathrm{g}_y(u,u) \le \mathrm{c}_2\|u\|^2. \tag{B3.38}$$

Straight segments give the upper distance bound, and a path from the center that leaves a smaller ball of radius $\delta_0$ has length at least $\sqrt{\mathrm{c}_1}\delta_0$, so small length distance forces the endpoint into every prescribed neighborhood; no minimizing geodesic is used.

For $c_\xi(t) = \exp(t\xi)\cdot Q$ and $\mathscr{A}_g(Q) = g\cdot Q$, the group law and isometry give

$$\dot c_\xi(t) = T_Q\mathscr{A}_{\exp(t\xi)}A_Q\xi, \qquad \|\dot c_\xi(t)\|_{\mathfrak{g}} = \|A_Q\xi\|_{\mathfrak{g}_Q}, \qquad d_{\mathfrak{g}}\big(Q,\exp(\xi)\cdot Q\big) \le \|A_Q\xi\|_{\mathfrak{g}_Q}. \tag{B3.39}$$

If $A_Q\xi = 0$, the curve is constant, freeness gives $\exp(t\xi) = e$ for every $t$, and hence $\xi = 0$.

Suppose that unit vectors $\xi_n$ in the Banach norm of $\mathfrak{g}$ satisfied $\|A_Q\xi_n\|_{\mathfrak{g}_Q} \to 0$. Choose $\epsilon > 0$ whose closed ball in $\mathfrak{g}$ lies in an exponential chart, and set $g_n = \exp(\epsilon\xi_n)$. Then (B3.39) and the metric topology give $g_n\cdot Q \to Q$. Properness over the compact set $\{(g_n\cdot Q, Q)\}\cup\{(Q,Q)\}$ gives a subsequence $(g_n, Q)\to(g, Q)$ with $g\cdot Q = Q$, freeness gives $g = e$, and the continuous local logarithm gives $\epsilon\xi_n \to 0$, a contradiction. Therefore, for some $\mathrm{c}_Q > 0$,

$$\|A_Q\xi\|_{\mathfrak{g}_Q} \ge \mathrm{c}_Q\|\xi\|_{\mathfrak{g}} \quad \text{for every } \xi\in\mathfrak{g}. \tag{B3.40}$$

This is vacuous if $\mathfrak{g} = \{0\}$; otherwise a convergent sequence $A_Q\xi_n$ makes $\xi_n$ Cauchy, and completeness puts its limit in the range, which is closed, in agreement with Jotz and Neeb [15, Theorem 2.1].

Fix $Q_0$. The inner product $\langle\xi,\eta\rangle_{\mathfrak{g},H} = \mathrm{g}_{Q_0}(A_{Q_0}\xi, A_{Q_0}\eta)$ has a complete norm equivalent to the Banach norm by (B3.40), so $\mathfrak{g}$ is Hilbertizable; adjoints are taken for this fixed Hilbert norm. In a tangent chart write $\mathrm{g}_y(v,u) = \langle v, M_y u\rangle$ and define

$$B_y = A_y^*M_yA_y, \qquad P_y = A_yB_y^{-1}A_y^*M_y, \qquad \xi_*(y,v) = B_y^{-1}A_y^*M_yv. \tag{B3.41}$$

The action makes $A_y$ smooth in operator norm. Near each point, (B3.40) and continuity of the metric give $B_y \ge \mathrm{c}_0 I$ for one $\mathrm{c}_0 > 0$; such a self-adjoint operator has closed range, since $\|B_y\xi\| \ge \mathrm{c}_0\|\xi\|$, and dense range, since its orthogonal complement is $\ker B_y = \{0\}$, so it is invertible, and the local Neumann series gives smoothness of the inverse.

Direct multiplication gives $P_y^2 = P_y$, $\operatorname{ran}P_y = V_y$, and $A_y^*M_y(v - P_yv) = 0$, so $P_y$ is the metric-orthogonal vertical projection. Its ranges and kernels are smooth subbundles: near a point $y_0$ the smooth operator

$$S_y = P_yP_{y_0} + (I-P_y)(I-P_{y_0}), \qquad P_yS_y = S_yP_{y_0}, \tag{B3.42}$$

equals $I$ at $y_0$ and remains invertible nearby, which identifies the splitting at $y_0$ with that at $y$. Orthogonality gives

$$\|v - A_y\xi\|^2_{\mathrm{g}_y} = \|(I - P_y)v\|^2_{\mathrm{g}_y} + \|A_y(\xi - \xi_*(y, v))\|^2_{\mathrm{g}_y}, \tag{B3.43}$$

which proves attainment and uniqueness of the minimizer before any quotient chart is used.

For the quotient charts, properness and freeness imply that for each identity neighborhood $\mathcal{W} \subseteq \mathbb{G}$ there is a neighborhood $\mathcal{O}$ of $Q$ with

$$g\mathcal{O} \cap \mathcal{O} \neq \varnothing \quad \Longrightarrow \quad g \in \mathcal{W}. \tag{B3.44}$$

Otherwise there are $y_n, z_n \to Q$ and $g_n \notin \mathcal{W}$ with $g_n y_n = z_n$; properness applied to the compact set of pairs $(z_n, y_n)$ and its limit gives a convergent subsequence of $(g_n, y_n)$ whose limiting group element fixes $Q$, hence is $e$, a contradiction. Choose a local submanifold $\mathcal{S}_0$ through $Q$, obtained in a chart from the closed complemented space $H_Q$. The map $(g, s) \mapsto g \cdot s$ on $\mathbb{G} \times \mathcal{S}_0$ has derivative $(\xi, u) \mapsto A_Q\xi + u$ at $(e, Q)$, a bounded isomorphism by the orthogonal splitting and (B3.40), so the Banach inverse function theorem gives a product $\mathcal{W} \times \mathcal{S}_1$ on which it is a diffeomorphism onto an open set. Shrink to $\mathcal{S} \subseteq \mathcal{S}_1 \cap \mathcal{O}$ using (B3.44) for this $\mathcal{W}$. If $g \cdot s = t$ for $s, t \in \mathcal{S}$, then $g \in \mathcal{W}$, and injectivity of the local product gives $g = e$ and $s = t$. Consequently

$$\mathbb{G} \times \mathcal{S} \longrightarrow \mathbb{G} \cdot \mathcal{S}, \qquad (g, s) \longmapsto g \cdot s, \tag{B3.45}$$

is bijective onto an open set, being a union of translates of the local product, and a local diffeomorphism at every point by translation; it is a diffeomorphism. These are principal trivializations, with smooth transition maps given by the slice components of their inverses.

The orbit relation is closed by the closedness of (B3.12), and $\pi$ is open because saturations of open sets are open, so a product neighborhood disjoint from the orbit relation separates two distinct orbits, and the quotient is Hausdorff. Truncating the strong length distance at $1$, and giving points in different components distance $1$, gives an invariant metric $d_0$ inducing the topology of $\mathcal{M}$, and

$$\bar{d}\big([Q], [R]\big) = \inf_{g \in \mathbb{G}} d_0(Q, g \cdot R) \tag{B3.46}$$

is positive for distinct orbits because every orbit is closed; symmetry follows by inversion, and the triangle inequality by isometry and multiplication in $\mathbb{G}$. Its balls give the quotient topology, since their inverse images are open and an open saturated neighborhood of $Q$ contains a $d_0$-ball at $Q$. The quotient is therefore metrizable, hence paracompact, as the standard definitions of Hilbert manifolds require; $\bar{d}$ is not asserted to be the Riemannian quotient distance.

The action transports $V_Q$ by $T\mathscr{A}_g A_Q\xi = A_{gQ}\,\mathrm{Ad}_g\,\xi$, and isometry transports $H_Q$ as well. Define

$$\overline{\mathrm{g}}_{[Q]}(d\pi_Q v, d\pi_Q u) = \mathrm{g}_Q\big((I - P_Q)v, (I - P_Q)u\big). \tag{B3.47}$$

Adding vertical vectors to the lifts leaves this unchanged, and equivariance of the orthogonal projection gives independence of $Q$ within its orbit. In each slice it is smooth, and $(I - P_s)|_{T_s\mathcal{S}} : T_s\mathcal{S} \to H_s$ is a bounded isomorphism because the local product gives $T_s\mathcal{M} = V_s \oplus T_s\mathcal{S}$, so the pulled-back inner product is strong. Equations (B3.43) and (B3.47) give (B3.14).

Finally, for $\widetilde{Q}(t) = g(t) \cdot Q(t)$,

$$\dot{\widetilde{Q}}(t) = T\mathscr{A}_{g(t)}\dot{Q}(t) + A_{\widetilde{Q}(t)}\delta^r g(t), \tag{B3.48}$$

where $\delta^r g$ is the right logarithmic derivative of $g$. The second term is vertical and the first preserves the horizontal norm, which gives representative independence of speed and duration. The unique coefficient $\xi_*(Q, v)$ also defines a smooth principal connection, since it sends $A_Q\xi$ to $\xi$ and transforms by $\mathrm{Ad}_g$ by uniqueness and equivariance of $P$; no global gauge section is needed. □

# Chapter B4

# Covers and Connection Descent

This chapter asks how the pointwise parent construction of the main manuscript, whose datum at one context is corollary 10.11 and whose free energy obeys theorem 10.7, extends over a neighborhood of contexts, and when fine sections, laws, and connections descend along a base map. The substantive answer is the descent criterion with its two obstructions (theorem B4.20): a section descends exactly when its image is constant on the fibers of the base map, and a connection form exactly when its two pullbacks to the fiber product agree. Compatibility of local laws, gluing of availability parents, and descent of pulled-back data hold by construction once one ambient law is adopted (proposition B4.8). The geometric results show that a coarse map with context-dependent weights keeps a first-order connection defect (theorem B4.11), that a metric projection can create curvature from a flat source and that staged projections compose only when the intermediate quotient metric is propagated (theorems B4.17 and B4.18), and that a normalized record distinguishes two base holonomy classes (theorem B4.13). That record is the base-loop companion of the agent-network witness theorem 11.30 of the main manuscript, which its interpretation chapter uses in open problem 15.8; the covariant defect used throughout is that of definition B1.8.

The chapter declines to claim that the ambient law is inferred from independently specified local marginals; that the full fine Gaussian family forms one fixed-rank statistical manifold across a change of active set, or that a parent extends through a context at which all its children are unavailable; that a comparison across coarse levels gives Fisher contraction for derivatives in the context, an inference-time decrease, or increased evidence, since the channels depend on the context and differentiating in $c$ also differentiates the channel; that any construction here recovers a connection from data, identifies an arbitrary graph-link product with base transport, or infers nontrivial bundle topology from connection holonomy, which a connection on a trivial bundle can already carry; that the change formulas of proposition B2.4 or the passive covariance of theorem B2.2 select a canonical connection; and that the holonomy-sensitive record is a canonical observable or has an empirical realization. [NOT-CLAIMED] The open constructions are the gluing of independently supplied local data (open problem B4.9), quotients beyond the chosen metric splitting (open problem B4.19), informative base aggregation (open problem B4.21), graph-to-base identification (open problem B4.14), and connection recovery and record-class sensitivity (open problem B4.15).

## B4.1 A cover without a common intersection

**Definition B4.1 (Four-agent cover and availability weights).** Take the contextual base $\mathcal{U} = (-3/2, 3/2)^2$ with coordinates $c = (c^1, c^2)$ and four agents labeled $0, 1, 2, 3$, a local indexing used only in this example. Their domains are

$$\begin{aligned} &\mathscr{I}_0 = (-8/5, -1/10), \quad \mathscr{I}_1 = (-8/5, 4/5), \\ &\mathscr{I}_2 = (-4/5, 8/5), \qquad \mathscr{I}_3 = (1/10, 8/5), \\ &\mathcal{U}_i = \mathcal{U} \cap (\mathscr{I}_i \times \mathbb{R}). \end{aligned} \tag{B4.1}$$

No probability measure is placed on the base. With $\mathrm{m}_i$ and $\mathrm{h}_i$ the midpoint and half-width of $\mathscr{I}_i$, the availability weights are

$$w_i(c) = \begin{cases} \exp(1 - 1/t_i(c)), & t_i(c) > 0, \\ 0, & t_i(c) \le 0, \end{cases} \qquad t_i(c) = 1 - \left(\frac{c^1 - \mathrm{m}_i}{\mathrm{h}_i}\right)^2. \tag{B4.2}$$

They encode availability, not posterior source probabilities: $w_i(c)$ is a per-endpoint retention probability, applied conditionally independently to every record endpoint of agent $i$, so one agent can be retained in one record and replaced by the null endpoint $\perp$ in another at the same context. A single presence variable per agent, applied to all of its endpoints, would be a different mechanism.

**Theorem B4.2 (The overlap pattern).** The sets $\mathcal{U}_i$ cover $\mathcal{U}$. Their nonempty pairwise intersections have indices $01, 02, 12, 13, 23$, their nonempty triple intersections have indices $012, 123$, and their fourfold intersection is empty. In particular,

$$\mathcal{U}_{012} = \mathcal{U} \cap \{-4/5 < c^1 < -1/10\}, \qquad \mathcal{U}_{123} = \mathcal{U} \cap \{1/10 < c^1 < 4/5\}. \tag{B4.3}$$

The cover has two nonempty triple overlaps and no common intersection. [ESTABLISHED]

*Proof.* An intersection of the intervals is nonempty exactly when its largest lower endpoint is smaller than its smallest upper endpoint, which gives the listed intersections. The intervals $\mathscr{I}_1, \mathscr{I}_2$ alone cover the first coordinate of $\mathcal{U}$, and $\mathcal{U}_0 \cap \mathcal{U}_3 = \varnothing$ excludes the fourfold intersection. □

**Theorem B4.3 (Smooth availability with an explicit null event).** Each $w_i$ is smooth, lies in $[0, 1]$, and is positive exactly on $\mathcal{U}_i$. Adjoin the null endpoint $\perp$ and define the endpoint channel

$$\mathsf{K}_c^{\mathrm{av}}(i, j) = w_i(c)\mathbf{1}_{j=i} + (1 - w_i(c))\mathbf{1}_{j=\perp}, \qquad \mathsf{K}_c^{\mathrm{av}}(\perp, j) = \mathbf{1}_{j=\perp}. \tag{B4.4}$$

For any fixed finite number of directed records, conditionally independent applications of this channel to their endpoints give a smooth normalized channel on the complete record state, so smooth ambient event probabilities give smooth visible joint event probabilities, including at agent entry and exit. [ESTABLISHED]

*Proof.* On its positive domain every derivative of $\exp(-1/t)$ is that exponential times a polynomial in $1/t$, and all such products tend to zero as $t \downarrow 0$, so extension by zero is smooth; since $0 < t_i \leq 1$ inside the interval, the range follows. Each row of $\mathsf{K}_c^{\mathrm{av}}$ sums to one, products of endpoint rows remain normalized, and their entries and finite sums against smooth ambient probabilities are smooth. The channel preserves any dependence it does not remove; it does not replace correlated records by the product of their marginals. □

**Proposition B4.4 (Occupancy and conditional attention on the cover).** For each channel $x \in \{b, m\}$ let $\eta_{ij}^x(c)$ be the marginal law of one visible directed record, including $\perp$, and set

$$\alpha_i^x(c) = \sum_j \eta_{ij}^x(c), \qquad \beta_{ij}^x(c) = \frac{\eta_{ij}^x(c)}{\alpha_i^x(c)} \quad \text{only when } \alpha_i^x(c) > 0. \tag{B4.5}$$

If every ambient receiver has positive mass, then $\alpha_i^x > 0$ exactly on $\mathcal{U}_i$, because the visible occupancy of receiver $i$ equals its ambient receiver mass times $w_i$, and the conditional row $\beta_{i\cdot}^x$ is smooth there. At zero occupancy every entry of the event-law row of $i$ is zero, so the event law imposes no conditional row, whatever conditional completion is chosen; smoothness of $\eta^x$ and $\alpha^x$ therefore supplies no identifiable attention at zero occupancy. [ESTABLISHED]

*Proof.* The receiver endpoint of a record survives with probability $w_i(c)$, independently of the rest of the record, so summing the visible record law over sources, $\perp$ included, gives the ambient receiver mass times $w_i(c)$, which is positive exactly where $w_i$ is. A quotient of smooth functions with positive denominator is smooth. If $\alpha_i^x(c) = 0$, each nonnegative summand $\eta_{ij}^x(c)$ vanishes. No null outcome is removed or renormalized in this calculation. □

The joint event law, the occupancy, and the conditional row remain different objects, as at one context; a generative posterior row is obtained from its own event law by the same operation.

## B4.2 Compatible laws, availability parents, and pullback descent

**Hypothesis B4.5 (Ambient law on the cover).** For the compatibility statements of this chapter, one smooth normalized ambient joint $\mathbb{P}_c(dY, do, de \mid X)$, a smooth selected recognition $\mathbb{Q}_{c,o,X}(dY, de)$, and a reference law are given on one finite product of agent fibers and record states for every $c \in \mathcal{U}$, with fixed structural datum $X$, positive finite evidence at the admitted observation, and selected posterior versions pushed with these same laws. The continuous coordinates may be correlated Gaussian coordinates with positive-definite ambient covariance, and records have finite laws; the section of agent $i$ is exposed only on $\mathcal{U}_i$. This excludes local laws supplied on the members of the cover without a common ambient extension; those are the subject of open problem B4.9.

**Definition B4.6 (Availability parent).** For agent subsets $V_1 \subseteq V_2$ let $\mathrm{pr}_{V_1 \leftarrow V_2}$ project continuous coordinates to $V_1$ and send every record endpoint outside $V_1$ to $\perp$, with null endpoints kept null; restriction of the base variable to a smaller open set is a separate operation. For a block $I_A$ the

fixed-child domain, the availability domain, and the availability mass are

$$\mathcal{U}_A^{\cap} = \bigcap_{i \in I_A} \mathcal{U}_i, \qquad \mathcal{U}_A^{\cup} = \bigcup_{i \in I_A} \mathcal{U}_i, \qquad s_A(c) = \sum_{i \in I_A} w_i(c). \tag{B4.6}$$

The pointwise parent of the main manuscript lives on $\mathcal{U}_A^{\cap}$. On $\mathcal{U}_A^{\cup}$ the availability parent uses the convex coefficients $\widehat{w}_{Ai} = w_i / s_A$ and all locally available joint child data supplied by the ambient law; these coefficients are a separate choice, not derived from the thinned record law of (B4.4). For the explicit geometric example take the standard two-dimensional representation of $\mathrm{SO}(2)$ separately for belief and model, let $O_{\mathrm{f}}, O_A$ be the resulting orthogonal frame matrices of the active fine and parent coordinates, and set

$$L_A = (\widehat{w}_{Ai} I_4)_{i \in I_A}, \qquad F_A = O_A L_A O_{\mathrm{f}}^{-1}, \tag{B4.7}$$

a deterministic normalized channel on the continuous coordinates whose event channel sends a child endpoint to its block label and keeps $\perp$. The example uses equal standard representations; an inequivalent pair needs its own intertwiners.

**Definition B4.7 (Pullback presentation along a base submersion).** Let $f : \widetilde{\mathcal{U}} \to \mathcal{U}$ be a smooth surjective submersion, let the principal and associated bundles over both bases be trivial, and let $O : \widetilde{\mathcal{U}} \to \mathrm{SO}(2)$ be a smooth passive frame field, with $\mathbf{O}$ repeating its representation in each separately typed associated coordinate. The principal and associated maps

$$\Psi_{\mathscr{P}}\big(\tilde{c}, u\big) = \big(f(\tilde{c}), O(\tilde{c})^{-1} u\big), \qquad \Psi\big(\tilde{c}, v\big) = \big(f(\tilde{c}), \mathbf{O}(\tilde{c})^{-1} v\big) \tag{B4.8}$$

cover $f$. Given target sections, laws, connection potentials $A_{\mathrm{t}}$, and channels $F_{\mathrm{t}}$ on $\mathcal{U}$, their source presentations are defined by $\mathbf{O}$: sections and laws are pulled back by $f$ and re-expressed by $\mathbf{O}$, and

$$A_{\mathrm{s}} = \mathbf{O} f^* A_{\mathrm{t}} \mathbf{O}^{-1} - d\mathbf{O}\mathbf{O}^{-1}, \qquad F_{\mathrm{s}} = \mathbf{O}_{\mathrm{c}} f^* F_{\mathrm{t}} \mathbf{O}_{\mathrm{f}}^{-1}, \tag{B4.9}$$

with observations and structural data pulled back unchanged.

**Proposition B4.8 (A global law induces compatible local data, and pullbacks descend).** Assume hypothesis B4.5.

(a) For $V_1 \subseteq V_2 \subseteq V_3$ and on nested open context domains,

$$\mathrm{pr}_{V_1 \leftarrow V_2} \mathrm{pr}_{V_2 \leftarrow V_3} = \mathrm{pr}_{V_1 \leftarrow V_3}, \qquad (\mathbb{Q}_{V_3} \mathrm{pr}_{V_2 \leftarrow V_3}) \mathrm{pr}_{V_1 \leftarrow V_2} = \mathbb{Q}_{V_3} \mathrm{pr}_{V_1 \leftarrow V_3}, \tag{B4.10}$$

and the same holds for the generative joint, the reference, and the selected posterior families; smooth local presentations that obey their transition maps on pairwise overlaps obey the corresponding triple-overlap cocycles.

(b) The availability parent of definition B4.6 is smooth on $\mathcal{U}_A^{\cup}$. For disjoint blocks the active rows of the combined channel are independent, so a positive-definite fine Gaussian covariance pushes

to a positive-definite covariance on the active parent fiber. If a second stage weights the active blocks by $s_A/\sum_B s_B$, the direct fine coefficient is

$$\frac{s_A}{\sum_B s_B}\frac{w_i}{s_A} = \frac{w_i}{\sum_j w_j} \quad (i \in I_A,\ s_A > 0), \tag{B4.11}$$

so the direct and staged complete law channels agree. For the blocks $01$ and $23$ the final availability parent exists on all of $\mathcal{U}$, whereas a fixed four-child parent has empty domain.

(c) For a pullback presentation (definition B4.7), sections, full laws, evaluators, and channels descend through $\Psi$. Both channel connections are compatible with descent, $d\Psi + f^* A_{\mathrm{t}}\Psi - \Psi A_{\mathrm{s}} = 0$, their curvatures are $\mathrm{Curv}(A_{\mathrm{s}}) = \mathbf{O} f^* \mathrm{Curv}(A_{\mathrm{t}})\mathbf{O}^{-1}$ with $\mathrm{Curv}(A) = dA + A \wedge A$, and the vertical coarse defect of $F_{\mathrm{s}}$ is the conjugate pullback of that of $F_{\mathrm{t}}$. The cover identities persist on the preimages $f^{-1}(\mathcal{U}_i)$.

Each clause holds by construction, from functoriality of pushforward, the frame cocycle, and the pullback definitions. [ESTABLISHED]

*Proof.* (a) Coordinate projections compose, and sending a removed endpoint to $\perp$ while keeping $\perp$ also composes, so pushforward through these deterministic channels gives (B4.10); restricting in $c$ twice is restriction to the smaller domain. For three local frame matrices $O_1, O_2, O_3$ of one associated fiber the coordinate transformations are $O_k O_j^{-1}$, and $(O_3 O_2^{-1})(O_2 O_1^{-1}) = O_3 O_1^{-1}$; pushing all coordinates and full covariances by these transformations gives the law cocycle. These are presentations of one law, not equalities between beliefs of different agents.

(b) On $\mathcal{U}_A^{\cup}$ we have $s_A > 0$, so every coefficient is smooth; at a child boundary its vanishing bump extends smoothly while another child keeps the denominator positive, and the formula is the same in every chart. Each active block row has nonzero coefficients on its own disjoint child block, which gives full row rank, and for a nonzero parent vector $v$, $v^\top F \Sigma F^\top v = (F^\top v)^\top \Sigma (F^\top v) > 0$. At two stages the coefficient product is (B4.11); an absent block contributes zero directly and needs no conditional coefficient, frame matrices cancel at intermediate fibers, endpoint maps compose with the same null convention, and equal complete channels push every law equally. By theorem B4.2 $\mathcal{U}_0 \cap \mathcal{U}_3 = \varnothing$, while $\mathcal{U}_0 \cup \mathcal{U}_1 = \{c^1 < 4/5\} \cap \mathcal{U}$ and $\mathcal{U}_2 \cup \mathcal{U}_3 = \{c^1 > -4/5\} \cap \mathcal{U}$ cover $\mathcal{U}$.

(c) The principal map is equivariant, since $O^{-1}(uk) = (O^{-1}u)k$. Pulling back the target principal connection form $u^{-1}A_{\mathrm{t}}u + u^{-1}du$ by $\Psi_{\mathscr{P}}$ gives (B4.9), and differentiating $\Psi = \mathbf{O}^{-1}$ gives $d\Psi = -\mathbf{O}^{-1}d\mathbf{O}\mathbf{O}^{-1}$, which cancels the remaining terms of $f^* A_{\mathrm{t}}\Psi - \Psi A_{\mathrm{s}}$. Expanding $dA_{\mathrm{s}} + A_{\mathrm{s}} \wedge A_{\mathrm{s}}$, the derivatives of $\mathbf{O}$ cancel and leave the conjugate pullback curvature; the same expansion of $dF_{\mathrm{s}} + A_{\mathrm{s,c}}F_{\mathrm{s}} - F_{\mathrm{s}}A_{\mathrm{s,f}}$ leaves the conjugate pullback of the target defect. A section $s_{\mathrm{s}} = \mathbf{O} f^* s_{\mathrm{t}}$ satisfies $\Psi \circ s_{\mathrm{s}} = s_{\mathrm{t}} \circ f$, which is constant on the fibers of $f$. For a target kernel $C_c$, the source kernel undoes its input frame, applies $C_c$, and restores its output frame; evaluating on a target measurable set gives

$$\Psi_{\mathrm{c}\#}(\mathbb{Q}_{\mathrm{s}} C_{\mathrm{s}}) = (\Psi_{\mathrm{f}\#}\mathbb{Q}_{\mathrm{s}})C_{\mathrm{t}}. \tag{B4.12}$$

The same change of variables holds for the full generative joint, the reference, and the selected posterior versions. Keeping observation coordinates unchanged preserves evidence, common

invertible presentation maps preserve relative entropy because the density Jacobians cancel, and the normalized evaluator transforms by its separately typed input-output square. Restrictions to $\mathcal{U}_i$ pull back to restrictions to $f^{-1}(\mathcal{U}_i)$. □

At each fixed admitted context, every normalized recognition-independent state channel of this section is covered by theorem 10.7.

**Open Problem B4.9 (Gluing of independently supplied local data).** Proposition B4.8 starts from one ambient law and therefore cannot establish the converse. What is not settled is which local laws, channels, and presentations supplied on the members of a cover without an ambient law admit one joint extension compatible with every restriction and selected conditional version. The same question is open for the descent of independently specified source data along an informative base map and for a canonical choice of ambient dependence, channel, or membership. The problem is settled by a gluing theorem with hypotheses on the local data alone, or by an obstruction computed from them; a compatible ambient law supplies one extension but does not reconstruct every marginal family.

## B4.3 Nonflat connections and metric-dependent projections

**Definition B4.10 (Reference potentials).** On the trivial common principal $\mathrm{SO}(2)$ bundle over $\mathcal{U}$, choose two connections independently, with local reference potentials

$$A_*^x = \frac{\kappa_x}{2}\big(c^1 dc^2 - c^2 dc^1\big)\mathbb{J}, \qquad \mathbb{J} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad x \in \{b, m\}. \tag{B4.13}$$

The convention is $\nabla = d + A$, so the potential in a frame $O$ is $A = OA_*O^{-1} - dOO^{-1}$; this is the represented form of (B1.8) and (B1.9), with $O$ the represented inverse of the passive reframing of the channel. The connections are supplied geometric data, not quantities determined by attention.

**Theorem B4.11 (A nonflat parent retains its connection defect).** The reference curvatures are $\mathrm{Curv}(A_*^x) = \kappa_x \mathbb{J} dc^1 \wedge dc^2$. Wherever the weights in $L$ are smooth, in particular on $\mathcal{U}_A^{\cup}$, and with the same channel connection repeated in its child fibers, the channel $F = O_{\mathrm{c}} L O_{\mathrm{f}}^{-1}$ has covariant defect

$$\mathcal{D}F = dF + A_{\mathrm{c}}F - FA_{\mathrm{f}} = O_{\mathrm{c}}(dL)O_{\mathrm{f}}^{-1}, \tag{B4.14}$$

and for composable maps

$$\mathcal{D}(F_2F_1) = (\mathcal{D}F_2)F_1 + F_2(\mathcal{D}F_1). \tag{B4.15}$$

Nonzero curvature is therefore compatible with gauge covariance, and a coarse map with context-dependent weights is not parallel. [ESTABLISHED]

*Proof.* The abelian potential has $A_* \wedge A_* = 0$, and differentiating its coefficient gives $\kappa_x \mathbb{J} dc^1 \wedge dc^2$. Differentiate $F = O_{\mathrm{c}} L O_{\mathrm{f}}^{-1}$ and substitute the frame potentials; the frame derivatives cancel, and

the repeated reference connection commutes through the scalar block weights, leaving $dL$. The product rule gives (B4.15) after adding and subtracting $F_2 A_1 F_1$. □

**Definition B4.12 (Operational base-holonomy witness).** An operational base-holonomy witness consists of a named principal bundle and a family of connections, an assigned base loop, a gauge-invariant statistic of a normalized population record, a typed map from the connection data to the record law, and two admissible connection data for which every input other than the connection is fixed, the represented loop holonomies have distinct conjugacy classes, and the induced laws of the statistic differ. It is strong when the record-law map factors nontrivially through the conjugacy class of the represented loop holonomy.

**Theorem B4.13 (An operational record sensitive to a declared base holonomy).** Fix the base point $c_\circ = (7/20, 1/5)$ and the closed rectangular path with successive vertices $(7/20, 1/5)$, $(13/20, 1/5)$, $(13/20, 1/2)$, $(7/20, 1/2)$, and $c_\circ$, which lies in the domain $\mathcal{U}_2$ of the probe agent; only its fibers at $c_\circ$ enter the record. Choose a selector $P_{\mathrm{pr}} \neq 0$ onto its separately typed belief and model coordinates and a positive-definite noise covariance $R_{\mathrm{pr}}$. Let a finite label $\iota$ prescribe connection parameters $(\kappa_{b,\iota}, \kappa_{m,\iota})$ in (B4.13), with positive prior masses $\mathsf{p}_\iota$. Keep one normalized joint $p_0(Y, e, o_0)$, independent of $\iota$, with matter posterior $\mathbf{\Pi}_0(dY \mid o_0)$ at an admitted $o_0$, let $\mathbf{O}_\circ$ be the represented probe frame at $c_\circ$ in both channels, and declare the additional record $o_1$ and joint by

$$\begin{aligned} \mathrm{Hol}_\iota &= \mathbf{O}_\circ \exp\left[-\mathcal{A}\,\mathrm{diag}(\kappa_{b,\iota}\mathbb{J}, \kappa_{m,\iota}\mathbb{J})\right] \mathbf{O}_\circ^{-1}, \qquad \mathcal{A} = 9/100, \\ B_\iota &= (\mathrm{Hol}_\iota - I_4) P_{\mathrm{pr}}, \\ p(\iota, Y, e, o_0, o_1) &= \mathsf{p}_\iota\, p_0(Y, e, o_0)\, \mathcal{N}(o_1; B_\iota Y, R_{\mathrm{pr}}). \end{aligned} \tag{B4.16}$$

This is a normalized statistical experiment, and given $\iota$ and $o_0$ the record is distributed as $B_\iota Y + \varepsilon$, with $Y \sim \mathbf{\Pi}_0(\cdot \mid o_0)$ and independent $\varepsilon \sim \mathcal{N}(0, R_{\mathrm{pr}})$. The record laws of two labels coincide exactly when $B_\iota Y$ and $B_{\iota'} Y$ have the same law under $\mathbf{\Pi}_0(\cdot \mid o_0)$. For the parameter pairs $(0, 0)$ and $(7/10, -2/5)$, labeled $\iota = 0, 1$, the two record laws differ exactly when the posterior law of $P_{\mathrm{pr}} Y$ is not the point mass at $0$, in particular whenever $\mathbf{\Pi}_0(\cdot \mid o_0)$ is Gaussian with positive-definite covariance. Under the same condition the frame-invariant statistic $\mathsf{S}(o_1) = o_1^\top R_{\mathrm{pr}}^{-1} o_1$ has different laws under the two labels, with means $4$ and $4 + \mathbb{E}_{\mathbf{\Pi}_0}\left[(B_1 Y)^\top R_{\mathrm{pr}}^{-1} B_1 Y\right] \in (4, \infty]$, while the matter law, loop, selector, and noise are held fixed. Under that condition these data form an operational base-holonomy witness in the sense of definition B4.12. [ESTABLISHED]

*Proof.* The rectangle and the strip $\mathscr{I}_2 \times \mathbb{R}$ are convex, so every segment stays in $\mathcal{U}_2$. Integrating the reference one-form along a segment from $c_j$ to $c_{j+1}$ gives $\kappa \det(c_j, c_{j+1})/2$; summing the segments gives $\kappa\mathcal{A}$ with the signed area $\mathcal{A} = 9/100$, and the transport equation $\dot{v} = -A(\dot{\gamma})v$ gives the displayed exponential, which the endpoint frame conjugates at the base point. Each Gaussian factor integrates to one in $o_1$, and $p_0$ and $\mathsf{p}$ normalize the joint.

The supplied pairs give $\mathrm{Hol}_0 = I_4$ and $B_0 = 0$, while $\mathrm{Hol}_1$ is conjugate to two plane rotations by the angles $-63/1000$ and $9/250$; neither is a multiple of $2\pi$, so $\mathrm{Hol}_1 - I_4$ is invertible and the two holonomies have distinct conjugacy classes. Under the declared joint the label is independent of

$(Y, e, o_0)$, so the joint is the record joint of theorem 11.30 with latent $(Y, e)$, observation $o_0$, prior masses $\mathsf{p}_\iota$, probe coordinates $P_{\mathrm{pr}}Y \in \mathbb{R}^4$, represented holonomies $\mathrm{Hol}_\iota$, and noise covariance $R_{\mathrm{pr}}$, and S is its statistic q. The proofs of parts (i) and (ii) there use the holonomies only as fixed matrices and apply verbatim. They give the record kernel, the criterion for equal record laws, and that the two record laws differ exactly when the posterior law of $P_{\mathrm{pr}}Y$ is not the point mass at $0$, in which case the laws of S differ with the displayed means; otherwise both record laws, and hence both laws of S, coincide. A Gaussian posterior with positive-definite covariance gives $P_{\mathrm{pr}}Y$ a nondegenerate law, which is not a point mass. Under label $0$ we have $o_1 = \varepsilon$, so S has the chi-square law with four degrees of freedom. In general the label likelihood is $\mathrm{lik}_\iota(o_1) = \int \mathcal{N}(o_1; B_\iota y, R_{\mathrm{pr}})\mathbf{\Pi}_0(dy \mid o_0)$. If the matter posterior is $\mathcal{N}(\mu_0, \Sigma_0)$ with $\Sigma_0$ positive definite, Gaussian integration gives

$$\begin{gathered}\mathrm{lik}_\iota(o_1) = \mathcal{N}\big(o_1; B_\iota\mu_0, R_{\mathrm{pr}} + B_\iota\Sigma_0 B_\iota^\top\big), \qquad Z(o_1, o_0) = Z_0(o_0)\sum_\iota \mathsf{p}_\iota \mathrm{lik}_\iota(o_1), \\ \mathsf{p}(\iota \mid o_1, o_0) = \frac{\mathsf{p}_\iota \mathrm{lik}_\iota(o_1)}{\sum_{\iota'} \mathsf{p}_{\iota'} \mathrm{lik}_{\iota'}(o_1)},\end{gathered} \tag{B4.17}$$

and the mean of S under label $1$ is $4 + \mathrm{Tr}(R_{\mathrm{pr}}^{-1}B_1\Sigma_0B_1^\top) + \mu_0^\top B_1^\top R_{\mathrm{pr}}^{-1}B_1\mu_0$.

Under a passive change of fine and probe coordinates by represented orthogonal frame maps $O_{\mathrm{f}}, O_{\mathrm{pr}}$, send $P_{\mathrm{pr}}$ to $O_{\mathrm{pr}}P_{\mathrm{pr}}O_{\mathrm{f}}^{-1}$, $\mathrm{Hol}_\iota$ to $O_{\mathrm{pr}}\,\mathrm{Hol}_\iota\,O_{\mathrm{pr}}^{-1}$, $o_1$ to $O_{\mathrm{pr}}o_1$, and $R_{\mathrm{pr}}$ to $O_{\mathrm{pr}}R_{\mathrm{pr}}O_{\mathrm{pr}}^\top$. Then $B'_\iota = O_{\mathrm{pr}}B_\iota O_{\mathrm{f}}^{-1}$, so the whole observation kernel is pushed coherently and its evidence is invariant, and S is unchanged since $o_1^\top O_{\mathrm{pr}}^\top (O_{\mathrm{pr}}R_{\mathrm{pr}}O_{\mathrm{pr}}^\top)^{-1}O_{\mathrm{pr}}o_1 = o_1^\top R_{\mathrm{pr}}^{-1}o_1$. A zero-area loop gives $B_\iota = 0$ for every label and no label reweighting. □

The proof uses the base loop only to compute $\mathrm{Hol}_\iota$; the distinguishability argument needs only a represented group element acting on the probe fibers at one point and conjugated by the probe frame under a passive change. At $c_*$ the represented loop product of independent graph links around a closed walk of the interaction graph plays that role, which is theorem 11.30 of the main manuscript, with no base curve, connection, or neighborhood. A specified connection is thereby associated with a normalized population record; reading that association as a statement about physical geometry is an interpretation, not a result of this chapter.

Identifying a graph link $\Theta_e$ with a base transport $\Omega_{\gamma_e}$ requires an embedded interaction complex, assigned piecewise-smooth base curves, and equality with the path-ordered exponential of a specified connection. Hypothesis B1.13 is satisfiable by declaration, by setting each represented link equal to a chosen base transport, but that is a definition, not a theorem; the cover of this chapter contains no interaction graph and no graph links, and its base witness uses one declared loop for one probe agent. Even an exact assignment concerns connection holonomy, not topological nontriviality.

**Open Problem B4.14 (Graph-to-base identification).** No realization of independently specified graph links as base transports is known. The problem is to choose vertex contexts, an embedding, and oriented edge curves for given graph data and a connection, and to prove that the independently specified represented links equal the corresponding base transports as in (B1.26), together with

hypotheses on $\mathcal{C}$, $G$, and the curve class under which the realization is unique up to gauge, or to exhibit graph data for which no such choice exists.

**Open Problem B4.15 (Connection recovery and record-class sensitivity).** Theorem B4.13 supplies one witness for Gaussian record noise and one pair of connection labels. What is not settled is whether a connection, or its holonomy class, can be recovered from population records beyond such a controlled family, and for which record classes holonomy sensitivity persists. The problem is settled positively by a recovery theorem for a declared record class, and negatively by a theorem that every statistic in a declared admissible controlled class is insensitive to the holonomy class.

**Definition B4.16 (Minimum-norm splitting and projected connection).** Let $E$ carry a Euclidean fiber metric and a compatible connection $d + A$, with $A^\top = -A$ in orthonormal frames, and let $F : E \to E_{\mathrm{c}}$ be a smooth surjective vector-bundle map of fixed rank. Set

$$\begin{gathered}\bar{\mathrm{g}} = (FF^\top)^{-1}, \qquad F^\dagger = F^\top \bar{\mathrm{g}}, \qquad P_F = F^\dagger F, \qquad P_F^\perp = I - P_F, \\ A^{\mathrm{quot}} = F(dF^\dagger + AF^\dagger) = (FA - dF)F^\dagger.\end{gathered} \tag{B4.18}$$

Here $P_F$ is a fiberwise orthogonal projector, $\bar{\mathrm{g}}$ is the quotient metric of the chosen metric, $F^\dagger$ is the minimum-norm lift, and $d + A^{\mathrm{quot}}$ is the projected connection.

**Theorem B4.17 (Projected connection versus exact connection descent).** The projected connection preserves $\bar{\mathrm{g}}$. With the second fundamental form $\mathrm{II} = P_F^\perp(dF^\dagger + AF^\dagger)$ and the defect $\mathcal{D}F = dF + A^{\mathrm{quot}}F - FA$,

$$d\bar{\mathrm{g}} = (A^{\mathrm{quot}})^\top \bar{\mathrm{g}} + \bar{\mathrm{g}} A^{\mathrm{quot}}, \qquad \mathcal{D}F = (dF - FA)P_F^\perp = \bar{\mathrm{g}}^{-1}\mathrm{II}^\top, \tag{B4.19}$$

and $F$ is parallel exactly when $\ker F$ is preserved by the fine connection. The curvature is

$$\mathrm{Curv}(A^{\mathrm{quot}}) = F\,\mathrm{Curv}(A)F^\dagger + \bar{\mathrm{g}}^{-1}\mathrm{II}^\top \wedge \mathrm{II}, \qquad \mathrm{Curv}(A) = dA + A \wedge A, \tag{B4.20}$$

so a projected connection can be curved with a flat source: for

$$A = 0, \qquad F(c^1, c^2) = \begin{pmatrix} 1 & 0 & -c^1 \\ 0 & 1 & -c^2 \end{pmatrix}, \qquad \mathrm{Curv}(A^{\mathrm{quot}})(0) = -\mathbb{J}dc^1 \wedge dc^2 \neq 0. \tag{B4.21}$$

For the scalar-weight parent $L = (\widehat{w}_i I)_i$ with a common internal connection $A_*$, the construction instead gives

$$\bar{\mathrm{g}} = |\widehat{w}|^{-2} I, \qquad A^{\mathrm{quot}} = A_* - \tfrac{1}{2} d \log |\widehat{w}|^2 I, \qquad \mathrm{Curv}(A^{\mathrm{quot}}) = \mathrm{Curv}(A_*), \qquad |\widehat{w}|^2 = \sum_i \widehat{w}_i^2. \tag{B4.22}$$

[ESTABLISHED]

*Proof.* Direct multiplication gives $FF^\dagger = I$, $P_F = P_F^\top = P_F^2$, and $(F^\dagger)^\top F^\dagger = \bar{\mathrm{g}}$. A fine representative of $y$ is uniquely $F^\dagger y + v$ with $v \in \ker F$, of squared norm $y^\top \bar{\mathrm{g}} y + \|v\|^2$, which identifies the quotient metric; the two expressions for $A^{\mathrm{quot}}$ agree by differentiating $FF^\dagger = I$. Multiplying by $\bar{\mathrm{g}}$ gives $\bar{\mathrm{g}} A^{\mathrm{quot}} = (F^\dagger)^\top (dF^\dagger + AF^\dagger)$, and adding its transpose cancels the skew $A$ terms and gives $d\bar{\mathrm{g}}$. Differentiating $FF^\dagger = I$ gives the first formula for $\mathcal{D}F$. With $D_A P_F = dP_F + [A, P_F]$, differentiating $P_F^2 = P_F$ shows that $D_A P_F$ has only off-diagonal blocks, metric compatibility makes its coefficients symmetric, so $(D_A P_F)F^\dagger = \mathrm{II}$ and $\mathcal{D}F = F(D_A P_F)P_F^\perp = \bar{\mathrm{g}}^{-1}\mathrm{II}^\top$. Its vanishing is equivalent to preservation of the retained subbundle and, by differentiating orthogonal inner products, of its orthogonal complement $\ker F$. On a retained section $u = P_F u$ the projected connection is $P_F \nabla$; expanding its square gives $P_F \operatorname{Curv}(A) u + P_F(D_A P_F) \wedge (D_A P_F) u$, since the term with $P_F(D_A P_F) P_F$ vanishes and $P_F^\perp \nabla u = (D_A P_F) u$, and transporting through $F^\dagger$ gives (B4.20). The second term is an endomorphism-valued two-form, not a nonnegative scalar energy. In the example $\bar{\mathrm{g}}(0) = I_2$ and the only nonzero row of $\mathrm{II}(0)$ is $(-dc^1, -dc^2)$, whose exterior square gives $-\mathbb{J} dc^1 \wedge dc^2$; all bundles there are trivial, so the curvature does not indicate nontrivial topology. For the scalar-weight parent insert $F^\dagger = \mathrm{col}_i(\widehat{w}_i I)/|\widehat{w}|^2$: the additional scalar one-form is exact, its square vanishes, and its mixed wedge terms with $A_*$ cancel. The quotient-orthonormal coordinates multiply $y$ by $|\widehat{w}|^{-1}$ and remove this term; the variable-weight defect of theorem B4.11 need not vanish. □

**Theorem B4.18 (Composition requires propagation of the quotient metric).** For smooth surjections $F_1 : E_0 \to E_1$ and $F_2 : E_1 \to E_2$, start with the metric $I$ on $E_0$, give $E_1$ its induced metric $\bar{\mathrm{g}}_1 = (F_1 F_1^\top)^{-1}$, and use it in the second projection, whose minimum-norm lift is $F_2^{\dagger_1} = \bar{\mathrm{g}}_1^{-1} F_2^\top (F_2 \bar{\mathrm{g}}_1^{-1} F_2^\top)^{-1}$. Then

$$F_1^\dagger F_2^{\dagger_1} = (F_2 F_1)^\top \big((F_2 F_1)(F_2 F_1)^\top\big)^{-1} = (F_2 F_1)^\dagger, \tag{B4.23}$$

and the quotient metrics and projected connections of the staged and direct projections agree, whether or not either map is parallel. Resetting the intermediate metric to $I$ is a different construction and can change the connection class: for $A_0 = 0$, $F_1 = \left(\begin{smallmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{smallmatrix}\right)$, and $F_2 = (1, c^1)$, the reset connection is $-c^1 dc^1/(1 + (c^1)^2)$ and the direct one is $-c^1 dc^1/(4 + (c^1)^2)$, which differ by the exact form $d\big(\tfrac{1}{2} \log[(4 + (c^1)^2)/(1 + (c^1)^2)]\big)$, so both are flat and related by a rescaling of the quotient frame; for $F_1 = \left(\begin{smallmatrix} 2 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{smallmatrix}\right)$ and $F_2 = \left(\begin{smallmatrix} 1 & c^1 & 0 \\ 0 & 1 & c^2 \end{smallmatrix}\right)$, the determinant of the $dc^1 \wedge dc^2$ coefficient of the curvature at $(c^1, c^2) = (1/3, 1/5)$ is $15625/61731$ for the propagated metric and $3645/103823$ for the reset metric, so the reset connection is not gauge equivalent to the direct one. [ESTABLISHED]

*Proof.* Multiplication gives (B4.23), and the final quotient metric has the same denominator. The product rule and $F_1 F_1^\dagger = I$ give $F_2(d + A_1^{\mathrm{quot}}) F_2^{\dagger_1} = F_2 F_1 (d + A_0)(F_1^\dagger F_2^{\dagger_1})$, which is the direct connection. The two examples are direct computations; gauge equivalent connections have pointwise conjugate curvatures and hence equal curvature determinants. □

**Open Problem B4.19 (Quotients beyond the chosen metric splitting).** Theorems B4.17 and B4.18 close the chosen minimum-norm construction for a declared metric and fixed rank. Not settled are a canonical effective-support quotient for arbitrary positive-semidefinite edge weights, an

intertwiner into an arbitrarily prescribed associated representation, and quotient geometry across changes of rank. The problem is settled by constant-rank support data, compatible represented actions, and an admitted quotient or bundle map for which these objects are constructed, or by an obstruction to them.

## B4.4 Base descent with inherited nonzero curvature

The pullback presentation of definition B4.7 with $\widetilde{\mathcal{U}} = \mathcal{U} \times (-1, 1)$, extra coordinate $z$, $f(c, z) = c$, and $O(z) = \exp(\mathrm{k} z \mathbb{J})$ removes an explicitly redundant base coordinate; it does not average contexts. By proposition B4.8 its source data descend. The following criterion concerns data that are given on the source rather than defined as pullbacks.

**Theorem B4.20 (The descent criterion and its obstructions).** Let $f : \mathcal{M} \to \overline{\mathcal{M}}$ be a smooth surjective submersion and $\Psi : E \to \overline{E}$ a smooth bundle map over $f$. A smooth section $s$ of $E$ has a unique descended section exactly when $\Psi \circ s$ is constant on every fiber of $f$. For a smooth surjective principal submersion $\Psi_{\mathscr{P}} : \mathscr{P} \to \overline{\mathscr{P}}$, a connection form descends exactly when its pullbacks by the two projections of $\mathscr{P} \times_{\overline{\mathscr{P}}} \mathscr{P}$ agree. In the example above both criteria have concrete failures: the source law $\mathcal{N}(O(z)(1 + z/4, 0)^\top, I_2)$ is pushed by $\Psi$ to the fiberwise law $\mathcal{N}((1 + z/4, 0)^\top, I_2)$, whose mean varies with $z$, so it does not descend, and no passive reframing makes it constant on the forgotten fiber; and adding $\mathrm{s} z \mathbb{J} dc^1$, $\mathrm{s} \neq 0$, to the pullback potential excludes compatible connection descent through this $f$. By contrast the pullback potentials $A^x_{\mathrm{s}} = f^* A^x_* - \mathrm{k} \mathbb{J} dz$ of (B4.13) have zero mixed curvature and inherited $dc^1 \wedge dc^2$ curvature $\kappa_x \mathbb{J}$, so a nonzero coefficient in the forgotten direction is compatible with exact descent. These are criteria for given maps, not the existence of a canonical quotient. [ESTABLISHED]

*Proof.* Necessity follows by evaluating a descended object at two lifts of the same base point or tangent vector. Conversely, a submersion has local smooth sections: in local coordinates, augment its full-rank derivative by complementary source coordinates and apply the inverse function theorem. Composing $\Psi \circ s$ with such a local section defines a smooth target section; fiber constancy makes the definition independent of the local section and hence consistent on overlaps, and surjectivity gives uniqueness. For forms, choose a local section $\varsigma$ of $\Psi_{\mathscr{P}}$ and set $\overline{\omega} = \varsigma^* \omega$; the fiber-product equality applied to $(v, d\varsigma d\Psi_{\mathscr{P}}(v))$ gives $\omega = \Psi^*_{\mathscr{P}} \overline{\omega}$ and independence of $\varsigma$. Equivariance and reproduction of the principal vertical generators descend by surjectivity and the equivariant group-fiber isomorphisms.

In the example the pushed mean is $O(z)^{-1} O(z)(1 + z/4, 0)^\top$. Its relative entropy to $\mathcal{N}(0, I_2)$ is $(1 + z/4)^2/2$, which every passive reframing preserves, so no reframing makes the pair constant in $z$. The potential $A^x_{\mathrm{s}} = f^* A^x_* - \mathrm{k} \mathbb{J} dz$ follows from (B4.9); its $dz \wedge dc^j$ curvature components vanish because $\mathbb{J}$ commutes with itself and the $dz$ coefficient is constant. Adding $\mathrm{s} z \mathbb{J} dc^1$ adds $\mathrm{s} \mathbb{J} dz \wedge dc^1$ to the curvature, whose contraction with $\partial_z$ is nonzero, whereas every conjugate pullback curvature annihilates $\partial_z$. □

**Open Problem B4.21 (Informative base aggregation).** Choosing a probability measure on a varying base fiber and averaging its laws defines a new mixture operation, which requires a base prior or reference, a joint base-fiber model, and the induced observation weighting; neither fiber constancy nor a canonical such measure follows from the cover. The problem is to construct informative base aggregation of laws, channels, and connections along a base map that collapses contexts, with a probability law on smooth random sections where the construction needs one (compare open problem B1.22), and to state its descent and free-energy identities; it is settled by such a construction or by an obstruction for a declared base map.

# Part B-II

# Infinite Networks and Wilsonian Asymptotics

# Chapter B5

# Comparison Data, Modes, and Interaction Beta Functions

The main manuscript coarse-grains a finite agent network at $c_*$ by exact channels, retaining an effective action, a closure residual, finite-step multipliers, and a native growth bound (chapters 10 and 12). A renormalization flow in the Wilsonian sense adds maps that compare each blocked theory with the original units, and only then speaks of modes, exponents, relevance, beta functions, and fixed points. This part asks which comparison data those words require, what they license, and which of them a finite network at $c_*$ can carry.

Comparison data are declared structure independent of the channels, and the finite networks of the main manuscript do not inhabit the asymptotic classification: finite levels with different numbers of configurations admit no stochastic comparison, blocking terminates, and the sign of an exponent depends on the declared block schedule (proposition B5.15). Where the classification is available, its canonical content is the forward Lyapunov filtration (proposition B5.10); exponents are invariant only under tempered comparisons (theorem B5.12 and proposition B5.13); and the exact interaction cocycle has no relevant direction under tempered comparisons (corollary B5.18), so a relevant direction needs an extensive normalization, as in the Gaussian block spectrum (definition B5.19). A beta function is a difference quotient on one reference space. It depends on the reference trajectory (proposition B5.24) and on the comparison scheme (proposition B5.27), and a retained beta differs from the exact one by the transported closure residual of the main manuscript (proposition B5.26). These results are used in chapter B6, which adds continuous scale, fixed points, spectra, and two-index limits.

Throughout this part $b_k > 1$ are declared block ratios, $\varsigma$ is the cumulative log block scale, $J_\ell$ are comparison isomorphisms into the scale-$\ell$ objects, $I_b$ and $K_b$ are identification and rescaled kernels, and beta functions are written $\mathfrak{b}$, so that they are not confused with the attention rows $\beta^x_{ij}$ of the main manuscript. The measure pairs $(\varrho, \mathrm{m})$, the reverse conditional expectations $\mathsf{U}$, the finite-step multipliers $D_\ell$ and $D_{n\leftarrow\ell}$ on Banach spaces $\mathscr{X}_\ell$, and the interaction objects $\mathcal{G}_\ell$, $\mathsf{g}$, $\mathcal{T}_\ell$, $\mathrm{asm}_\ell$, $\mathrm{hd}_\ell$, $\mathsf{P}^{\mathrm{ret}}_\ell$, and $r^{\mathcal{G}}_{\ell+1}$ are those of the main manuscript.

**Scope.** This part asserts no existence theorem for exponents. No Oseledets splitting of a scale cocycle is claimed: a multiplicative ergodic theorem [24] requires a measure-preserving base flow, measurability of the cocycle in the base variable, integrability of $\log^+\|D\|$ and, for the two-sided splitting, of $\log^+\|D^{-1}\|$, and either finite dimension [2, Chapters 3 and 4] or, in infinite dimensions,

compactness of the operators on a Hilbert space [27] or quasi-compactness on a separable Banach space [21], and a deterministic scale sequence supplies none of these. No autonomous spectrum, mode, or exponent is asserted for a nonautonomous sequence; no continuous beta function is derived from a discrete blocking sequence; no universality class is identified; and none of the invariant or fixed objects of this part is identified with a physical law, which would need the apparatus of open problem 15.7. [NOT-CLAIMED] The open constructions are open problems B5.6, B5.14, B5.21 and B5.22 in this chapter and open problems B6.4, B6.19 and B6.23 in chapter B6.

## B5.1 Comparison data and dynamical types

**Definition B5.1 (Comparison data and reference cocycle).** Let a scale diagram in the sense of definition 10.21 have objects $\mathfrak{X}_\ell$ and adjacent arrows $C_{\ell,\ell+1}$ in a declared target category $\mathscr{K}$. Comparison data are a reference object $\mathfrak{X}_\star$ and declared isomorphisms $J_\ell : \mathfrak{X}_\star \xrightarrow{\sim} \mathfrak{X}_\ell$ in $\mathscr{K}$. Field, base, unit, and reference-measure rescalings are components of $J_\ell$, and so is the parameter rescaling of a separately supplied dynamical model. The reference cocycle is

$$\widehat{\mathfrak{R}}_\ell = J_{\ell+1}^{-1} \circ C_{\ell,\ell+1} \circ J_\ell : \mathfrak{X}_\star \longrightarrow \mathfrak{X}_\star, \tag{B5.1}$$

where $\circ$ is composition in $\mathscr{K}$; in a stochastic component it is represented by right-acting kernel juxtaposition (10.10). A positive scalar $\zeta_\ell$ can be one component of $J_\ell$ but not all of it, since multiplication by $\zeta_\ell$ maps each object to itself and identifies no two distinct objects.

**Definition B5.2 (Dynamical type).** The reference cocycle (B5.1) is autonomous when $\widehat{\mathfrak{R}}_\ell = \widehat{\mathfrak{R}}$ for every $\ell$, periodic of period $p$ when $\widehat{\mathfrak{R}}_{\ell+p} = \widehat{\mathfrak{R}}_\ell$, stationary random when it is a measurable stationary cocycle over a measure-preserving base, and general otherwise. The invariant object depends on the type. An autonomous fixed object satisfies $\widehat{\mathfrak{R}}x = x$; a periodic scheme has a fixed object of its monodromy $\widehat{\mathfrak{R}}_{\ell+p-1} \circ \cdots \circ \widehat{\mathfrak{R}}_\ell$ or an attracting $p$-cycle; a stationary random scheme may have a random invariant section; a general cocycle may have pullback or forward attracting sections. A fixed ray additionally requires a positive cone and its projectivization. An ordered segment $\widehat{\mathfrak{R}}_{n-1} \circ \cdots \circ \widehat{\mathfrak{R}}_\ell$ is a two-parameter family, a power of one map only in the autonomous case; writing it as a power otherwise loses the base scale and, for noncommuting factors, the order, since $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ give $BA = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right) \neq AB = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. For a mixed state the composition law is owed in every declared component category.

**Proposition B5.3 (Uniform contraction need not give one fixed ray).** If every $\widehat{\mathfrak{R}}_\ell$ is a positive map on the interior of a proper cone that contracts Hilbert's projective metric $d_H$ by a factor at most $\mathrm{c} < 1$, then

$$d_H\big(\widehat{\mathfrak{R}}_{\ell-1} \cdots \widehat{\mathfrak{R}}_0 x, \widehat{\mathfrak{R}}_{\ell-1} \cdots \widehat{\mathfrak{R}}_0 y\big) \leq \mathrm{c}^\ell d_H(x, y), \tag{B5.2}$$

so initial rays are forgotten, but the attracting section need not be one scale-independent ray: for the alternating cocycle $A, B, A, B, \ldots$ on $\mathbb{R}^2_{>0}$ with $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, even iterates converge projectively to the ray $[1, \sqrt{2}]$ and odd iterates to the distinct ray $A[1, \sqrt{2}]$. [ESTABLISHED]

*Proof.* The bound is the iterated one-step contraction [4, 23]. Both matrices have strictly positive entries, hence finite projective diameter and a common contraction factor below one. The product $BA = \left(\begin{smallmatrix} 3 & 2 \\ 4 & 3 \end{smallmatrix}\right)$ has eigenvalues $3 \pm 2\sqrt{2}$ with Perron eigenvector $(1, \sqrt{2})$, so even iterates $(BA)^k x$ converge projectively to $[1, \sqrt{2}]$ for every $x$ in the open cone, and odd iterates, their images under $A$, converge to $(2 + \sqrt{2}, 1 + \sqrt{2})$, which is not proportional to $(1, \sqrt{2})$. □

### B5.1.1 The rescaled measure-pair step

**Definition B5.4 (Rescaled kernel and measure-pair step).** For a blocking ratio $b > 1$, a rescaled step consists of a coarse channel $C_b$ and a declared identification kernel $I_b$ that returns the coarse space to one common standard Borel space; the rescaled kernel is $K_b = C_b I_b$. On measure pairs $(\varrho, \mathsf{m})$ in the sense of definition 12.1 the step is

$$\mathfrak{R}_b(\varrho, \mathsf{m}) = \big((\varrho C_b) I_b, (\mathsf{m} C_b) I_b\big) = (\varrho K_b, \mathsf{m} K_b), \tag{B5.3}$$

with reference component $\mathfrak{R}_b^{\varrho} \varrho = \varrho K_b$. The family is compatible when, after the declared identifications of the ratios,

$$K_{b_1 b_2} = K_{b_1} K_{b_2}. \tag{B5.4}$$

Under compatibility $\mathfrak{R}_{b_1 b_2} = \mathfrak{R}_{b_2} \circ \mathfrak{R}_{b_1}$, because $(\varrho K_{b_1}) K_{b_2} = \varrho K_{b_1 b_2}$; otherwise the steps form a cocycle in the sense of definition B5.2. Compatibility is the rule semigroup in which the functor axiom of definition 10.21 has content.

**Remark B5.5 (A declared family that fails compatibility).** For one declared rescaling family on nested-cycle instances of six categorical agents, whose identification kernels compose a Wilson-line coarse connection from the retained dressed boundary generators with a cyclic graph identification and the identity state identification, blocking by the full ratio and blocking in declared stages gave coarse laws at total variation distance $0.042$ and relative entropy $0.0050$ (check `CHK-RG-RESCALING-COMPATIBILITY`, table BD.3). This family therefore fails (B5.4), and its scale maps form a cocycle; the computation decides nothing about other families. [NUMERICAL]

**Open Problem B5.6 (Compatible rescaling kernels on agent networks).** The measure-pair steps compose exactly under (B5.4), the family of remark B5.5 fails that law, and the compatible cover and path-blocking constructions of chapters B4 and B7 live on their own domains and supply no identification kernels for the agent network. It is not known whether identification kernels acting on the gauge-retained network state of the main manuscript satisfy (B5.4) on a declared class of finite networks at $c_*$. The problem is settled by such a family with a proof of (B5.4), or by a proof that none exists for a stated class, whose composed statements are then stated in cocycle form.

## B5.2 Generalized cross-scale modes and exponents

A scale sequence supplies no eigenvalue problem. Along an exact orbit it supplies the finite-step multipliers $D_\ell \in \mathcal{L}(\mathscr{X}_\ell, \mathscr{X}_{\ell+1})$ and their ordered products $D_{n \leftarrow \ell}$ of definition 12.18, which compose

by (12.32) between different Banach spaces $(\mathscr{X}_\ell, \|\cdot\|_\ell)$. This section fixes the mode and exponent types relative to those declared norms.

**Definition B5.7 (Compatible cross-scale mode line).** A rank-one mode line of index $a$ along the orbit is a family of nonzero vectors $v_{\ell,a} \in \mathscr{X}_\ell$ (in the complexification when $\mathscr{X}_\ell$ is real) and scalars $\lambda_{\ell,a} \in \mathbb{C}$ with

$$D_\ell v_{\ell,a} = \lambda_{\ell,a} v_{\ell+1,a}. \tag{B5.5}$$

This compatibility equation relates adjacent fibers. The expression $D_\ell v = \lambda v$ equates vectors of $\mathscr{X}_{\ell+1}$ and $\mathscr{X}_\ell$ and is not used unless an identification is declared: with $\mathscr{X}_0 = \mathbb{R}$, $\mathscr{X}_1 = \mathbb{R}^2$, and $D_0 x = (x, 0)$, the line $v_0 = 1$, $v_1 = (1, 0)$, $\lambda_0 = 1$ is well typed while $D_0 v = \lambda v$ is not. For a mode subbundle of finite rank $d_a$, choose injective frames $V_{\ell,a} : \mathbb{C}^{d_a} \to \mathscr{X}_\ell$ and matrices $A_{\ell,a}$ with

$$D_\ell V_{\ell,a} = V_{\ell+1,a} A_{\ell,a}. \tag{B5.6}$$

**Proposition B5.8 (Ordered product law for mode coefficients).** Under (B5.5),

$$D_{n\leftarrow\ell} v_{\ell,a} = \lambda_{n\leftarrow\ell,a} v_{n,a}, \qquad \lambda_{n\leftarrow\ell,a} = \prod_{k=\ell}^{n-1} \lambda_{k,a}, \tag{B5.7}$$

with the empty product equal to one, and under (B5.6) the finite-step matrix is the ordered product $A_{n-1,a} \cdots A_{\ell,a}$. A vanishing factor annihilates the propagated mode from that step on, and the phase or sign of each factor is retained separately from any growth exponent. Conversely, the definition constrains nothing by itself: for every $u \in \mathscr{X}_\ell$ whose forward images $D_{k\leftarrow\ell} u$ are nonzero, the unit vectors $v_k = D_{k\leftarrow\ell} u / \|D_{k\leftarrow\ell} u\|_k$ and scalars $\lambda_k = \|D_{k+1\leftarrow\ell} u\|_{k+1} / \|D_{k\leftarrow\ell} u\|_k$ form a mode line. The index $a$ therefore acquires content only from an additional declared constraint, such as invariance of a declared subbundle, equivariance under a declared symmetry, or a comparison identification under which the mode section is constant. [ESTABLISHED]

*Proof.* Induction on $n - \ell$ with (12.32): $D_{n+1\leftarrow\ell} v_{\ell,a} = D_n D_{n\leftarrow\ell} v_{\ell,a} = \lambda_{n\leftarrow\ell,a} \lambda_{n,a} v_{n+1,a}$. The frame case is the same induction with matrix factors in the order fixed by (B5.6). For the converse, $D_k v_k = D_{k+1\leftarrow\ell} u / \|D_{k\leftarrow\ell} u\|_k = \lambda_k v_{k+1}$. □

The object carried by the multiplier cocycle alone is the forward filtration of proposition B5.10.

**Definition B5.9 (Cumulative log block scale, growth rates, and relevance).** Let $b_k > 1$ be declared adjacent block ratios and set

$$\Delta\varsigma_k = \log b_k, \qquad \varsigma_{n\leftarrow\ell} = \sum_{k=\ell}^{n-1} \Delta\varsigma_k, \tag{B5.8}$$

with $\varsigma_{n\leftarrow\ell}\to\infty$. For $u\in\mathscr{X}_\ell\setminus\{0\}$ the finite-time growth rate relative to the declared norms is

$$\chi_{n\leftarrow\ell}(u)=\frac{\log\|D_{n\leftarrow\ell}u\|_n-\log\|u\|_\ell}{\varsigma_{n\leftarrow\ell}},\tag{B5.9}$$

equal to $-\infty$ once the image vanishes. The upper rate $\bar\chi_\ell(u)$ and the lower rate are its limit superior and limit inferior as $n\to\infty$, with $\bar\chi_\ell(0)=-\infty$; a Lyapunov exponent is asserted only when they agree. For a mode line the scalar finite-time exponent is

$$\upsilon_{n\leftarrow\ell,a}=\frac{\log|\lambda_{n\leftarrow\ell,a}|}{\varsigma_{n\leftarrow\ell}}=\frac{\sum_{k=\ell}^{n-1}\log|\lambda_{k,a}|}{\varsigma_{n\leftarrow\ell}},\qquad\log 0=-\infty.\tag{B5.10}$$

A nonzero direction $u$ is relevant, marginal, or irrelevant according as $\bar\chi_\ell(u)$ is positive, zero, or negative, and a mode line according as the limit superior of $\upsilon_{n\leftarrow\ell,a}$ is positive, zero, or negative. The classification is relative to the declared norms, the declared extensive normalization of the mode sections, the declared block ratios, and the declared comparison maps; it is not a property of the scale sequence alone. In the autonomous case, where a fixed point, one ratio $b_k=b$, and bounded isometric identifications make the linearization one bounded endomorphism $T$ of one complex Banach space, an isolated eigenvector $T\psi_a=\lambda_a\psi_a$ is a constant mode line with exponent $\log|\lambda_a|/\log b$, its sign or phase is retained, and generalized eigenspaces are used only in finite dimension or for isolated points of the spectrum with their spectral projections.

**Proposition B5.10 (Forward Lyapunov filtration).** Under definition B5.9, set, for $c\in[-\infty,+\infty]$,

$$V_\ell^{\le c}=\{u\in\mathscr{X}_\ell:\bar\chi_\ell(u)\le c\}.\tag{B5.11}$$

Then every $V_\ell^{\le c}$ is a linear subspace, $V_\ell^{\le c}\subseteq V_\ell^{\le c'}$ for $c\le c'$, and

$$\bar\chi_{\ell+1}(D_\ell u)=\bar\chi_\ell(u)\quad(D_\ell u\neq 0),\qquad D_\ell V_\ell^{\le c}\subseteq V_{\ell+1}^{\le c}.\tag{B5.12}$$

If $\mathscr{X}_\ell$ is finite dimensional, $\bar\chi_\ell$ takes at most $\dim\mathscr{X}_\ell$ distinct values on $\mathscr{X}_\ell\setminus\{0\}$. The irrelevant directions are the nonzero vectors of the forward-invariant subspace $V_\ell^{<0}=\bigcup_{c<0}V_\ell^{\le c}$, the marginal ones those of $V_\ell^{\le 0}\setminus V_\ell^{<0}$, and relevant directions are canonical only as nonzero classes of $\mathscr{X}_\ell/V_\ell^{\le 0}$; a complementary subspace of relevant directions is not determined by the forward cocycle. [ESTABLISHED]

*Proof.* Since $\varsigma_{n\leftarrow\ell}\to\infty$, the term $\log\|u\|_\ell/\varsigma_{n\leftarrow\ell}$ vanishes in the limit. The triangle inequality gives $\log\|D(u+w)\|\le\log 2+\max\{\log\|Du\|,\log\|Dw\|\}$ for $D=D_{n\leftarrow\ell}$; after division by $\varsigma_{n\leftarrow\ell}$ the constant disappears, and the limit superior of a maximum of two sequences is the maximum of their limit superiors, so $\bar\chi_\ell(u+w)\le\max\{\bar\chi_\ell(u),\bar\chi_\ell(w)\}$. With $\bar\chi_\ell(tu)=\bar\chi_\ell(u)$ for $t\neq 0$, each $V_\ell^{\le c}$ is a subspace, and monotonicity in $c$ is immediate. For invariance, $D_{n\leftarrow\ell+1}D_\ell u=D_{n\leftarrow\ell}u$ and $\varsigma_{n\leftarrow\ell}/\varsigma_{n\leftarrow\ell+1}\to 1$, so the two limit superiors agree, including infinite values. If $c_1<\cdots<c_k$ are attained at $u_1,\dots,u_k$, then $u_j\in V_\ell^{\le c_j}\setminus V_\ell^{\le c_{j-1}}$, so the subspaces increase strictly and $k\le\dim\mathscr{X}_\ell$. A nested union of subspaces is a subspace, and invariance passes to it. For the last assertion

take $D_k = \mathrm{diag}(2, 1/2)$ on $\mathbb{R}^2$ with $b_k = e$: the vector $e_2$ has rate $-\log 2$ and every vector outside $\mathrm{span}(e_2)$ has rate $\log 2$, so every line other than $\mathrm{span}(e_2)$ is a complement of relevant directions. □

**Proposition B5.11 (Scalar and norm growth agree for tempered mode sections).** For a mode line and every $n > \ell$ with $\lambda_{n\leftarrow\ell,a} \neq 0$,

$$\chi_{n\leftarrow\ell}(v_{\ell,a}) = \upsilon_{n\leftarrow\ell,a} + \frac{\log\|v_{n,a}\|_n - \log\|v_{\ell,a}\|_\ell}{\varsigma_{n\leftarrow\ell}}. \tag{B5.13}$$

The two rates therefore have the same upper and lower limits whenever the mode sections have unit norm, or more generally whenever $\log\|v_{n,a}\|_n$ is sublinear in $\varsigma_{n\leftarrow\ell}$. Under a scalar regauging $v'_{\ell,a} = c_{\ell,a}v_{\ell,a}$ with $c_{\ell,a} \neq 0$,

$$\lambda'_{\ell,a} = \lambda_{\ell,a}\frac{c_{\ell,a}}{c_{\ell+1,a}}, \qquad \lambda'_{n\leftarrow\ell,a} = \lambda_{n\leftarrow\ell,a}\frac{c_{\ell,a}}{c_{n,a}}, \tag{B5.14}$$

so the upper and lower limits of the scalar exponent are unchanged whenever $|\log|c_{n,a}||/\varsigma_{n\leftarrow\ell} \to 0$. This tempering condition is sufficient but not necessary for preserving those two values; when the original scalar exponent converges to a finite limit, it is necessary for the regauged exponent to converge to that same finite limit. [ESTABLISHED]

*Proof.* Taking norms in (B5.7) and substituting into (B5.9) gives the remainder as the endpoint norm difference divided by $\varsigma_{n\leftarrow\ell}$. Substituting $v'_{\ell,a} = c_{\ell,a}v_{\ell,a}$ into (B5.5) gives the first equality in (B5.14), and telescoping gives the second, so $\upsilon'_{n\leftarrow\ell,a} = \upsilon_{n\leftarrow\ell,a} + (\log|c_{\ell,a}| - \log|c_{n,a}|)/\varsigma_{n\leftarrow\ell}$, which proves sufficiency; if $\upsilon$ and $\upsilon'$ converge to the same finite limit, their difference forces $\log|c_{n,a}|/\varsigma_{n\leftarrow\ell} \to 0$. For an infinite limit this need not hold: with $b_k = e$, $\lambda_{k,a} = e^{-(2k+1)}$, and $c_{k,a} = e^k$, the rates at $\ell = 0$ are $-n$ and $-n-1$, both tending to $-\infty$, while $|\log c_{n,a}|/n = 1$. Necessity also fails for nonconvergent exponents: take $\ell = 0$, $b_k = e$, $c_{0,a} = 1$, $\log|\lambda_{n\leftarrow 0,a}| = nu_n$ and $\log|c_{n,a}| = nd_n$, where $(u_n, d_n)$ is $(0,0)$, $(-1,0)$, $(-1/2,1/2)$, $(-1/2,-1/2)$ for $n \equiv 0,1,2,3 \pmod 4$. The regauged exponent $u_n - d_n$ runs through $0,-1,-1,0$, with the same limit superior $0$ and limit inferior $-1$ as $u_n$, while $|\log|c_{n,a}||/n = 1/2$ for $n \equiv 2,3 \pmod 4$. □

For a unit-norm mode line the scalar exponent is exactly the rate of $v_{\ell,a}$, so the two clauses of definition B5.9 agree. A splitting of $\mathscr{X}_\ell$ into relevant, marginal, and irrelevant subspaces requires backward data, such as an exponential dichotomy or the hypotheses of a multiplicative ergodic theorem (open problem B5.14).

**Theorem B5.12 (Exponent invariance under tempered comparison trivializations).** Let $J_k : \mathscr{X}_\star \to \mathscr{X}_k$ be bounded linear isomorphisms with bounded inverses, and set

$$\widehat{D}_k = J_{k+1}^{-1}D_kJ_k, \qquad \widehat{D}_{n\leftarrow\ell} = J_n^{-1}D_{n\leftarrow\ell}J_\ell. \tag{B5.15}$$

If the trivializations and their inverses are tempered relative to the cumulative log block scale,

$$\frac{\log^+\|J_n\| + \log^+\|J_n^{-1}\|}{\varsigma_{n\leftarrow\ell}} \longrightarrow 0, \tag{B5.16}$$

then for every $u \in \mathscr{X}_\ell \setminus \{0\}$ the upper and lower rates of $u$ in the native norms coincide with those of $J_\ell^{-1}u$ computed for $\widehat{D}_{n\leftarrow\ell}$ in the reference norm. Together with the scalar tempering condition of proposition B5.11, this preserves both the scalar mode exponent and the norm growth rate. [ESTABLISHED]

*Proof.* Write $x_n = D_{n\leftarrow\ell}u$ and $\widehat{x}_n = J_n^{-1}x_n = \widehat{D}_{n\leftarrow\ell}J_\ell^{-1}u$, the second identity in (B5.15). Bounded invertibility gives

$$\frac{\|x_n\|_n}{\|J_n\|} \le \|\widehat{x}_n\|_\star \le \|J_n^{-1}\|\|x_n\|_n. \tag{B5.17}$$

Taking logarithms changes the numerator of (B5.9) by at most $\log^+\|J_n\| + \log^+\|J_n^{-1}\|$, and the fixed initial scale contributes the constant $|\log\|J_\ell\|| + |\log\|J_\ell^{-1}\||$. Dividing by $\varsigma_{n\leftarrow\ell}$ and using (B5.16) makes the difference of the two finite-time rates tend to zero. The scalar statement is (B5.14). □

**Proposition B5.13 (Superexponential distortion changes the apparent exponent).** Tempering the scalar mode representatives alone is not sufficient. Let $\mathscr{X}_k = \mathbb{R}$ with the absolute value, every $D_k$ the identity, $b_k = e$, and every scalar gauge factor one, so that the native exponent is zero and the scalar normalization is tempered. With $J_k u = e^{k^2}u$, the transported cocycle is $\widehat{D}_{n\leftarrow 0} = e^{-n^2}I$ and the apparent reference rate is $-n \to -\infty$. The comparison trivialization and its inverse, not only the mode normalization, must satisfy (B5.16). [ESTABLISHED]

*Proof.* $\widehat{D}_k = J_{k+1}^{-1}J_k = e^{-2k-1}$, whose product over $k = 0, \dots, n-1$ is $e^{-n^2}$; with $\varsigma_{n\leftarrow 0} = n$, (B5.9) gives $-n$, and (B5.16) fails because $\log\|J_n\| = n^2$. □

**Open Problem B5.14 (Multiplicative ergodic structure of scale cocycles).** For every deterministic multiplier cocycle with $\varsigma_{n\leftarrow\ell} \to \infty$, the forward filtration of proposition B5.10 is canonical, and exponents are invariant under tempered comparisons (theorem B5.12). Existence of exact exponents and of a splitting into exponent subspaces is not established for any cocycle of this part, because a deterministic blocking sequence supplies none of the hypotheses of a multiplicative ergodic theorem listed in the scope paragraph. The problem is settled by a declared stationary family of blocking data over a measure-preserving base whose multiplier cocycle meets those hypotheses, or by a proof that no such family exists for the exact interaction cocycle of the main manuscript.

**Proposition B5.15 (Finite single-point networks do not inhabit the asymptotic classification).** Let $V_0$ be a finite set of agents at $c_*$. (i) No stochastic comparison isomorphism exists between finite levels with different numbers of configurations. (ii) If every coarse agent is a block of at least two agents, the depth is at most $\lfloor \log_2 |V_0| \rfloor$, so the limits $n \to \infty$ and the hypothesis $\varsigma_{n\leftarrow\ell} \to \infty$ of definition B5.9 are unavailable whatever ratios are declared, and only the finite-time rates (B5.9) over the declared depth are defined; if the spaces $\mathscr{X}_\ell$ are finite dimensional and $\dim \mathscr{X}_\ell$ is not constant along the sequence, no linear isomorphisms $J_\ell : \mathscr{X}_\star \to \mathscr{X}_\ell$ exist, and theorem B5.12 has no instance. (iii) The signs of the asymptotic rates depend on the declared schedule unless it is two-sided bounded: if $0 < \log b \le \Delta\varsigma_k, \Delta\varsigma'_k \le \log B < \infty$, the upper rates of the two schedules

have the same sign and so do the lower rates, whereas a mode with $\lambda_k = 2$ has exponent $1$ under $b_k = 2$ and exponent $2/(n+1) \to 0$ under $b_k = 2^{k+1}$. [ESTABLISHED]

*Proof.* (i) Mutually inverse Markov kernels between finite sets are permutation matrices (proposition 10.22). (ii) If $|V_{\ell+1}| \le |V_\ell|/2$ and $|V_L| \ge 1$, then $2^L \le |V_0|$; linear isomorphisms preserve dimension. For the interaction spaces of the main manuscript the dimension count and the termination of vertex-ratio blocking are proposition 12.22. (iii) For two bounded schedules, $\varsigma_{n\leftarrow\ell}/\varsigma'_{n\leftarrow\ell} \in [\log b/\log B, \log B/\log b]$, and the two rates have the same numerator. Multiplying a sequence by factors in a fixed compact subinterval of $(0,\infty)$ keeps a positive limit superior positive and a negative one negative; the relation is symmetric, so a zero limit superior stays zero, and the same holds for limits inferior. In the example, $\varsigma_{n\leftarrow 0} = \frac{1}{2}n(n+1)\log 2$ while $\log|\lambda_{n\leftarrow 0}| = n\log 2$. □

For a finite agent network at $c_*$, only finite-time rates over the declared depth are therefore used, and the asymptotic classification of definition B5.9, proposition B5.10, and theorem B5.12 is reserved for the two-index limits of section B6.4, in which system size grows with depth. An inhabited instance needs unboundedly many sites and a normalization that identifies the spaces, as in definition B5.19, where $n$ steps consume $b^n$ independent coordinates. Whether a family of finite networks at $c_*$ has a two-index limit with nonvacuous comparison maps and nontrivial exponents is part of open problem B6.19.

### B5.2.1 Contraction certificates and the exact interaction cocycle

**Proposition B5.16 (Dobrushin contraction is a sufficient cocycle certificate).** For a reverse-kernel version of theorem 12.5 with Dobrushin coefficient $\delta$ of (12.16), the reverse conditional expectation induces an operator $\overline{\mathsf{U}}$ on the quotient by constants with, in the oscillation norm (12.15),

$$\|\overline{\mathsf{U}}\|_{\mathrm{osc}\to\mathrm{osc}} \le \delta. \tag{B5.18}$$

For a sequence of such arrows with coefficients $\delta_k$ and declared ratios $b_k > 1$, write

$$\overline{\mathsf{U}}_{n\leftarrow\ell} = \overline{\mathsf{U}}_{n-1}\cdots\overline{\mathsf{U}}_{\ell}, \qquad \varsigma_{n\leftarrow\ell} = \sum_{k=\ell}^{n-1} \log b_k. \tag{B5.19}$$

If $\varsigma_{n\leftarrow\ell} \to \infty$ and

$$\limsup_{n\to\infty} \frac{\sum_{k=\ell}^{n-1} \log \delta_k}{\varsigma_{n\leftarrow\ell}} < 0, \qquad \log 0 = -\infty, \tag{B5.20}$$

then every nonzero bounded class has negative upper rate in the oscillation norms, so every direction is irrelevant in the sense of definition B5.9; for one ratio $b$ and $\delta_k \le \delta < 1$ the upper rate is at most $\log\delta/\log b$. The condition is sufficient only: one-step coefficients equal to one do not prevent complete two-step forgetting, one-step coefficients below one do not give a negative normalized rate, and failure of (B5.20) proves neither marginality nor relevance. [ESTABLISHED]

*Proof.* The one-step bound (B5.18) and its iteration are the Dobrushin clauses of theorem 12.5:

$$\|\overline{\mathrm{U}}_{n\leftarrow\ell}\|_{\mathrm{osc}\to\mathrm{osc}} \le \prod_{k=\ell}^{n-1} \delta_k, \tag{B5.21}$$

and taking logarithms with the zero convention proves the rate bound. For the first witness take the reverse matrices

$$R_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1/2 & 1/2 \end{pmatrix}, \qquad R_1 = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{B5.22}$$

compatible with the laws $(1/2, 1/2)$, $(1/3, 1/3, 1/3)$, and $(2/3, 1/3)$. Both one-step coefficients are one, but both rows of $R_1 R_0$ are $(1/2, 1/2)$, so the composed coefficient is zero. For the second, take the law $(1/2, 1/2)$ at every scale, let $J$ have both rows $(1/2, 1/2)$, and set

$$R_k = (1 - a_k) I + a_k J, \qquad a_k = 2^{-k-2}. \tag{B5.23}$$

Each $R_k$ is a compatible reverse kernel with $\delta_k = 1 - a_k < 1$, but $\prod_k \delta_k > 0$, so with fixed $b_k = b$ the normalized rate is zero. The identity channel has $\delta = 1$ and leaves every bounded direction marginal, and a two-cycle has eigenvalue $-1$, so $\delta$ carries no phase information. □

**Corollary B5.17 (Temperedness of interaction comparison trivializations).** Suppose declared comparison isomorphisms $J_\ell : \mathcal{G}_\star \to \mathcal{G}_\ell$ obey

$$\log^+\|J_n\| + \log^+\|J_n^{-1}\| \le \mathrm{c}\,(1 + |V_n|) \tag{B5.24}$$

for a constant $\mathrm{c}$ independent of $n$. This bound holds for every trivialization assembled from the maps of (12.25) together with a bounded gauge realization, because $\|\mathrm{asm}_\ell\| \le 1$ and $\|\mathrm{hd}_\ell\| \le 3^{|V_\ell|} - 1$. Then the tempering hypothesis (B5.16) of theorem B5.12 holds whenever $|V_n|/\varsigma_{n\leftarrow\ell} \to 0$, in particular for every infinite blocking sequence whose vertex counts do not increase and whose declared ratios satisfy $b_k \ge b > 1$. If those ratios are the vertex ratios $|V_k|/|V_{k+1}|$, a finite network admits only finitely many such steps (proposition 12.22), so no such infinite sequence exists and the clause is vacuous; it has asymptotic content only for ratios declared independently of the vertex counts, and every exponent computed with them scales inversely with that declaration. [ESTABLISHED]

*Proof.* Insert (B5.24) into the left side of (B5.16) and divide by $\varsigma_{n\leftarrow\ell}$; the constant term vanishes because $\varsigma_{n\leftarrow\ell} \to \infty$. If the vertex counts do not increase, $|V_n| \le |V_\ell|$ while $\varsigma_{n\leftarrow\ell} \ge (n - \ell)\log b \to \infty$. The operator bounds are (12.25). □

The extraction bound is exponential in $|V_n|$, so the corollary gives no tempering when the network grows at least linearly in $\varsigma_{n\leftarrow\ell}$, and proposition B5.13 shows that an untempered comparison can change the reported exponent.

**Corollary B5.18 (No relevant exact interaction mode under tempered comparisons).** Assume $\varsigma_{n\leftarrow\ell} \to \infty$ and $|V_n|/\varsigma_{n\leftarrow\ell} \to 0$. Along an exact interaction orbit, with the multipliers (12.33) and the native norms of proposition 12.21, every finite-time rate obeys

$$\chi_{n\leftarrow\ell}(u) \le \frac{|V_n| \log 3}{\varsigma_{n\leftarrow\ell}}, \tag{B5.25}$$

so every native upper rate is at most zero and no direction is relevant in the sense of definition B5.9. If the comparison isomorphisms satisfy (B5.16), the same holds in the reference norm, and every mode line of definition B5.7 for these multipliers whose sections satisfy the tempering condition of proposition B5.11 has scalar exponent with limit superior at most zero. A positive exponent can arise only from a declared comparison family or mode normalization that violates these tempering conditions, that is, from a declared extensive normalization; the exact step supplies none. [ESTABLISHED]

*Proof.* If $D_{n\leftarrow\ell}u = 0$ the rate is $-\infty$. Otherwise $\chi_{n\leftarrow\ell}(u) \le \log\|D_{n\leftarrow\ell}\|/\varsigma_{n\leftarrow\ell}$, and (12.35) with $\log(3^{|V_n|} - 1) < |V_n| \log 3$ gives (B5.25), whose right side tends to zero. The reference-norm statement is theorem B5.12. For a mode line, (B5.13) writes the scalar exponent as the native rate of $v_{\ell,a}$ minus the endpoint-norm correction, which the tempering condition sends to zero. □

### B5.2.2 An inhabited spectrum: the Gaussian block

**Definition B5.19 (Relevance conventions for the Gaussian block spectrum).** Take the replication sum (G6.6), the block statistic (G6.18), the Fisher norm $\|\cdot\|_{L^2(\nu)}$ on $L^2_0(\nu)$ at every scale, and one ratio $b_k = b$, so that $\Delta\varsigma_k = \log b$ in (B5.8). By theorem G6.8, each normalized Hermite function $\widehat{\mathrm{He}}_k$ is a constant mode line of the score operator (G6.19) with coefficient $b^{1-k/2}$, and its exponent in the sense of definition B5.9 is $v_k = 1 - k/2$: degree one is relevant with $v_1 = \frac{1}{2}$, degree two is marginal with $v_2 = 0$, and every degree $k \ge 3$ is irrelevant. The constant mode $k = 0$ would have coefficient $b$ on $L^2(\nu)$ but is absent from the centered tangent space.

The degree-one growth comes from the extensive normalization alone: the score operator is $\mathsf{U}_b\mathscr{I}_b$ with $\mathsf{U}_b$ a contraction (theorem 12.5) and $\|\mathscr{I}_b\| = \sqrt{b}$ by (G6.7), while no exact arrow supplies growth (proposition 12.21).

**Proposition B5.20 (Convention dependence and forbidden extensions).** The spectrum (G6.21) is specific to the scalar independent Gaussian realization and its normalization, and four boundaries hold. (i) If the coordinates are jointly centered Gaussian with unit variances and common pairwise correlation $\mathrm{r} > -1/(b-1)$, and the statistic is $\bar{\xi}_{\mathrm{r}} = (\sum_i \xi_i)/\sqrt{b[1+(b-1)\mathrm{r}]}$, then $\mathrm{Corr}(\xi_i, \bar{\xi}_{\mathrm{r}}) = \mathrm{c}_{\mathrm{r}} = \sqrt{[1+(b-1)\mathrm{r}]/b}$. With $\mathsf{U}_{b,\mathrm{r}}$ the reverse conditional expectation under the correlated block law, the operator $\mathscr{L}_{b,\mathrm{r}} = \mathsf{U}_{b,\mathrm{r}}\mathscr{I}_b$, applied to the replicated sum $\sum_i h(\xi_i)$, which is a centered direction of that law but not the score of the independent replication of proposition G6.3, satisfies

$$\mathscr{L}_{b,\mathrm{r}}\widehat{\mathrm{He}}_k = b\mathrm{c}_{\mathrm{r}}^k\widehat{\mathrm{He}}_k = b^{1-k/2}\,[1+(b-1)\mathrm{r}]^{k/2}\,\widehat{\mathrm{He}}_k, \tag{B5.26}$$

which differs from (G6.20) for every $\mathrm{r} \neq 0$; keeping the uncorrelated normalization instead changes the output law away from $\nu$, and the operator is then not an endomorphism. (ii) For independent $\mathcal{N}(0, I_d)$ coordinates and componentwise normalized sums the tensor Hermite functions satisfy $\mathscr{L}_b \widehat{\mathrm{He}}_\alpha = b^{1-|\alpha|/2} \widehat{\mathrm{He}}_\alpha$, so degree $k$ has multiplicity $\binom{d+k-1}{k}$. (iii) A common orthogonal action commutes with the componentwise normalized sum and preserves each degree space, whereas a general $\mathrm{GL}(d)$ transformation changes the reference covariance and needs an explicitly transported family of laws and tangent norms; no gauge link, holonomy, or curvature datum enters the scalar theorem. (iv) (G6.6) is the score of an independent product perturbation, a restricted direction in $L^2_0(\nu^{\otimes b})$; the theorem says nothing about interaction scores or higher Hoeffding sectors (section 12.3). [ESTABLISHED]

*Proof.* (i) The pair $(\xi_i, \bar{\xi}_\mathrm{r})$ is jointly centered Gaussian with unit variances and correlation $\mathrm{c_r}$, so $\xi_i \mid \bar{\xi}_\mathrm{r} = z \sim \mathcal{N}(\mathrm{c_r} z, 1 - \mathrm{c_r}^2)$; the generating-function computation (G6.22) with $b^{-1/2}$ replaced by $\mathrm{c_r}$ gives $\mathbb{E}[\mathrm{He}_k(\xi_i) \mid \bar{\xi}_\mathrm{r}] = \mathrm{c_r}^k \mathrm{He}_k(\bar{\xi}_\mathrm{r})$, and summing over $i$ gives (B5.26). (ii) Tensor Hermite functions factor over coordinates and each factor obeys the scalar identity with the same $b^{-1/2}$, so the eigenvalue depends only on the total degree, and $\binom{d+k-1}{k}$ multi-indices have total degree $k$ in $d$ variables. (iii) An orthogonal map preserves $\mathcal{N}(0, I_d)$ and commutes with the componentwise sum, while a general invertible $Q$ sends $\mathcal{N}(0, I_d)$ to $\mathcal{N}(0, QQ^\top)$. (iv) is the definition of $\mathscr{I}_b$. □

Jona-Lasinio [14, Sections 2, 5 and 7] obtains the binary case $b = 2$ of (G6.20) by linearizing the induced map on laws at the centered unit-variance Gaussian, identifies the linearized renormalization at a self-similar Gaussian fixed point with a conditional expectation, and gives the two-tangent-space form of the eigenvalue equation with its composition law across scales, leaving the nonlinear terms of a limit theorem unpursued. The arbitrary-integer eigenvalues, lemma G6.2, the Fisher budget (G6.10), the exact spectrum (G6.21), the correlated boundary (B5.26), and the typed cocycle of section B5.2 are not attributed to that source.

**Open Problem B5.21 (Extensive relevance beyond the scalar Gaussian block).** Definition B5.19 inhabits all three relevance classes for independent scalar Gaussian coordinates under the replication sum, the block statistic, and a fixed Fisher norm, and proposition B5.20 shows that correlated coordinates, multivariate covariance, and general $\mathrm{GL}(d)$ actions each leave that setting. The corresponding spectrum for a correlated, anisotropic, or gauge-mixed block is not known. The problem is settled by a declared transported family of laws and tangent norms for such a block, together with a diagonalization of its score operator.

**Open Problem B5.22 (Nonlinear attraction at the Gaussian block law).** The spectrum of theorem G6.8 controls the derivative of the normalized block map at $\nu$. The normalized sum maps each $\mathcal{N}(0, \sigma^2)$ to itself, and the tangent of that curve at $\sigma^2 = 1$ is $\frac{1}{2}\mathrm{He}_2$, so the marginal degree-two coefficient is carried by a curve of fixed laws. No remainder estimate for the nonlinear block map in the score-tangent topology of this part is proved here, so the spectrum does not by itself give a basin of attraction or the behavior of the map transverse to that curve. The problem is settled by such an estimate on a declared neighborhood of $\nu$.

## B5.3 Beta functions and scheme dependence

A beta function is a difference quotient formed on one space. It therefore exists only after the comparison data of definition B5.1 or the rescaled kernel of definition B5.4 have been declared.

**Definition B5.23 (Reference-dependent action map and discrete action beta).** Fix a rescaled kernel $K_b$. On a dominated tier the action map keeps its reference argument,

$$\mathfrak{R}_b^H[H;\varrho] := -\log \frac{d\big((e^{-H}\varrho)K_b\big)}{d(\varrho K_b)}, \tag{B5.27}$$

so that $\mathfrak{R}_b(\varrho, e^{-H}\varrho) = (\varrho K_b, e^{-\mathfrak{R}_b^H[H;\varrho]}\varrho K_b)$. On a declared vector space of finite-valued actions that is closed under this map along a declared reference trajectory and carries subtraction, the discrete action beta is

$$\mathfrak{b}_b^H[H;\varrho] = \frac{\mathfrak{R}_b^H[H;\varrho] - H}{\log b}. \tag{B5.28}$$

If the evidence mass is deliberately forgotten and finite actions are quotiented by additive constants, the same formula is read in the quotient. For extended-valued actions, including points at which the numerator would be $+\infty - (+\infty)$, no beta is defined and the measure-pair step (B5.3) is the primary object. A beta function on a truncated coupling vector requires invariance of that finite family; otherwise the omitted hyperedges contribute an explicit residual (proposition B5.26).

**Proposition B5.24 (Change of reference is an inhomogeneous law for the action beta).** Let $H$ be an action with $\mathsf{m} = e^{-H}\varrho$ finite and positive, let $\Delta$ be bounded and measurable with $\varrho(e^{-\Delta}) = 1$, and set

$$\varrho' = e^{-\Delta}\varrho, \qquad H' = H - \Delta, \qquad \text{so that} \qquad e^{-H'}\varrho' = e^{-H}\varrho = \mathsf{m}. \tag{B5.29}$$

Then $\varrho'$ is a probability reference for the same measure, and $(\varrho' K_b)$-almost everywhere

$$\mathfrak{R}_b^H[H';\varrho'] = \mathfrak{R}_b^H[H;\varrho] - \mathfrak{R}_b^H[\Delta;\varrho]. \tag{B5.30}$$

On a declared finite-valued action vector space containing $H$, $\Delta$, and their images, the discrete action beta therefore obeys the inhomogeneous law

$$\mathfrak{b}_b^H[H';\varrho'] = \mathfrak{b}_b^H[H;\varrho] - \mathfrak{b}_b^H[\Delta;\varrho]. \tag{B5.31}$$

The added term is not a linear reparameterization, and it vanishes exactly when the increment is itself fixed, $\mathfrak{R}_b^H[\Delta;\varrho] = \Delta$. A beta function is therefore a joint statement about the action and the declared reference trajectory. [ESTABLISHED]

*Proof.* Since $\varrho(e^{-\Delta}) = 1$ and $\Delta$ is bounded, $\varrho'$ is a probability measure with $e^{-\sup\Delta}\varrho \le \varrho' \le e^{-\inf\Delta}\varrho$, and (B5.29) holds by construction. Pushing these inequalities through $K_b$ shows that $\varrho K_b$ and $\varrho' K_b$ are equivalent and that $d(\varrho' K_b)/d(\varrho K_b)$ lies in $[e^{-\sup\Delta}, e^{-\inf\Delta}]$; by (B5.27) it equals $\exp(-\mathfrak{R}_b^H[\Delta;\varrho])$. Both references dominate $\mathsf{m}K_b$, and the chain rule for Radon–Nikodym derivatives

gives $d(\mathrm{m}K_b)/d(\varrho' K_b) = \big[d(\mathrm{m}K_b)/d(\varrho K_b)\big]\big[d(\varrho' K_b)/d(\varrho K_b)\big]^{-1}$ almost everywhere. Taking $-\log$ gives (B5.30); subtracting $H' = H - \Delta$ and dividing by $\log b$ gives (B5.31), and the vanishing criterion is $\mathfrak{R}_b^H[\Delta;\varrho] - \Delta = 0$. □

**Definition B5.25 (Typed discrete interaction beta).** Let $\mathcal{G}_\star$ be a declared reference interaction space and let

$$J_\ell : \mathcal{G}_\star \xrightarrow{\sim} \mathcal{G}_\ell \tag{B5.32}$$

be bounded isomorphisms with bounded inverses, the comparison data of definition B5.1 for the interaction tier. The pulled exact interaction map and the exact discrete interaction beta are

$$\widehat{\mathcal{T}}_\ell = J_{\ell+1}^{-1}\mathcal{T}_\ell J_\ell : \mathcal{G}_\star \longrightarrow \mathcal{G}_\star, \qquad \mathfrak{b}_\ell^{\mathrm{ex}}(\mathrm{g}) = \frac{\widehat{\mathcal{T}}_\ell(\mathrm{g}) - \mathrm{g}}{\Delta\varsigma_\ell}, \qquad \Delta\varsigma_\ell = \log b_\ell, \tag{B5.33}$$

with $\mathcal{T}_\ell$ the exact step (12.27). Both terms of the numerator lie in $\mathcal{G}_\star$; the difference $\mathcal{T}_\ell(\mathrm{g}) - \mathrm{g}$ of elements of $\mathcal{G}_{\ell+1}$ and $\mathcal{G}_\ell$ is not formed. If the coupling object is a manifold, the difference is replaced by a declared chart difference, logarithm, or retraction into one tangent space.

The definition has no instance on a finite-alphabet step that reduces the support cardinality, because the two interaction spaces then have different dimensions (proposition 12.22); on such steps the untransported residual (12.37) is used.

**Proposition B5.26 (Exact, retained, and residual interaction beta).** Let $\mathsf{P}_\ell^{\mathrm{ret}}$ be the bounded idempotent retained projections of (12.37), put $\widehat{\mathsf{P}}_\ell = J_\ell^{-1}\mathsf{P}_\ell^{\mathrm{ret}}J_\ell$, and define, for $\mathrm{g} \in \operatorname{Ran}\widehat{\mathsf{P}}_\ell$,

$$\mathfrak{b}_\ell^{\mathrm{ret}}(\mathrm{g}) = \frac{\widehat{\mathsf{P}}_{\ell+1}\widehat{\mathcal{T}}_\ell(\mathrm{g}) - \mathrm{g}}{\Delta\varsigma_\ell}, \qquad \delta\mathfrak{b}_\ell(\mathrm{g}) = \mathfrak{b}_\ell^{\mathrm{ex}}(\mathrm{g}) - \mathfrak{b}_\ell^{\mathrm{ret}}(\mathrm{g}). \tag{B5.34}$$

Then the omitted part of the flow is the transported coordinate residual divided by the log block step,

$$\delta\mathfrak{b}_\ell(\mathrm{g}) = \frac{(I - \widehat{\mathsf{P}}_{\ell+1})\widehat{\mathcal{T}}_\ell(\mathrm{g})}{\Delta\varsigma_\ell} = \frac{J_{\ell+1}^{-1} r_{\ell+1}^{\mathcal{G}}(J_\ell \mathrm{g})}{\Delta\varsigma_\ell}, \tag{B5.35}$$

and its action-quotient partner is

$$\delta\bar{\mathfrak{b}}_{\ell+1}^{\mathcal{A}}(\mathrm{g}) = \frac{\mathrm{asm}_{\ell+1}\, r_{\ell+1}^{\mathcal{G}}(J_\ell \mathrm{g})}{\Delta\varsigma_\ell}, \qquad \mathrm{hd}_{\ell+1}\, \delta\bar{\mathfrak{b}}_{\ell+1}^{\mathcal{A}}(\mathrm{g}) = \frac{r_{\ell+1}^{\mathcal{G}}(J_\ell \mathrm{g})}{\Delta\varsigma_\ell}. \tag{B5.36}$$

Consequently $\delta\mathfrak{b}_\ell$ vanishes on the retained sector if and only if the exact image of the retained sector is retained,

$$\delta\mathfrak{b}_\ell\big|_{\operatorname{Ran}\widehat{\mathsf{P}}_\ell} = 0 \quad \Longleftrightarrow \quad \mathcal{T}_\ell(\operatorname{Ran}\mathsf{P}_\ell^{\mathrm{ret}}) \subseteq \operatorname{Ran}\mathsf{P}_{\ell+1}^{\mathrm{ret}}, \tag{B5.37}$$

which is (12.39), and otherwise $\delta\mathfrak{b}_\ell$ obeys the two-sided norm control of (12.40). Boundedness and idempotence do not imply this invariance, and a retained fixed point need not be an exact one: on $\mathcal{G}_\star = \mathbb{R}^2$ with $b = e$, $\widehat{\mathsf{P}}(x,y) = (x,0)$, and $\widehat{\mathcal{T}}(x,y) = (x,x)$, every retained input $(x,0)$ has $\mathfrak{b}^{\mathrm{ret}} = (0,0)$ while $\mathfrak{b}^{\mathrm{ex}} = (0,x)$. [ESTABLISHED]

*Proof.* Subtracting the two displays of (B5.34) cancels $\mathrm{g}$ and leaves $(I-\widehat{\mathrm{P}}_{\ell+1})\widehat{\mathcal{T}}_\ell(\mathrm{g})/\Delta\varsigma_\ell$. Substituting the conjugated forms of $\widehat{\mathrm{P}}_{\ell+1}$ and $\widehat{\mathcal{T}}_\ell$ gives $J_{\ell+1}^{-1}(I-\mathrm{P}_{\ell+1}^{\mathrm{ret}})\mathcal{T}_\ell(J_\ell\mathrm{g})/\Delta\varsigma_\ell$, which is (B5.35) by the definition of $r_{\ell+1}^{\mathcal{G}}$. Applying $\mathrm{asm}_{\ell+1}$ and $\mathrm{hd}_{\ell+1}$ and using the inverse identities of theorem 12.12 with (12.38) gives (B5.36). Since $J_{\ell+1}^{-1}$ is injective and $\Delta\varsigma_\ell>0$, $\delta\mathfrak{b}_\ell(\mathrm{g})=0$ exactly when $r_{\ell+1}^{\mathcal{G}}(J_\ell\mathrm{g})=0$, and $J_\ell$ maps $\operatorname{Ran}\widehat{\mathrm{P}}_\ell$ onto $\operatorname{Ran}\mathrm{P}_\ell^{\mathrm{ret}}$; quantifying over that sector and using (12.39) gives (B5.37). The witness is direct substitution: $\widehat{\mathcal{T}}(x,0)=(x,x)$ and $\widehat{\mathrm{P}}(x,x)=(x,0)$. □

### B5.3.1 Scheme dependence and invariant candidates

**Proposition B5.27 (Scheme dependence of beta functions).** (i) Let $g(t)$ be running coordinates with beta function $\mathfrak{b}(g)=dg/dt$. Under a smooth reparameterization $g'=f(g)$ and a scale change $t'=\mathrm{c}t$ with $\mathrm{c}>0$,

$$\mathfrak{b}'(g')=\frac{1}{\mathrm{c}}Df(g)\mathfrak{b}(g). \tag{B5.38}$$

(ii) If the comparison isomorphisms of definition B5.25 are replaced by $J'_\ell=J_\ell\mathcal{S}_\ell$ for bounded isomorphisms $\mathcal{S}_\ell$ of $\mathcal{G}_\star$, then $\widehat{\mathcal{T}}'_\ell=\mathcal{S}_{\ell+1}^{-1}\widehat{\mathcal{T}}_\ell\mathcal{S}_\ell$ and

$$\mathfrak{b}'_\ell(\mathrm{g})=\frac{\mathcal{S}_{\ell+1}^{-1}\widehat{\mathcal{T}}_\ell(\mathcal{S}_\ell\mathrm{g})-\mathrm{g}}{\Delta\varsigma_\ell}, \tag{B5.39}$$

which reduces to the pushforward relation $\mathfrak{b}'_\ell(\mathrm{g})=\mathcal{S}^{-1}\mathfrak{b}_\ell^{\mathrm{ex}}(\mathcal{S}\mathrm{g})$ only for a scale-independent $\mathcal{S}_\ell\equiv\mathcal{S}$; in a moving comparison frame even the identity step has nonzero beta: on $\mathcal{G}_\ell=\mathbb{R}$ with $\mathcal{T}_\ell=\mathrm{id}$, $b_\ell=e$, and $J_\ell u=a_\ell u$, (B5.33) gives $\mathfrak{b}_\ell(\mathrm{g})=(a_\ell/a_{\ell+1}-1)\mathrm{g}$. (iii) For a differentiable positive endomorphism homogeneous of degree one with a fixed ray, radial eigenvalue $\mu_0$, and transverse eigenvalue $\mu_a$, define for a declared block factor $b$

$$r_a=\frac{\mu_a}{\mu_0},\qquad \upsilon_a=\frac{\log|r_a|}{\log b}. \tag{B5.40}$$

The ratios $r_a$ are the eigenvalues at the fixed ray of the derivative of the induced map on rays, and $r_a$ and $\upsilon_a$ are invariant under conjugation of the induced map by a $C^1$ diffeomorphism near the fixed ray with invertible derivative there, provided the same $b$ is used and, when a declared gauge group has an orbit of fixed rays through the ray, the conjugacy maps that orbit to the corresponding orbit and transverse eigenvalues are computed on the quotient by its tangent space. Invertibility of the derivative is needed: $x\mapsto x^3$ conjugates $x\mapsto 2x$ to $y\mapsto 8y$. [ESTABLISHED]

*Proof.* (i) By the chain rule, $dg'/dt'=Df(g)(dg/dt)(dt/dt')=\mathrm{c}^{-1}Df(g)\mathfrak{b}(g)$. (ii) Substitution gives $\widehat{\mathcal{T}}'_\ell=\mathcal{S}_{\ell+1}^{-1}J_{\ell+1}^{-1}\mathcal{T}_\ell J_\ell\mathcal{S}_\ell$ and then (B5.39); for constant $\mathcal{S}$ the numerator is $\mathcal{S}^{-1}(\widehat{\mathcal{T}}_\ell(\mathcal{S}\mathrm{g})-\mathcal{S}\mathrm{g})$. In the example $\widehat{\mathcal{T}}_\ell=J_{\ell+1}^{-1}J_\ell=a_\ell/a_{\ell+1}$ and $\Delta\varsigma_\ell=1$. (iii) Differentiating $F(tx)=tF(x)$ in $t$ gives Euler's relation $DF(x)x=F(x)$, so at a fixed ray $F(x_*)=\mu_0x_*$ the ray is an eigendirection of $DF(x_*)$ with eigenvalue $\mu_0$, and in an affine chart of rays the derivative of the induced map is $DF(x_*)/\mu_0$ on the quotient by the ray, with eigenvalues $r_a$. Under a $C^1$ conjugacy with invertible derivative at the ray, the chain rule makes the conjugated derivative similar to the original; a conjugacy mapping the orbit of fixed rays to the corresponding orbit carries the tangent space of the orbit to its counterpart, so

the induced quotient maps are similar as well. For the last claim, $(2y^{1/3})^3 = 8y$, and the derivative of $x^3$ vanishes at the fixed point. □

Raw couplings, beta components, and a chosen blocking statistic are therefore scheme-dependent coordinates; they are diagnostics once their scheme is stated. Candidate cross-scheme quantities are conjugacy classes, invariant manifolds, projective ratios, dimensionless amplitude ratios, and matched spectral or heat exponents. Their existence, convergence, and completeness in a given realization, on which any universality statement rests, are part of open problem B6.19.

# Chapter B6

# Continuous Scale, Fixed Points, Spectra, and Two-Index Limits

Given the comparison data of chapter B5, a renormalization flow usually carries four further constructions: a continuous scale with its beta function, fixed points with their linearization, spectra, and limits in which system size and depth grow together. This chapter asks what each requires and what survives for the finite networks of the main manuscript. Over an interval, a continuous beta function is a $C^1$ family of comparison maps that discrete endpoints do not determine (proposition B6.3). A fixed measure pair of a rescaled kernel with a unique invariant law has constant action (proposition B6.6), and fixedness does not transfer between the law, action, interaction, configuration, and attention tiers (proposition B6.10). The positive-unital essential-spectrum theorem has a hypothesis that no exact arrow satisfies (theorem B6.12), and one exact kernel has different spectral radii in two topologies (proposition B6.13). Fixed-depth and maximal-depth limits need not agree (proposition B6.15). On Gaussian operator hierarchies, dense scaling leads to a boundary operator ray (proposition B6.20), and under one primitive identification Perron–Frobenius attraction fixes the fiber matrix without selecting it (proposition B6.22). The open problems of the chapter state what an infinite-network theory of the agent network would have to supply.

Beyond the symbols of chapter B5, this chapter writes $t$ for a scale coordinate, $\mathfrak{b}^{\mathrm{scale}}$ for the continuous beta, $(\varrho_*, \mathrm{m}_*)$ for a fixed measure pair, $\mathrm{T}$ for a tier label, $\mathcal{N}$ for spectral counting functions, $\mathcal{W}$ for operator cones, and $\xi$ for growth exponents of block sums.

**Scope.** The statements this part declines are listed in the scope paragraph of chapter B5. The open constructions of this chapter are a canonical continuous beta (open problem B6.4), two-index limits and universality (open problem B6.19), and a primitive hierarchical identification for Gaussian operator hierarchies (open problem B6.23); the continuum theory of laws is open problem B1.22.

## B6.1 Continuous scale

**Definition B6.1 (Smooth scale bundle, scale connection, and continuous beta).** A continuous beta function lives on a separately declared tier. Declare an oriented one-dimensional $C^1$ scale manifold $S$ with coordinate $t$, a $C^1$ Banach vector bundle $\pi_S : \mathscr{G} \to S$ whose fibers are coupling spaces, and a scale connection, a covariant derivative $\nabla^{\mathrm{scale}}$ on sections of $\mathscr{G}$ with the usual linearity

and Leibniz rules. For a $C^1$ running coupling section $g(t) \in \mathscr{G}_t$, the continuous beta is

$$\mathfrak{b}^{\mathrm{scale}}(t) = \nabla^{\mathrm{scale}}_{\partial_t} g(t) \in \mathscr{G}_t. \tag{B6.1}$$

In a local $C^1$ trivialization $J_t : \mathcal{G}_\star \to \mathscr{G}_t$ with connection form $A_t \in \mathcal{L}(\mathcal{G}_\star)$ and $\widetilde{g}(t) = J_t^{-1} g(t)$,

$$J_t^{-1}\mathfrak{b}^{\mathrm{scale}}(t) = \partial_t \widetilde{g}(t) + A_t \widetilde{g}(t), \qquad A'_t = \mathcal{S}_t A_t \mathcal{S}_t^{-1} - (\partial_t \mathcal{S}_t)\mathcal{S}_t^{-1} \ \text{ under } J'_t = J_t \mathcal{S}_t^{-1}. \tag{B6.2}$$

A raw coordinate derivative does not obey this law: on the trivial line bundle with $g = 1$ parallel in a frame with $A = 0$, the frame change $\mathcal{S}_t = e^{t^2}$ gives raw derivative $2te^{t^2} \neq 0$, while $A'_t = -2t$ and the covariant derivative $\partial_t(e^{t^2}) - 2te^{t^2}$ vanishes. The scale connection compares coupling fibers at different resolutions. It is distinct from the principal connections of chapter B1, which compare statistical fibers along the base at one resolution, and from the inference-orbit parameter and the Fisher duration of chapter 7.

**Proposition B6.2 (Density realization of the continuous beta).** Suppose scale is a continuous semigroup and that, relative to one common dominating measure, positive densities $r_t, p_t \in \mathrm{Dom}(\mathcal{A}^*)$ are differentiable in a topology that permits pointwise ratios and the dual pairings below, with $p_t = r_t \exp[-H_t]$, $\dot{r}_t = \mathcal{A}^* r_t$, and $\dot{p}_t = \mathcal{A}^* p_t$. Then, almost everywhere wherever the displayed ratios are defined,

$$\dot{H}_t = -\frac{\mathcal{A}^*(r_t \exp[-H_t])}{r_t \exp[-H_t]} + \frac{\mathcal{A}^* r_t}{r_t}, \tag{B6.3}$$

and the second term vanishes for a stationary reference. If, on a declared Banach action space with a Schauder basis $\{\psi_A\}$ and continuous coordinate functionals $\{\psi_A^*\}$, or a Hilbert space with a Riesz basis, the expansion $H_t = \sum_A g_A(t)\psi_A$ converges and differentiation and coefficient extraction can be interchanged, then the component beta functions are

$$\dot{g}_A = \left\langle \psi_A^*, -\frac{\mathcal{A}^*(r_t e^{-H_t})}{r_t e^{-H_t}} + \frac{\mathcal{A}^* r_t}{r_t} \right\rangle, \tag{B6.4}$$

and a truncated basis defines a projection with an explicit residual. [ESTABLISHED]

*Proof.* Differentiate $H_t = -\log(p_t/r_t)$ and substitute the two forward equations; apply $\psi_A^*$ to both sides and use the interchange hypothesis. □

**Proposition B6.3 (Discrete endpoints do not determine a continuous beta).** (i) On an interval of $S$ carrying a $C^1$ trivialization $J_t$ whose connection form $t \mapsto A_t$ is continuous in operator norm, the equation $\partial_t \mathcal{S}_t = \mathcal{S}_t A_t$, $\mathcal{S}_{t_0} = I$, has a unique $C^1$ solution by bounded invertible operators, and in the frame $J'_t = J_t \mathcal{S}_t^{-1}$ the connection form vanishes and $\mathfrak{b}^{\mathrm{scale}}(t) = J'_t \partial_t \big(J'^{-1}_t g(t)\big)$; over an interval the scale tier is therefore equivalent to a $C^1$ family of comparison isomorphisms. (ii) A scale evolution family

$$\mathsf{V}(t,t') : \mathscr{G}_{t'} \longrightarrow \mathscr{G}_t, \qquad \mathsf{V}(t,t') \circ \mathsf{V}(t',t'') = \mathsf{V}(t,t''), \qquad \mathsf{V}(t,t) = I_{\mathscr{G}_t}, \tag{B6.5}$$

the continuous analogue of (12.32) and again a two-parameter law, is not determined by its restriction to integer endpoints, and neither is (B6.1): on $S = \mathbb{R}$ with the trivial line bundle and zero connection form, the families

$$\mathsf{V}^{(0)}(t,t') = 1, \qquad \mathsf{V}^{(\epsilon)}(t,t') = \exp\{\epsilon[\sin(2\pi t) - \sin(2\pi t')]\} \tag{B6.6}$$

both satisfy (B6.5) and agree at every pair of integer endpoints, while their generators are $0$ and $2\pi\epsilon\cos(2\pi t)$. The one-parameter form, in which $\mathsf{V}(t,t')$ depends only on $t-t'$, is not a consequence of (B6.5). [ESTABLISHED]

*Proof.* (i) On a compact subinterval, $\sup_t\|A_t\| < \infty$ and $\|\mathcal{S}A_t - \mathcal{S}'A_t\| \le \|A_t\|\|\mathcal{S}-\mathcal{S}'\|$, so Picard iteration for the linear equation in the Banach algebra $\mathcal{L}(\mathcal{G}_\star)$ converges to a unique $C^1$ solution, and the solutions on nested compact subintervals agree. The solution $\mathcal{T}_t$ of $\partial_t\mathcal{T}_t = -A_t\mathcal{T}_t$, $\mathcal{T}_{t_0} = I$, satisfies $\partial_t(\mathcal{S}_t\mathcal{T}_t) = \mathcal{S}_tA_t\mathcal{T}_t - \mathcal{S}_tA_t\mathcal{T}_t = 0$, so $\mathcal{S}_t\mathcal{T}_t = I$; and $\mathcal{U}_t = \mathcal{T}_t\mathcal{S}_t$ solves $\partial_t\mathcal{U}_t = \mathcal{U}_tA_t - A_t\mathcal{U}_t$ with $\mathcal{U}_{t_0} = I$, whose unique solution is $I$, so $\mathcal{T}_t\mathcal{S}_t = I$. By (B6.2), $A_t' = \mathcal{S}_tA_t\mathcal{S}_t^{-1} - \mathcal{S}_tA_t\mathcal{S}_t^{-1} = 0$. (ii) Both families have the form $f(t)/f(t')$ with $f$ positive, so they satisfy the composition and identity laws. At integer $t$ and $t'$ the sine terms vanish. Differentiating $\mathsf{V}^{(\epsilon)}(t,t')$ in $t$ at $t=t'$ gives $2\pi\epsilon\cos(2\pi t')$, which is not identically zero for $\epsilon \neq 0$, so $\mathsf{V}^{(\epsilon)}$ is not a function of $t-t'$. □

The smooth scale base, the bundle, the interpolation, and the connection are therefore declared input; a discrete blocking sequence supplies the discrete beta (B5.33) and nothing more.

**Open Problem B6.4 (Canonical continuous beta).** Definition B6.1 declares the tier a continuous beta requires, and proposition B6.3 shows that over an interval the tier is a $C^1$ comparison family which a discrete blocking sequence does not determine. No canonical interpolation of a discrete sequence is known, and no reported continuous beta has been shown independent of the interpolation inside a declared class. The problem is settled by a canonical interpolation with a proof of its uniqueness in a declared class, or by a proof that the continuous beta is the same for every member of that class.

## B6.2 Fixed points, flows, and scaling operators

**Definition B6.5 (Fixed measure pair).** For a declared family of ratios $b$ and rescaled kernels $K_b$ on one common standard Borel space Y (definition B5.4), a fixed theory is a measure pair $(\varrho_*, \mathsf{m}_*)$ with $0 < \mathsf{m}_*(\mathsf{Y}) < \infty$ and

$$\mathfrak{R}_b(\varrho_*, \mathsf{m}_*) = (\varrho_*, \mathsf{m}_*) \quad \text{for every declared } b. \tag{B6.7}$$

On the dominated tier, $\mathsf{m}_* = e^{-H_*}\varrho_*$, and by uniqueness of Radon–Nikodym derivatives and (B5.27) this is equivalent to

$$\mathfrak{R}_b^{\varrho}\varrho_* = \varrho_*, \qquad \mathfrak{R}_b^{H}[H_*;\varrho_*] = H_*, \tag{B6.8}$$

$\varrho_*$-almost everywhere. If the evidence mass is discarded and actions are taken projectively, the fixed-ray equation is

$$\mathfrak{R}_b^H[H_*;\varrho_*] = H_* + c_{\mathrm{ray}}(b); \tag{B6.9}$$

with an invariant reference it forces $c_{\mathrm{ray}}(b) = 0$, since $K_b$ preserves mass and $\mathsf{m}_*(\mathsf{Y}) = (\mathsf{m}_* K_b)(\mathsf{Y}) = e^{-c_{\mathrm{ray}}(b)}\mathsf{m}_*(\mathsf{Y})$. A fixed attention law is a fixed rescaled event measure $\eta_*$; by uniqueness of disintegration it is fixed exactly when its receiver marginal $\alpha$ is fixed and its row $\beta$ is fixed $\alpha$-almost everywhere.

**Proposition B6.6 (Fixed measure pairs of a uniquely ergodic kernel have constant action).** Suppose that for some declared $b$ the rescaled kernel $K_b = C_b I_b$ on the common standard Borel space has exactly one invariant probability law. If $(\varrho_*, \mathsf{m}_*)$ satisfies (B6.7) with $0 < M_* = \mathsf{m}_*(\mathsf{Y}) < \infty$, then $\mathsf{m}_* = M_* \varrho_*$, so on the dominated tier $H_* = -\log M_*$ is constant $\varrho_*$-almost everywhere. On a finite space the hypothesis holds whenever $K_b$ is irreducible, in particular whenever some power of $K_b$ has all entries positive, and $K_b$ has all entries positive whenever every row of $I_b$ charges every point. Consequently a fixed theory with nonconstant action requires a rescaled kernel with at least two invariant probability laws: on a finite space a reducible rescaled kernel, for example a reducible identification, and otherwise an infinite space on which uniqueness of the invariant law fails. [ESTABLISHED]

*Proof.* The normalized kernel preserves total mass, so $\mathsf{m}_* K_b = \mathsf{m}_*$ makes $\mathsf{m}_*/M_*$ an invariant probability law, and uniqueness gives $\mathsf{m}_*/M_* = \varrho_*$. For the finite case, first let some power $K_b^p$ have all entries positive. For probability vectors $\mu, \nu$, write $\mu - \nu = s(\mu_1 - \nu_1)$ with $s = \|\mu - \nu\|_{\mathrm{TV}}$ and probability vectors $\mu_1, \nu_1$; convexity gives $\|\mu K_b^p - \nu K_b^p\|_{\mathrm{TV}} \le \delta \|\mu - \nu\|_{\mathrm{TV}}$ with $\delta = \max_{y,y'} \|K_b^p(y,\cdot) - K_b^p(y',\cdot)\|_{\mathrm{TV}} \le 1 - \sum_z \min_y K_b^p(y,z) < 1$, so two invariant laws of $K_b$, which are invariant for $K_b^p$, coincide. If $K_b$ is only irreducible on $n$ states, the lazy kernel $(I + K_b)/2$ has the same invariant laws, and $(I + K_b)^{n-1} = \sum_{k=0}^{n-1} \binom{n-1}{k} K_b^k$ has all entries positive, because irreducibility gives a positive-probability path of length at most $n-1$ between any two states; the previous case applies. Finally, $(C_b I_b)(y,y') = \sum_z C_b(y,z) I_b(z,y') > 0$ when every $I_b(z,y') > 0$, because each row of $C_b$ is a probability vector. □

At a fixed pair with $H_*$ in the declared finite-valued action Banach space, set

$$\mathsf{m}_*(dy) = e^{-H_*(y)}\varrho_*(dy), \qquad M_* = \mathsf{m}_*(\mathsf{Y}), \qquad \bar{\mathsf{m}}_* = M_*^{-1}\mathsf{m}_*, \qquad \bar{\mathsf{m}}_*^c = \bar{\mathsf{m}}_* K_b, \tag{B6.10}$$

fix a standard Borel reverse-kernel version $K_b^\dagger$ with

$$\bar{\mathsf{m}}_*(dy) K_b(y, dz) = \bar{\mathsf{m}}_*^c(dz) K_b^\dagger(z, dy), \tag{B6.11}$$

and require

$$0 < \mathbb{E}_{\varrho_*}\left[e^{-H_*} \mid Z = z\right] < \infty \quad \varrho_* K_b\text{-almost everywhere.} \tag{B6.12}$$

For bounded perturbations $\varphi \in L^\infty(\bar{\mathsf{m}}_*)$, theorem 12.4, applied to the pair $(\varrho_*, \mathsf{m}_*)$ and the composite kernel $K_b$, which already includes the rescaling, gives $\bar{\mathsf{m}}_*^c$-almost everywhere

$$D_H\mathfrak{R}_b^H\big|_{(H_*,\varrho_*)}[\varphi](z) = \mathbb{E}_{K_b^\dagger(z,dy)}\varphi(Y), \qquad D_H^2\mathfrak{R}_b^H\big|_{(H_*,\varrho_*)}[\varphi,\varphi](z) = -\operatorname{Var}_{K_b^\dagger(z,dy)}\varphi(Y). \quad \text{(B6.13)}$$

**Remark B6.7 (A declared passive projection scheme).** On a homogeneous six-cycle with ratio two and a homogeneous nine-cycle with ratio three, a declared projected map was iterated: tile one site table and one pair table onto the cycle, apply the blocking step with attention frozen as declared data, and read back one coarse site table and one pair table in anchored Möbius coordinates. This is a projection scheme on a finite-dimensional coupling space, not the exact step (12.27) with comparison isomorphisms, and no exact-image invariance (12.39) is proved for its read-back. At both ratios the iteration converged to fixed structures of the scheme whose pairwise block vanished to machine precision (check `CHK-RG-PASSIVE-BLOCKING`, table BD.3); these are fixed points of the scheme, not fixed objects of the exact step. [NUMERICAL]

**Remark B6.8 (A declared row-regeneration scheme).** On the same instances, a declared row-regeneration rule, which at each level recomputes attention rows as softmax rows of transported divergences averaged over the flow, with $\tau = 1$ and uniform receiver occupancies, and folds the alignment energy into the coarse action, gave fixed structures of the scheme with nonzero pairwise block at both ratios (check `CHK-RG-REGENERATED-ATTENTION`, table BD.3). The rule is a declared approximate channel, not the exact event-law pushforward (12.50): its rows are softmax rows of averaged energies, the form of the constant-row optimum of proposition 6.3, which differs from an average of posterior rows by the gap (6.11), and its error relative to the exact pushforward is not bounded. The passive triviality of remark B6.7 is therefore a property of the frozen-attention scheme on these instances; nothing here shows that the exact pushforward maintains interaction. [NUMERICAL]

**Definition B6.9 (Typed invariant and fixed objects).** Write $Y_\ell^{\mathrm{T}}$ for the object of one declared tier $\mathrm{T}$ and $F_\ell^{\mathrm{T}} : Y_\ell^{\mathrm{T}} \to Y_{\ell+1}^{\mathrm{T}}$ for its declared step. The general invariant object of a nonautonomous scale diagram is an invariant section, a sequence with

$$y_{\ell+1} = F_\ell^{\mathrm{T}}(y_\ell), \quad \text{(B6.14)}$$

which needs no identification. After declared isomorphisms $J_\ell : Y_\star^{\mathrm{T}} \to Y_\ell^{\mathrm{T}}$, the reference step is

$$\widehat{F}_\ell^{\mathrm{T}} = J_{\ell+1}^{-1} \circ F_\ell^{\mathrm{T}} \circ J_\ell : Y_\star^{\mathrm{T}} \longrightarrow Y_\star^{\mathrm{T}}, \quad \text{(B6.15)}$$

and a reference fixed object is a $y_\star \in Y_\star^{\mathrm{T}}$ with

$$\widehat{F}_\ell^{\mathrm{T}}(y_\star) = y_\star \qquad \text{for every } \ell, \quad \text{(B6.16)}$$

the ordinary fixed-point equation in an autonomous scheme. If the identified maps are $p$-periodic, the base-$\ell$ reference monodromy is

$$\widehat{\mathcal{M}}_\ell^{\mathrm{T}} = \widehat{F}_{\ell+p-1}^{\mathrm{T}} \circ \cdots \circ \widehat{F}_\ell^{\mathrm{T}} : Y_\star^{\mathrm{T}} \longrightarrow Y_\star^{\mathrm{T}}, \tag{B6.17}$$

a reference monodromy fixed object satisfies $\widehat{\mathcal{M}}_\ell^{\mathrm{T}}(y_{\star,0}) = y_{\star,0}$, and its reference $p$-cycle $y_{\star,j+1} = \widehat{F}_{\ell+j}^{\mathrm{T}}(y_{\star,j})$, $y_{\star,p} = y_{\star,0}$, lies in $Y_\star^{\mathrm{T}}$, with points that need not be fixed by any one-step reference map. Without a common object the equation $F_\ell^{\mathrm{T}}(y) = y$ is ill typed, and a fixed ray additionally requires a declared cone and projectivization. The tiers are the law tier (the normalized law, the measure pair of definition 12.1, and the event law (6.13), with fixed objects as in definition B6.5); the action tier ($H_\ell$ relative to $\varrho_\ell$ and, separately, its additive class, related by (B6.8) and (B6.9) and meaningful only with a reference trajectory by proposition B5.24); the interaction tier ($\mathsf{g}_\ell \in \mathcal{G}_\ell$, step $\mathcal{T}_\ell$, reference step $\widehat{\mathcal{T}}_\ell$); the retained tier ($\operatorname{Ran} \mathsf{P}_\ell^{\mathrm{ret}}$, with exact fixed objects only under (B5.37)); the gauge-retained state tier (11.1); and the configuration tier of chapter 7.

**Proposition B6.10 (Fixedness does not transfer between tiers).** None of the following implications holds: a fixed normalized law gives a fixed reference-dependent action; a fixed action class gives a fixed tracked measure pair; a fixed retained interaction gives a fixed exact interaction; a fixed law gives a fixed configuration; a fixed conditional attention row gives a fixed event law; a monodromy fixed object is a one-step fixed object. [ESTABLISHED]

*Proof.* On a two-point set, hold $\mu = (1/2, 1/2)$ fixed while the reference alternates between $(1/4, 3/4)$ and $(3/4, 1/4)$; the densities of $\mu$ are $(2, 2/3)$ and $(2/3, 2)$, so the actions are interchanged. The actions $H$ and $H + c$ have the same class, while $e^{-H}\varrho$ and $e^{-c}e^{-H}\varrho$ have evidence masses differing by the factor $e^{-c}$, which is why (B6.9) is weaker than (B6.7) until mass preservation forces $c_{\mathrm{ray}}(b) = 0$. The witness of proposition B5.26 has zero retained beta at every $(x, 0)$ and exact image $(x, x)$ for $x \neq 0$. On the configuration manifold $S^1$ with the antipodal step $x \mapsto -x$ and a constant configuration-to-law extraction, the law is fixed while the configuration step has no fixed point. Holding $\beta$ fixed while the receiver occupancy $\alpha_\ell$ alternates makes $\eta_\ell = \alpha_\ell\beta$ alternate, which is the disintegration clause of definition B6.5. Finally, $F_0(x) = x + 1$ and $F_1(x) = x - 1$ on $\mathbb{R}$ have monodromy $F_1 \circ F_0 = \mathrm{id}$, which fixes every point, while neither one-step map has a fixed point. □

**Proposition B6.11 (Exact fixed and invariant sectors).** (i) The identity channel with the identity identification fixes every measure pair. (ii) A terminal channel followed by its one-point identification has the constant fixed theory. (iii) On $\mathsf{Y} = (\mathbb{R}^d)^{\mathbb{Z}}$, let $K_b$ be the deterministic map $(y_i)_i \mapsto \big(b^{-1/\alpha} \sum_{i \in B_j} y_i\big)_j$ over consecutive blocks $B_j$ of size $b$, with the blocks relabeled as sites. The product of a strictly $\alpha$-stable law with constant likelihood $\mathsf{m}_* = M_*\varrho_*$ is a fixed measure pair for every integer $b \geq 2$; invariance of a law for every integer $b \geq 2$ under this normalization is the definition of strict $\alpha$-stability, and at one integer $b$ the list is not exhaustive, because a law can be invariant for that $b$ without being strictly stable. In (ii) and (iii) the action is constant, whereas the identity channel, which has every law invariant, fixes pairs with nonconstant action. (iv) Graph-flat link data, with trivial holonomy (2.30) on every closed walk, stay graph flat under rooted contraction: every dressed edge (11.13) internal to a block is the identity, and every coarse closed walk has

trivial holonomy. (v) One-hot rows are invariant under the softmax parameter-path derivative (6.21). [ESTABLISHED]

*Proof.* (i) and (ii) are immediate, and in (ii) the one-point space carries a constant action. (iii) If $Y_i$ are independent with a strictly $\alpha$-stable law, $b^{-1/\alpha}\sum_{i\in B_j} Y_i$ has the same law for every block, and the blocks are independent, so $\varrho_* K_b = \varrho_*$ and $\mathsf{m}_* K_b = M_* \varrho_* K_b = \mathsf{m}_*$. For the last clause take $d = 1$, $0 < \alpha < 1$, and the infinitely divisible law on $[0,\infty)$ with Lévy measure $\nu = \sum_{k\in\mathbb{Z}} b^k \delta_{b^{-k/\alpha}}$ and no drift, which exists because $\int \min\{1, x\}\nu(dx) < \infty$ [28, Theorem 8.1]. The image of $\nu$ under $x \mapsto b^{1/\alpha}x$ is $b\nu$, so the sum of $b$ independent copies has the law of $b^{1/\alpha}$ times one copy. If the sum of $n$ copies had the law of $c_n$ times one copy for every $n$, uniqueness of the Lévy measure [28, Theorem 8.1] would give $n\nu$ equal to the image of $\nu$ under $x \mapsto c_n x$; matching atoms forces $c_n = b^{j/\alpha}$ and $n = b^j$ for an integer $j$, which fails for $n = b + 1$. (iv) For an edge $e : j \to i$ inside a block $I$, the dressed edge $\Theta^x_{I\leftarrow i}\Theta^x_e(\Theta^x_{I\leftarrow j})^{-1}$ is the holonomy of the closed walk from the root to $j$ along the tree, across $e$, and back to the root, hence the identity. A product of dressed edges along a coarse closed walk is, after the inner tree factors $(\Theta^x_{J\leftarrow j})^{-1}\Theta^x_{J\leftarrow j'}$ are read as tree paths inside $J$, the holonomy of a fine closed walk, hence trivial. (v) A row with $\beta_J = \delta_{JK}$ has $\dot\beta_J = \beta_J(u_J - \sum_L \beta_L u_L) = 0$ for every $J$. □

Flat link data and uniform attention rows are invariant forms; they are fixed objects only after a self-similar identification of graphs, block sizes, receiver occupancies, and source sets is declared (definition B6.9).

## B6.3 Spectra and topology dependence

**Theorem B6.12 (Qualified positive-unital essential-spectrum theorem).** Let $\mathcal{V}$ be a complex Banach lattice with quasi-interior unit $\mathbf{1} \in \mathcal{V}_+$, let $U \in \mathcal{L}(\mathcal{V})$ be positive and unital, and write

$$r_{\mathrm{ess}}(U) = r\big([U]\big) \tag{B6.18}$$

for the spectral radius of the class $[U]$ of $U$ in the Calkin algebra $\mathcal{L}(\mathcal{V})/\mathcal{K}(\mathcal{V})$. Assume that whenever $r(U) > r_{\mathrm{ess}}(U)$, the value $r(U)$ is a pole of the resolvent and $U^*$ has a nonzero positive eigenfunctional $f^*$ with $U^* f^* = r(U) f^*$. Then

$$r(U) > 1 \quad\Longrightarrow\quad r(U) = r_{\mathrm{ess}}(U). \tag{B6.19}$$

The hypothesis $r(U) > 1$ fails for every reverse conditional expectation of the exact arrows, which is a contraction on each $L^p$, $1 \le p \le \infty$ (theorem 12.5); the theorem concerns positive unital operators outside the exact arrow class, and it does not assert quasi-compactness or a Perron theorem under weaker hypotheses. [ESTABLISHED]

*Proof.* The quotient map is contractive, so $r_{\mathrm{ess}}(U) \le r(U)$. Suppose $r(U) > 1$ and that the inequality is strict. The hypothesis gives $f^* \ge 0$, $f^* \ne 0$, and $U^* f^* = r(U) f^*$. Quasi-interiority implies $f^*(\mathbf{1}) > 0$: otherwise positivity makes $f^*$ vanish on the principal ideal generated by $\mathbf{1}$, hence on its

dense closure. Unitality gives

$$r(U)f^*(\mathbf{1}) = (U^*f^*)(\mathbf{1}) = f^*(U\mathbf{1}) = f^*(\mathbf{1}), \tag{B6.20}$$

a contradiction, so the two radii agree. A contraction has spectral radius $r(U) = \lim_n \|U^n\|^{1/n} \le 1$. □

**Proposition B6.13 (Circle witness for norm-dependent radius).** Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ carry its geodesic metric and Haar probability $\mathrm{Leb}_{\mathbb{T}}$, let $\mathsf{T}_2(z) = 2z \bmod 1$, and set

$$K(y, A) = \frac{1}{2} \sum_{z \in \mathsf{T}_2^{-1}\{y\}} \mathbf{1}_A(z) \qquad (A \subseteq \mathbb{T} \text{ Borel}). \tag{B6.21}$$

The kernel preserves Haar measure and its reverse is $K^\dagger(z, dy) = \delta_{\mathsf{T}_2(z)}(dy)$, so the reverse conditional expectation is $\mathsf{U}f = f \circ \mathsf{T}_2$. Its spectral radius is one on $L^\infty(\mathrm{Leb}_{\mathbb{T}})$, and for every $0 < \alpha \le 1$ it is $2^\alpha$ on complex periodic $C^\alpha(\mathbb{T})$ with the sup-plus-Hölder norm, which at $\alpha = 1$ is the Lipschitz space $C^{0,1}$. The Hölder space with this norm is not the Banach lattice of theorem B6.12; the same exact kernel has different spectral radii in two declared topologies. [ESTABLISHED]

*Proof.* Haar preservation gives $\|f \circ \mathsf{T}_2^n\|_\infty = \|f\|_\infty$, so $\|\mathsf{U}^n\| = 1$ on $L^\infty$. With $[f]_\alpha$ the Hölder seminorm,

$$[f \circ \mathsf{T}_2^n]_\alpha \le 2^{\alpha n}[f]_\alpha, \qquad \|\mathsf{U}^n f\|_\infty \le \|f\|_\infty, \tag{B6.22}$$

so the radius on $C^\alpha$ is at most $2^\alpha$. For $f(s) = e^{2\pi i s}$, the values of $f \circ \mathsf{T}_2^n$ at $0$ and $2^{-(n+1)}$ differ by two while their distance is $2^{-(n+1)}$, so

$$[f \circ \mathsf{T}_2^n]_\alpha \ge 2^{1+\alpha(n+1)}, \tag{B6.23}$$

and division by the fixed $C^\alpha$ norm of $f$ gives the matching lower bound. □

## B6.4 Two-index limits and infinite index sets

**Definition B6.14 (Two-index family).** A finite-volume scale family has two indices,

$$X_{n,\ell}, \qquad 0 \le \ell \le L(n), \tag{B6.24}$$

where $n$ controls system size and $\ell$ depth. Its fixed-depth, maximal-depth, and diagonal limits are

$$\lim_{\ell\to\infty} \lim_{n\to\infty} X_{n,\ell}, \qquad \lim_{n\to\infty} X_{n,L(n)}, \qquad \lim_{n\to\infty} X_{n,\ell(n)}, \tag{B6.25}$$

each in a declared mode of convergence. A thermodynamic scale statement specifies a triangular family, projective consistency at fixed depth, uniform control of the coarse error, and a diagonal scaling $\ell(n)$; operator, law, connection, and cross-morphism convergence are separate obligations.

**Proposition B6.15 (Fixed-depth and maximal-depth limits need not agree).** Fix an integer $b \geq 2$, put $N = b^n$, and let $\mathrm{Path}_N$ be the unweighted path on $\{1, \dots, N\}$ with combinatorial Laplacian $L_N$. At depth $0 \leq \ell \leq n$ form the consecutive blocks

$$B_j^{(\ell)} = \{(j-1)b^\ell + 1, \dots, jb^\ell\}, \qquad 1 \leq j \leq b^{n-\ell}; \tag{B6.26}$$

joining two blocks exactly when a path edge crosses between them gives the simple quotient $\mathrm{Path}_{b^{n-\ell}}$. For $N$ divisible by $b$, the unnormalized block-indicator prolongation

$$S_{N,b} \in \mathbb{R}^{N \times (N/b)}, \qquad (S_{N,b})_{ij} = \mathbf{1}_{\{(j-1)b+1,\dots,jb\}}(i), \tag{B6.27}$$

realizes the operator quotient

$$S_{N,b}^{\mathsf{T}} L_N S_{N,b} = L_{N/b}, \tag{B6.28}$$

and with $S_{N,1} = I_N$ and, for $1 \leq \ell \leq n$,

$$S_{N,b^\ell} := S_{N,b} S_{N/b,b} \cdots S_{N/b^{\ell-1},b}, \tag{B6.29}$$

the indicator of the blocks (B6.26),

$$S_{N,b^\ell}^{\mathsf{T}} L_N S_{N,b^\ell} = L_{N/b^\ell}. \tag{B6.30}$$

This is a precomposition of the quadratic form, not the pushforward of a law; with orthonormalized indicators $b^{-1/2} S_{N,b}$ the quotient is $b^{-1} L_{N/b}$, and at depth $\ell$ the factor is $b^{-\ell}$. The eigenvalues of $L_N$, with multiplicity, are

$$\lambda_{N,k} = 2 - 2\cos\left(\frac{\pi k}{N}\right), \qquad 0 \leq k \leq N-1. \tag{B6.31}$$

With the right-continuous full and zero-mode-excluded counts

$$\mu_N = \frac{1}{N} \sum_{k=0}^{N-1} \delta_{\lambda_{N,k}}, \qquad \mathcal{N}_N(\lambda) = \mu_N((-\infty, \lambda]), \qquad \mathcal{N}_N^+(\lambda) = \frac{1}{N} \#\{1 \leq k \leq N-1 : \lambda_{N,k} \leq \lambda\}, \tag{B6.32}$$

so that $\mathcal{N}_N^+ = \mathcal{N}_N - N^{-1} \mathbf{1}_{[0,\infty)}$, one has for $0 \leq \lambda \leq 4$, with $\mathrm{k}_N(\lambda) = \min\{N-1, \lfloor N \arccos(1 - \lambda/2)/\pi \rfloor\}$,

$$\mathcal{N}_N(\lambda) = \frac{1 + \mathrm{k}_N(\lambda)}{N}, \qquad \mathcal{N}_N^+(\lambda) = \frac{\mathrm{k}_N(\lambda)}{N}, \tag{B6.33}$$

both counts vanish for $\lambda < 0$, and for $\lambda > 4$ they equal $1$ and $1 - N^{-1}$. For every fixed $\ell$, $b^{n-\ell} \to \infty$ as $n \to \infty$, and both counts converge pointwise to

$$\mathcal{N}_\infty(\lambda) = \mathcal{N}_\infty^+(\lambda) = \begin{cases} 0, & \lambda < 0, \\ \dfrac{1}{\pi} \arccos\left(1 - \dfrac{\lambda}{2}\right), & 0 \leq \lambda \leq 4, \\ 1, & \lambda > 4, \end{cases} \tag{B6.34}$$

whose lower edge obeys

$$\mathcal{N}_\infty^+(\lambda) \sim \frac{1}{\pi}\sqrt{\lambda} \qquad (\lambda \downarrow 0). \tag{B6.35}$$

At maximal depth $\ell = n$ the quotient is $\mathrm{Path}_1$ for every $n$, with $\mu_1 = \delta_0$, $\mathcal{N}_1 = \mathbf{1}_{[0,\infty)}$, and $\mathcal{N}_1^+ \equiv 0$, so the fixed-depth thermodynamic limit and the maximal-depth limit differ. Because (B6.30) is exact, every observable at $(n, \ell)$ depends only on $b^{n-\ell}$, and the diagonal $\ell(n) = n - k$ gives $\mathrm{Path}_{b^k}$ for every $n$; the example separates the limits only through this degeneracy. [ESTABLISHED]

*Proof.* Each block is connected and exactly one original edge joins block $j$ to block $j+1$, which proves the quotient statement. For $z \in \mathbb{R}^{N/b}$,

$$z^{\mathsf{T}} S_{N,b}^{\mathsf{T}} L_N S_{N,b} z = \sum_{i=1}^{N-1} \big((S_{N,b}z)_{i+1} - (S_{N,b}z)_i\big)^2 = \sum_{j=1}^{N/b-1} (z_{j+1} - z_j)^2 = z^{\mathsf{T}} L_{N/b} z,$$

and equality of the symmetric quadratic forms proves (B6.28); iteration proves (B6.30), and scalar normalization gives the stated factors. The discrete cosine eigenvectors of the path Laplacian give (B6.31), and monotonicity of $k \mapsto \lambda_{N,k}$ gives (B6.33). For bounded continuous $f$, $\int f d\mu_N = N^{-1}\sum_k f(2 - 2\cos(\pi k/N)) \to \pi^{-1}\int_0^\pi f(2-2\cos\theta)d\theta$; the limit has the continuous distribution function (B6.34), and removing the zero atom changes a count by $N^{-1}$. Finally $\arccos(1-\lambda/2) \sim \sqrt{\lambda}$, while $L_1 = 0$. The dependence on $b^{n-\ell}$ alone is (B6.30). □

**Proposition B6.16 (Heat-moment susceptibility).** Let $\mathcal{N}^+$ be nondecreasing and right continuous with $\mathcal{N}^+(0+) = 0$, set

$$Z_+(t) = \mathrm{M}_0(t), \qquad \mathrm{M}_k(t) = \int_{(0,\infty)} \lambda^k e^{-t\lambda} d\mathcal{N}^+(\lambda),$$

suppose $Z_+(t_0) < \infty$ for some $t_0 > 0$, and suppose $\mathcal{N}^+(\lambda) \sim \mathrm{c}\lambda^\alpha \mathfrak{l}(1/\lambda)$ as $\lambda \downarrow 0$, with $\mathrm{c} > 0$, $\alpha > 0$, and $\mathfrak{l}$ slowly varying at infinity. Then $\mathrm{M}_k(t) \sim \mathrm{c}\alpha\Gamma(\alpha+k)t^{-(\alpha+k)}\mathfrak{l}(t)$ as $t \to \infty$ for $k = 0, 1, 2$, and the entropy $S_+(t) = \log Z_+(t) + t\mathrm{M}_1(t)/Z_+(t)$ of the spectral Gibbs law $e^{-t\lambda}d\mathcal{N}^+(\lambda)/Z_+(t)$ satisfies

$$-\frac{dS_+}{d\log t} = t^2\left(\frac{\mathrm{M}_2}{Z_+} - \left(\frac{\mathrm{M}_1}{Z_+}\right)^2\right) \longrightarrow \alpha. \tag{B6.36}$$

Dividing $S_+$ by $\log|V_n|$ divides this susceptibility by the same factor, so zero-mode atoms and the entropy normalization must be declared before a plateau is read as a spectral dimension. [ESTABLISHED]

*Proof.* Write $\mathcal{N}_k^+(\lambda) = \int_{(0,\lambda]} \mu^k d\mathcal{N}^+(\mu)$. Integration by parts and the regular variation of $\mathcal{N}^+$ at zero give $\mathcal{N}_k^+(\lambda) \sim \mathrm{c}\alpha(\alpha+k)^{-1}\lambda^{\alpha+k}\mathfrak{l}(1/\lambda)$, and the Abelian direction of the Laplace–Stieltjes regular-variation theorem [3, §1.7.1] gives $\mathrm{M}_k(t) \sim \Gamma(1+\alpha+k)\mathrm{c}\alpha(\alpha+k)^{-1}t^{-(\alpha+k)}\mathfrak{l}(t) = \mathrm{c}\alpha\Gamma(\alpha+k)t^{-(\alpha+k)}\mathfrak{l}(t)$. Since $\mathrm{M}_1 = -Z_+'$ and $\mathrm{M}_2 = Z_+''$ for $t > t_0$, differentiation gives $dS_+/dt = -t(\mathrm{M}_2/Z_+ - (\mathrm{M}_1/Z_+)^2)$, the identity in (B6.36). The asymptotics give $t^2\mathrm{M}_2/Z_+ \to \alpha(\alpha+1)$ and $t\mathrm{M}_1/Z_+ \to \alpha$, so the limit is $\alpha(\alpha+1) - \alpha^2 = \alpha$. □

This is the Abelian direction; the converse Tauberian direction is not used.

### B6.4.1 Projective laws and refinement

**Theorem B6.17 (Cylinder-level projective law).** Let $\mathcal{I}$ be an index set and each coordinate space $\mathsf{Z}_i$ standard Borel. For every finite $D \subset \mathcal{I}$ let $P_D$ be a probability law on $\mathsf{Z}_D = \prod_{i\in D} \mathsf{Z}_i$. If

$$(\pi_{E,D})_{\#} P_E = P_D \qquad (D \subset E), \tag{B6.37}$$

then there is a unique law $P$ on the product cylinder sigma-algebra with finite-dimensional marginals $P_D$. Let a group act on $\mathsf{Z}_{\mathcal{I}} = \prod_{i\in\mathcal{I}} \mathsf{Z}_i$ by maps compatible with the finite projections: either each $g$ satisfies $\pi_D \circ g = g_D \circ \pi_D$ for bimeasurable bijections $g_D$ of $\mathsf{Z}_D$, as coordinatewise product actions do, or $g$ is an index permutation $\sigma$, for which $\pi_D \circ \sigma = \sigma_D \circ \pi_{\sigma^{-1}D}$ with the relabeling bijection $\sigma_D : \mathsf{Z}_{\sigma^{-1}D} \to \mathsf{Z}_D$. Then $g_{\#}P$ is the unique law with marginals $(g_D)_{\#}P_D$, respectively $(\sigma_D)_{\#}P_{\sigma^{-1}D}$; in particular, invariance of every finite-dimensional marginal under this rule implies $g_{\#}P = P$. An action that mixes infinitely many coordinates is not covered, because the marginals of $g_{\#}P$ are then not determined by one $P_D$. [ESTABLISHED]

*Proof.* The first statement is the Kolmogorov extension theorem for standard Borel coordinate spaces [16]. For a compatible action, $(\pi_D)_{\#}(g_{\#}P) = (g_D)_{\#}(\pi_D)_{\#}P = (g_D)_{\#}P_D$ for every finite $D$, and the permutation case is the same computation with $\pi_D \circ \sigma = \sigma_D \circ \pi_{\sigma^{-1}D}$; uniqueness of the extension identifies $g_{\#}P$. □

Consistency alone does not put the limit on smooth sections, define a continuum reference density, justify an ELBO limit, or produce a scale flow (open problem B1.22), and an energy recursion does not supply compatible laws: a probabilistic scale flow needs normalized kernels with $\mu_k = \mu_\ell K_{\ell k}$ and $K_{\ell k}K_{kr} = K_{\ell r}$, with corresponding rules for the two bundle sectors and their reference measures. For a countable agent or time index the extension supplies the normalized joint, and the finite identities of the main manuscript hold for every regular finite-cylinder observation and finite block. Conditioning on an infinite record needs a chosen regular conditional at an almost-everywhere record or a proved DLR, Doob-transform, or finite-volume posterior limit, and an infinite-law relative-entropy pushforward needs absolute continuity of the recognition law with respect to the posterior on the infinite product and an infinite coarse channel; chapters B7 and B8 prove such hypotheses on their own domains, whose generality is open problem B7.17.

**Proposition B6.18 (Infinite records and refinement).** (i) A projectively consistent family can give a prescribed infinite record probability zero: for independent binary successes of probability $p \in (0,1)$ each, the record of all successes has probability zero. (ii) If a coarse map $Q$ is noninjective, no refinement kernel $R$ recovers every realized microstate, and a refinement kernel compatible with $Q$ is a stochastic right section,

$$Q_{\#}R(c,\cdot) = \delta_c, \tag{B6.38}$$

which redraws a compatible microstate. [ESTABLISHED]

*Proof.* (i) The cylinder events of the first $n$ successes decrease to the full record and have probabilities $p^n \to 0$. (ii) If $Q(x) = Q(x')$ with $x \neq x'$, recovery of both would require $R(Q(x), \cdot) = \delta_x$ and $R(Q(x), \cdot) = \delta_{x'}$. □

**Open Problem B6.19 (Two-index limits and universality).** Fixed-depth and maximal-depth limits can differ (proposition B6.15), finite networks at $c_*$ do not inhabit the asymptotic classification (proposition B5.15), and the lattice part fixes every finite blocking word before its volume limit on its own domains (chapter B8 and open problem B7.17). It is not known whether a triangular family of finite agent networks at $c_*$ has fixed-depth or diagonal limits with nonvacuous comparison maps and nontrivial exponents, whether the two limits can be exchanged or a nontrivial diagonal limit exists, or whether the spectral labels of definition B6.24 converge for a declared Gaussian hierarchy. A universality statement needs scheme-invariant observables (proposition B5.27) with proved convergence; one matched exponent does not give it, since the limiting measure, amplitudes, corrections to scaling, invariant faces, basin, and blocking scheme may still differ. The problem is settled by a triangular family with projective consistency at fixed depth, uniform control of the coarse error, a diagonal scaling $\ell(n)$, and a proof of convergence of a declared invariant observable, or by an impossibility theorem for a stated class.

## B6.5 Gaussian operator flow on infinite hierarchies

This section applies the preceding types to the Gaussian interaction family of definition G3.1, with the hard-aggregation map (G9.2) and the positive rescaling (G9.3), whose finite exact properties are established in chapter G9. Along a hierarchy the source and target dimensions differ, so (G9.3) is a cocycle of positive maps in the sense of definition B5.2; a fixed ray is typed only after comparison maps identify a repeated hierarchy with one parameter space.

### B6.5.1 Dense scaling and the homogeneous sector spectrum

**Proposition B6.20 (Dense scaling and topology-dependent exponents).** (i) On a homogeneous complete graph with equal blocks of size $b$, common self term $A$, and common coupling $W$, the unrescaled aggregation acts on the pair $(A, W)$ by

$$A \longmapsto bA, \qquad W \longmapsto b^2 W. \tag{B6.39}$$

On $\mathbb{S}^K \oplus \mathbb{S}^K$ its eigenvalues are $b$ on the self sector and $b^2$ on the coupling sector, each of multiplicity $K(K+1)/2$, so the spectral ratio outside the dominant eigenspace is

$$\frac{b}{b^2} = \frac{1}{b}. \tag{B6.40}$$

(ii) The dense normalization $\zeta = b^2$ in (G9.3) fixes the coupling sector and contracts the self sector by $1/b$, so the candidate limiting ray has $A = 0$. It is a boundary operator: the consensus directions lie in its kernel, so it defines no normalized Gaussian density on the full space. (iii) For

a typical block $B(b)$ and coarse cut $\partial B(b)$, let $W_\partial(b) = \sum_{e\in\partial B(b)} W_e$ and $A_B(b) = \sum_{i\in B(b)} A_i$. If $\|W_\partial(b)\| \asymp b^{\xi_W}$ and $\|A_B(b)\| \asymp b^{\xi_A}$ in a declared matrix norm, a scalar normalization that keeps the coupling sector nondegenerate scales as $b^{\xi_W}$, and the normalized self sector as $b^{\xi_A-\xi_W}$. If moreover the cut has $n_{\mathrm{cut}}(b) \asymp b^{\xi}$ edges, $W_\partial(b)/n_{\mathrm{cut}}(b) \to \overline{W} \neq 0$, and $A_B(b)/b \to \overline{A} \neq 0$, then $\xi_W = \xi$ and $\xi_A = 1$, so the coupling and normalized self exponents are $\xi$ and $1-\xi$; a cut between two blocks of the complete graph has $\xi = 2$, and consecutive blocks of a nearest-neighbor chain have $\xi = 0$. [ESTABLISHED]

*Proof.* (i) By (G9.2), a coarse block $I$ has self term $\sum_{i\in I} A_i = bA$, and two coarse blocks $I \neq J$ have coupling $\sum_{i\in I, j\in J} W_{ij} = b^2 W$; the linear map is diagonal on the two sectors. (ii) Under $\zeta = b^2$ the self term scales by $b/b^2$. With $A = 0$, the block row $i$ of the assembled precision applied to $\mathbf{1}\otimes v$ is $(A_i + \sum_{j\neq i} W_{ij})v - \sum_{j\neq i} W_{ij} v$, which equals $A_i v = 0$, so every consensus vector lies in the kernel and $\exp(-x^{\mathsf{T}}\Lambda x/2)$ is not integrable along those directions. (iii) With $\zeta(b) \asymp \|W_\partial(b)\|$, the normalized sectors are of orders $1$ and $b^{\xi_A-\xi_W}$; the count assumptions give $\|W_\partial(b)\| \sim n_{\mathrm{cut}}(b)\|\overline{W}\| \asymp b^{\xi}$ and $\|A_B(b)\| \sim b\|\overline{A}\|$. A cut between two blocks of size $b$ of the complete graph has $b^2$ edges, and a chain cut has one. □

Pinning, quotienting the kernel, and retaining an external positive mass are different repairs of the boundary ray and define different probability problems; inserting a pseudoinverse into the original Lebesgue density is not a further repair. Edge counts alone do not determine the exponents when the fine weights vary with $b$, and relevance of the self term is a property of the declared weighted graph family and normalization, not of (G9.3) alone.

### B6.5.2 Cone dynamics

Let

$$\mathcal{W}_N = \left(\prod_i \mathbb{S}_+^K\right) \times \left(\prod_{i<j} \mathbb{S}_+^K\right) \tag{B6.41}$$

be the full parameter cone. On the interior of a proper cone $\mathcal{W}$, Hilbert's projective metric is

$$d_H(x,y) = \log \frac{t_+(x,y)}{t_-(x,y)}, \qquad t_+ = \inf\{t>0 : x \preceq_{\mathcal{W}} ty\}, \quad t_- = \sup\{t>0 : ty \preceq_{\mathcal{W}} x\}. \tag{B6.42}$$

A positive endomorphism of finite projective diameter contracts this metric; a matrix with strictly positive entries on the nonnegative orthant is the familiar case, and for a primitive nonnegative matrix the contraction applies to a strictly positive power, since the matrix itself can map boundary rays to boundary rays and then has infinite projective diameter [4, 23]. These hypotheses fail for sum-only aggregation on the full cone.

**Proposition B6.21 (The full-cone route is blocked).** After a hierarchical identification makes $\mathcal{A}_S$ an endomorphism, sum-only aggregation is not primitive on $\mathcal{W}_N$, and the dense candidate ray with $A = 0$ of proposition B6.20 lies on $\partial\mathcal{W}_N$, whereas a full-cone Birkhoff contraction has an interior Perron ray; Birkhoff's full-cone theorem therefore cannot prove attraction to that candidate. On the coupling cone alone, if an identified spatial map $B \geq 0$ has a positive eigenvector $c = (c_e)_e$, then for

every $M \in \mathbb{S}_+^K$

$$W_e = c_e M \tag{B6.43}$$

is an eigenray with the same spatial eigenvalue, so sum-only aggregation does not mix matrix rays and cannot select a fiber direction. [ESTABLISHED]

*Proof.* An absent coarse cut remains a zero edge coordinate, so the map can meet the boundary even from positive data on the represented fine edges. For a subspace $U \subseteq \mathbb{R}^K$, the face

$$\mathcal{F}_U = \{(W_e)_e : \operatorname{ran} W_e \subseteq U \text{ for every } e\} \tag{B6.44}$$

is invariant, because the range of a sum of positive semidefinite matrices with ranges in $U$ lies in $U$, and proper invariant faces rule out primitivity. The coordinate $A = 0$ is a boundary coordinate of $\prod_i \mathbb{S}_+^K$, while the Perron ray of a strict interior contraction is interior. Finally, aggregation adds weights, so $(B \otimes \mathrm{id})(c \otimes M) = (Bc) \otimes M$. □

**Proposition B6.22 (Fixed-$B$ attraction under a primitive identification).** Suppose a hierarchical identification makes one aggregation step act on coupling coordinates as $B \otimes \mathrm{id}$, that is $W'_f = \sum_e B_{fe} W_e$, for one fixed nonnegative matrix $B$ on edge indices with $B^q$ strictly positive for some $q \geq 1$. Let $r_B > 0$ be its spectral radius and $u, v > 0$ its left and right Perron vectors with $u^\top v = 1$. (i) Fix $M_0 \succ 0$. On the scalarized cone

$$\mathcal{W}_{M_0} = \{(c_e M_0)_e : c_e \geq 0\}, \tag{B6.45}$$

every nonzero orbit converges projectively to the ray $(v_e M_0)_e$, which is therefore the unique attracting ray; if $v_{ij} = x_i x_j$, that ray is the biadditive coupling $W_{ij} = x_i x_j M_0$ of (G9.6). (ii) On the full coupling cone, for every initial datum $(W_e^{(0)})_e$ with $W_e^{(0)} \succeq 0$,

$$r_B^{-\ell} \big(B^\ell W^{(0)}\big)_f \longrightarrow v_f M_\infty, \qquad M_\infty = \sum_e u_e W_e^{(0)}, \tag{B6.46}$$

and $\operatorname{ran} M_\infty = \sum_e \operatorname{ran} W_e^{(0)}$. Every nonzero orbit converges projectively to the product ray through $v \otimes M_\infty$: aggregation fixes the fiber matrix as the left-Perron average of the initial weights and does not select it. [ESTABLISHED]

*Proof.* A nonnegative matrix with a strictly positive power is primitive, and then $r_B^{-\ell} B^\ell \to v u^\top$ [11, §8.5]. Hence $r_B^{-\ell} (B^\ell W^{(0)})_f \to v_f \sum_e u_e W_e^{(0)}$. Because every $u_e > 0$ and every $W_e^{(0)} \succeq 0$, $\ker M_\infty = \bigcap_e \ker W_e^{(0)}$, which gives the range identity and $M_\infty \neq 0$ for every nonzero datum, so the normalized orbit converges to the product ray. On $\mathcal{W}_{M_0}$ the limit reads $v(u^\top c) M_0$, with $u^\top c > 0$ for every nonzero $c \geq 0$. Birkhoff's contraction applied to the strictly positive matrix $B^q$ gives the scalar conclusion with a geometric rate in Hilbert's metric [4]. □

The proposition covers one fixed identification acting on edge indices. It says nothing about a nonautonomous sequence $B_\ell$, whose projective contraction can approach a periodic orbit

(proposition B5.3), or about an identification that also acts on fiber directions by a level-dependent congruence, for which the step is not $B \otimes \mathrm{id}$.

**Open Problem B6.23 (Primitive hierarchical identification).** Proposition B6.22 proves attraction for one fixed primitive identification $B \otimes \mathrm{id}$. It is not known whether sum-only aggregation on a declared family of graphs admits a hierarchical identification that yields one fixed primitive $B$ and, for the biadditive reading, one whose Perron vector factorizes as $v_{ij} = x_i x_j$; attraction, uniqueness, and scheme independence of the dense ray of proposition B6.20 likewise need a declared invariant cone, hierarchy, and limiting protocol. The problem is settled by constructing such an identification for a stated family of graphs, or by proving that none exists.

### B6.5.3 Thermodynamic spectral labels

**Definition B6.24 (Thermodynamic spectral labels).** For a declared sequence $(V_n, L_n, R_n)$ of interaction Laplacians and positive reference forms, with generalized eigenvalues $\lambda_k^{(n)}$ of the pencil $(L_n, R_n)$ of section G5.3, the integrated density of states is

$$\mathcal{N}_n(t) = \frac{1}{|V_n|K} \#\{k : \lambda_k^{(n)} \le t\}. \tag{B6.47}$$

A cumulative exponent, $\mathcal{N}(t) - \mathcal{N}(0) \sim c t^{\alpha}$, is a label of the sequence only after convergence $\mathcal{N}_n \to \mathcal{N}$ in a stated mode has been proved.

A fixed finite graph has an atomic spectral measure, so no density exponent is defined there. Depth and size form the two-index family of definition B6.14, and neither the existence nor the commutation of its limits follows from the finite congruence identity (open problem B6.19).

# Part B-III

# The Lattice Cluster

# Chapter B7

# The Lattice Cluster: Typing Question and Constructive Spatial RG

This chapter asks whether the coarse-channel calculus of the main manuscript can be carried out on an infinite-volume law with local interactions so that every coarse law, every generated interaction, and every Fisher density is constructed rather than assumed. For the square-lattice $\mathbb{Z}_2$ gauge–matter law at high temperature it can. Under straight-path blocking the pushed law has a unique coarse specification at every exterior, on a coupling domain that does not shrink with the block size (theorem B7.7); its canonical interaction is complete and exponentially small in the block size (theorem B7.9); on an open ball of character coefficients the coarse-graining map is Fréchet differentiable and composes exactly (theorem B7.11), and the source law enters that ball at every sufficiently large block size (theorem B7.14); at each fixed depth the Fisher information per cell is a regular metric on an open parameter region (theorem B7.15). The proofs are in appendix BB. The every-exterior specification and the interaction bound are the inputs of the pressure and specific-VFE chapter, chapter B8, and the response, jet, and chart estimates of appendix BB are the inputs that chapter B9 and appendix BC transfer to free boundaries.

The lattice law realizes the coarse-channel layer of the main manuscript (chapter 10), not its gauge-variational agent theory: it carries no agent sections, belief or model channels, recognition law, or ELBO, and it imposes no restriction on that theory. Whether its sites are agents at $c_*$ or points of a discretized base is open problem B7.1, which heads this part. The two chapters that follow place a finite catalog of such laws on one cell lattice. Chapter B8 separates branchwise normalization from one global label factor, transports the fine observation experiment and its scalar normalization, and proves pressure and specific-VFE limits for retained labels; there, erasing a label can destroy Gibbs regularity, and a disappearing label can make the relative entropy to a limiting joint infinite while the finite-ensemble VFE density stays finite. Chapter B9 proves the Fisher geometry of these retained ensembles in volume limits.

The source family is classical. With hopping coupling $h$ and plaquette coupling $j$, the identities $e^{hz} = \cosh h(1 + bz)$ and $e^{j\chi} = \cosh j(1 + t\chi)$ for signs $z, \chi \in \{-1, 1\}$ show that definition B7.2 is the $\mathbb{Z}_2$ lattice gauge theory of Wegner [32] coupled to a $\mathbb{Z}_2$ matter (Higgs) field, as in Fradkin and Shenker [9], written in the gauge-invariant link variables $z_e$ of unitary gauge. Osterwalder and Seiler [25] proved existence and analyticity of the infinite-volume limit of strongly coupled lattice gauge theories, which is the small-$t$ regime here, and treated the lattice Higgs mechanism; Fradkin and Shenker [9] used those results to show that the Higgs and confinement regions are analytically connected for a

matter field in the fundamental representation. Dobrushin uniqueness and comparison, Kolmogorov extension, Kozlov's Gibbs representation, and Israel's high-temperature renormalization maps are cited where they are used.

Relative to these antecedents, the statements proved here for this family and its straight-path channel are the five results named in the first paragraph, the path semigroup and reference square of proposition B7.5, and the finite-amplitude approximation of theorem B7.13. The block-size uniformity of the domain $12t < 1$ in theorem B7.7 is the principal candidate for a new result: the high-temperature renormalization results summarized by van Enter et al. [31, Sec. 6.1.1] concern decimation, Kadanoff, and averaging maps of Ising-spin interactions, not this gauge–matter path channel. The literature check behind this paragraph is limited to the works cited in this chapter, so priority for these statements is not claimed. The limiting law and its region of attraction are not claimed to identify a continuum theory, and no statement about agents is made while the typing of the sites is open. The open items are open problems B7.1 and B7.17. [NOT-CLAIMED]

## B7.1 The typing question and the notation of this part

**Open Problem B7.1 (Typing of the lattice sites).** The lattice family of this part is a law on the sign configurations of the edge set $E = \mathbb{Z}^2 \times \{1, 2\}$, together with its matter signs, and nothing in its construction declares what a site $x \in \mathbb{Z}^2$ is in the bundle-geometric model of the main manuscript. Two readings are available. In the first, the sites are agents at $c_*$, the structure group is $G = \mathbb{Z}_2$, the matter sign $\phi_x$ is the frame coordinate $U_x$ of proposition 2.12, the raw link $\psi_e$ is an independent link $\Theta_e$ of definition 2.11 on the edge copy $e$, the dressed sign $z_e$ is the per-edge twist of eq. (2.32), which here is invariant under every vertex and reference change because $\mathbb{Z}_2$ is abelian, and a plaquette character is the link holonomy of eq. (2.30) around an agent-network four-cycle; the straight-path channel then keeps the agents at the sites $Bx$ and composes links along paths. In the second, the sites are points of a discretized base, $\psi_e$ is the parallel transport of a $\mathbb{Z}_2$ connection along an edge curve, a plaquette character is a discrete curvature, and the channel $C_B$ is a scale map over a base map from the fine to the coarse lattice, which is deferred base material of the kind treated in chapters B1 and B4. Every statement of this part holds under either reading, since none uses the typing, and the unique raw lift of theorem B7.3 makes the matter signs independent and uniform, so the law depends on them only through the dressed signs. What is unproved is either identification. The first requires a countably infinite agent set, whereas the interaction multigraph of definition 2.11 is finite, and it requires belief and model channels, a recognition law, and an ELBO, none of which the family carries. The second requires a declared base manifold, edge curves, and a connection whose transports are the raw links. The question is settled by declaring one reading and proving it: for the first, by realizing the lattice law as the link marginal of a normalized joint with belief and model channels on a countable agent set at $c_*$ that satisfies the countable-index projective and DLR hypotheses of chapter 12, which would make this family an instance of the infinite networks of chapter B5; for the second, by constructing the base discretization and a connection whose holonomies are the plaquette characters and showing that $C_B$ is a scale map over a declared base map.

The symbols of table B7.1 are local to this part, which comprises this chapter, chapter B8, chapter B9, and the proof appendices appendices BB and BC. Neither the belief channel label nor the relative frame occurs here. A constant introduced inside one proof is local to that proof.

**Table B7.1:** Local notation of the lattice part.

| Symbol | Meaning |
|---|---|
| $h, j$; $b = \tanh h$, $t = \tanh j$ | field and plaquette couplings and strengths; $0 < b < 1$, $0 < t < 1/12$ on the constructive domain; branch subscripts $b_\gamma, t_\gamma$ in catalog chapters |
| $E = \mathbb{Z}^2 \times \{1,2\}$, $e = (x, \iota)$, $e_\iota$ | edges; $e$ joins the cell $x$ to $x + e_\iota$ along the axis $\iota$ |
| $\phi_x$, $\psi_e$, $z_e = \phi_x \psi_e \phi_{x+e_\iota}$ | matter sign, raw link, dressed sign |
| $\chi_S(z) = \prod_{e \in S} z_e$ | character of a finite edge set $S$ |
| $\Lambda$; $\Lambda_L$, $L^2$ | finite free region; the $L$-square of cells and its volume |
| $\varsigma_\Lambda^\eta$; $\varsigma^{(B)}$ | finite specification with dressed exterior $\eta$; coarse specification at block size $B$ |
| $\mu_{b,t}$; $C_B$, $\mathcal{P}_B(x, \iota)$; $\mu_B = (C_B)_* \mu_{b,t}$ | dressed Gibbs law; straight-path channel and its retained path; pushed law |
| $\rho = \tanh(h + 2j)$; $\tau$, $\mathrm{q} = 12te^\tau$ | bound on one conditional sign mean; decay rate and contraction number of the every-exterior comparison |
| $\mathscr{U}_B$, $\mathcal{V}_B$ | canonical vacuum interaction of $\mu_B$ and its norm bound |
| $J = (J_S)$, $\| \cdot \|_{\kappa,\lambda}$, $\mathscr{R}_a$ | character coefficients, character norm, coarse-graining map on coefficients for the integer factor $a$ |
| $\mathcal{I}_B$ | Fisher density per coarse cell at block size $B$ |
| $\gamma \in \mathfrak{G}$; $\mathfrak{p}$ | finite catalog label; pressure (theorem B8.10) |

## B7.2 The lattice law and its geometric channels

**Definition B7.2 (Dressed square-lattice source).** Let $E = \mathbb{Z}^2 \times \{1,2\}$, with $e = (x, \iota)$ joining $x$ to $x + e_\iota$, and retain the four distinct edge copies of every unit plaquette. The matter sign is $\phi_x \in \{-1, 1\}$, the raw link is $\psi_e \in \{-1, 1\}$, and the dressed sign is $z_e = \phi_x \psi_e \phi_{x+e_\iota}$. For a finite free edge set $\Lambda$ and fixed exterior dressed signs $\eta$, define

$$r_\Lambda^\eta(z) = \prod_{e \in \Lambda} (1 + b z_e) \prod_{p : p \cap \Lambda \neq \varnothing} (1 + t \chi_p(z\eta)),$$
$$Z_\Lambda^\eta = 2^{-|\Lambda|} \sum_z r_\Lambda^\eta(z), \qquad \varsigma_\Lambda^\eta(z) = \frac{2^{-|\Lambda|} r_\Lambda^\eta(z)}{Z_\Lambda^\eta}, \tag{B7.1}$$

where $\chi_S(z) = \prod_{e \in S} z_e$ is the character of a finite edge set and $z\eta$ agrees with $z$ on $\Lambda$ and with $\eta$ outside. All touching plaquettes, including crossing plaquettes, occur, and the reference is normalized counting. The finite algebra allows $0 \le b, t < 1$; the theorems below impose smaller domains explicitly. Periodic versions retain all sectors and require side at least three, with distinct edge and plaquette copies.

**Theorem B7.3 (An infinite-volume source).** For $0 < b < 1$ and $0 < t < 1/6$, the specification eq. (B7.1) has a unique dressed Gibbs law $\mu_{b,t}$, without a translation-invariance premise, and this

law is translation invariant. Every finite cylinder is the boundary-independent limit of expanding boxes and of locally embedded full tori. If $\mathrm{dist}_{\square}(e, \Lambda^c)$ is co-plaquette graph distance, then, for $A \subset \Lambda$,

$$\mathrm{TV}((\varsigma_\Lambda^\eta)_A, (\varsigma_\Lambda^{\eta'})_A) \le \min\left\{1, \sum_{e\in A} (6t)^{\mathrm{dist}_{\square}(e,\Lambda^c)}\right\}. \tag{B7.2}$$

Independent uniform matter signs and $\psi_e = \phi_x z_e \phi_{x+e_\iota}$ give the unique Gibbs lift for the raw gauge–matter factors. [ESTABLISHED]

The proof in appendix BB.1 constructs a countably additive law, as in the Kolmogorov extension theorem for a countable product of finite spaces [16], and establishes its DLR equations separately from uniqueness. Uniqueness is Dobrushin's theorem for an influence matrix with row sums at most $6t < 1$ [7][10, Sec. 8.1], and eq. (B7.2) is a boundary comparison of the kind derived from that condition [10, Ch. 8].

**Definition B7.4 (Straight-path blocking).** For an integer $B \ge 1$, the deterministic channel $C_B$ retains $\phi'_x = \phi_{Bx}$ and multiplies the raw links along $\mathcal{P}_B(x, \iota) = \{(Bx + ke_\iota, \iota) : 0 \le k < B\}$. Its dressed output is

$$z'_{(x,\iota)} = \prod_{e\in\mathcal{P}_B(x,\iota)} z_e, \qquad \mu_B = (C_B)_* \mu_{b,t}. \tag{B7.3}$$

Every output lattice is identified with $\mathbb{Z}^2$, and its cell alphabet is $\{-1, 1\}^2$; this identification makes fixed-law statements meaningful. On a torus of side $L$, require $B \mid L$ and $L/B \ge 3$.

**Proposition B7.5 (Geometry and reference square of the path channel).** Distinct retained paths have disjoint edge supports, although their endpoints may coincide. The channel is gauge equivariant, pushes normalized counting to normalized counting, and satisfies $C_a C_B = C_{aB}$, including $C_2 C_3 = C_3 C_2 = C_6$. For any positive finite density $f$, its coarse density is $\mathbb{E}_\nu[f \mid C_B]$, where $\nu$ is the fine counting reference. A coarse plaquette holonomy equals the product of all fine plaquette holonomies in its square, and a retained winding Wilson loop retains the corresponding fine winding loop. [ESTABLISHED]

*Proof.* Along a path every internal matter sign occurs twice and cancels. Distinct axes distinguish edge copies, and along each grid line consecutive intervals of length $B$ partition its edges. Each prescribed parity has exactly half the assignments on its path, so disjoint parities and the retained uniform frames have the asserted reference. Concatenating intervals gives the indices $Bk + r$, once each from zero to $aB - 1$, which proves composition. Interior edges occur twice in the plaquette product and cancel. Conditional summation over counting fibers proves the density formula and its composition without any change of reference. □

**Proposition B7.6 (Large blocks converge as full cylinder laws).** Put

$$\rho = \tanh(h + 2j) = \frac{b(1 + t^2) + 2t}{1 + t^2 + 2bt} < 1. \tag{B7.4}$$

For any fixed $0 \le b, t < 1$, any Gibbs law of the specification eq. (B7.1), and any $M$ distinct retained $B$-paths, the law $\mu_{B,M}$ of their parities satisfies

$$\mathrm{TV}(\mu_{B,M}, \nu_M) \le \min\left\{1, \frac{1}{2}\sum_{s=1}^{M}\binom{M}{s}\rho^{\lceil sB/7\rceil}\right\}, \tag{B7.5}$$

where $\nu_M$ is the fair product law. The same bound is uniform over finite volumes and dressed boundaries when all counted path edges are free. In the source uniqueness domain, fixed blocking commutes with the thermodynamic cylinder limit. Both iterated limits and the joint box or torus limit $B_n \to \infty$, $L_n/B_n \to \infty$, under the stated embedding and divisibility conditions, yield the fair product law on each fixed coarse cylinder. The raw limit has independent uniform matter and links, and negative plaquette holonomy has probability $1/2$. [ESTABLISHED]

*Proof.* The single-edge conditional field has absolute value at most $h + 2j$. A greedy independent subset of an $sB$-edge support has at least $\lceil sB/7\rceil$ members, because co-plaquette degree is at most six. Conditional on its complement the selected signs are independent, and each conditional mean has magnitude at most $\rho$. This bounds every nonempty Walsh moment, and Fourier inversion with the triangle inequality gives eq. (B7.5). The exponent is not sharp: no plaquette contains two edges of one straight path, so for $s = 1$ the whole path is independent and its parity mean is at most $\rho^B$. A fixed channel pulls a cylinder back to finitely many fine signs, so eq. (B7.2) gives the fixed-block passage. The volume-uniform moment bound gives the increasing-block and joint passages directly; it does not amplify a fixed-support boundary estimate to growing support. Independent uniform frames preserve the total-variation bound, and a nonempty plaquette character is fair at the product limit. □

## B7.3 Every-exterior coarse laws and all generated interactions

**Theorem B7.7 (A common high-temperature domain at every depth).** Fix $0 < b < 1$, $0 < t < 1/12$, and choose $\tau > 0$ with $\mathrm{q} = 12te^{\tau} < 1$. Put $\mathrm{c}_1 = 16\mathrm{q}/(1-\mathrm{q})$ and $p_{\min} = (1-\rho)^2/4$. For every integer $B \ge 1$, $\mu_B$ has a unique positive, translation-covariant, continuous specification $\varsigma_A^{(B)}$ defined at every coarse exterior, and $\mu_B$ is the unique Gibbs law of that specification. For exterior patterns $y, y'$ agreeing in the radius-$R$ neighborhood of a finite cell set $A$,

$$\mathrm{TV}\big(\varsigma_A^{(B)}(\cdot \mid y), \varsigma_A^{(B)}(\cdot \mid y')\big) \le \min\{1, \mathrm{c}_1|A|B^2(R+1)e^{-\tau(R+1)}\}. \tag{B7.6}$$

Each one-cell atom lies between $(1-\rho^B)^2/4$ and $(1+\rho^B)^2/4$, hence is at least $p_{\min}$. The same modulus compares a whole-block finite conditional calculation against its matching infinite kernel. These statements include infinite exterior prescriptions of probability zero, constructed by normalized finite limits. [ESTABLISHED]

The proof in appendix BB.2 uses fine Hamming transport and column incidence rather than a block total-variation row estimate, whose constant grows in proportion to $Bt$. This is the source of the block-size uniformity of the domain $12t < 1$.

**Definition B7.8 (Canonical vacuum interaction).** Fix the reference cell state $* = (1,1)$ and a translation-invariant lexicographic order on cells. At the cell $x$, let $\eta^{x,r}$ use $\sigma_y$ for past cells $y < x$ within distance $r$, and $*$ elsewhere. Define

$$\mathrm{e}^{(B)}_{x,r}(\sigma) = -\log \frac{\varsigma^{(B)}_x(\sigma_x \mid \eta^{x,r})}{\varsigma^{(B)}_x(* \mid \eta^{x,r})}, \qquad A_{x,r} = \{x\} \cup \{y < x : |y - x|_\infty \le r\}. \tag{B7.7}$$

Set $\mathscr{U}_{B,\{x\}} = \mathrm{e}^{(B)}_{x,0}$, $\mathscr{U}_{B,A_{x,r}} = \mathrm{e}^{(B)}_{x,r} - \mathrm{e}^{(B)}_{x,r-1}$ for $r \ge 1$, and all other terms of $\mathscr{U}_B$ to zero. This convention fixes an interaction representative; it does not set the finite partition function to one.

**Theorem B7.9 (Complete interaction and norm attraction).** Under the hypotheses of theorem B7.7, define

$$\begin{aligned} &\mathrm{c}_2 = \frac{4}{1-\rho^2}, \qquad \mathrm{c}_3 = \frac{2\mathrm{c}_1}{p_{\min}}, \qquad \lambda_v = \tau/8, \qquad z_v = e^{-\tau/4}, \\ &\mathcal{V}_B = \mathrm{c}_2\rho^B + \sqrt{2\mathrm{c}_2\mathrm{c}_3}B\rho^{B/2}\frac{z_v(9+14z_v+z_v^2)}{(1-z_v)^4}. \end{aligned} \tag{B7.8}$$

Then the canonical interaction $\mathscr{U}_B$ generates exactly $\varsigma^{(B)}$, and

$$\|\mathscr{U}_B\|_{\lambda_v} := \sum_{A \ni 0} e^{\lambda_v \operatorname{diam} A}\|\mathscr{U}_{B,A}\|_\infty \le \mathcal{V}_B \longrightarrow 0, \qquad \sum_{A\ni 0}\operatorname{diam}(A)\|\mathscr{U}_{B,A}\|_\infty \le \mathcal{V}_B/\lambda_v. \tag{B7.9}$$

No generated shell is removed. Canonical reconstruction after $C_a$ defines an integer-block action $\mathscr{U}_B \mapsto \mathscr{U}_{aB}$ on this orbit, which respects composition and fixes the zero interaction. [ESTABLISHED]

The proof and the explicit spatial tail are in appendix BB.3. The existence of some absolutely summable interaction for a quasilocal, uniformly nonnull specification is the Gibbs representation theorem, proved by Kozlov [18]; for a finite single-site space nonnullity suffices, as observed by Sullivan [30] and by Kozlov [31, Sec. 2.3.3, Thm. 2.12]. The canonical interaction is one explicit translation-invariant representative, with the displayed exponential bound. Its norm is a cell-interaction norm; the character norm introduced next is a different topology (proposition B7.12), and entry into its ball requires the separate argument of theorem B7.14.

## B7.4 An RG map on an open interaction ball

**Definition B7.10 (Consolidated character coordinates).** Let $J = (J_S)$ be a translation-invariant real coefficient family over nonempty finite edge supports, with exponential density convention $\exp(\sum_S J_S\chi_S)$. Its finite conditional energies retain all terms touching the free set. For $\kappa, \lambda > 0$, use the Banach norm

$$\|J\|_{\kappa,\lambda} = \sup_e \sum_{S\ni e} |S| e^{\kappa|S| + \lambda \operatorname{diam} S}|J_S|. \tag{B7.10}$$

Diameters use fine-edge basepoints and Chebyshev distance. The empty character is a separate scalar; it is absent from these coordinates, while every finite law retains its own normalizer.

**Theorem B7.11 (Same-norm differentiability, remainder, and composition).** For $a \in \{2,3,6\}$, put

$$\begin{aligned} \delta_a &= \kappa(1-1/a), \qquad \mathcal{K}_a = \frac{8ae^{\lambda(1-1/a)}}{\delta_a}, \\ \mathfrak{c}_a &= a^{-1}e^{-\kappa(a-1)}, \qquad M_a = 11\mathcal{K}_a/a, \\ \varepsilon &= \min_a \left\{ \frac{1}{48}, \frac{1}{2\mathcal{K}_a}, \frac{1-\mathfrak{c}_a}{2M_a} \right\} > 0. \end{aligned} \tag{B7.11}$$

On the open $\varepsilon$-ball, the path push of the unique Gibbs law has exactly one scalar-free representative $\mathscr{R}_a J$ in the same ball, and its specification is continuous at every exterior. The map $\mathscr{R}_a$ is Fréchet $C^1$, and, writing $L_a = D\mathscr{R}_a(0)$,

$$\begin{aligned} (L_a J)_T &= J_{\mathrm{lift}_a(T)}, \qquad \|L_a\| \le \mathfrak{c}_a < 1, \\ \|\mathscr{R}_a J - L_a J\| &\le M_a \|J\|^2, \qquad \|D\mathscr{R}_a(J) - L_a\| \le 2M_a\|J\|. \end{aligned} \tag{B7.12}$$

On the ball $\|\mathscr{R}_a J\| \le \frac{1+\mathfrak{c}_a}{2}\|J\|$, so zero is the only fixed point of $\mathscr{R}_a$ there, and $\mathscr{R}_2\mathscr{R}_3 = \mathscr{R}_3\mathscr{R}_2 = \mathscr{R}_6$. There is no area multiplier in the coefficient map. [ESTABLISHED]

The coefficient construction and the law passage are proved in appendix BB.4; the cumulant step is the bounded dependency-graph inequality of Féray et al. [8, Sec. 9.4.2, Eq. (47)]. The classical precedent is Israel's construction of decimation and Kadanoff maps at high temperature on a Banach space of absolutely summable interactions [13][31, Sec. 6.1.1]. Since $\|L_a\| < 1$, the linearization at zero contracts every direction; interacting fixed points and scaling exponents lie outside this construction (open problem B7.17), and the asymptotic apparatus of this supplement is in chapter B5.

**Proposition B7.12 (The two interaction norms are not interchangeable).** Arbitrarily small vacuum-vanishing interactions in the diameter norm can have infinite norm eq. (B7.10). Thus theorem B7.9 alone does not imply entry into the ball of theorem B7.11. [ESTABLISHED]

For interaction spaces weighted by support cardinality, van Enter et al. [31, Sec. 2.6.7] survey the corresponding dichotomy: an exponential weight gives analyticity of the pressure near zero interaction, whereas a weight not bounded below by an exponential admits complex zeros of finite-volume partition functions arbitrarily close to every real interaction. The factor $e^{\kappa|S|}$ in eq. (B7.10) is of the exponential kind.

*Proof.* On an $R \times R$ cell square put a translated term $-a_R(z_{x_R,1} - 1)\prod_{x \ne x_R} z_{x,1}$, anchored at its maximal cell, with $a_R = \eta_0 e^{-2\lambda R}/R^4$, $\eta_0 > 0$, and $R \ge 2$. Its diameter-norm sum is at most $2\eta_0 \sum_{R \ge 2} e^{-\lambda(R+1)}/R^2$, which is arbitrarily small with $\eta_0$. Its full-square character has coefficient $a_R$, distinct from every removed-corner character, so it cannot cancel. Its contribution at a horizontal edge to eq. (B7.10) is $\eta_0 e^{\kappa R^2 - \lambda R - \lambda}$, which does not tend to zero, while the ordinary anchored absolute coefficient sums still converge. □

**Theorem B7.13 (Complete finite-amplitude approximation).** Let $J$ lie in the ball of theorem B7.11 and have finitely many nonzero translation orbits. For every order $\bar{n}$, the Taylor polyno-

mial $\widehat{J}_{\bar{n}}$ of $\mathscr{R}_a J$, the sum of its terms of orders one through $\bar{n}$, has finitely many nonzero orbits and contains every support generated through order $\bar{n}$; its coefficients are obtained by finitely many exact rational operations and are rational when those of $J$ are. With $\omega = \mathcal{K}_a\|J\| < 1$, its error satisfies

$$\|\mathscr{R}_a J - \widehat{J}_{\bar{n}}\| \le \frac{\|J\|}{a}\frac{\omega^{\bar{n}}[(\bar{n}+1) - \bar{n}\omega]}{(1-\omega)^2}, \tag{B7.13}$$

which tends to zero as $\bar{n} \to \infty$. For a finite word $a_1, a_2, \ldots$, choose one radius $r < \varepsilon$; then $\mathrm{Lip}_a(r) = \mathfrak{c}_a + 2M_a r < 1$ is a Lipschitz constant of $\mathscr{R}_a$ on the $r$-ball, and if every approximate input remains in that ball, the errors obey $e_{k+1} \le \mathrm{Lip}_{a_k}(r)e_k + e'_k$, where $e'_k$ is the Taylor error of step $k$. [ESTABLISHED]

The proof in appendix BB.5 counts all translations, repeated source occurrences, and omitted orders.

**Theorem B7.14 (Constructive entry of the spatial source into any prescribed norm).** Fix a compact box $0 < b_- \le b \le b_+ < 1$, $0 < t_- \le t \le t_+ < 1/12$ and $\kappa, \lambda > 0$, all rational. There is an integer $B_0$, determined by the explicit gates of appendix BB.6 and found by a finite rational search, such that for every integer $B \ge B_0$ the law $\mu_B$ has a unique translation-invariant scalar-free character representative $J^{(B)}$ in the open ball of theorem B7.11, with a norm bound that decays exponentially in $B$ uniformly on the box. For such $B$, $\mathscr{R}_a J^{(B)} = J^{(aB)}$. For rational source parameters, a finite rational library approximates $J^{(B)}$ to any prescribed positive error in the norm $\|\cdot\|_{\kappa,\lambda}$. If an input has error $\delta$ and a finite-word Taylor approximation has error $e$, with both exact trajectories in one ball of radius $r < \varepsilon$, the total error after the word is at most $e + \delta\prod_k \mathrm{Lip}_{a_k}(r)$. [ESTABLISHED]

The construction and its constants are given in appendix BB.6. The approximation concerns the interaction of the path-pushed law; scalar-free coefficients determine neither a discarded partition scalar nor an ELBO.

## B7.5 Regular statistical geometry at fixed depths

**Theorem B7.15 (Fisher density and regular finite hierarchies).** In the natural parameters $(h, j)$, fix a compact rectangle in $h > 0$, $0 < j < \mathrm{atanh}(1/12)$, and an integer $B$. The coarse finite marginals and every constrained local row are $C^3$ functions. The canonical interaction is $C^3$ in one exponential diameter norm, and its derivatives through order three tend to zero as $B$ increases, uniformly on each such compact rectangle. Let $I_{B,\Lambda_L}$ be the Fisher matrix of the $L$-square marginal, and $I_{\mathrm{cell},B}(h, j \mid \mathrm{past})$ the predictive row Fisher matrix with its complete past fixed and its future integrated. Then

$$\mathcal{I}_B(h,j) = \mathbb{E}_{\mu_B} I_{\mathrm{cell},B}(h, j \mid \mathrm{past}) = \lim_{L\to\infty} L^{-2} I_{B,\Lambda_L}(h,j) \quad \text{in } C^2 \text{ on compact parameter sets.} \tag{B7.14}$$

For each fixed depth, and for every prescribed finite list of depths, there is a nonempty open region with positive $j$ on which these densities are positive-definite metrics. Their Levi–Civita connections

are $C^1$, their curvatures are continuous, and the connections and curvatures of the area-normalized finite-marginal metrics converge to them on compact regular subcharts. [ESTABLISHED]

The response, moving-expectation, and positive-floor arguments are in appendix BB.7. The predictive row in eq. (B7.14) integrates its future, whereas the canonical interaction row of eq. (B7.7) fixes its future to $*$; substituting the latter for the former would change the Fisher-density construction.

**Proposition B7.16 (Area-qualified information contraction).** For integer blocking by $a$,

$$\mathcal{I}_{aB} \preceq a^2 \mathcal{I}_B, \qquad \frac{\mathcal{I}_{aB}}{(aB)^2} \preceq \frac{\mathcal{I}_B}{B^2}. \tag{B7.15}$$

[ESTABLISHED]

*Proof.* By proposition B7.5, $\mu_{aB} = (C_a)_* \mu_B$, and a coarse $L$-square is the image under a parameter-independent deterministic map of the fine $(aL+1)$-square marginal, which contains every path it uses. The finite marginals are positive $C^3$ families, hence differentiable in quadratic mean, so by theorem 7.9 the Fisher matrix of the image is at most that of the fine marginal, the defect being the expected conditional covariance of the fine score. Divide by $L^2$. The fixed-depth marginal limit in eq. (B7.14), together with $(aL+1)^2/L^2 \to a^2$, gives the factor $a^2$, and dividing by the fine-cell area $(aB)^2$ gives the second inequality. □

**Open Problem B7.17 (Limits beyond the admitted high-temperature construction).** The constructions of this part hold on the high-temperature domain $0 < t < 1/12$, at fixed blocking depth, with Fisher floors on finitely many depths and every blocking word fixed before volume limits. They do not supply an interacting critical fixed point, blocking-independent critical exponents, a continuum gauge-field measure, or a continuous-scale generator; the finite-depth Fisher floors may vanish as depth increases, so neither a regular Fisher inverse nor a curvature limit uniform in unbounded depth is available; unrestricted simultaneous volume and depth limits of the full labeled experiment of chapter B9 are not controlled; and a Fisher-curvature action computed from the same unknown law that it reweights is not shown to have a solution that exists, is unique, is normalizable, or is regular. Settling these requires, in the same order, new parameter domains with the corresponding tightness, interaction, and scaling estimates; positive Fisher floors uniform in depth; joint volume and depth estimates for the labeled experiment; and an existence and regularity theorem for the self-consistent normalized law, whose curvature term needs its own normalized generative model and scalar normalizer.

# Chapter B8

# Transport, Label Pressure, and Specific VFE

This chapter asks what becomes of a finite label ensemble of definition 13.15 when every branch is a native lattice law of chapter B7 and the volume grows: which label survives, at what rates the evidence and the variational free energy scale, and how these rates move along a fixed blocking word. The answers are the pressure with explicit surface rates of theorem B8.10, strict label selection with the exact tie condition of theorem B8.13, the specific VFE of theorem B8.20 with its minimum, minus the evidence density, in proposition B8.21, and the transport of the specific VFE along a fixed word in theorem B8.26. The chapter opens with the finite transport of the labeled experiment, the anchor scalar, the curvature factors that may enter an ensemble, and the failure of Gibbs regularity after a global label is erased. The label rates found here are the bulk scores of the Fisher-density statements in chapter B9.

The finite ensembles, their two normalization modes, and their finite VFE identities are those of definition 13.15 and proposition 13.16 in the main manuscript; this chapter adds only what the lattice and the volume limit require. That definition types its catalog as values of the structural mark $R_\ell$, that is, as partitions of an agent set; here a label $\gamma$ is instead a coupling label, the strengths $(b_\gamma, t_\gamma)$ of a law on one common carrier, and the lattice ensemble uses only the finite-mixture data of the definition (carrier, reference, raw factor, weight, and likelihood). The lattice notation of chapter B7 is used throughout, with branch subscripts: $b_\gamma = \tanh h_\gamma$ and $t_\gamma = \tanh j_\gamma$ are the field and plaquette strengths of branch $\gamma \in \mathfrak{G}$, $\tau$ is the decay exponent of theorem B7.7, and $L$ is the side of the square $\Lambda_L = \{0, \dots, L-1\}^2$ of cells. For a cell interaction $\mathscr{U}$, $\mathrm{m}_0(\mathscr{U})$ and $\mathrm{m}_1(\mathscr{U})$ are its norm and first diameter moment and $\mathfrak{p}(\mathscr{U})$ its pressure; the specific entropy, energy, relative entropy, and VFE are $\mathfrak{s}$, $\mathfrak{e}_{\mathscr{U}}$, $\mathfrak{h}$, and $\mathfrak{f}$. Record kernels of the finite ensemble are written $K_\gamma(o \mid x)$.

The lattice is a realization, and whether its sites are agents at $c_*$ or points of a base is open problem B7.1. The chapter does not claim any observation statement for records other than the constant all-positive record of hypothesis B8.15, any conclusion drawn by conditioning on a null infinite event, or any derivative statement, which belongs to chapter B9. The quenched evidence density of general records is open problem B8.17. Uniform invertibility at unbounded depth, simultaneous volume and depth limits of the labeled experiment, self-consistent curvature actions, critical universality, and continuum field measures are collected in open problem B7.17. [NOT-CLAIMED]

## B8.1 Transport of reference, scalar, and experiment

**Theorem B8.1 (Finite transport of the labeled experiment).** In the ensemble of definition 13.15, let $C_\gamma(x,y)$ be, for each label, a normalized channel that is independent of recognition and of parameters and retains $\gamma$; suppress the label. On targets reachable from the prior define

$$\nu'(y)=\sum_x \nu(x)C(x,y), \qquad r'(y)=\frac{\sum_x \nu(x)r(x)C(x,y)}{\nu'(y)}, \qquad p'(y)=\frac{\nu'(y)r'(y)}{Z}. \tag{B8.1}$$

Then $\sum_y \nu'(y)r'(y)=Z$, so $p'=pC$. The likelihood induced through the prior reverse row,

$$\overleftarrow{C}_p(x\mid y)=\frac{p(x)C(x,y)}{p'(y)}, \qquad K'(o\mid y)=\sum_x \overleftarrow{C}_p(x\mid y)K(o\mid x), \tag{B8.2}$$

preserves the evidence of every record, the selected posterior is pushed by the same channel, and reference, prior, posterior, recognition, and the observation joint compose along finite channel words. The reference reverse row $\nu(x)C(x,y)/\nu'(y)$ induces $K'(\cdot\mid y)$ for every likelihood $K$ exactly when $r$ is constant on the reverse fiber of $y$. [ESTABLISHED]

*Proof.* The reference rows sum to one, and strict positivity of $\nu$ and $r$ makes $\nu'(y)>0$ exactly on the reachable support of $pC$. Multiplying and summing in eq. (B8.1) proves the partition identity. The row $\overleftarrow{C}_p$ is normalized and $p'(y)K'(o\mid y)=\sum_x p(x)C(x,y)K(o\mid x)$; summing over $y$ preserves the evidence, and the posterior statement is the finite case of corollary 10.8. For a following channel $D$, interchanging the finite sums over $x$ and $y$ gives the composite $CD$. The ratio of the two reverse rows at $x$ is $r(x)/r'(y)$, which is one on the reverse fiber exactly when $r$ is constant there; if the rows differ at some $x_0$, the two-outcome likelihood with $K(o_1\mid x)=\mathbf{1}_{\{x=x_0\}}$ receives different values from them. On a null target the raw quotient and the reverse rows are undefined; that column contributes zero, even if the next channel has a nonzero row there. □

**Proposition B8.2 (Anchor scalar cocycle).** Choose a reachable reference anchor $x_*$ and write $r=\mathsf{z}_0\widehat{r}$ with $\widehat{r}(x_*)=1$ and a physical scalar $\mathsf{z}_0>0$. Denote the raw push of $\widehat{r}$ relative to $\nu C$ by $C_\nu\widehat{r}$, and choose a reachable target anchor $y_*$. Then

$$\mathsf{z}_C=(C_\nu\widehat{r})(y_*), \qquad \widehat{r}'=\frac{C_\nu\widehat{r}}{\mathsf{z}_C}, \qquad \mathsf{z}_0'=\mathsf{z}_0\mathsf{z}_C, \tag{B8.3}$$

and, with the same final reference and anchor,

$$\mathsf{z}_{CD}(\nu,\widehat{r})=\mathsf{z}_C(\nu,\widehat{r})\mathsf{z}_D(\nu C,\widehat{r}'). \tag{B8.4}$$

The global-factor weight must absorb $\mathsf{z}_0'$ if only $\widehat{r}'$ is used as the branch factor, whereas the scalar cancels within a hierarchical branch. [ESTABLISHED]

*Proof.* The numerator of the two-stage raw push is $\sum_{x,y}\nu(x)\widehat{r}(x)C(x,y)D(y,z)$ and its denominator is $(\nu CD)(z)$. The one-stage push relative to $\nu C$ applied to $\widehat{r}'$ has the same numerator divided by

$z_C$, so linearity pulls the first anchor scalar outside the second push, and evaluation at the final anchor gives the cocycle. The product $z_0' \widehat{r}'$ recovers the raw factor, and substitution into eqs. (13.20) and (13.21) gives the two weight rules. The identity channel has scalar one only for an unchanged anchored representative. □

Anchoring changes a representative of the raw factor; it does not change a principal frame.

## B8.2 Curvature factors and the gauge action

**Definition B8.3 (Admitted curvature factor).** Gauge curvature is the curvature $F_\omega$ of a principal connection $\omega$, base curvature that of a nondegenerate metric on $\mathcal{C}$, and Fisher curvature that of a regular statistical family with its Fisher metric. A curvature factor of an ensemble is a function of one of these objects, together with its pairing, metric contraction, measure, coupling, and a finite normalizer, entering the ensemble through a raw factor, a label weight, or a likelihood. For a unitary representation of dimension $n$, a Riemannian metric on $\mathcal{C}$ that raises both base indices, its volume measure, and a coupling $\kappa_{\mathrm{g}} \geq 0$, the gauge action is

$$\mathscr{S}_{\mathrm{g}} = -\frac{\kappa_{\mathrm{g}}}{4n} \int_{\mathcal{C}} \mathrm{Re}\,\mathrm{Tr}(F_{\mu\nu} F^{\mu\nu}) \, d\,\mathrm{vol}_{\mathcal{C}} \,. \tag{B8.5}$$

On the lattice of chapter B7, a gauge-curvature factor is typed only under the base reading of open problem B7.1. Under the agent reading the sites are agents at $c_*$, over which every connection vanishes, and a plaquette character is the link holonomy of eq. (2.30), whose triviality is graph flatness, notion (b) of proposition B1.15, and not a curvature $F_\omega$.

**Remark B8.4 (Sign of the gauge action).** For anti-Hermitian represented curvature and a Riemannian contraction, $\mathrm{Re}\,\mathrm{Tr}(F_{\mu\nu} F^{\mu\nu}) \leq 0$ pointwise, so the integrand of eq. (B8.5) is nonnegative and $\mathscr{S}_{\mathrm{g}} \geq 0$. [ESTABLISHED]

*Proof.* In an orthonormal frame of the base metric, $F^{\mu\nu} = F_{\mu\nu}$, and for an anti-Hermitian matrix $X$ one has $-\mathrm{Re}\,\mathrm{Tr}(XX) = \mathrm{Tr}(X^* X) \geq 0$, with equality only at $X = 0$. Summing over $\mu, \nu$ proves the sign. □

A nonnegative action normalizes a law on connections only together with a reference measure and a finite nonzero partition function. The three curvatures have different underlying spaces and are not identified here.

## B8.3 Erasing a global label can leave the Gibbs class

**Theorem B8.5 (Product-branch erasure can destroy Gibbs regularity).** On a countably infinite lattice choose $0 < p < q < 1$ and $0 < w < 1$, and let $\mu_p, \mu_q$ be independent Bernoulli product laws. The mixture $\mu = w\mu_p + (1-w)\mu_q$ has no singleton conditional version that is continuous in the product topology at any exterior. In particular it has no absolutely summable finite-alphabet Gibbs interaction, and both branches can lie in an arbitrarily small neighborhood of the fair product

law. This is the Bernoulli-product case of the convex-combination theorem of van Enter et al. [31, Sec. 4.5.1, Prop. 4.12 and Cor. 4.13], which gives the absence of a Feller, and for a finite alphabet of a quasilocal, specification; the discontinuity at every exterior is the sharper form proved here. [ESTABLISHED]

*Proof.* Enumerate the exterior sites and let $T_r$ be the event that their empirical mean tends to $r$. For centered Bernoulli variables $\mathbb{E}(\sum_{k=1}^{n}(X_k - r))^4 \le n^2/4$, so a mean deviation above $\epsilon$ has probability at most $1/(4n^2\epsilon^4)$, which is summable; hence $\mu_r(T_r) = 1$, also after removing finitely many sites. The exterior therefore identifies the branch almost surely, and by independence within a branch a singleton conditional version equals $p$ on $T_p$ and $q$ on $T_q$. Every nonempty finite exterior cylinder has positive mass under both branches, so it meets both $T_p$ and $T_q$ in positive mixture mass, and any other version agrees off a null set; every cylinder therefore carries both values, and continuity at any exterior would require a cylinder with oscillation below $q - p$. An absolutely summable interaction gives a uniformly convergent singleton energy sum of cylinder functions, whose normalized exponential would be a continuous version. The one-site fields $\frac{1}{2}\log[r/(1-r)]$ tend to zero as $r \to 1/2$, which gives the last assertion. □

The labeled joint itself remains normalized, and its finite erasure is an ordinary stochastic push with the posterior-fiber loss of theorem 10.7. What fails is membership of the erased mixture in the Gibbs class of each branch.

## B8.4 Carrier, boundary convention, and pressure

**Definition B8.6 (Native cell model).** A cell $x \in \mathbb{Z}^2$ carries its outgoing signs $z_{x,1}, z_{x,2}$. On $\Lambda_L$ use the normalized counting reference $4^{-L^2}$, all $2L^2$ outgoing signs, and the $(L-1)^2$ plaquettes whose four edges are outgoing edges of cells in $\Lambda_L$. Branch $\gamma$ has raw factor

$$r_{\gamma,L}(z) = \prod_{e \in \Lambda_L \times \{1,2\}} (1 + b_\gamma z_e) \prod_{\square \subset \Lambda_L} (1 + t_\gamma z_{\partial\square}), \qquad 0 < b_\gamma < 1, \quad 0 < t_\gamma < 1/12, \tag{B8.6}$$

where $z_{\partial\square}$ is the product of the four signs of the plaquette, and the finite catalog of strengths lies in a compact subdomain. An all-touching fixed-exterior model includes every plaquette meeting the free cells and is a different finite boundary convention. Independent uniform gauge frames contribute reference mass one. For local limits the coarse square is centered at $-\lfloor L/2 \rfloor$ in both coordinates and the fine square of side $BL$ at $-B\lfloor L/2 \rfloor$.

**Remark B8.7 (Centered local limits).** With this centering every fixed cylinder has diverging distance to the boundary, and the free, fixed-boundary, and infinite-law-marginal approximations of a branch converge on every fixed cylinder to its unique Gibbs law. A box with a fixed corner does not exhaust $\mathbb{Z}^2$ and gives no such statement. [ESTABLISHED]

*Proof.* Condition each of the three laws on the signs outside a centered inner square one interaction layer inside $\Lambda_L$. Every plaquette meeting the inner square is present, so each conditional is the specification of the inner square with the conditioning exterior. The estimate of theorem B7.7 is

uniform over exteriors; integrating it over the three exterior laws and letting the distance from a fixed cylinder to the boundary diverge proves convergence to the unique Gibbs law of theorem B7.3. □

**Remark B8.8 (The native free model is a plaquette Ising lattice gas).** Let $\square_L$ be the set of the $(L-1)^2$ plaquettes of definition B8.6, and for $S \subseteq \square_L$ let $\partial_2 S$ be the set of edges that lie in an odd number of plaquettes of $S$. For all real strengths $b, t$, the free partition of the raw factor eq. (B8.6) relative to normalized counting is

$$Z_L(b,t) = 4^{-L^2} \sum_{z} \prod_{e \in \Lambda_L \times \{1,2\}} (1 + b z_e) \prod_{\square \in \square_L} (1 + t z_{\partial \square}) = \sum_{S \subseteq \square_L} t^{|S|} b^{|\partial_2 S|}. \tag{B8.7}$$

For $0 < b < 1$ and $0 < t < 1$, in the plaquette spins $\sigma_\square = 2\mathbf{1}_{\{\square \in S\}} - 1$,

$$Z_L(b,t) = \sum_{\sigma \in \{-1,1\}^{\square_L}} \exp\Big( \mathsf{k} \sum_{\{\square,\square'\}} (\sigma_\square \sigma_{\square'} - 1) + \sum_{\square} \mathsf{u}_\square (\sigma_\square + 1) \Big), \qquad \mathsf{k} = \tfrac{1}{2} \log \tfrac{1}{b}, \quad \mathsf{u}_\square = \tfrac{1}{2} \log t + \tfrac{n_\square}{2} \log b, \tag{B8.8}$$

where the first sum runs over the unordered pairs of plaquettes that share an edge and $n_\square$ is the number of edges of $\square$ that lie in no other plaquette of $\square_L$. The coupling $\mathsf{k}$ is positive and every field $\mathsf{u}_\square$ is negative, so the native free model is the nearest-neighbor ferromagnetic Ising model on the $(L-1) \times (L-1)$ plaquette grid in a nonzero field of one sign, equivalently a two-dimensional lattice gas with attractive pair interaction and bulk activity $t$. [ESTABLISHED]

*Proof.* Expanding both products gives $\sum_{F,S} b^{|F|} t^{|S|} \prod_{e \in F} z_e \prod_{\square \in S} z_{\partial \square}$ over edge sets $F \subseteq \Lambda_L \times \{1,2\}$ and plaquette sets $S \subseteq \square_L$. The monomial is $\prod_e z_e^{m_e}$, where $m_e$ counts membership of $e$ in $F$ plus the number of plaquettes of $S$ that contain $e$, and its average under the normalized counting reference is one if every $m_e$ is even and zero otherwise. For fixed $S$ the only such $F$ is $\partial_2 S$, which proves eq. (B8.7). An edge lies in at most two plaquettes of $\square_L$. An edge shared by $\square$ and $\square'$ lies in $\partial_2 S$ exactly when one of them lies in $S$ and the other does not, an event with indicator $(1 - \sigma_\square \sigma_{\square'})/2$; an edge of $\square$ alone lies in $\partial_2 S$ exactly when $\square \in S$, with indicator $(1 + \sigma_\square)/2$; and an edge of no plaquette never does. Hence $|\partial_2 S| = \sum_{\{\square,\square'\}} (1 - \sigma_\square \sigma_{\square'})/2 + \sum_\square n_\square (1 + \sigma_\square)/2$ and $|S| = \sum_\square (1 + \sigma_\square)/2$, and writing $t^{|S|} b^{|\partial_2 S|}$ as an exponential gives eq. (B8.8). □

The dual sum holds for all real strengths and the Ising form on the whole square $0 < b, t < 1$; the bound $12t < 1$ of definition B8.6 enters neither. Through the identity the analyticity theory of the Ising ferromagnet in a nonzero field applies to the native free pressure, as follows.

**Proposition B8.9 (The native free pressure is real-analytic on the whole square).** For $0 < b < 1$ and $0 < t < 1$ the limit $\mathfrak{p}(b,t) = \lim_{L \to \infty} L^{-2} \log Z_L(b,t)$ of the free partition eq. (B8.7) exists, and $\mathfrak{p}$ is a real-analytic function of $(b,t)$ on the open square $(0,1)^2$. In the natural coordinates $b = \tanh h$, $t = \tanh j$ of chapter B9, the functions $\mathfrak{p}$ and $\mathsf{A} = \mathfrak{p} + 2 \log \cosh h + \log \cosh j$ are real-analytic on the quadrant $h > 0$, $j > 0$, and so is the Hessian of $\mathsf{A}$; on the domain $0 < j < \operatorname{atanh}(1/12)$ they coincide with the limits of theorem B9.3. [ESTABLISHED]

*Proof.* Write $\mathsf{u} = \frac{1}{2} \log t$ for the bulk field, $P_L = 2(L-1)(L-2)$ for the number of adjacent plaquette pairs, and $\Xi_L(\mathsf{k}, \mathsf{v}) = \sum_\sigma \exp\big( \mathsf{k} \sum_{\{\square,\square'\}} \sigma_\square \sigma_{\square'} + \sum_\square \mathsf{v}_\square \sigma_\square \big)$ for the free-boundary Ising

partition on the $(L-1)\times(L-1)$ plaquette grid with fields $\mathsf{v}=(\mathsf{v}_\square)$. By eq. (B8.8), $\log Z_L = -\mathsf{k}P_L+\sum_\square \mathsf{u}_\square+\log\Xi_L(\mathsf{k},(\mathsf{u}_\square))$. Every plaquette has $\mathsf{u}_\square-\mathsf{u}=\frac{n_\square}{2}\log b$, and $\sum_\square n_\square=4(L-1)$ counts the edges on the outer boundary of the grid, so replacing the fields $\mathsf{u}_\square$ by the uniform field $\mathsf{u}$ changes $\log\Xi_L$ and $\sum_\square \mathsf{u}_\square$ by at most $2(L-1)|\log b|$ each, which is $o(L^2)$. Hence $L^{-2}\log Z_L$ and $-2\mathsf{k}+\mathsf{u}+L^{-2}\log\Xi_L(\mathsf{k},\mathsf{u})$ have the same limit, if either exists. With exchange constant and magnetic moment set to one, $\Xi_L(\mathsf{k},\mathsf{u})$ is the partition function of the nearest-neighbor Ising ferromagnet at reciprocal temperature $\beta=\mathsf{k}>0$ in the field $H=\mathsf{u}/\mathsf{k}<0$. For that model the free energy per site exists in the infinite-lattice limit and is analytic in the two variables $\beta$ and $H$ whenever $\beta>0$ and $H$ is not a limit point of zeros of the partition function, which by the Lee–Yang theorem holds for every real $H\neq 0$ [20, abstract and Section IV]. The free energy per site times $-\beta$ is $\lim(L-1)^{-2}\log\Xi_L$, so $\psi(\mathsf{k},\mathsf{u})=\lim(L-1)^{-2}\log\Xi_L(\mathsf{k},\mathsf{u})$ exists and is real-analytic on $\{\mathsf{k}>0,\mathsf{u}<0\}$, because $(\mathsf{k},\mathsf{u})\mapsto(\beta,H)=(\mathsf{k},\mathsf{u}/\mathsf{k})$ is a real-analytic diffeomorphism of that set onto $\{\beta>0,H<0\}$. Since $(L-1)^2/L^2\to 1$, $\mathfrak{p}(b,t)=\psi(\mathsf{k},\mathsf{u})-2\mathsf{k}+\mathsf{u}$ with $\mathsf{k}=\frac12\log(1/b)$ and $\mathsf{u}=\frac12\log t$, a composition of real-analytic maps from $(0,1)^2$ into $\{\mathsf{k}>0,\mathsf{u}<0\}$. The map $(h,j)\mapsto(\tanh h,\tanh j)$ is real-analytic from the quadrant onto $(0,1)^2$, and $\log\cosh$ is real-analytic, which gives the statements for $\mathfrak{p}$, $\mathsf{A}$, and its Hessian in natural coordinates. On $0<j<\operatorname{atanh}(1/12)$ the limit of eq. (B9.5) is the same sequence $L^{-2}\log Z_L$, so the two limits agree there. □

The proposition concerns the free pressure and its derivatives as functions. The Fisher statements of chapter B9, which identify volume-normalized Fisher tensors of the finite laws and of their pushes with these derivatives, are proved only on the constructive domain.

**Theorem B8.10 (Constructive pressure and surface comparison).** Let $\mathscr{U}$ be a translation-invariant cell interaction with

$$\mathsf{m}_0(\mathscr{U})=\sum_{A\ni 0}\|\mathscr{U}_A\|_\infty<\infty,\qquad \mathsf{m}_1(\mathscr{U})=\sum_{A\ni 0}\operatorname{diam}(A)\|\mathscr{U}_A\|_\infty<\infty, \tag{B8.9}$$

using the Chebyshev diameter of cells. Write $Z_L$ for its free partition relative to normalized counting and $\mu_{\Lambda_L}$ for the marginal of a Gibbs law with that interaction. Then

$$\mathfrak{p}(\mathscr{U})=\lim_{L\to\infty}L^{-2}\log Z_L,\qquad \left|\mathfrak{p}(\mathscr{U})-n^{-2}\log Z_n\right|\le\frac{4\mathsf{m}_1(\mathscr{U})}{n}, \tag{B8.10}$$

$$\left|\log\frac{\mu_{\Lambda_L}(z)}{p_L^{\mathrm{free}}(z)}\right|\le 8\mathsf{m}_1(\mathscr{U})L, \tag{B8.11}$$

and the same mass-ratio bound holds for any fixed exterior. The native interaction, with singleton energy $-\log[(1+bz_1)(1+bz_2)]$ and plaquette energy $-\log(1+tz_{\partial\square})$, satisfies

$$\mathsf{m}_0\le\frac{2b}{1-b}+\frac{3t}{1-t},\qquad \mathsf{m}_1\le\frac{3t}{1-t}. \tag{B8.12}$$

Existence of the pressure for absolutely summable translation-invariant interactions is standard [10, Chs. 15 and 16]; the first-moment hypothesis supplies the explicit rates $4\mathsf{m}_1/n$ and $8\mathsf{m}_1L$, which are proved here without an infinite partition function. [ESTABLISHED]

*Proof.* Put $T(d) = \sum_{A \ni 0, \operatorname{diam}(A) \geq d} \|\mathscr{U}_A\|_\infty$. A support crossing the square and containing $x$ has diameter at least $\operatorname{dist}(x, \Lambda_L^c)$, and at most $4L$ cells lie at each positive boundary distance, so the crossing energy is at most $4L \sum_{d \geq 1} T(d) = 4L\mathrm{m}_1$. Adding the exterior changes the free Hamiltonian by at most $4L\mathrm{m}_1$, its log partition by the same amount, and its normalized log mass by at most $8L\mathrm{m}_1$; integration over an exterior law preserves the multiplicative bounds. Tile a large square by $n$-squares and a leftover region $D$. Interactions touching $D$ have total norm at most $\mathrm{m}_0|D|$, and crossings of the small squares cost at most $4\mathrm{m}_1 n$ per tile. Assigning independent normalized counting to $D$ and comparing partition sums exponentially, with $n$ fixed and the leftover fraction tending to zero, puts both limiting upper and lower pressures within $4\mathrm{m}_1/n$ of $n^{-2} \log Z_n$; their difference is at most $8\mathrm{m}_1/n$ for every $n$, so they agree. For the native interaction a cell lies in its singleton support, of diameter zero, and in three plaquette supports of diameter one. The bounds $-\log(1-u) \leq u/(1-u)$, obtained by integrating $(1-u)^{-1}$, and $\log(1+u) \leq -\log(1-u)$ for $0 \leq u < 1$ give eq. (B8.12). □

## B8.5 Pressure of the coarse interaction

Fix an integer block size $B$ before any volume limit, and let $\mathscr{U}_B$ be the canonical vacuum interaction of the pushed law $\mu_B$ of definition B7.8 and theorem B7.9, with vacuum $(+,+)$. Its pressure exists by theorem B8.10, because $\mathrm{m}_1(\mathscr{U}_B)$ is bounded by eq. (B7.9). The following explicit shell bounds use one positive-coupling domain for all blocks and all branches, including their observation-shifted fields.

**Proposition B8.11 (Explicit coarse shell bounds).** Let $b_+ < 1$ and $t_+ < 1/12$ bound the field and plaquette strengths of the branches and of their shifted fields in proposition B8.16. Take the decay exponent and constants

$$e^\tau = \frac{1+12t_+}{24t_+}, \qquad \mathrm{q} = 12t_+ e^\tau = \frac{1+12t_+}{2}, \qquad \mathrm{c}_1 = \frac{16\mathrm{q}}{1-\mathrm{q}}, \tag{B8.13}$$

$$\rho = \frac{b_+(1+t_+^2)+2t_+}{1+t_+^2+2b_+t_+}, \qquad p_{\min} = \frac{(1-\rho)^2}{4}, \qquad \mathrm{c}_3 = \frac{2\mathrm{c}_1}{p_{\min}}. \tag{B8.14}$$

Then $\tau > 0$, $0 < \mathrm{q} < 1$, $0 < \rho < 1$, and for every $B \geq 1$ the canonical interaction satisfies

$$\|\mathscr{U}_{B,(x,0)}\|_\infty \leq \frac{4\rho^B}{1-\rho^2}, \qquad \|\mathscr{U}_{B,(x,r)}\|_\infty \leq \mathrm{c}_3 B^2 r e^{-\tau r} \quad (r \geq 1), \tag{B8.15}$$

where $(x,r)$ indexes the radius-$r$ support anchored at $x$, and

$$\mathrm{m}_1(\mathscr{U}_B) \leq \overline{\mathrm{m}}_B := 2\mathrm{c}_3 B^2 \sum_{r \geq 1} r^2 (2r+1)^2 e^{-\tau r} < \infty. \tag{B8.16}$$

[ESTABLISHED]

*Proof.* Since $t_+ < 1/12$, the displayed exponent is positive and $\mathrm{q} = (1+12t_+)/2 < 1$; as $\mathrm{q}$ and $\mathrm{c}_1$ increase with the plaquette strength, the modulus eq. (B7.6) holds with these constants for every branch, and bounds the target variation by $\mathrm{c}_1|A|B^2(R+1)e^{-\tau(R+1)}$ when two exteriors agree within radius $R$. Conditional on the other fine signs, a path consists of independent signs with field magnitude at most $\operatorname{atanh} b_+ + 2\operatorname{atanh} t_+$, so revealing the two target paths successively puts every one-cell atom in $[(1-\rho^B)^2/4, (1+\rho^B)^2/4]$, hence above $p_{\min}$. The singleton term of the canonical interaction is a conditional log-odds of two such atoms, bounded by $4\operatorname{atanh}\rho^B \le 4\rho^B/(1-\rho^2)$. The radius-$r$ term is the difference of two successive log-odds whose exteriors agree within radius $r-1$; each of the two probabilities moves by at most $\mathrm{c}_1 B^2 r e^{-\tau r}$, and the logarithm is $1/p_{\min}$-Lipschitz on $[p_{\min}, 1]$, which gives the second bound. At most $(2r+1)^2$ translated radius-$r$ supports contain a fixed cell, each of diameter at most $2r$; summing proves eq. (B8.16). □

The explicit bound $\overline{\mathsf{m}}_B$ grows with $B$, whereas the bound of eq. (B7.9) decays; either supplies the first-moment hypothesis of theorem B8.10.

## B8.6 Label weights, selection, and ties

**Hypothesis B8.12 (Controlled label growth).** For the global-factor ensemble, $\log a_{\gamma,L} = L^2\alpha_\gamma + \delta_{\gamma,L}$ with $|\delta_{\gamma,L}| \le U_\gamma(L) = o(L^2)$ and a supplied modulus for this $o(L^2)$ bound, and each branch uses its matching free partition and law, or its matching fixed-exterior pair. This excludes volume-dependent weights without a controlled subextensive part. It is the value-level lattice form of the growth hypothesis eq. (10.12) of the finite-mixture Fisher asymptotics in section 10.8; the derivative statements of chapter B9 impose that $C^3$ form instead.

**Theorem B8.13 (Strict selection and the exact tie condition).** Assume hypothesis B8.12. Write $\mathsf{s}_\gamma = \alpha_\gamma + \mathfrak{p}_\gamma$ and $W_{\gamma,L} = a_{\gamma,L}Z_{\gamma,L}/\sum_{\gamma'} a_{\gamma',L}Z_{\gamma',L}$, and let $\overline{\mathsf{m}}_\gamma$ be a finite bound for the first moment $\mathsf{m}_1(\mathscr{U}_\gamma)$ of the interaction of branch $\gamma$, for example $3t_\gamma/(1-t_\gamma)$ from eq. (B8.12) for a native branch or $\overline{\mathsf{m}}_B$ of eq. (B8.16) for a canonical coarse branch. For free boundaries put $\Delta_\gamma(L) = U_\gamma(L) + 4\overline{\mathsf{m}}_\gamma L$, and replace $4$ by $8$ for fixed exteriors. If $\gamma_*$ uniquely maximizes $\mathsf{s}_\gamma$, then

$$1 - W_{\gamma_*,L} \le \sum_{\gamma\neq\gamma_*} \exp\left[-L^2(\mathsf{s}_{\gamma_*} - \mathsf{s}_\gamma) + \Delta_\gamma(L) + \Delta_{\gamma_*}(L)\right] \longrightarrow 0. \tag{B8.17}$$

If instead $S$ is the full maximal-score set and $\widehat{\delta}_{\gamma,L} = \delta_{\gamma,L} + \log Z_{\gamma,L} - L^2\mathfrak{p}_\gamma$, the labeled local law converges exactly when the finite vector

$$\left(\frac{e^{\widehat{\delta}_{\gamma,L}}}{\sum_{\gamma'\in S} e^{\widehat{\delta}_{\gamma',L}}}\right)_{\gamma\in S} \tag{B8.18}$$

converges; pressure equality alone does not determine it. [ESTABLISHED]

*Proof.* The pressure error bounds give $|\log(a_{\gamma,L}Z_{\gamma,L}) - L^2\mathsf{s}_\gamma| \le \Delta_\gamma(L)$. Dividing each competing weight by that of $\gamma_*$ and bounding the common denominator below by its largest term proves

eq. (B8.17). For a tied maximal set the complement, if nonempty, has a positive gap and hence vanishing mass, and within $S$ the relative weights are exactly eq. (B8.18). Branch cylinder limits exist by remark B8.7. Finite summation proves sufficiency, and testing the cylinder consisting only of the label proves necessity. □

The Fisher-tensor counterparts of the strict and tie statements are the lattice-free statements of section 10.8; their lattice instances are collected in chapter B9.

**Remark B8.14 (Ties are not fixed by pressures).** For two identical positive-coupling branches with $a_{1,L} = v_L/Z_L$ and $a_{2,L} = 1/Z_L$, the choice of $v_L$ alternating between $1$ and $2$ gives identical bulk scores and label weights alternating between $1/2$ and $2/3$, and $v_L = e^{-L}$ removes one label with zero bulk gap. A fixed hierarchy $w_\gamma \mu_\gamma$ is a finite sum of probability laws, reproduced at each finite volume by the factor weights $a_{\gamma,L} = w_\gamma/Z_{\gamma,L}$ of proposition 13.16, but compensation by $e^{-L^2 \mathfrak{p}_\gamma}$ alone leaves the surface term $\log Z_{\gamma,L} - L^2 \mathfrak{p}_\gamma$: multiplying one of two identical raw branches by $e^L$ leaves both pressures unchanged and makes this term select that label. A bulk counterterm therefore does not restore prescribed tied weights in general. [ESTABLISHED]

*Proof.* The weights are proportional to $a_{\gamma,L} Z_{\gamma,L}$, which gives $v_L/(1+v_L) \in \{1/2, 2/3\}$ and $e^{-L}/(1+e^{-L}) \to 0$, while $L^{-2} \log v_L \to 0$ leaves both bulk scores equal. The branch $e^L$ times a raw factor has partition $e^L Z_{\gamma,L}$ and the same pressure, and its relative weight $e^L \to \infty$ selects it. The example concerns an explicit scalar factor, not the sign of a native boundary correction. □

## B8.7 Observation, entropy, and specific VFE

**Hypothesis B8.15 (Constant all-positive observation record).** Each branch has the independent edge sensor $K_\gamma^{\mathrm{sens}}(o \mid z) = (1 + \eta_\gamma o z)/2$ with $0 < \eta_\gamma < 1$, and the observation conditioned on is the single finite record in which all $2L^2$ outgoing edge sensors of $\Lambda_L$ report $o = +$. This record is a choice, and it excludes generic and typical records. Under every branch it has probability at most $[(1+\eta_+)/2]^{2L^2}$, with $\eta_+ = \max_\gamma \eta_\gamma$. A negative entry $o_e = -$ multiplies the prior factor by $(1+bz_e)(1-\eta z_e)/2 = A_-(1+b'_- z_e)$ with $A_- = (1-b\eta)/2$ and $b'_- = (b-\eta)/(1-b\eta)$, which is negative whenever $b < \eta$ and then leaves the positive-field domain of definition B8.6 and theorem B7.7; a general record produces a site-dependent field of mixed sign, whose evidence density is the subject of open problem B8.17. The hypothesis is used in proposition B8.16 and in theorems B8.20 and B8.26 and proposition B8.21.

**Proposition B8.16 (Positive sensor and posterior branch).** Under hypothesis B8.15, define

$$A_\gamma = \frac{1 + b_\gamma \eta_\gamma}{2}, \qquad b'_\gamma = \frac{b_\gamma + \eta_\gamma}{1 + b_\gamma \eta_\gamma}, \qquad Z_{\gamma,L}^{\mathrm{obs}} = A_\gamma^{2L^2} Z_L(b'_\gamma, t_\gamma). \tag{B8.19}$$

The selected free or fixed-boundary posterior is the matching shifted branch, and reweighting infinite-law marginals gives the same centered local posterior limit. The evidence density of branch

$\gamma$ is

$$e_\gamma = 2\log A_\gamma + \mathfrak{p}(b'_\gamma, t_\gamma) - \mathfrak{p}(b_\gamma, t_\gamma). \tag{B8.20}$$

For fixed positive hierarchical weights the total evidence density is $\bar{e} = \max_\gamma e_\gamma$ and the posterior label scores are $\mathfrak{s}^o_\gamma = e_\gamma$. For global weights under hypothesis B8.12, with $\mathfrak{p}^{\mathrm{obs}}_\gamma = \mathfrak{p}_\gamma + e_\gamma$, they are

$$\bar{e} = \max_\gamma(\alpha_\gamma + \mathfrak{p}^{\mathrm{obs}}_\gamma) - \max_\gamma(\alpha_\gamma + \mathfrak{p}_\gamma), \qquad \mathfrak{s}^o_\gamma = \alpha_\gamma + \mathfrak{p}^{\mathrm{obs}}_\gamma. \tag{B8.21}$$

These are limits of finite observations, not conditioning on the null infinite all-positive event. [ESTABLISHED]

*Proof.* The identity $(1+bz)(1+\eta z)/2 = A(1+b'z)$ follows from $z^2 = 1$; multiplying it over the observed free edges leaves the plaquette factors unchanged. The shifted field remains below one and the plaquette strength below $1/12$. For an infinite-law marginal, the finite reweighted full-space law conditioned on the exterior of the window is the shifted all-touching specification, and integrating the uniform interior comparison over that reweighted exterior proves local convergence; a surface log-ratio bound alone would not. For densities, the prior log-ratio between marginal and free law is at most $8\mathfrak{m}_1 L$; multiplying by the same positive likelihood bounds the evidence log-ratio by $8\mathfrak{m}_1 L$ and the posterior log-ratio by $16\mathfrak{m}_1 L$, and the comparison for the shifted interaction, of first moment $\mathfrak{m}'_1$, adds $8\mathfrak{m}'_1 L$. These vanish after division by $L^2$ and give eq. (B8.20). The unnormalized posterior label masses are $w_\gamma E_{\gamma,L}$ for the hierarchy and $a_{\gamma,L} Z^{\mathrm{obs}}_{\gamma,L}$ for global factors, and the logarithm of a finite exponential sum differs from its largest term by at most $\log|\mathfrak{G}|$, which gives the label rates. The infinite all-positive event is null because the finite likelihoods are bounded by $[(1+\eta_+)/2]^{2L^2} \to 0$. □

Posterior ties require their own subextensive ratios, as in eq. (B8.18).

**Open Problem B8.17 (Quenched evidence density for general records).** Let the record $o = (o_e)$ on all outgoing edges be drawn from a stationary ergodic law, for example from the model itself, and condition on its restriction to $\Lambda_L$. The problem is whether $L^{-2}\log E_L(o)$, the posterior label rates, and the specific VFE of theorem B8.20 converge almost surely to record-independent quenched limits, and whether proposition B8.21 holds for those limits. Settling it requires an every-exterior comparison and a pressure identity for the site-dependent shifted fields in $(-1, 1)$, of both signs, that a general record produces, together with an ergodic theorem for the record-dependent evidence. The obstruction is that theorem B7.7 and every construction of this chapter are stated for one constant field $0 < b < 1$ and use translation invariance of the branch law, which a fixed realization of a random record breaks.

**Theorem B8.18 (Stationary specific relative entropy).** For a stationary law $Q$ on the four-state cell lattice with square entropy $\mathrm{Ent}(Q_{\Lambda_L})$, using $0\log 0 = 0$, and a Gibbs law $\mu$ whose interaction

satisfies eq. (B8.9),

$$\mathfrak{s}(Q) = \lim_{L\to\infty} L^{-2}\,\mathrm{Ent}(Q_{\Lambda_L}) = \inf_{n\geq 1} n^{-2}\,\mathrm{Ent}(Q_{\Lambda_n}), \tag{B8.22}$$

$$\mathfrak{e}_{\mathscr{U}}(Q) = \sum_{A\ni 0} \frac{\mathbb{E}_Q \mathscr{U}_A}{|A|}, \tag{B8.23}$$

$$\mathfrak{h}(Q\|\mu) := \lim_{L\to\infty} L^{-2} D_{\mathrm{KL}}(Q_{\Lambda_L}\|\mu_{\Lambda_L}) = \log 4 - \mathfrak{s}(Q) + \mathfrak{e}_{\mathscr{U}}(Q) + \mathfrak{p}(\mathscr{U}). \tag{B8.24}$$

The limits need neither ergodicity nor positive recognition atoms, and they concern specific relative entropy, not relative entropy between infinite laws. The formula is the finite-alphabet form of the specific entropy, specific energy, and specific free energy of Georgii [10, Secs. 15.2 and 15.3, Thm. 15.30]; the first-moment hypothesis supplies the explicit surface rate used in the proof. [ESTABLISHED]

*Proof.* Tile a large square by $n$-squares. Subadditivity of entropy and stationarity bound its entropy by the number of tiles times $\mathrm{Ent}(Q_{\Lambda_n})$ plus $|D|\log 4$, so the limiting upper density is at most every $n$-square density, whereas their infimum is at most the limiting lower density. Allocating each interaction equally among its sites, absolute summability and stationarity give eq. (B8.23), with omitted boundary allocations bounded by $4\mathsf{m}_1 L$. The free relative entropy equals $-\mathrm{Ent}(Q_{\Lambda_L}) + L^2\log 4 + \mathbb{E}_Q \mathscr{H}_L + \log Z_L$, where $\mathscr{H}_L$ is the free Hamiltonian, and changing the comparison to the Gibbs marginal costs at most $8\mathsf{m}_1 L$. Dividing by $L^2$ and applying theorem B8.10 proves the formula. □

**Remark B8.19 (No uniform finite-query entropy bound).** Stationarity alone admits no procedure that, after finitely many cylinder queries, returns the specific entropy with error below $\log 4/2$ for every stationary law. [ESTABLISHED]

*Proof.* A uniformly random $K$-square four-symbol array repeated periodically is stationary, has specific entropy zero, and has the fair independent marginal on every square of side at most $K$. A terminating procedure receives the same answers, and so returns the same value, for it and for the fair independent field once $K$ exceeds every queried coordinate span, whereas their specific entropies differ by $\log 4$; one value cannot lie within $\log 4/2$ of both. □

**Theorem B8.20 (Label-retaining specific VFE).** Fix a recognition law $\mathbb{Q} = q_\gamma Q_\gamma$ with each $Q_\gamma$ stationary, and put $\mathfrak{h}_\gamma = \mathfrak{h}(Q_\gamma\|\mu_\gamma)$ and $\mathfrak{l}_\gamma^Q = \mathbb{E}_{Q_\gamma}\sum_{a=1}^{2}\log K_\gamma^{\mathrm{sens}}(+\mid z_{0,a})$. With $\mathsf{s}_* = \max_\gamma \mathsf{s}_\gamma$, the specific VFE of the two ensembles are

$$\mathfrak{f}_{\mathrm{h}} = \sum_\gamma q_\gamma(\mathfrak{h}_\gamma - \mathfrak{l}_\gamma^Q), \qquad \mathfrak{f}_{\mathrm{f}} = \sum_\gamma q_\gamma(\mathfrak{h}_\gamma - \mathfrak{l}_\gamma^Q + \mathsf{s}_* - \mathsf{s}_\gamma). \tag{B8.25}$$

They exist under hypothesis B8.15 and, for global factors, hypothesis B8.12. [ESTABLISHED]

*Proof.* Disintegrate the finite relative entropy over the label. For the hierarchy $D_{\mathrm{KL}}(q\|w)/L^2 \to 0$, and the branch limits follow from theorem B8.18; stationarity evaluates the likelihood sum per cell exactly. For global factors $-L^{-2}\log W_{\gamma,L} \to \mathsf{s}_* - \mathsf{s}_\gamma$ by the finite exponential bounds, and the

recognition label entropy again vanishes per cell. Summing finitely many branch terms proves the formulas; matching fixed-exterior or compared infinite-law-marginal versions have the same density by the surface bounds. □

**Proposition B8.21 (Minimum of the specific VFE).** Assume the hypotheses of theorem B8.20. Let $\mu_\gamma^{\mathrm{post}}$ be the Gibbs law of the shifted branch $(b'_\gamma, t_\gamma)$ of proposition B8.16, unique and translation invariant by theorem B7.3 because $0 < b'_\gamma < 1$ and $t_\gamma < 1/12$, and put $\mathfrak{h}_\gamma^{\mathrm{post}} = \mathfrak{h}(Q_\gamma \| \mu_\gamma^{\mathrm{post}})$. With the posterior label scores $\mathsf{s}_\gamma^o$ and total evidence density $\bar{e}$ of proposition B8.16 and $\mathsf{s}_*^o = \max_\gamma \mathsf{s}_\gamma^o$, every recognition $\mathbb{Q} = q_\gamma Q_\gamma$ with stationary branches satisfies, for $\mathfrak{f} = \mathfrak{f}_{\mathrm{h}}$ or $\mathfrak{f} = \mathfrak{f}_{\mathrm{f}}$,

$$\mathfrak{f} = -\bar{e} + \sum_\gamma q_\gamma \left[ (\mathsf{s}_*^o - \mathsf{s}_\gamma^o) + \mathfrak{h}_\gamma^{\mathrm{post}} \right], \qquad \mathfrak{h}_\gamma^{\mathrm{post}} = \mathfrak{h}_\gamma - \mathfrak{l}_\gamma^Q + e_\gamma. \tag{B8.26}$$

Consequently

$$\min_{\mathbb{Q}} \mathfrak{f} = -\bar{e}, \tag{B8.27}$$

and the minimizers are exactly the recognitions whose label law is supported on the maximal posterior-score set $\{\gamma : \mathsf{s}_\gamma^o = \mathsf{s}_*^o\}$ and whose charged branches are $Q_\gamma = \mu_\gamma^{\mathrm{post}}$. When several labels attain $\mathsf{s}_*^o$, every label law on them is a minimizer; the density does not determine limits of the finite posterior label weights, which require the posterior analogue of eq. (B8.18). At each finite volume the unique minimizer of the finite VFE is the finite posterior, whose branch conditionals converge locally to $\mu_\gamma^{\mathrm{post}}$ by proposition B8.16. [ESTABLISHED]

*Proof.* Fix a boundary convention of theorem B8.20 and let $\mathbb{Q}_L$ and $Q_{\gamma,L}$ be the $\Lambda_L$ marginals. Write $E_{\gamma,L}$ for the branch evidence of the record, $E_L$ for the total evidence, $W_L^o$ for the posterior label law, $\Pi_{\gamma,L}$ for the posterior branch, and $F_L(\mathbb{Q}_L)$ for the finite VFE. The identity $F_L(\mathbb{Q}_L) = -\log E_L + D_{\mathrm{KL}}(\mathbb{Q}_L \| \mathbf{\Pi}_L)$ of corollary 5.7 and the chain rule of relative entropy over the label give

$$F_L(\mathbb{Q}_L) = -\log E_L + D_{\mathrm{KL}}(q \| W_L^o) + \sum_\gamma q_\gamma D_{\mathrm{KL}}(Q_{\gamma,L} \| \Pi_{\gamma,L}). \tag{B8.28}$$

Because the prior branch law $p_{\gamma,L}$ and the record likelihood $K_L$ are strictly positive,

$$D_{\mathrm{KL}}(Q_{\gamma,L} \| \Pi_{\gamma,L}) = D_{\mathrm{KL}}(Q_{\gamma,L} \| p_{\gamma,L}) - \mathbb{E}_{Q_\gamma} \log K_L + \log E_{\gamma,L}. \tag{B8.29}$$

Divide by $L^2$. The first term on the right tends to $\mathfrak{h}_\gamma$ by theorem B8.18, stationarity makes the second exactly $-L^2 \mathfrak{l}_\gamma^Q$, and $L^{-2} \log E_{\gamma,L} \to e_\gamma$ by proposition B8.16. For free boundaries $\Pi_{\gamma,L}$ is the free law of the shifted branch, and for the other conventions its log mass differs from that law by $O(L)$; hence theorem B8.18, applied to $\mu_\gamma^{\mathrm{post}}$, identifies the limit of the left side with $\mathfrak{h}_\gamma^{\mathrm{post}}$, which proves the second identity in eq. (B8.26). By eq. (B8.19) and theorem B8.10 and, for global factors, hypothesis B8.12, the logarithms of the unnormalized posterior label masses are $L^2 \mathsf{s}_\gamma^o + o(L^2)$, so $L^{-2} \log E_L \to \bar{e}$ and $L^{-2} D_{\mathrm{KL}}(q \| W_L^o) \to \sum_\gamma q_\gamma (\mathsf{s}_*^o - \mathsf{s}_\gamma^o)$. Since $L^{-2} F_L(\mathbb{Q}_L) \to \mathfrak{f}$ by theorem B8.20, dividing eq. (B8.28) by $L^2$ proves the first identity.

Every bracket in eq. (B8.26) is nonnegative, because $\mathsf{s}_*^o \geq \mathsf{s}_\gamma^o$ and specific relative entropy is a limit of nonnegative quantities. Hence $\mathfrak{f} \geq -\bar{e}$, with equality exactly when $q$ charges only maximal posterior scores and $\mathfrak{h}_\gamma^{\mathrm{post}} = 0$ on every charged label. The law $\mu_\gamma^{\mathrm{post}}$ attains $\mathfrak{h}(\mu_\gamma^{\mathrm{post}} \| \mu_\gamma^{\mathrm{post}}) = 0$. Conversely, a stationary law whose specific relative entropy with respect to a translation-invariant Gibbs law of a finite-range specification vanishes along squares is a Gibbs law of the same specification [10, Thm. 15.37], and uniqueness gives $Q_\gamma = \mu_\gamma^{\mathrm{post}}$. At finite volume $D_{\mathrm{KL}}(\mathbb{Q}_L \| \mathbf{\Pi}_L) \geq 0$, with equality only at $\mathbf{\Pi}_L$, whose branch conditionals are the posterior branches of proposition B8.16. □

**Remark B8.22 (Density convergence and the limiting joint).** If $W_{\gamma,L} \to 0$ while $q_\gamma > 0$, the limiting labeled joint gives zero mass to a recognition-positive label, so the relative entropy to that limiting joint is infinite even on a finite state window, whereas eq. (B8.25) can remain finite; the $e^{-L}$ example of remark B8.14 has zero label-rate penalty. Convergence of the density therefore neither identifies the VFE against a limiting joint nor requires convergence of the label weights. [ESTABLISHED]

*Proof.* A recognition-positive atom of zero reference mass makes the relative entropy infinite. In the $e^{-L}$ example both labels have score $\mathsf{s}_*$, so the penalty $\mathsf{s}_* - \mathsf{s}_\gamma$ in eq. (B8.25) vanishes, and the alternating example of remark B8.14 has a finite density with nonconvergent weights. □

Erasing the label instead remains subject to theorem B8.5.

## B8.8 Scalar composition and the induced VFE density

**Theorem B8.23 (Normalization density along fixed blocking words).** Let $\mathscr{U}$ and $\mathscr{R}_a^{\mathrm{vac}}\mathscr{U}$ be the canonical vacuum representatives of a law and of its $a$-path push, each with finite first moment, and let $\mathsf{z}_{a,L}$ be the vacuum anchor scalar of the finite raw push. For a word fixed before the volume limit, with coherent references, anchors, and centered embeddings,

$$\mathfrak{n}_a(\mathscr{U}) := \lim_{L\to\infty} L^{-2} \log \mathsf{z}_{a,L} = a^2 \mathfrak{p}(\mathscr{U}) - \mathfrak{p}(\mathscr{R}_a^{\mathrm{vac}}\mathscr{U}), \tag{B8.30}$$

$$\mathfrak{n}_{ad}(\mathscr{U}) = d^2 \mathfrak{n}_a(\mathscr{U}) + \mathfrak{n}_d(\mathscr{R}_a^{\mathrm{vac}}\mathscr{U}). \tag{B8.31}$$

[ESTABLISHED]

*Proof.* Disjoint parity paths push normalized counting to normalized counting, so $Z_{\mathrm{fine},aL} = \mathsf{z}_{a,L}\widehat{Z}_L$ exactly. The fine comparison between free law and marginal costs $8\mathsf{m}_1 aL$ in log mass, the push preserves the multiplicative bound, and the coarse comparison adds $8\mathsf{m}_1' L$; thus the pushed free law and the canonical coarse free law differ by at most $8(a\mathsf{m}_1 + \mathsf{m}_1')L$ in log mass. Both anchored factors equal one at the vacuum, so evaluating there compares $\log \widehat{Z}_L$ with the canonical $\log Z_L'$ by the same amount. Dividing the partition identity by $L^2$ and using the pressure limits proves eq. (B8.30); substitution proves eq. (B8.31), where $d^2$ counts intermediate cells per final cell, and an unchanged canonical identity has $\mathfrak{n}_1 = 0$. □

**Definition B8.24 (Coarse label rates).** For a fixed blocking word of area factor $B^2$ and a native label rate $\alpha$, the native scalar density and the coarse label rate are

$$\widetilde{\mathfrak{n}}_B = B^2\mathfrak{p}_{\mathrm{native}} - \mathfrak{p}(\mathscr{U}_B), \qquad \alpha'_B = B^2\alpha + \widetilde{\mathfrak{n}}_B. \tag{B8.32}$$

**Remark B8.25 (Native conversion and coarse label scores).** The native raw factor has vacuum logarithm per cell $2\log(1+b)+\log(1+t)$, and $\mathfrak{p}(\mathscr{U}_1) = \mathfrak{p}_{\mathrm{native}} - 2\log(1+b) - \log(1+t)$; at $t=0$ the native pressure is zero and the canonical coarse pressure is $-2\log(1+b^B)$. By definition B8.24, $\alpha'_B + \mathfrak{p}(\mathscr{U}_B) = B^2(\alpha + \mathfrak{p}_{\mathrm{native}})$ identically: the coarse label score is the fine score times the area factor, which is the rate per coarse cell of the labeled push, since that push leaves every label weight unchanged at each finite volume and a coarse cell carries $B^2$ fine cells. [ESTABLISHED]

*Proof.* Dividing the native raw factor by its vacuum value gives a finite-range vacuum representative whose free partition differs from that of the canonical representative $\mathscr{U}_1$ only by a surface term, as both compare to their common infinite-law marginal at the vacuum; dividing by $L^2$ gives the conversion. At $t=0$ the raw partition is one, and the pushed coarse cell law is a product of two independent signs of mean $b^B$, whose canonical vacuum representative has free partition $(1+b^B)^{-2L^2}$. The last assertion is algebra. Equality of infinite laws alone would specify neither a raw normalizer nor a representative conversion. □

**Theorem B8.26 (Specific VFE transport of the labeled experiment).** Retain the label and the fine observation alphabet through a fixed word of product block size $B$. Let $Q_{\gamma,B}$ be the stationary path push of $Q_\gamma$ and $\mathfrak{h}^{\mathrm{post}}_{\gamma,B}$ its specific relative entropy to the pushed shifted branch. Then

$$B^2\mathfrak{f}_{\mathrm{fine}} = \mathfrak{f}_{\mathrm{coarse,induced}} + \mathfrak{d}_{\mathrm{fib}}, \qquad \mathfrak{d}_{\mathrm{fib}} = \sum_\gamma q_\gamma \left[B^2\mathfrak{h}^{\mathrm{post}}_{\gamma,1} - \mathfrak{h}^{\mathrm{post}}_{\gamma,B}\right] \geq 0, \tag{B8.33}$$

for each of the two ensembles, using limits of finite quantities rather than differences of infinite-law relative entropies. [ESTABLISHED]

*Proof.* On a finite window the induced likelihood is $K'_\gamma(o \mid y) = E_{\gamma,L}(o)p^{\mathrm{post,push}}_{\gamma,L}(y)/p^{\mathrm{prior,push}}_{\gamma,L}(y)$ on its positive support, by theorem B8.1, and the finite VFE difference is the posterior-fiber loss of theorem 10.7, which is nonnegative. The coarse recognition marginal is the push of its fine marginal on the square of side $BL$. The comparison between posterior and marginal in proposition B8.16 is $O(BL)$ uniformly in the state and is preserved by the push, so after division by $L^2$ theorem B8.18 gives both posterior limits. For the hierarchy the coarse density is $\sum_\gamma q_\gamma(\mathfrak{h}^{\mathrm{post}}_{\gamma,B} - B^2 e_\gamma)$, and for global factors it is $\sum_\gamma q_\gamma[\mathfrak{h}^{\mathrm{post}}_{\gamma,B} + B^2(\mathsf{s}_* - \alpha_\gamma - \mathfrak{p}^{\mathrm{obs}}_\gamma)]$. Subtracting these separately finite limits from $B^2$ times the fine formulas of proposition B8.21 cancels the evidence and label rates and leaves the fiber term, a limit of nonnegative finite losses. □

The identity concerns the likelihood induced from the fine record; a new local sensor on coarse edges defines a different experiment, to which it does not apply.

**Remark B8.27 (Label actions and bulk rates).** A prescribed order-one label action has zero bulk rate but can change tied weights, and an extensive factor $e^{-L^2\mathscr{A}_\gamma}$ shifts $\alpha_\gamma$ by $-\mathscr{A}_\gamma$. A state-dependent curvature action changes the branch specification and requires its own admission proof. [ESTABLISHED]

*Proof.* The bulk rate is $\lim L^{-2}\log a_{\gamma,L}$, which an order-one factor leaves unchanged and an extensive factor shifts by $-\mathscr{A}_\gamma$; an order-one factor multiplies the corresponding entry of eq. (B8.18) and changes the tied weights unless it is common to all tied labels. A state-dependent factor multiplies the raw factor, not the weight. □

# Chapter B9

# Thermodynamic Fisher Geometry of Retained Ensembles

This chapter asks whether the Fisher tensors of the native lattice experiment have volume densities that are differentiable in the source parameters, and whether a fixed blocking word preserves them. The answer is affirmative on compact positive high-temperature source boxes: the native pressure is $C^4$ and its Fisher density $C^2$ (theorem B9.3 and proposition B9.4), and the finite free law pushed along a fixed word has a $C^2$ Fisher density per coarse cell, equal to that of the infinite-law marginals although the finite laws differ (theorem B9.6). With these branch inputs the finite-mixture Fisher asymptotics of section 10.8 in the main manuscript hold for the lattice with $N = L^2$ cells, in the strict phase, at a tie, and in the reciprocal-volume window (corollary B9.8). These statements refine the label rates of chapter B8; their proofs, and the lattice-side label results that accompany them, are in appendix BC. Every Fisher tensor here belongs to the complete-data experiment that observes the label and the configuration (section 10.7), and no result uses one as the metric of a dynamics.

The chapter keeps the lattice notation of chapter B7 and adds the following. The square $\Lambda_L$ has $N = L^2$ cells, and a fixed word of total linear factor $B$ maps $B^2N$ fine cells to $N$ coarse cells. The source point in natural coordinates is $\mathsf{g} = (h, j)$, with $b = \tanh h$ and $t = \tanh j$; a statistical parameter $\theta$ ranges over an open $\Theta \subset \mathbb{R}^d$, source maps are $\mathsf{g}_\gamma : \Theta \to \mathbb{R}^2$, and $\mathcal{K}$ denotes a compact parameter or source set. The natural log-partition density is $\mathsf{A}$, its Hessian $\mathsf{H}$, and $\mathcal{I}_B$ is the Fisher density per coarse cell of a push along a word of factor $B$. The bulk log-mass densities $\mathsf{s}_\gamma$, tie slopes $\mathsf{v}_\gamma$, branch Fisher densities $\mathcal{I}_\gamma$, limiting label law $\bar{\mathsf{p}}$, and window coordinate $\mathsf{w}$ are those of definition 10.23 and the statements that follow it.

The chapter and its proof appendix do not claim a solution of a self-consistent curvature action, a regular inverse at unbounded depth, a critical or continuum limit, a simultaneous limit in depth and volume, or a derivative theorem for an arbitrary positive likelihood or for the free law of a canonical coarse interaction. The constructions that would settle the first four are part of open problem B7.17, and the typing of the lattice sites is open problem B7.1. [NOT-CLAIMED]

## B9.1 Source box, carriers, and fixed words

**Hypothesis B9.1 (Compact native source box).** Fix a compact rectangle $\mathcal{K}$ inside $h > 0$, $0 < j < \operatorname{atanh}(1/12)$, and an enlarged rectangle $\widehat{\mathcal{K}} = [h_-, h_+] \times [0, j_+]$, with $h_- > 0$ and $0 < j_+ <$

$\operatorname{atanh}(1/12)$, whose positive-$j$ interior contains $\mathcal{K}$. Choose $\tau > 0$ with $12\tanh(j_+)e^{\tau} < 1$ and put

$$\mathrm{q} = 12\tanh(j_+)e^{\tau}, \qquad \mathrm{c}_1 = \frac{16\mathrm{q}}{1-\mathrm{q}}, \qquad \rho_* = \tanh(h_+ + 2j_+), \qquad d_* = 1 - \rho_* > 0. \tag{B9.1}$$

These are the constants of theorem B7.7 at the plaquette strength $\tanh j_+$, fixed on $\widehat{\mathcal{K}}$. The edge $j = 0$ of $\widehat{\mathcal{K}}$ is used only for one-sided continuation and for the product control. The hypothesis excludes nonpositive fields, plaquette couplings with $j \geq \operatorname{atanh}(1/12)$, and noncompact source sets, and it is standing for this chapter and appendix BC.

**Definition B9.2 (Native free law and its exact pushes).** On $\Lambda_L$ retain all $2N$ signs $z_{x,a}$, $a = 1, 2$, write $P_x = z_{x,1}z_{x+e_1,2}z_{x+e_2,1}z_{x,2}$ for the plaquette products, $x \in \{0, \dots, L-2\}^2$, and let $r_L(\mathrm{g}, z)$ be the raw factor eq. (B8.6). The finite free law is

$$\nu^{\mathrm{f}}_{L,\mathrm{g}}(z) = \frac{4^{-N} r_L(\mathrm{g}, z)}{Z_L(\mathrm{g})}, \qquad Z_L(\mathrm{g}) = \sum_z 4^{-N} r_L(\mathrm{g}, z), \tag{B9.2}$$

and $\mu_{\mathrm{g}}$ is the unique infinite native Gibbs law of theorem B7.3. For a fixed integer $B \geq 1$, retain the paths $\mathcal{P}_B(x, a) = \{(Bx + ke_a, a) : 0 \leq k < B\}$, complete them to a partition of the fine edges with unused singleton blocks, and let $C_B$ record their parities, as in definition B7.4. The three families are

$$P^{\mathrm{f}}_{B,L,\mathrm{g}} = (C_B)_{\#}\nu^{\mathrm{f}}_{BL,\mathrm{g}}, \qquad \mu_{B,\mathrm{g}} = (C_B)_{\#}\mu_{\mathrm{g}}, \qquad \mu_{B,L,\mathrm{g}} = \mu_{B,\mathrm{g}}|_{\Lambda_L}, \tag{B9.3}$$

each with its own normalized reference. A coarse translation by $x$ is a fine translation by $Bx$, and local limits use the centered squares of definition B8.6. A fixed finite word of integer factors with total linear factor $B$ acts by $C_B$ on the same fine square, by proposition B7.5, and is fixed throughout every volume limit. Restricted parity conditionals use the product of uniform references on their prescribed fibers, and unrevealed paths and unused singleton blocks remain unconstrained.

## B9.2 Volume limits and exact pushforward

The finite free laws, their pushes along a fixed word, and the marginals of the infinite law share their limiting Fisher densities but not their finite laws. The statements below keep the three families apart.

**Theorem B9.3 (Native free pressure in $C^4$).** Define the natural statistics and pressures by

$$\mathsf{T}_{h,L} = \sum_{x\in\Lambda_L}\sum_{a=1}^{2} z_{x,a}, \qquad \mathsf{T}_{j,L} = \sum_{x\in\{0,\dots,L-2\}^2} P_x, \qquad \mathsf{A}_L(\mathrm{g}) = \frac{1}{N}\log \mathbb{E}_{\mathrm{ref}} e^{h\mathsf{T}_{h,L} + j\mathsf{T}_{j,L}}, \tag{B9.4}$$

$$\mathfrak{p}_L(\mathrm{g}) = \frac{1}{N}\log Z_L(\mathrm{g}). \tag{B9.5}$$

There are $C^4$ functions $\mathsf{A}$ and $\mathfrak{p}$ on the domain $h > 0$, $0 < j < \operatorname{atanh}(1/12)$ such that $\mathsf{A}_L \to \mathsf{A}$ and $\mathfrak{p}_L \to \mathfrak{p}$ in $C^4(\mathcal{K})$ for every compact source box of hypothesis B9.1, with the finite and limiting

conversions

$$\begin{aligned} \mathfrak{p}_L &= \mathsf{A}_L - 2\log\cosh h - \frac{(L-1)^2}{N}\log\cosh j, \\ \mathfrak{p} &= \mathsf{A} - 2\log\cosh h - \log\cosh j. \end{aligned} \tag{B9.6}$$

[ESTABLISHED]

**Proposition B9.4 (Native Fisher density and a source pullback).** The finite free family satisfies

$$\frac{1}{N} I(\nu^{\mathrm{f}}_{L,\mathsf{g}}) = \nabla^2_{\mathsf{g}} \mathsf{A}_L(\mathsf{g}) \longrightarrow \mathsf{H}(\mathsf{g}) := \nabla^2_{\mathsf{g}} \mathsf{A}(\mathsf{g}) \quad \text{in } C^2(\mathcal{K}), \qquad \mathsf{H} = \nabla^2_{\mathsf{g}} \mathfrak{p} + \operatorname{diag}(2\operatorname{sech}^2 h, \operatorname{sech}^2 j). \tag{B9.7}$$

If $\mathsf{g} : \Theta \to \mathbb{R}^2$, $\mathsf{g}(\theta) = (h(\theta), j(\theta))$, is $C^4$ and sends a neighborhood of a compact parameter set into an admitted source box, then

$$\frac{1}{N} I(\nu^{\mathrm{f}}_{L,\mathsf{g}(\theta)}) = D\mathsf{g}(\theta)^\top \nabla^2_{\mathsf{g}} \mathsf{A}_L(\mathsf{g}(\theta)) D\mathsf{g}(\theta) \longrightarrow D\mathsf{g}(\theta)^\top \mathsf{H}(\mathsf{g}(\theta)) D\mathsf{g}(\theta) \quad \text{in } C^2, \tag{B9.8}$$

whereas the Hessian of $\mathsf{A}_L \circ \mathsf{g}$ differs from this pullback by $\sum_{i=1}^2 (\partial_i \mathsf{A}_L \circ \mathsf{g}) \nabla^2_\theta \mathsf{g}_i$, and the Hessian of the raw pressure differs from $\mathsf{H}$ by the diagonal term above. [ESTABLISHED]

In natural coordinates the native law is an exponential family, so its Fisher density is the Hessian of the log-partition density, the Hessian structure of exponential families [1, Ch. 3] and the thermodynamic metric of Crooks [6].

**Remark B9.5 (The raw pressure is not convex).** For every $h > 0$, with $b = \tanh h$,

$$\lim_{j \downarrow 0} \partial_j^2 \mathfrak{p}(h, j) = 4b^6 - 5b^8, \tag{B9.9}$$

which is negative for $b^2 > 4/5$. For such $h$ the raw pressure has $\partial_j^2 \mathfrak{p} < 0$ at every point $(h, j)$ with $j > 0$ small enough, so $\nabla^2_{\mathsf{g}} \mathfrak{p}$ is not positive semidefinite there and is not a metric, whereas $\mathsf{H} = \nabla^2_{\mathsf{g}} \mathsf{A}$ is positive semidefinite on the whole domain of theorem B9.3. [ESTABLISHED]

*Proof.* Fix a source box of hypothesis B9.1 whose enlarged rectangle contains $(h, 0)$ in its edge $j = 0$. The proof of theorem B9.3 gives $\partial_j \mathsf{A} = m_j$ with $m_j(\mathsf{g}) = \mathbb{E}_{\mu_{\mathsf{g}}} P_0$, and lemma BC.2 with $B = 1$ and $f = P_0$ shows that $\partial_j m_j$ is the uniform limit on $\widehat{\mathcal{K}}$ of the continuous functions $\partial_j \mathbb{E}_{\nu^{\mathrm{f}}_{L,\mathsf{g}}} P_0$, taken on squares whose inventory contains the neighborhood $A_R$ of the support of $P_0$ with $R \to \infty$. Hence $\partial_j m_j$ is continuous on $\widehat{\mathcal{K}}$, and its value at $(h, 0)$ is the limit of the finite derivatives there. At $j = 0$ the free law is the product law with sign means $b$, and in the exponential family eq. (B9.4) the finite derivative is $\operatorname{Cov}(P_0, \mathsf{T}_{j,L}) = \sum_x \operatorname{Cov}(P_0, P_x)$. The term $x = 0$ is $1 - b^8$. Each of the four plaquettes that share exactly one edge with $P_0$ contributes $b^6 - b^8$, since the shared sign squares to one and the remaining six signs are independent, and every other plaquette shares no edge with $P_0$ and contributes zero. Thus $\partial_j m_j(h, 0) = 1 + 4b^6 - 5b^8$, and the conversion eq. (B9.6) gives $\partial_j^2 \mathfrak{p} = \partial_j m_j - \operatorname{sech}^2 j$ for $j > 0$, whose limit as $j \downarrow 0$ is eq. (B9.9). The sign claims follow

from $4b^6 - 5b^8 = b^6(4 - 5b^2)$ and continuity. Finally $\nabla^2_{\mathrm{g}}\mathsf{A}_L = N^{-1}I(\nu^{\mathrm{f}}_{L,\mathrm{g}})$ is a covariance matrix by proposition B9.4, so its limit H is positive semidefinite. □

**Theorem B9.6 (Exact pushed-free Fisher density).** For every fixed integer $B \geq 1$ and compact positive source box,

$$\frac{1}{N}I(P^{\mathrm{f}}_{B,L,\mathrm{g}}) \longrightarrow \mathcal{I}_B(\mathrm{g}) := \mathbb{E}_{\mu_{B,\mathrm{g}}}\mathsf{I}_{\mathcal{P}}(\mathrm{g}, Y) \quad \text{in } C^2(\mathcal{K}), \tag{B9.10}$$

where $\mathsf{I}_{\mathcal{P}}$ is the Fisher matrix of the four-state predictive row of a coarse cell with the complete lexicographic past $\mathcal{P}$ of the coarse configuration $Y$ prescribed and every future path free. This is the density eq. (B7.14) of the infinite-law marginals, although the finite laws of the two limits need not agree, and at $B = 1$

$$\mathcal{I}_1 = \mathsf{H} = \nabla^2_{\mathrm{g}}\mathsf{A}. \tag{B9.11}$$

[ESTABLISHED]

For $B > 1$ the push need not be a two-statistic exponential family, and $\mathcal{I}_B$ is not identified with the Hessian of a native or canonical coarse pressure.

**Proposition B9.7 (Branch inputs from the native theorems).** Let each $\mathrm{g}_\gamma : \Theta \to \mathbb{R}^2$, $\mathrm{g}_\gamma(\theta) = (h_\gamma(\theta), j_\gamma(\theta))$, be $C^4$ on a neighborhood of a compact $\mathcal{K} \subset \Theta$ and map it into a common compact admitted source box, and let $p_{\gamma,L,\theta} = \nu^{\mathrm{f}}_{L,\mathrm{g}_\gamma(\theta)}$. Then the branch raw pressure

$$\mathfrak{p}_{\gamma,L} := \frac{1}{N}\log Z_{\gamma,L} = \mathsf{A}_L \circ \mathrm{g}_\gamma - 2\log\cosh h_\gamma - \frac{(L-1)^2}{N}\log\cosh j_\gamma \tag{B9.12}$$

converges in $C^4(\mathcal{K})$ to a limit $\mathfrak{p}_\gamma$, and the branch Fisher densities satisfy

$$\frac{1}{N}I(p_{\gamma,L}) = (D\mathrm{g}_\gamma)^\top(\nabla^2_{\mathrm{g}}\mathsf{A}_L \circ \mathrm{g}_\gamma)D\mathrm{g}_\gamma, \tag{B9.13}$$

$$\frac{1}{N}I(p_{\gamma,L}) \longrightarrow \mathcal{I}_\gamma = (D\mathrm{g}_\gamma)^\top(\mathsf{H} \circ \mathrm{g}_\gamma)D\mathrm{g}_\gamma \quad \text{in } C^2(\mathcal{K}), \tag{B9.14}$$

with bounded $C^2$ norms. For the push along a fixed word of total factor $B$, from $B^2N$ fine cells to $N$ coarse cells,

$$I_\theta(p_{\gamma,BL}C_B) = (D\mathrm{g}_\gamma)^\top I_{\mathrm{g}}(P^{\mathrm{f}}_{B,L,\mathrm{g}_\gamma(\theta)})D\mathrm{g}_\gamma, \tag{B9.15}$$

$$\frac{1}{N}I_\theta(p_{\gamma,BL}C_B) \longrightarrow (D\mathrm{g}_\gamma)^\top(\mathcal{I}_B \circ \mathrm{g}_\gamma)D\mathrm{g}_\gamma \quad \text{in } C^2(\mathcal{K}). \tag{B9.16}$$

No rank assumption is needed for these semidefinite pullbacks. [ESTABLISHED]

**Corollary B9.8 (Lattice instances of the finite-mixture asymptotics).** Let the branches be the free laws of proposition B9.7, with raw label weights satisfying eq. (BC.35), in either normalization mode, and let $\mathsf{s}_\gamma$ be the bulk limits of proposition BC.8. Write $P_{L,\theta}$ for the labeled law on $\Lambda_L$. If a

label $\gamma_*$ has a uniform positive gap in $\mathsf{s}_\gamma$ on $\mathcal{K}$, then

$$\frac{I(P_{L,\theta})}{N} \longrightarrow \mathcal{I}_{\gamma_*}(\theta) \quad \text{in } C^2(\mathcal{K}). \tag{B9.17}$$

At a tie $\theta_0$, if $W_{\gamma,L}(\theta_0) \to w_\gamma$ and $\mathsf{v}_\gamma = \nabla \mathsf{s}_\gamma(\theta_0)$, then

$$\frac{I(P_{L,\theta_0})}{N^2} \longrightarrow \mathrm{Cov}_w(\mathsf{v}_\gamma). \tag{B9.18}$$

If the center ratios of eq. (10.22) converge to positive $\kappa_\gamma$ on the tied set $S$, then in $\theta = \theta_0 + \mathsf{w}/N$ the label law and the Fisher tensor satisfy, in $C^3$ and $C^2$ on compact $\mathsf{w}$-sets,

$$\bar{\mathsf{p}}_\gamma(\mathsf{w}) = \frac{\kappa_\gamma e^{\mathsf{w}\cdot\mathsf{v}_\gamma}}{\sum_{\gamma'\in S} \kappa_{\gamma'} e^{\mathsf{w}\cdot\mathsf{v}_{\gamma'}}}, \qquad \gamma \in S, \tag{B9.19}$$

$$N^{-2} I(P_{L,\theta_0+\mathsf{w}/N}) \longrightarrow \mathrm{Cov}_{\bar{\mathsf{p}}(\mathsf{w})}(\mathsf{v}_\gamma). \tag{B9.20}$$

The same conclusions hold for the pushes along a fixed word of total factor $B$ in the coarse volume $N$, with the branch densities of eq. (B9.16) and with $\mathsf{s}_\gamma$ and $\mathsf{v}_\gamma$ replaced by $B^2\mathsf{s}_\gamma$ and $B^2\mathsf{v}_\gamma$, and for the selected posterior of the record of hypothesis B8.15 under the sensor hypotheses of proposition BC.8, with its posterior representatives and the shifted branches. [ESTABLISHED]

*Proof.* Proposition BC.8 gives the growth hypothesis eq. (10.12) with bulk limits $\mathsf{s}_\gamma$ in both modes, and proposition B9.7 gives $C^2$ convergence of the branch densities, hence $I(p_{\gamma,L})(\theta_0) = O(N)$. With $N = L^2$ these are the hypotheses of theorem 10.29, of proposition 10.30, and, with the center ratios, of theorem 10.31, whose conclusions are the three displays. For a push, the label weights are those of the fine volume $B^2N$, so the representatives divided by $N$ converge to $B^2\mathsf{s}_\gamma$ in $C^3$ with the same winner and center ratios, and eq. (B9.16) supplies the branch densities. For the posterior, the same proposition supplies the posterior representatives and proposition B9.7 applies to the shifted source maps. □

The strict density is a semimetric unless it has a positive lower bound on the chart in use; on such charts its inverse, Levi-Civita connection, and curvature converge by proposition BC.9, and corollary BC.11 exhibits nonempty charts. A native pullback has rank at most two by proposition BC.5. The order-$N^2$ tie information is information about the label: in the native realization of proposition BC.15 the experiment that forgets the label has zero Fisher information, in line with proposition 10.26. Residual log-weight gradients and rare labels enter before score squaring (proposition BC.12 and proposition 10.32). The reciprocal-volume scale and the logistic label law have the form of the finite-size rounding at a first-order phase transition, where Borgs and Kotecký [5, Thm. 3.2] derive, for two coexisting phases, a hyperbolic-tangent crossover of width of order $N^{-1}$ in the field from the lattice model itself; here the phase mixture is prescribed through the effective log masses, which makes the statement elementary. Parameter-independent comparison kernels between the window experiment and its limit, its identifiable quotient, and a moving center for a transverse two-label tie are propositions BC.13 and BC.14 and theorem BC.16. Along a fixed word an extensive density carries the factor $B^2$ and a density normalized by squared volume the

factor $B^4$, provided the denominator, the parameter coordinate, and the observed experiment agree on both sides (proposition BC.17). Prescribed label actions change the raw weights, and their relative center values, volume orders, and derivative bounds govern the resulting experiments by propositions BC.19 and BC.20; curvature actions keep the types of definition B8.3.

# Appendix BA

# Proofs for the Base-Direction Classifications

This appendix proves the rank and quotient statements and the intrinsic defect classification of Chapter B2, and collects four witnesses that mark the boundaries of the classification. The notation is that of Section B2.4 and Section B2.11.

## BA.1 Rank and quotient geometry

*Proof of Theorem B2.7.* At each $c$ the jet $D^\omega s$ is a linear map $T_c\mathcal{U} \to V_{s(c)}E$ and $h^\omega_s = (D^\omega s)^* g^F$. The radical of the pullback of a positive-semidefinite form by a linear map is the preimage of its radical, which is the pointwise algebra of Theorem 7.15; since $g^F$ is positive definite, the preimage is $\ker D^\omega s$, and the rank statement follows. The constant-rank statement is the one for morphisms of smooth vector bundles over a fixed base, not the constant-rank theorem for maps of manifolds: $D^\omega s$ is a morphism $T\mathcal{U} \to s^*VE$ over the identity, and a morphism of locally constant rank has smooth kernel and image subbundles by local adaptation of frames. Replacing an argument of (B2.16) or (B2.17) by a representative that differs by an element of $K^\omega_s$ leaves its vertical jet unchanged, so the quotient tensors are well defined. The factored map is injective onto the image of $D^\omega s$ and preserves the displayed tensors, so the quotient metric is positive definite. □

*Proof of Proposition B2.8.* (i) If $q_K$ exists with $\ker Tq_K = K^\omega_s$, the distribution is tangent to the fibers of $q_K$, which are submanifolds, so brackets of its sections are again its sections; since $K^\omega_s = \ker D^\omega s$, this is (B2.18). For the witness, take the torus $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$, the trivial principal translation bundle with fiber $\{\mathcal{N}(\mu, 1)\}$, the constant section $\mu = 0$, and the connection form $A = \mathrm{d}y - \lambda \mathrm{d}x$ with $\lambda$ irrational. Then $D^\omega s = A$ has constant rank one, its kernel is spanned by $\partial_x + \lambda\partial_y$ and is involutive, and every leaf is the image of a line of irrational slope, which is dense in $\mathbb{T}^2$ and not all of it. The fibers of a continuous map into a Hausdorff space are closed, and a dense proper leaf is not, so no surjective submersion onto a Hausdorff manifold has the leaves as its fibers.

(ii) If $\tau$ is a tensor on $\overline{\mathcal{U}}$ with $q_K^*\tau = h^\omega_s$, every $Z \in \Gamma(K^\omega_s)$ is $q_K$-related to the zero field, so its flow preserves the fibers and $\mathcal{L}_Z q_K^*\tau = 0$; the same holds for the cubic tensor. For the witness, take $\mathbb{R}^2$, the trivial translation bundle with fiber $\{\mathcal{N}(\mu, 1)\}$, the zero section, and the connection form $A = \varphi(y)\mathrm{d}x$ with $\varphi$ smooth, positive, and nonconstant. Then $D^\omega s = \varphi(y)\mathrm{d}x$, the null leaves

are the vertical lines, $q_K(x,y) = x$ is a smooth Hausdorff quotient, and $h^\omega_s = \varphi(y)^2\mathrm{d}x^2$ has $\mathcal{L}_{\partial_y} h^\omega_s = 2\varphi\varphi'\mathrm{d}x^2 \neq 0$ where $\varphi' \neq 0$; no tensor $\tau(x)\mathrm{d}x^2$ on $\mathbb{R}$ pulls back to it.

(iii) For $Z \in \Gamma(K^\omega_s)$, $(\mathcal{L}_Z h^\omega_s)(X,Y) = Z(h^\omega_s(X,Y)) - h^\omega_s([Z,X],Y) - h^\omega_s(X,[Z,Y])$. Metric compatibility gives $Z\big(g^F(D^\omega sX, D^\omega sY)\big) = g^F(\nabla_Z(D^\omega sX), D^\omega sY) + g^F(D^\omega sX, \nabla_Z(D^\omega sY))$, and subtracting the bracket terms gives (B2.21). If $\nabla' = \nabla + B$ is another $g^F$-compatible connection, $B_Z$ is $g^F$-skew-adjoint, so the changes of the two summands cancel; the left side is the ordinary Lie derivative and does not depend on $\nabla$. □

## BA.2 Proofs for the intrinsic classification

*Proof of Theorem B2.39.* The coarse jet is $\mathsf{p} + \mathsf{J}\eta = \mathsf{p}_\perp + C$ with $C = \mathsf{p}_{\mathcal{O}} + \mathsf{J}\eta$ taking values in $\mathcal{O}$. By orthogonality the left side of (B2.72) is $\mathsf{p}_\perp^* g_1 \mathsf{p}_\perp + C^* g_1 C$, so the problem is $C^* g_1 C \preceq \mathsf{H}$. This forces $\mathsf{H} \succeq 0$ and $Cx = 0$ whenever $\mathsf{H}(x,x) = 0$. For a positive-semidefinite form the diagonal null set is the radical, since positivity of $\mathsf{H}(x+ty, x+ty)$ for both signs of small $t$ forces $\mathsf{H}(x,y) = 0$. Hence $\mathsf{k}([x]) = Cx$ is well defined, and the inequality says exactly that $\mathsf{k}$ is a contraction; conversely every contraction satisfies it. Since $\mathsf{k}\,\mathrm{pr}_{\mathsf{H}} - \mathsf{p}_{\mathcal{O}}$ takes values in $\mathcal{O}$, a linear right inverse of $\mathsf{J}$ on its image gives a lift, and two lifts differ by a map into $\ker \mathsf{J}$. If $\mathsf{H} \succeq 0$, $\mathsf{k} = 0$ proves existence. □

*Proof of Corollary B2.40.* Fiberwise, Corollary 7.12 gives $g_0 - \Psi_*^* g_1 \Psi_* \succeq 0$, hence $\delta \succeq 0$. Orthogonality gives $\mathsf{p}^* g_1 \mathsf{p} = \mathsf{p}_\perp^* g_1 \mathsf{p}_\perp + \mathsf{p}_{\mathcal{O}}^* g_1 \mathsf{p}_{\mathcal{O}}$, so $\mathsf{H} = \mathsf{D}^* g_0 \mathsf{D} - \mathsf{p}^* g_1 \mathsf{p} + \mathsf{p}_{\mathcal{O}}^* g_1 \mathsf{p}_{\mathcal{O}}$, which is (B2.77). The zero anomaly corresponds to $C = \mathsf{p}_{\mathcal{O}}$, and $\mathsf{p}_{\mathcal{O}}^* g_1 \mathsf{p}_{\mathcal{O}} \preceq \mathsf{H}$. For feasible $C_0, C_1$ and $0 \le t \le 1$, $(1-t)\|C_0 x\|^2 + t\|C_1 x\|^2 - \|(1-t)C_0 x + tC_1 x\|^2 = t(1-t)\|(C_0 - C_1)x\|^2 \ge 0$, so the feasible set is convex, and the set of lifts is the preimage of a convex set under a linear map. □

*Proof of Theorem B2.41.* In the quotient inner product, $(\mathsf{p} + \mathsf{J}\eta)^* \Delta (\mathsf{p} + \mathsf{J}\eta) = (\mathsf{p}_\Delta + \mathsf{J}_\Delta \eta)^* (\mathsf{p}_\Delta + \mathsf{J}_\Delta \eta)$. The component $(I - P_\Delta)\mathsf{p}_\Delta$ is unchanged by $\mathsf{J}_\Delta \eta$ and is orthogonal to $\mathcal{O}_\Delta$, so removing its identical contribution from both sides reduces the cocycle to

$$(\mathsf{w}_\Delta + \mathsf{J}_\Delta \eta)^* (\mathsf{w}_\Delta + \mathsf{J}_\Delta \eta) = \mathsf{w}_\Delta^* \mathsf{w}_\Delta. \tag{BA.1}$$

If $\mathsf{w}_\Delta x = 0$, equality of squared norms forces $(\mathsf{w}_\Delta + \mathsf{J}_\Delta \eta)x = 0$, so $\mathsf{T}(\mathsf{w}_\Delta x) = (\mathsf{w}_\Delta + \mathsf{J}_\Delta \eta)x$ is well defined and linear, and polarization shows that it is isometric. Conversely an isometric $\mathsf{T}$ gives (BA.1). Every right side of (B2.79) lies in the image of $\mathsf{J}_\Delta$, so it lifts with the stated kernel ambiguity. The choice $\mathsf{T} = \iota$ gives $\mathsf{J}_\Delta \eta = 0$, that is $\mathsf{J}\eta \in \mathrm{rad}\,\Delta$, which contains the isotropy case $\mathsf{J}\eta = 0$; $\mathsf{T} = -\iota$ gives $-2\mathsf{w}_\Delta$; if $\mathsf{w}_\Delta = 0$ the only map is on the zero space; and $\Delta = 0$ makes $E_\Delta = 0$. □

*Proof of Theorem B2.42.* If $o \in \mathcal{O}_1$ is $\Delta$-orthogonal to $\mathcal{O}_1$, it is orthogonal to $\mathcal{O}_0$ as well, since $\mathcal{O}_0$ is orthogonal to all of $\mathcal{O}$; thus $o \in \mathcal{O}_0 \cap \mathcal{O}_1 = 0$, which proves nondegeneracy of $\Delta_1$ and uniqueness of $\mathsf{w}$. Decompose $\mathsf{J}\eta = A_1 + \mathsf{n}$ with $A_1 \in \mathcal{O}_1$ and $\mathsf{n} \in \mathcal{O}_0$, and put $\mathsf{k} = \mathsf{w} + A_1$. Pairings within $\mathcal{O}_0$ and between $\mathcal{O}_1$ and $\mathcal{O}_0$ vanish, and each pairing of $\mathsf{p}$ with $\mathcal{O}_1$ may be replaced by the pairing of $\mathsf{w}$. Expanding the cocycle therefore gives exactly

$$\mathsf{k}^* \Delta_1 \mathsf{k} = \mathsf{G}_{\mathsf{n}}. \tag{BA.2}$$

The terms with $\mathsf{l}$ remain because the radical of $\Delta|_{\mathcal{O}}$ need not be orthogonal to $\mathrm{p}$. The restriction of $\mathsf{G}_\mathsf{n}$ to $\mathcal{V}_\mathsf{n}$ is nondegenerate, so a Gram factor restricts there to an injective isometry, whose image has signature $(m_+, m_-)$ inside a space of signature $(n_+, n_-)$, and the inertia inequalities are necessary. Its restriction to $\mathcal{K}_\mathsf{n}$ is orthogonal to that image and totally isotropic. Conversely these properties give the correct block on $\mathcal{V}_\mathsf{n}$ and zero cross terms and block on $\mathcal{K}_\mathsf{n}$. The inertia inequalities suffice for $\mathsf{I}_\mathsf{n}$: diagonalize the two nondegenerate forms, rescale the eigenvectors to squared norms $\pm 1$, and map like-signed basis vectors to each other; the spectral step maximizes a symmetric quadratic form on a unit sphere and inducts on the orthogonal complement. Taking $\mathsf{Z} = 0$ gives existence. In a space of signature $(r_+, r_-)$ the projections of a totally isotropic subspace onto the positive and the negative summands are injective, since a vector with either projection zero and zero norm vanishes, so its dimension is at most $\min(r_+, r_-)$; applied to the nondegenerate complement of $\mathsf{I}_\mathsf{n}(\mathcal{V}_\mathsf{n})$, of signature $(n_+ - m_+, n_- - m_-)$, this bounds the rank of $\mathsf{Z}$. Lifting through $\mathsf{J}$ completes the parameterization. In the example $\Delta(e_1, e_1) = 0$, so $\mathcal{O}_0 = \mathcal{O}$, $\mathcal{O}_1 = 0$, and $\mathsf{w} = 0$; then $\mathsf{l}_X(e_1) = X\Delta(e_2, e_1) = X$, so $\mathsf{G}_\mathsf{n}(X, X) = -2\mathrm{c}X^2$, and $n_\pm = 0$ forces $\mathsf{G}_\mathsf{n} = 0$; the cocycle difference is $2\mathrm{c}X^2\Delta(e_1, e_2) = 2\mathrm{c}X^2$, which confirms $\mathrm{c} = 0$, and for $\mathsf{p} = 0$ the form $\mathsf{l}$ vanishes. □

*Proof of Proposition B2.43.* A nonzero minor of maximal size of a constant-rank smooth bundle map stays invertible locally; its columns give a smooth image frame, and solving with it gives smooth preimages and kernel frames, so the kernels, images, and quotients are smooth bundles. Positive bundle metrics are obtained by patching local metrics with a partition of unity, orthogonal projections are smooth because their local matrices invert positive Gram matrices, and restriction to the orthogonal complement of a kernel gives a smooth isomorphism onto the image, hence a smooth right inverse of $\mathsf{J}$ or $\mathsf{J}_\Delta$. These constructions also prove the converse: the maps $\mathsf{k}([x]) = (\mathsf{p}_\mathcal{O} + \mathsf{J}\eta)x$ and $\mathsf{T}(\mathsf{w}_\Delta x) = (\mathsf{w}_\Delta + \mathsf{J}_\Delta\eta)x$ induced by a smooth solution are smooth. In the signed case $\mathsf{I}_\mathsf{n}$ and $\mathsf{Z}$ are restrictions of the smooth $\mathsf{k}$ to the chosen smooth splitting, and local frames for a nondegenerate form of fixed signature are obtained by retaining a basis with nonzero successive squared norms near a point, orthogonalizing, and dividing by the square roots of their absolute norms; mapping like-signed vectors gives local smooth embeddings under the inertia bounds, with no claim that independent local isometries glue. The intrinsic statements follow because every construction uses only the tensor transformation laws of the listed data, under which contraction, isometry, inertia, and total isotropy are preserved. □

*Proof of Theorem B2.44.* The difference of two principal connections is horizontal and equivariant, hence the tensorial form of a unique $\bar{\eta}$ on the target base; conversely adding such a form preserves the values of a connection on fundamental vectors and its equivariance. Pullback by $\mathcal{P}$ therefore gives exactly (B2.83). A pullback annihilates $\ker Tf$ and satisfies the equality on each fiber. Conversely define $\bar{\eta}_y(Y) = \delta\eta_c(X)$ using any $c$ over $y$ and any $X$ with $T_c f X = Y$; the two conditions make this independent of both choices and linear in $Y$. In local submersion coordinates there is a smooth local section $\mathsf{r}$ of $f$, and $\bar{\eta} = \mathsf{r}^*\delta\eta$ proves smoothness; surjectivity gives uniqueness. For a diffeomorphism both conditions are automatic. A different lift changes $\eta$ by a $\ker \mathsf{J}$-valued one-form, which may restore the two conditions. □

## BA.3 Witnesses for the classification

**Proposition BA.1 (Witnesses for the defect classification).** (i) Action rank can obstruct a continuous lift. For trivial $\mathrm{SO}(2)$ principal bundles over the identity map of $\mathbb{R}$, fine and coarse normal-location families on $\mathbb{R}^2$ with covariances $I$ and $2I$, the rotation-equivariant channel adding an independent $\mathcal{N}(0, I)$ vector, zero source connection, and related mean sections $\mu(x) = (x^2, 0)$, one has on the unit base vector

$$\mathsf{D} = (2x, 0), \qquad \Psi_* = I, \qquad g_0 = I, \qquad g_1 = \tfrac{1}{2}I, \qquad \mathsf{J}_x(t) = (0, x^2 t). \tag{BA.3}$$

The smooth anomaly $(0, x)$ lies in $\operatorname{im} \mathsf{J}_x$ at every point and satisfies the positivity problem, but its unique lift at $x \neq 0$ is $\eta_x = 1/x$, so no continuous connection defect realizes it across zero, although both Fisher metrics are positive.

(ii) A smooth defect need not have an ambient isometry extension. Let $a(x) = e^{-1/x^2}$ for $x \neq 0$ and $a(0) = 0$, $\mu(x) = \int_0^x a(t)\mathrm{d}t$, scalar unit-variance normal-location families at the first two levels, an identity first arrow, translation group $\mathbb{R}$, identity base maps, zero source connection, and an additive unit-normal noise second arrow, so that $\Delta = \frac{1}{2}$, $\mathsf{p} = a$, and $\mathsf{J} = 1$ relative to $\mathrm{d}x$. The defect

$$\eta_x = \begin{cases} -2a(x), & x > 0, \\ 0, & x \leq 0, \end{cases} \tag{BA.4}$$

is smooth and satisfies the cocycle and the first-arrow positivity with equality, while the isometry $\mathsf{T}$ of (B2.79) is $-1$ on the positive and $+1$ on the negative half-line and has no continuous extension as a field in $\mathrm{End}(\mathbb{R})$. On the other hand, if continuous $\mathsf{W}, \mathsf{Y}$ obey $\mathsf{Y}^*\mathsf{Y} = \mathsf{W}^*\mathsf{W}$, the induced isometry $\mathsf{T}(\mathsf{W}x) = \mathsf{Y}x$ is continuous on the image total space: in local bundle coordinates, if $(c_n, z_n) \to (c_0, z_0)$ with $z_n \in \operatorname{im} \mathsf{W}_{c_n}$ and $\mathsf{W}_{c_0} u_0 = z_0$, then

$$\|\mathsf{T}_{c_n} z_n - \mathsf{T}_{c_0} z_0\| \leq C_0 \|z_n - \mathsf{W}_{c_n} u_0\| + \|\mathsf{Y}_{c_n} u_0 - \mathsf{Y}_{c_0} u_0\| \longrightarrow 0, \tag{BA.5}$$

with $C_0 < \infty$ from the uniform local equivalence of continuous positive metrics with coordinate norms.

(iii) Smooth algebraic solutions can fail connection descent. For $f(x, y) = x$ on $\mathbb{R}^2$, trivial translation principal bundles with the principal map $(x, y, g) \mapsto (x, g)$, identical scalar unit-variance normal-location fibers, an identity first fiber map, zero mean sections, source connection $-\mathrm{d}y$, and target reference connection zero, one has $\mathsf{p} = -\mathrm{d}y$ and $\eta_0 = \mathrm{d}y$; the candidate $\eta = 2\mathrm{d}y$ satisfies positivity with equality and, with an additive unit-normal noise second arrow, the cocycle, but violates (B2.74) and is realized by no target connection. For $\mathcal{C} = \mathbb{R} \times \{+, -\}$, $f(x, \epsilon) = x$, translation-equivariant normal-location arrows with variances $1 \to 2 \to 3$, zero source and target reference connections, and related mean sections $\mu = x$, the choice $\eta = 0$ on the positive copy and $\eta = -2\mathrm{d}x$ on the negative copy solves both problems with constant ranks and $\ker Tf = 0$, and is realized by no target connection.

(iv) Signed isotropic solutions occur outside the Markov domain. For unit-covariance normal-location fibers on $\mathbb{R}^2$, full translation action, a constant related mean section, zero source connection, an identity first arrow, and a second arrow whose mean map has derivative $L_2 = \operatorname{diag}(0, \sqrt{2})$, one has $\Delta = I - L_2^\top L_2 = \operatorname{diag}(1, -1)$, $\mathsf{p} = 0$, and $\mathsf{J} = I$, and the constant target connection inducing the anomaly $X \mapsto (X, X)$ is a nonzero solution of the cocycle problem in the totally isotropic term of Theorem B2.42, although the ambient radical of $\Delta$ is zero; the second arrow is equivariant for the translation homomorphism $L_2$ and is not fiberwise Markov. [ESTABLISHED]

*Proof.* (i) The fundamental field of the rotation generator at the mean $(x^2, 0)$ is $t(0, x^2)$, the isotropic covariances are rotation invariant, and $g_1 = \frac{1}{2} I$ is the Fisher metric of the mean of $\mathcal{N}(\mu, 2I)$. The anomaly is $\mathsf{J}_x$ applied to $1/x$ for $x \neq 0$ and is zero at $x = 0$, where $\mathsf{J}_0 = 0$, and $\frac{1}{2}\|(2x, x)\|^2 = \frac{5}{2}x^2 \leq 4x^2 = \|(2x, 0)\|^2$. A continuous realizing defect would equal $1/x$ away from zero.

(ii) Every one-sided derivative of $\eta$ at zero is a polynomial in $1/x$ times $e^{-1/x^2}$ and tends to zero, so $\eta$ is smooth. The coarse jet is $\mathsf{p} + \eta = -a$ for $x > 0$ and $a$ for $x < 0$, so $(\mathsf{p} + \eta)^2 = \mathsf{p}^2$ gives both problems with equality, since the first arrow is the identity with unit Fisher metric. The isometry $\mathsf{T}$ sends $a$ to $-a$ for $x > 0$ and to $a$ for $x < 0$. For the continuity estimate, $\mathsf{T}_{c_n} z_n - \mathsf{T}_{c_0} z_0 = \mathsf{T}_{c_n}(z_n - \mathsf{W}_{c_n} u_0) + \mathsf{Y}_{c_n} u_0 - \mathsf{Y}_{c_0} u_0$, the isometry bounds the first term, and continuity of $\mathsf{W}$ and $\mathsf{Y}$ sends both terms to zero.

(iii) With the sign convention of Definition B2.3, the jet of a constant zero mean section is the local connection form, so $\mathsf{p} = -\mathrm{d}y$ and $\eta_0 = \mathcal{P}^* 0 - (-\mathrm{d}y) = \mathrm{d}y$. The coarse algebraic jet of $\eta = 2\mathrm{d}y$ is $\mathrm{d}y$, with $(\mathrm{d}y)^2 = \mathsf{p}^2$ and $\Delta = 1 - \frac{1}{2} = \frac{1}{2}$, but it is nonzero on $\partial_y \in \ker Tf$, and $\eta - \eta_0 = \mathrm{d}y$ does not annihilate $\partial_y$; the translation action is injective, so there is no other lift. In the second example $\mathsf{p} = \mathrm{d}x$, $\mathsf{J} = 1$, $\Delta = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$, both branches satisfy $(\mathsf{p} + \eta)^2 = \mathsf{p}^2$ and $\frac{1}{2}(\mathsf{p} + \eta)^2 \leq \mathsf{p}^2$, and (B2.84) fails because the two copies over one $x$ carry $0$ and $-2$.

(iv) The coarse Fisher metric of a unit-covariance mean is $I$, so the pulled-back metric is $L_2^\top L_2 = \operatorname{diag}(0, 2)$, and $(X, X)\Delta(X, X)^\top = X^2 - X^2 = 0$. Here $\mathcal{O} = \mathbb{R}^2$ with $\Delta_1$ of signature $(1, 1)$, $\mathsf{w} = 0$, and $\mathsf{G}_\mathsf{n} = 0$, so the solution is a totally isotropic $\mathsf{Z}$. A fiberwise Markov arrow has $\Delta \succeq 0$ by Corollary 7.12, which $\operatorname{diag}(1, -1)$ violates. All four witnesses use proper normal laws and normalized channels, and a one-point observation of likelihood one may be appended with evidence one. □

# Appendix BB

# Proofs for the Constructive Spatial RG Family

This appendix proves the constructions of chapter B7. It uses that chapter's inventories, reference measures, and parameters, and the local notation of table B7.1.

## BB.1 The source measure and boundary comparison

*Proof of theorem B7.3.* For fixed matter signs the coordinate change $\psi \mapsto z$ is a bijection preserving normalized counting, with inverse $\psi_e = \phi_x z_e \phi_{x+e_\iota}$. The finite raw law is therefore independent uniform matter times its dressed law in these coordinates. The exterior $\eta$ is dressed data; varying an exterior matter frame while holding $\eta$ fixed requires changing its reconstructed raw links as well.

For $\Delta \subset \Lambda$, conditioning on $z_{\Lambda\setminus\Delta}$ cancels the edge factors outside $\Delta$ and the plaquettes not meeting $\Delta$. The remaining factors are exactly $W_\Delta$ with the completed exterior, so the finite specification is consistent. Its one-edge conditional plus probability is

$$q_e = \frac{1+\tanh \widetilde{h}_e}{2}, \qquad \widetilde{h}_e = h + j \sum_{p \ni e} \prod_{f \in p\setminus\{e\}} z_f, \qquad |\widetilde{h}_e| \le h + 2j. \tag{BB.1}$$

Every conditional sign probability is at least $\epsilon_f = (1-\rho)/2 > 0$. Averaging over unrevealed variables preserves this bound, so a specified assignment on $A$ has probability at least $\epsilon_f^{|A|}$.

Let $n(e,f)$ count the plaquettes containing both of the distinct edges $e, f$. Each edge lies in two plaquettes with three other edges in each, whence $\sum_f n(e,f) = 6$. Changing one plaquette contribution changes the conditional mean by

$$\frac{|\tanh(s+j) - \tanh(s-j)|}{2} = \frac{\sinh(2j)}{\cosh(2s)+\cosh(2j)} \le t. \tag{BB.2}$$

Telescoping such changes gives the influence matrix $D_{ef} = t \cdot n(e,f)$, with row sums at most $\alpha = 6t < 1$.

A random-coordinate heat bath on the finite free set preserves its Gibbs law by conditional factorization. For $n = |\Lambda|$, updating each coordinate in a prescribed order to any target has probability at least $(\epsilon_f/n)^n$, so every entry of the $n$-step transition matrix has that positive lower bound; removing

the common row mass gives a strict total-variation contraction and a unique stationary law. Couple two such chains by the same chosen edge and the same uniform number in their conditional Bernoulli rows. Cesàro averages of the finite pair chain have a convergent subsequence, whose limit is stationary because one application changes the average by a difference divided by its length; its marginals are the two Gibbs laws. If $d_e$ is its probability of disagreement at $e$, stationarity cancels the unchanged update probability and gives

$$d_e \le \sum_{f\in\Lambda} D_{ef} d_f + \beta_e, \qquad \beta_e = \sum_{f\notin\Lambda} D_{ef} \mathbf{1}_{\{\eta_f \neq \eta'_f\}}. \tag{BB.3}$$

Iteration removes the remainder $D_\Lambda^n d$, which is bounded by $\alpha^n$. The surviving sum runs over paths making their first exit from $\Lambda$ at a differing exterior edge. Dividing each transition weight by $\alpha$, and adding a cemetery state for missing row mass, turns their weights into disjoint first-exit probabilities with sum at most one. Every exit path from $e$ has at least $\mathrm{dist}_\square(e,\Lambda^c)$ steps, so $d_e \le \alpha^{\mathrm{dist}_\square(e,\Lambda^c)}$, and a union bound over target disagreements proves eq. (B7.2).

A larger finite-volume restriction is a mixture of the smaller specifications, by finite consistency, so integrating the bound over two boundary mixtures compares volumes containing a common interior. On growing boxes each fixed cylinder is therefore Cauchy, independently of all exterior choices, and its limit is normalized, consistent, and has every atom at least $\epsilon_f^{|A|}$. To construct the measure, enumerate the countable edges, split $[0,1)$ into half-open intervals with the probabilities of the first edge, and split each interval into the joint probabilities of the next edge; consistency makes every subdivision exact. Each point specifies a unique infinite sign sequence whose cylinder preimages are finite unions of intervals, so the pushforward of Lebesgue measure is a countably additive law with these cylinders. Cylinders form a $\pi$-system, and the $\pi$–$\lambda$ theorem makes the extension unique.

For a cylinder function $f$, the finite update $\varsigma_\Lambda f$ remains a cylinder function because the interaction boundary is finite. Passing finite compatibility to the cylinder limit gives $\mu f = \mu\varsigma_\Lambda f$, and cylinder determination extends the identity to bounded measurable functions. For bounded exterior-measurable $w$, $\varsigma_\Lambda(wf) = w\varsigma_\Lambda f$, so this is the conditional DLR identity. Any two DLR laws have target marginals that are mixtures of the same finite specifications; the boundary bound tending to zero makes all their cylinders, hence the laws, equal. A translate of the unique law is a law for the same specification and therefore agrees with it. The conditional law in an ordinary box embedded in a full torus has the same factors and a random dressed exterior, so the same comparison proves the torus cylinder limit while retaining all winding sectors.

For the raw lift, append independent uniform matter signs and apply the inverse coordinate map. A finite raw window with free matter vertices and all their incident links converts to independent free frames and the correct dressed conditional; such windows are cofinal, and conditioning further proves the raw DLR equations for every finite window. Conversely, a local gauge flip of one matter vertex and its incident links preserves each raw conditional density and reference, so any raw Gibbs law is invariant under every finite product of these flips. In dressed coordinates these flip arbitrary finite sets of matter signs without changing $z$. Averaging over the flips shows that every assignment of the matter signs of $k$ given sites has probability $2^{-k}$, independently of every dressed cylinder.

The dressed marginal has the specification already identified, hence equals $\mu_{b,t}$, which proves uniqueness of the raw lift. □

## BB.2 Constrained parity laws uniformly in blocking depth

*Proof of theorem B7.7.* Partition the fine edges into the retained $B$-paths and unused singleton edges. No plaquette contains two edges of one retained path. A coarse anchor cell contains two retained paths and $2B(B-1)$ unused singleton blocks, hence $2B^2$ fine edges. A pattern prescribes each retained block to be unconstrained or to have one specified parity; singleton blocks remain unconstrained. Use the product of uniform probability references on these supports, multiply all edge and touching plaquette factors, and normalize; these are positive finite specifications on their supports. A block row with at most $B$ edges has every allowed atom at least

$$\xi_B = 2^{-B} \left(\frac{1-b}{1+b}\right)^B \left(\frac{1-t}{1+t}\right)^{2B} > 0. \tag{BB.4}$$

Two product sign laws obtained from one positive product law by conditioning on opposite total parities admit a coupling at Hamming distance exactly one. For one bit this is immediate. For $n$ bits, couple the last-bit marginals by any coupling. If those bits differ, the required prefix parities agree and the prefixes are coupled identically; if they agree, the induction coupling is applied to the opposite prefix parities. In either branch the prefix marginal on each side is its product law conditioned on the required parity, independently of the other branch, which proves the induction including both marginals.

Changing a single coordinate field from $+j$ to $-j$ in a product row or fixed-parity product row changes that coordinate's marginal by at most $t$, by eq. (BB.2) with the other-coordinate masses absorbed into $s$. Couple it maximally. If it differs in a fixed-parity row, the remaining prefix parities are opposite, so the preceding coupling adds exactly one further discrepancy. The expected Hamming cost is at most $2t$, and a one-bit fixed fiber has zero cost. Successive tilts are coupled by composing their conditional coupling kernels, and the endpoint Hamming distance is bounded by the sum of successive distances, which is the required triangle inequality.

For a target block $i$ and an exterior fine edge $e$, let $n(i,e)$ be the number of plaquettes meeting both. An exterior flip changes at most $n(i,e)$ single-coordinate fields, so the preceding coupling gives a row cost at most $2t\sum_e n(i,e)\mathbf{1}_{\{z_e \neq z'_e\}}$. The column identity, not a block-state row bound, is $\sum_i n(i,e) = 6$: each of the two plaquettes of $e$ meets three other blocks. Incident anchors differ by at most one.

Finite block heat baths have unique stationary laws by the common-mass argument using $\xi_B$, and a stationary pair coupling exists by the same Cesàro construction as above. Let $d_e$ be the fine-edge disagreement probabilities and $D_i = \sum_{e \in i} d_e$. Canceling the update probability separately in each block gives

$$D_i \leq 2t \sum_{e \text{ free}} n(i,e) d_e + 2t \sum_{e \text{ exterior}} n(i,e)\mathbf{1}_{\{\eta_e \neq \eta'_e\}}. \tag{BB.5}$$

For a target anchor set $A$, multiply by $w_i = e^{-\tau\,\mathrm{dist}_\infty(\mathrm{anchor}(i),A)}$ and sum. The column identity and $w_i \le e^{\tau} w_{\mathrm{block}(e)}$ when $n(i,e) > 0$ yield

$$S \le \mathrm{q} S + \mathrm{q} \sum_{e \in \partial_f \Lambda} w_{\mathrm{block}(e)} \mathbf{1}_{\{\eta_e \neq \eta'_e\}}, \qquad S = \sum_{i \text{ free}} w_i D_i. \tag{BB.6}$$

Use all blocks anchored in the radius-$R$ neighborhood of $A$. Every contributing exterior edge is anchored in its next shell, whose cardinality is at most $8|A|(R+1)$, because a square shell around one cell has exactly $8(R+1)$ cells, and there are $2B^2$ fine edges per anchor. The exterior sum is therefore at most $16|A|B^2(R+1)e^{-\tau(R+1)}$. Dividing by $1-\mathrm{q}$ and taking a union bound over target disagreements proves eq. (B7.6) for these finite constrained laws.

For every fixed full pattern, the bound makes all finite-cylinder limits Cauchy on expanding whole-block volumes. Interval subdivision constructs their measure, finite compatibility passes to the limit and gives its DLR law, and any two such laws have cylinders within the same vanishing boundary bound and hence agree. The argument also compares patterns agreeing in a large interior; different supports outside it are only exterior forcing. No conditioning on an infinite event is used.

Finite conditioning on previously free path parities has positive probability by eq. (BB.4). Restricting its finite conditional reference is precisely the updated support pattern, and uniqueness identifies the resulting infinite law. In the all-unconstrained pattern the law is the original fine Gibbs law, because its finite DLR identities also hold on unions of blocks, so its finite parity conditionals are these constrained laws. To define a target coarse kernel, leave its paths unconstrained, prescribe every exterior retained parity, and take the target marginal of the constructed constrained law.

Conditional on all other block states, the signs along a free path are independent, with conditional means bounded by $\rho$. Its parity mean has magnitude at most $\rho^B$, also after conditioning on any already revealed disjoint parities. Revealing the two target parities successively gives the asserted one-cell lower and upper probabilities. The boundary comparison makes every kernel continuous and measurable, and conditioning on a finite set of additionally revealed parities identifies the corresponding updated pattern, so finite Bayes identities give pointwise specification consistency at every exterior.

For an exterior cylinder event, the defining finite conditional integral holds under $\mu_B$. Increasing the revealed exterior set and using uniform convergence passes the integral to the constructed kernel, and cylinder determination extends this conditional identity to the full exterior $\sigma$-algebra. Thus $\mu_B$ is DLR for this everywhere-defined specification. Any two coarse DLR laws can be mixed over increasingly remote exteriors, and the same uniform comparison makes their target cylinders equal, which proves uniqueness. Translation covariance follows by transporting factors and patterns and invoking constrained uniqueness. At $t = 0$, independent product-fiber laws give a separate exact control with coarse edge mean $b^B$, and no formula dividing by $\mathrm{q}$ is needed. □

## BB.3 The complete canonical interaction

*Proof of theorem B7.9.* The one-cell bounds give log-odds amplitude $4\,\mathrm{atanh}(\rho^B) \le \mathrm{c}_2\rho^B$. Atom differences are bounded by total variation, and $|\log s - \log s'| \le |s - s'|/p_{\min}$ for $s, s' \ge p_{\min}$, so reference-state log odds have spatial variation at most $\mathrm{c}_3 B^2(R+1)e^{-\tau(R+1)}$. The canonical supports have diameter at most $2r$, and at most $(2r+1)^2$ translates contain the origin. Their maximal lexicographic cell uniquely specifies the anchor, so no assignments collide. Consequently

$$\|\mathscr{U}_{B,A_{x,r}}\|_\infty \le \min\{2\mathrm{c}_2\rho^B, \mathrm{c}_3 B^2 r e^{-\tau r}\}, \qquad r \ge 1. \tag{BB.7}$$

The inequality $\min(s, s') \le \sqrt{ss'}$, the bound $\sqrt{r} \le r$, and $\lambda_v = \tau/8$ reduce the weighted shell sum to $\sqrt{2\mathrm{c}_2\mathrm{c}_3}B\rho^{B/2}\sum_{r\ge1} r(2r+1)^2 z_v^r$. Differentiating the geometric series three times gives exactly $z_v(9 + 14z_v + z_v^2)/(1 - z_v)^4$, which proves the norm bound; its explicit tail after $N$ shells replaces that series with $\sum_{r>N} r(2r+1)^2 z_v^r$. Since $\mathrm{diam}\, A \le e^{\lambda_v \mathrm{diam}\, A}/\lambda_v$, it also proves the first moment. Both terms in $\mathcal{V}_B$ tend to zero; for an exponential bound, $B\rho^{B/4} \le \rho^{1/4}/(1 - \rho^{1/4})^2$.

For a configuration differing from $*$ on finitely many cells, change those cells in increasing lexicographic order. Positive finite specification consistency expresses its log probability ratio against the vacuum as the sum of the corresponding full-past log odds, and the absolutely convergent canonical shells telescope to exactly those odds. Terms anchored outside the changed set vanish, since their anchor stays at $*$. The interaction and the specification therefore give identical vacuum-exterior ratios. Comparing two finite-support configurations and canceling their common exterior gives equality for every finite-support exterior. Approximate an arbitrary exterior by its values in larger boxes and vacuum outside: quasilocality gives uniform convergence on the specification side, and absolute summability of all terms touching the finite changed set gives uniform convergence of interaction energy differences. Positive normalization then gives equality for every exterior. No infinite total Hamiltonian is used as an unnormalized density.

A continuous specification of a full-support law is unique everywhere: a difference at one exterior would persist on a cylinder of positive probability, contradicting almost-sure equality of conditional versions. The canonical interaction therefore depends only on the law. Path composition and this uniqueness give the stated action on its orbit, even if two block sizes happen to give the same interaction. At the fair product law all odds are zero, so the interaction is zero and fixed. At $t = 0$ the singleton interaction and its normalized-counting pressure are

$$\mathscr{U}_{B,\{x\}}(\sigma) = -\sum_{\iota=1}^{2} \log \frac{1 + b^B\sigma_\iota}{1 + b^B}, \qquad \mathfrak{p}(\mathscr{U}_B) = -2\log(1 + b^B), \tag{BB.8}$$

where $\sigma = (\sigma_1, \sigma_2) \in \{-1, 1\}^2$ is the cell state. This finite scalar is not zero despite vacuum normalization. □

## BB.4 The open-ball coefficient map and its law

*Proof of theorem B7.11.* The coefficient space is complete: a norm-Cauchy sequence converges in every anchored weighted $\ell^1$ space, its coordinate limits agree on overlaps, and the uniform Cauchy bound passes to the supremum over anchors. Translation invariance is a closed coordinate condition. Finite translation-orbit families of arbitrarily large supports give linearly independent members, so the positive-radius ball is infinite dimensional.

Condition the fair fine reference on all retained $a$-path parities. Its blocks are independent uniform parity fibers and unused fair singletons. A character has conditional expectation zero unless its support is a union of whole paths: a partially occupied path is killed by flipping one included and one excluded sign, and an occupied unused singleton by its own flip. A surviving support $\mathrm{lift}_a(T)$ has expectation $\chi_T$. For an ordered list of source occurrences $S_1, \dots, S_n$, repetitions allowed, every product of moments in the joint cumulant is therefore either zero or the character of $S_1 \triangle \cdots \triangle S_n$. Denote its constant coefficient by $c_a(S_1, \dots, S_n)$. The degree-$n$ map is

$$P_{a,n}(J_1, \dots, J_n)_T = \frac{1}{n!} \sum_{S_1 \triangle \cdots \triangle S_n = \mathrm{lift}_a(T)} c_a(S_1, \dots, S_n) \prod_{i=1}^{n} (J_i)_{S_i}, \qquad T \neq \varnothing. \tag{BB.9}$$

For finite source inventories this is differentiation of the finite conditional log partition, including its $1/n!$ and repeated occurrences.

Join two occurrence vertices when their supports meet a common reference block; disconnected groups then depend on independent reference variables. The bounded dependency-graph cumulant inequality [8, Sec. 9.4.2, Eq. (47)] gives $|c_a| \le 2^{n-1}\,\mathrm{ST}(G)$, where ST counts spanning trees of this occurrence graph. Its hypotheses hold uniformly in every parity assignment: the variables are sign characters of magnitude one, and the independence is at the block level. Convergence and the infinite-law identification are proved below.

Write $s_i = |S_i|$ and $d_i = \operatorname{diam} S_i$. For nonempty output $T$, disjoint paths give $a|T| \le \sum_i s_i$. Along an occurrence tree, travel within a source at cost at most $d_i$, and between two source edges in one reference block at cost at most $a - 1$. The first fine edges of two output paths realizing its diameter then give

$$a \operatorname{diam} T \le \sum_i d_i + (a-1)(n-1). \tag{BB.10}$$

Fix an output edge and its first fine edge. At least one occurrence contains that fine edge, so sum over $n$ possible roots. Pay $|T|$ by $a^{-1}\sum_i s_i$, summing over $n$ marked occurrences. A child support touching a parent reference block contains an edge in the union of blocks met by the parent, at most $as_i$ edges. Expanding this witness condition and summing from the leaves costs $a$ per tree edge and one parent support-size factor per child.

For an occurrence with $k_i$ children, the weighted anchored sum is bounded by its input norm times $k_i!\delta_a^{-k_i}$. Indeed the unmarked residual factor is $s_i^{k_i-1}e^{-\delta_a s_i}$ and the marked one is $s_i^{k_i}e^{-\delta_a s_i}$; both are at most $k_i!\delta_a^{-k_i}$, using $(\delta_a s_i)^{k_i}/k_i! \le e^{\delta_a s_i}$ and $s_i \ge 1$. The unused spatial weight only decreases

the bound. Summing $\prod_i k_i!$ over labeled trees with fixed root gives $(n-1)!\,\mathrm{Cat}_{n-1}$: ordering the children counts plane rooted trees, and the other $n-1$ labels are free. Balanced-parenthesis encoding gives $\mathrm{Cat}_{n-1} \le 4^{n-1}$. Combining the cumulant tree inequality, roots, mark, $1/n!$, and eq. (BB.10) proves

$$\|P_{a,n}(J_1,\dots,J_n)\| \le \frac{n}{a}\mathcal{K}_a^{n-1}\prod_{i=1}^{n}\|J_i\|. \tag{BB.11}$$

All rearrangements were of positive majorants, so they are justified also for infinite source inventories.

Define $\mathscr{R}_a J = \sum_{n\ge1} P_{a,n}(J,\dots,J)$ for $\mathcal{K}_a\|J\| < 1$. The majorant gives locally uniform convergence in the Banach norm and locally uniform convergence of $D\mathscr{R}_a(J)H = \sum_{n\ge1} nP_{a,n}(H,J,\dots,J)$ in operator norm. The multilinear increment expansion leaves a uniformly summable remainder with at least two increments, so this is the Fréchet derivative, and its locally uniform series is continuous. The first-order survival rule gives $L_a$. The lift is injective, has size $a|T|$, contains the first edge of every output path, and has diameter at least $a\,\mathrm{diam}\,T$, so comparing anchored weights gives $\|L_a\| \le \mathfrak{c}_a$. For $\omega = \mathcal{K}_a\|J\| \le 1/2$, $\sum_{n\ge2} n\omega^{n-2} \le 6$ and $\sum_{n\ge2} n^2\omega^{n-2} \le 22$, by differentiating the geometric series. These prove the quadratic bound with $6\mathcal{K}_a/a$ and the derivative bound with $22\mathcal{K}_a/a = 2M_a$, hence the stated bounds. On the ball, $\|\mathscr{R}_a J\| \le (\mathfrak{c}_a + M_a\|J\|)\|J\| \le \frac{1+\mathfrak{c}_a}{2}\|J\|$, because $\varepsilon \le (1-\mathfrak{c}_a)/(2M_a)$; a fixed point $J$ in the ball therefore satisfies $\|J\| \le \frac{1+\mathfrak{c}_a}{2}\|J\|$ and is zero.

It remains to identify a normalized law. For any real input in the ball, its one-edge half-log-odds are $\sum_{S\ni e} J_S\chi_{S\setminus\{e\}}$, absolutely and uniformly convergent. Flipping $f$ changes the Bernoulli row by at most $\sum_{S\ni e,f}|J_S|$, so the influence row sums are at most $\|J\| < 1$. Finite heat-bath coupling gives the resolvent comparison $d \le \sum_{m\ge0} D^m\beta$. For a fixed target, the contribution from a remote boundary tends to zero: each fixed power has summable rows, and the remaining powers have a common geometric tail. Finite-cylinder subsequences exist by finite-state compactness, and uniform convergence of finite local conditional energies passes their DLR equations to the limit. The comparison then makes the limit unique and makes the entire exhaustion converge.

Take finite edge boxes $E_m$ and source truncations $J_S^{[m]} = J_S\mathbf{1}_{\{S\subset E_m\}}$. Each is a finite inventory of no larger norm; no global norm convergence of these inhomogeneous truncations is asserted. Every fixed source tuple is eventually present, so the rooted majorant gives, for every fixed output edge, weighted incident convergence of $\mathscr{R}_a J^{[m]}$ to $\mathscr{R}_a J$. In the finite inventory the convergent scalar and nonempty Taylor series exponentiate to the conditional moment generating function near zero. The identity of the convergent analytic series extends over its real convergence interval, on which that function is strictly positive, so its real logarithm has precisely these coefficients throughout the ball. The scalar cancels only when the finite density is normalized.

The truncated fine law is the free-boundary Gibbs law on $E_m$, with fair signs outside. A path crossing its boundary retains those outside fair signs; incomplete characters are canceled rather than discarded by an altered channel. Its path push has exactly the finite coefficient expansion above. Incident convergence makes every finite coarse energy difference, hence every normalized conditional row, converge uniformly in the exterior. The fine laws converge locally to their unique

law, and the path map preserves convergence of cylinder observables. Passing the finite coarse DLR integral to this limit proves that the path-pushed law is Gibbs for $\mathscr{R}_a J$. Its norm lies in the ball, so it is the unique coarse Gibbs law by the same comparison.

Every prescribed parity exterior has a continuous conditional version. In the whole-reference-block representation, changing one outside block gives an influence row sum bounded by $2a\|J\| < 1$, since a block has at most $a$ fine edges. This estimate holds on every fixed-parity or unconstrained support, with all crossing sources present. Its summable resolvent constructs unique laws for every pattern and gives local continuity when remote patterns change. Free retained parity probabilities are at least $e^{-2a\|J\|}/2 > 0$, and finite parity conditioning corresponds exactly to these product-fiber supports. Testing the uniform limit of its finite conditional rows against exterior cylinders identifies it as a conditional version of the path-pushed law. The coefficient Gibbs kernel is also continuous, and since the law has full cylinder support the two versions agree everywhere.

Finally, integrating the conditional half-log-odds at $e$ against the fair exterior character $\chi_{S\setminus\{e\}}$ recovers $J_S$ by orthogonality and uniform convergence, which proves uniqueness of every nonempty coefficient. Path composition, unique laws, and this extraction give $\mathscr{R}_2\mathscr{R}_3 = \mathscr{R}_3\mathscr{R}_2 = \mathscr{R}_6$, and the same reasoning applies to any finite word with its directly defined path channel. The original field and plaquette source has norm $e^{\kappa}h + 8e^{4\kappa+\lambda}j$, so the ball contains a nonempty region with both couplings positive. □

## BB.5 Complete finite Taylor calculations and their errors

*Proof of theorem B7.13.* Represent each nonempty support once modulo integer translation, by moving its lexicographically first edge basepoint to zero and retaining its axis. If its shape is $S$, with size $|S|$ and diameter $\operatorname{diam} S$, and it contains $n_\iota(S)$ axis-$\iota$ edges, exactly $n_\iota(S)$ distinct translates contain a specified axis-$\iota$ edge. A nonempty finite set has no nonzero translation period, so a finite library's norm is

$$\|J\| = \max_{\iota=1,2} \sum_S n_\iota(S)|S|e^{\kappa|S|+\lambda \operatorname{diam} S}|J_S|. \tag{BB.12}$$

Duplicates are consolidated by exact addition before this formula is used. Only translations are identified; rotations and reflections are not.

At order $n$, let $d_{\max}$ and $s_{\max}$ be the largest input shape diameter and size. A connected source-occurrence list contributing an output with first edge at the origin stays within fine basepoint radius $R_n = nd_{\max} + (n-1)(a-1)$, by the occurrence-tree path argument, so every contributing literal instance belongs to the finite catalog of all input shapes translated by $[-R_n, R_n]^2 \cap \mathbb{Z}^2$. Reference-block ownership uses Euclidean division: a horizontal edge is retained exactly when its transverse coordinate is divisible by $a$, and its owner uses the floor of its parallel coordinate divided by $a$; vertical edges have the rotated rule, and negative coordinates follow the same floor rule. Unused singleton owners have a distinct type. A support is a union of complete paths exactly when every owner is retained and has all $a$ distinct members.

For a set $I$ of occurrence labels, let $m(I)$ be one if their symmetric-difference support is a union of complete paths, including the empty support, and zero otherwise; at all positive parities this is its conditional moment. With $r$ the first occurrence label in a nonempty $I$, the cumulant obeys

$$c(I) = m(I) - \sum_{\substack{D \subsetneq I \\ r \in D}} c(D) m(I \setminus D). \tag{BB.13}$$

The partition block containing $r$ uniquely decomposes the moment–cumulant formula, and summing all other blocks gives the second moment factor; this proves the recurrence and its singleton base. Occurrence labels remain distinct even when their supports repeat. Computing all subsets takes finitely many exact integer operations, and fewer than $3^n$ subset-pair contributions suffice.

For each order and each output first axis, enumerate every size-$n$ multiset of literal catalog instances once, and keep connected occurrence graphs with nonempty complete-path support whose output is already canonically anchored on that axis. A multiset with multiplicities $m_k$ contributes exactly $c(I) \prod_k J_k^{m_k} / \prod_k m_k!$, with $k$ running over the instances: its $n!/\prod_k m_k!$ ordered occurrences cancel the $n!$ in eq. (BB.9). Root choices and spanning trees are bounds, not extra multiplicities. Disconnected occurrences have zero cumulant by block independence, and the radius argument loses no connected term. Output canonicalization selects the already canonical output rather than accumulating all translated copies. Consolidation over all accepted terms gives the coefficient at every support through the specified order. Surviving supports satisfy $a|T| \le n s_{\max}$ and $a \operatorname{diam} T \le n d_{\max} + (a-1)(n-1)$, so the output is finite, and every finite catalog, multiset list, recurrence, and rational operation terminates at fixed order.

Summing eq. (BB.11) over orders above $\bar{n}$ gives eq. (B7.13) by differentiating the geometric tail $\omega^{\bar{n}+1}/(1-\omega)$. Its positive-series representation makes it monotone in the norm and in $\mathcal{K}_a$, so upper bounds for both give a valid bound despite the minus sign in the closed numerator. It tends to zero for $\omega < 1$, and the zero input gives the zero table and zero tail.

The transcendental constants admit convergent rational enclosures. For a nonnegative rational $\xi$, put $w_0 = 1$, $w_{k+1} = w_k \xi/(k+1)$, and $W_m = \sum_{k=0}^{m} w_k$. Once $v_m = \xi/(m+2) < 1$, positivity and decreasing successive term ratios give

$$W_m \le e^{\xi} \le W_m + \frac{w_{m+1}}{1 - v_m}. \tag{BB.14}$$

The width tends to zero, and reciprocal endpoints enclose $e^{-\xi}$. Applying these termwise to eq. (BB.12) and eq. (B7.11) gives convergent upper norm and lower radius bounds, so every strict admission or positive tolerance gap is decided at some finite precision, and the strict image margin and vanishing Taylor tail eventually admit the approximate image as well.

For one radius $r < \varepsilon$, the derivative estimate along a segment gives the Lipschitz constant $\operatorname{Lip}_a(r) < 1$, while the value estimate gives the radial factor $q_a(r) = \mathfrak{c}_a + M_a r < 1$. The triangle inequality separates one step's Taylor error from its input error and gives the stated recurrence. Choosing every tail below both a prescribed small fraction of the total tolerance and $(1 - q_a(r))r/2$

keeps all approximate states in the ball and achieves the requested error for a finite word. Two differently truncated direct and staged tables need not be equal; their difference is bounded by the sum of their errors relative to the same composed operator. The empty character is excluded only from the coefficient table: an unused fine field has a nontrivial $\cosh$ normalizer although its retained interaction is zero, so no evidence value can be inferred from that table. □

## BB.6 A constructive basin-entry theorem in the prescribed norm

*Proof of theorem B7.14.* The proof uses the compact box and retains the prescribed $\kappa, \lambda$. For $B \geq 3$, conditionally integrate all but the first and last edges of each retained path. No plaquette meets two eliminated edges, and the other three edges of such a plaquette lie outside every retained path, so these interiors are conditionally independent. For a horizontal path at the origin the conditional parity mean is

$$\bar{z}_p = z_{(0,0,1)} z_{(B-1,0,1)} \prod_{s=1}^{B-2} f(\chi_s^+, \chi_s^-), \qquad f(\chi, \chi') = \tanh(h + j(\chi + \chi')), \tag{BB.15}$$

where $\chi_s^{\pm}$ are the three-edge characters of the plaquettes above and below the edge $(s, 0, 1)$. Its strip consists of its endpoints, the horizontal edges at $(s, \pm 1)$, $1 \leq s \leq B-2$, and the vertical edges at $(s, 0), (s, -1)$, $1 \leq s \leq B-1$. It has $4B-4$ edges and diameter at most $B$. These strips and their rotations are pairwise disjoint: for horizontal edges, horizontal-strip interiors have first-coordinate residues $1, \ldots, B-2$, whereas vertical-strip interiors have residues $0, B-1$, and the transverse residues identify the unique adjacent grid line for $B \geq 3$. Retained endpoints belong only to their own path, and rotation proves the other orientation. This also proves conditional independence of all eliminated interiors.

Set $t_2 = \tanh 2j = 2t/(1+t^2)$. The four Walsh coefficients of $f$ are $\hat{f}_\varnothing, \hat{f}_1, \hat{f}_1, \hat{f}_{12}$, with

$$\hat{f}_\varnothing - \hat{f}_{12} = b, \qquad 2\hat{f}_1 = \frac{t_2(1-b^2)}{1-b^2 t_2^2}, \qquad \hat{f}_{12} = -\frac{b t_2^2 (1-b^2)}{2(1-b^2 t_2^2)}. \tag{BB.16}$$

Here $\hat{f}_\varnothing, \hat{f}_1 \geq 0$ and $\hat{f}_{12} \leq 0$, so the unweighted coefficient sum is $R(b, t_2) = b + t_2(1-b^2)/(1-b^2 t_2^2)$. It increases in $t_2$, and $\partial_b R \geq 1 - 2t_2/(1-t_2^2) > 0$ for $t_2 \leq 24/145$, the value at $t = 1/12$. Since $R(1, t_2) = 1$, its upper-endpoint value $R_*$ bounds the compact box with the explicit slack $1 - R_* \geq (13489/20449)(1 - b_+) > 0$. Put

$$\Delta = 1 - R_*, \qquad \alpha = \Delta/24, \qquad \bar{R} = \frac{R_*}{1 - \Delta/4} < 1, \qquad \mathbb{A}_B = \frac{\bar{R}^{B-2}}{1 - 2\alpha}. \tag{BB.17}$$

The fine Walsh norm with weight $e^{\alpha|S|}$, including its scalar, is submultiplicative under symmetric difference. Since $\chi, \chi'$ have three edges each, $\|f\|_\alpha \leq e^{6\alpha} R_* \leq \bar{R}$ and $\|\bar{z}_p\|_\alpha \leq \mathbb{A}_B$, using $e^x \leq 1/(1-x)$ for $0 \leq x < 1$, including the two endpoints.

For a finite path set $P$, the parity density relative to fair counting is

$$f_P(y) = \mathbb{E}_{\mu_{b,t}} \prod_{p \in P} (1 + y_p \bar{z}_p). \tag{BB.18}$$

Conditional independence of the eliminated interiors proves this identity, and averaging over $y$ gives one. The auxiliary fine law tilted by this product has the correct remaining-sign marginal for these observations; it is not the hard-parity posterior on all fine variables, which matters for its normalized conditional rows below.

For a root path $p_0$ with fine anchor $x_0$, let $D_{p_0}(S,T)$ be the maximum fine-coordinate distance from $x_0$ of any fine-edge anchor in $S$, or of $B$ times any coarse-edge anchor in $T$; an empty maximum is zero. Set $\beta = \kappa + 1$ and $\lambda_B = 2\lambda/B$, and complete the joint fine and coarse Walsh algebra in

$$\|F\|_{p_0} = \sum_{S,T} |F_{S,T}| e^{\alpha|S| + \beta|T| + \lambda_B D_{p_0}(S,T)}. \tag{BB.19}$$

Fine signs and coarse labels are distinct variables. Cardinalities and rooted radii are subadditive under multiplication, which proves the Banach algebra inequality. For a local update at a fine edge use the same norm centered there. Removing that edge lowers the fine weight by $e^{-\alpha}$, and replacing it by a local polynomial increases the rooted radius by at most that polynomial's local radius.

Put $\bar{p} = (1 + \bar{R})/2$, and increase $B$ until $2\lambda/B < 1$ and $\bar{R}/(1 - 2\lambda/B) \le \bar{p}$. The source conditional mean at an edge has local norm at most $\bar{p}$, because it reaches distance at most one. Choose rational upper bounds $\bar{e}_\beta \ge e^\beta$, $\bar{e}_{2\lambda} \ge e^{2\lambda}$, and $\bar{e}_\kappa \ge e^\kappa$, and set $H_B = \bar{e}_\beta \bar{e}_{2\lambda} \mathbb{A}_B$. Each observed strip has radius at most $B$ from any of its members, hence local activity at most $H_B$.

A fine edge belongs to at most one observed strip. If it does, write $\bar{z}_p = \bar{z}_p^{(0)} + z_e \bar{z}_p^{(1)}$ and let $f_e$ be its conditional mean under the source. Normalizing the two probabilities after multiplying by the observed factor gives

$$F_e = \frac{f_e + y_p(f_e \bar{z}_p^{(0)} + \bar{z}_p^{(1)})}{1 + y_p(\bar{z}_p^{(0)} + f_e \bar{z}_p^{(1)})} = f_e + \frac{y_p(1 - f_e^2)\bar{z}_p^{(1)}}{1 + y_p(\bar{z}_p^{(0)} + f_e \bar{z}_p^{(1)})}. \tag{BB.20}$$

If no observed strip touches the edge, $F_e = f_e$; the target strip is omitted when computing its own conditional mean. Both $y_p \bar{z}_p^{(0)}$ and $y_p \bar{z}_p^{(1)}$ have local norm at most $H_B$. Increase $B$ until

$$H_B \le \min\left\{ \frac{1}{2(1 + \bar{p})}, \frac{1 - \bar{p}}{4(1 + \bar{p}^2)} \right\}. \tag{BB.21}$$

The denominator perturbation then has norm at most $1/2$, so its geometric inverse has norm at most two, and the correction has norm at most $2(1 + \bar{p}^2)H_B \le (1 - \bar{p})/2$. Every normalized conditional mean therefore has local norm at most $r = (1 + \bar{p})/2 < 1$; the denominator and its even contribution are retained.

Let $\Pi$ project onto terms with empty fine support. For nonempty $S$, define

$$\mathcal{A}(\chi_S y_T) = \frac{1}{|S|}\sum_{e\in S}\chi_{S\setminus\{e\}} y_T F_e, \qquad \mathcal{A}_\perp = (1-\Pi)\mathcal{A}, \tag{BB.22}$$

and let $\mathcal{A}$ vanish on fine-constant terms. Removing the fine edge, using its local row norm, and averaging give $\|\mathcal{A}(1-\Pi)\| \le e^{-\alpha} r \le r$; the average avoids a support-size factor. Conditional expectation under the normalized finite tilted law satisfies $\mathbb{E}\chi_S y_T = \mathbb{E}\mathcal{A}(\chi_S y_T)$, and absorbing the fine-constant terms once and iterating proves

$$\mathbb{E}F = \Pi F + \sum_{k\ge 1} \Pi\mathcal{A}\mathcal{A}_\perp^{k-1}(1-\Pi)F. \tag{BB.23}$$

The unabsorbed remainder is bounded geometrically in the same norm. At any fixed substitution order only finitely many fine variables and coarse labels are reached, and the coefficients stabilize once the finite inventories contain that region. Fixing exterior signs decreases norms, and finite weights include every crossing original and observed factor, so the uniform geometric remainder proves boundary-independent continuous limits for arbitrary full observed patterns.

For a finite target set $A$ of coarse edges with target signs $y_A$, observing its complement gives the conditional density $\mathbb{E}_{\mu^{y,A^c}}\prod_{e\in A}(1+y_e\bar{z}_e)$ relative to fair counting on $A$. It is normalized and lies between $(1-\mathbb{A}_B)^{|A|}$ and $(1+\mathbb{A}_B)^{|A|}$. Increasing finite observed subsets gives this formula uniformly for every exterior by the preceding stabilization and tail, finite Bayes consistency passes to these limits, and testing their conditional integrals against exterior cylinders identifies them as the DLR kernels of the pushed law. This supplies the full finite kernels, not only one mean or an almost-sure version.

For a single target, start eq. (BB.23) at $\bar{z}_{p_0}$, whose rooted norm is at most $\bar{e}_{2\lambda}\mathbb{A}_B$. Its conditional mean $m_{p_0}$ therefore has norm at most $M_B = \bar{e}_{2\lambda}\mathbb{A}_B/(1-r)$. Increase $B$ until $M_B \le 1/2$. The norm-convergent series $\widetilde{h}_{p_0} = \operatorname{atanh} m_{p_0}$ gives the complete half-log-odds, with $\|\widetilde{h}_{p_0}\| \le \bar{h}_B = M_B/(1-M_B^2)$. Positive two-edge conditional kernels satisfy the conditional square identity

$$\widetilde{h}_e(y^{f,+}) - \widetilde{h}_e(y^{f,-}) = \widetilde{h}_f(y^{e,+}) - \widetilde{h}_f(y^{e,-}), \tag{BB.24}$$

since both sides are half the same log cross ratio. Taking Walsh coefficients, justified by absolute convergence, shows that the coefficient of $\chi_{T\setminus\{e\}}$ in $\widetilde{h}_e$ does not depend on the marked edge $e\in T$; define it to be $J_T^{(B)}$. It reproduces every single-edge ratio, and connectedness of the finite sign cube with normalization then reproduces all the conditional kernels. Translation covariance gives one canonical orbit representative.

For a size-$n$ orbit, its translated incidence and norm prefactor are at most $n^2$. The inequality $n^2 e^{\kappa n} \le 2e^{\kappa} e^{(\kappa+1)(n-1)}$ follows from $\sup_{s\ge 1} s^2 e^{-(s-1)} = 4/e < 2$. Because the root belongs to $T$, its diameter is at most twice its rooted radius. Selecting only supports whose literal first edge is the

root and summing the two root orientations therefore gives

$$\|J^{(B)}\|_{\kappa,\lambda} \le \mathcal{B}_B := 4\bar{e}_\kappa \frac{M_B}{1-M_B^2}. \tag{BB.25}$$

This estimate counts all supports and translated incidences in the prescribed weights; the fine weight $\alpha$ was auxiliary, and $\lambda$ has not been replaced by a smaller output weight.

Every displayed gate is eventually true and remains true for larger $B$: the base gate improves with $B$, and $\mathbb{A}_B$ decreases exponentially. Rational exponential enclosures as in eq. (BB.14) give a positive rational lower bound $\varepsilon_0 < \varepsilon$, and a finite rational search for $B_0$ ensuring the gates and $\mathcal{B}_B < \varepsilon_0/16$ terminates, uniformly over the compact box. The local theorem, theorem B7.11, now supplies uniqueness in its ball, and extraction of conditional coefficients with path composition identifies $\mathscr{R}_a J^{(B)} = J^{(aB)}$.

To obtain a finite rational interaction, keep absorbed substitution orders through $n_1$, obtaining $m_{p_0}^{[n_1]}$, and keep the odd powers of $\operatorname{atanh} m_{p_0}^{[n_1]}$ through $2n_2-1$. For rational $b, t$, every local row in eq. (BB.20) is a rational function of finitely many signs with positive denominator. Evaluate it on the complete cube and take its finite Walsh transform, retaining every coefficient. The cube may be reduced exactly to a basis for the $\mathbb{F}_2$-span of its character supports: the basis-character map is surjective by row rank, the function is constant on its fibers, and every outside-span coefficient vanishes by pairing with a kernel element that flips that character. The transform on the basis cube, lifted to literal supports, is therefore the entire function of the literal coordinates; the weights still use literal cardinalities and distances, not the number of basis generators. The local strip row's character rank is at most $2B$, from its $2(B-2)$ three-edge characters, endpoint character, possibly removed edge, and two original row characters.

All finite substitution orders and powers have finite inventories. The mean remainder is $d_{n_1} = M_B r^{n_1}$, and both exact and truncated absorbed sums have norm at most $M_B$. Telescoping the odd powers gives the $\operatorname{atanh}$ Lipschitz bound $(1-M_B^2)^{-1}$, and its omitted powers give the error

$$\|\widehat{J} - J^{(B)}\|_{\kappa,\lambda} \le 4\bar{e}_\kappa \left[\frac{d_{n_1}}{1-M_B^2} + \frac{M_B^{2n_2+1}}{(2n_2+1)(1-M_B^2)}\right]. \tag{BB.26}$$

The incidence argument applies also to omitted coefficients and to approximate rows that need not obey exact square identities, and canonical selection gives one valid approximate interaction. After $n_1$ substitutions its labels lie within coarse radius $n_1+1$, and the full $\operatorname{atanh}$ of that finite-variable mean introduces no new labels, so supports of diameter greater than $2(n_1+1)$ have total norm at most $4\bar{e}_\kappa d_{n_1}/(1-M_B^2)$. The finite odd polynomial pays its separate power tail in eq. (BB.26).

Increasing $n_1, n_2$ makes the error arbitrarily small, and each rational transform, substitution, and product terminates. For a finite word, choose the input error small enough that both the true interaction and its approximation lie in one ball of radius $r < \varepsilon$. Comparing their exact trajectories by the Lipschitz bounds gives $\delta \prod_k \operatorname{Lip}_{a_k}(r)$, and adding the separate Taylor word error proves the

stated total. The argument retains normalized fine tilts throughout and computes no discarded scalar partition. □

## BB.7 Higher response and the Fisher-density geometry

*Proof of theorem B7.15.* Enlarge the compact rectangle of natural parameters to $\widehat{K} = [h_-, h_+] \times [0, j_+]$, with $h_- > 0$ and $0 < j_+ < \operatorname{atanh}(1/12)$. Its positive-$j$ interior contains the original compact set, and the boundary $j = 0$ is used only for the one-sided continuation below. Write $t_+ = \tanh j_+$, choose $\tau > 0$ with $\mathrm{q} = 12 t_+ e^{\tau} < 1$, and put $\mathrm{c}_1 = 16\mathrm{q}/(1-\mathrm{q})$, $\rho = \tanh(h_+ + 2j_+)$, and $d = 1 - \rho > 0$. These constants work for every volume, depth, support pattern, and fixed fine exterior. Finite supports and references are parameter independent. The finite density is an exponential family in $(h, j)$; its field statistic sums free edge signs, and its plaquette statistic includes every touching plaquette with its exterior completion. Expectations and cumulants under the fine law at $(h, j)$ carry the subscript $h, j$.

### BB.7.1 Localized source cumulants and derivatives

Let $F$ be a finite vector of block observables supported on an anchor set $A$, and $F'$ another supported outside its radius-$R$ neighborhood. Condition on the fine blocks outside that neighborhood. The weighted Hamming proof compares any two resulting interior rows. If the finite whole-block volume is irregular, its inherited exterior is identical on both sides and contributes zero disagreement; only the newly revealed shell contributes. Thus

$$\mathrm{TV}(\mathrm{Law}(F, F'), \mathrm{Law}(F) \otimes \mathrm{Law}(F')) \le \min\{1, \mathrm{c}_1 |A| B^2 (R+1) e^{-\tau(R+1)}\}. \tag{BB.27}$$

Each conditional law of $F$ is within this bound of their mixture, and integrating with the exterior and marginalizing proves the group comparison. It passes to infinite constrained laws by finite-cylinder convergence, also at $j = 0$ by the exact product-fiber law.

For $m$ bounded observables define

$$\mathrm{cum}(F_1, \dots, F_m) = \sum_{\pi \in \mathcal{P}_m} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{D \in \pi} \mathbb{E} \prod_{i \in D} F_i, \tag{BB.28}$$

where $\mathcal{P}_m$ is the set of partitions of $\{1, \dots, m\}$. Its absolute value is at most $k_m \prod_i \|F_i\|_\infty$, and a change of the joint law by total variation $v$ changes it by at most $\bar{k}_m v \prod_i \|F_i\|_\infty$, where $k_m = \sum_\pi (|\pi| - 1)!$ and $\bar{k}_m = 2 \sum_\pi |\pi|!$. For $m = 2, 3, 4$, these pairs are $(2, 6), (6, 26), (26, 150)$. The second estimate telescopes each product of moments, each moment changing by at most twice the total variation. A cumulant of two independent nonempty groups vanishes, because its generating function factorizes and its logarithm has no mixed coefficient; expanding that finite logarithm also proves eq. (BB.28).

Let a local $f$ have anchor support $A$ of size $a_0 \ge 1$, and let $r = 1, 2, 3$ labeled edge or plaquette sources have base anchors $x_1, \dots, x_r$. Put $M = \max_s \mathrm{dist}_\infty(x_s, A)$. On the complete graph of

these labeled anchors and a root representing $A$, a minimum-weight spanning tree has an edge of length at least $M/r$. Removing a longest edge splits the graph into groups with all cross distances at least that length; otherwise exchanging a shorter cross edge would reduce the tree. Plaquette supports enlarge anchor sets by at most one on either side. For $M \geq 4r$, use $R = \lfloor M/r \rfloor - 3$ in eq. (BB.27); the root group has at most $a_0 + 4r$ anchors. For smaller $M$, use the individual cumulant bound. With

$$L_r(a_0) = e^{4\tau}[k_{r+1} + \bar{k}_{r+1}\mathrm{c}_1(a_0 + 4r)], \qquad S_r(a_0) = \frac{(8a_0)^r L_r(a_0)(2r+1)!}{(1 - e^{-\tau/r})^{2r+2}}, \tag{BB.29}$$

this proves the bound $L_r(a_0)B^2(M+1)e^{-\tau M/r}\|f\|_\infty$ for each cumulant. There are at most $2B^2$ choices of a source of one derivative type at each anchor, and at most $a_0(2M+1)^2$ anchors within distance $M$. Counting ordered tuples, including repeats, gives

$$\begin{aligned} \sum_{s_1,\dots,s_r} |\operatorname{cum}(f, T_{s_1}, \dots, T_{s_r})| &\leq S_r(a_0)B^{2r+2}\|f\|_\infty, \\ \sum_{\max \operatorname{dist}(x_s, A) > L} |\operatorname{cum}(f, T_{s_1}, \dots, T_{s_r})| &\leq S_r(a_0)B^{2r+2}(L+2)^{2r+1}e^{-\tau(L+1)/r}\|f\|_\infty. \end{aligned} \tag{BB.30}$$

Here $T_s$ is the edge or plaquette source for its labeled derivative direction. To sum the bound, use $(M+1)^k \leq k!\binom{M+k}{k}$ in the geometric series; for the tail, write $M = L + 1 + n$ and bound $(M+1)^k \leq (L+2)^k(n+1)^k$. Exterior constant sources have zero cumulant and may safely be overcounted.

Finite differentiation of the normalized exponential family gives these cumulant sums for each derivative of a parameter-independent observable. For an observable $f_{h,j}$ that depends on the parameters,

$$\partial^\alpha \mathbb{E}_{h,j} f_{h,j} = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \sum_{s_1,\dots,s_{|\alpha-\beta|}} \operatorname{cum}_{h,j}(\partial^\beta f_{h,j}, T_{s_1}, \dots, T_{s_{|\alpha-\beta|}}), \tag{BB.31}$$

where an empty source list means ordinary expectation. Differentiating the finite generating function or its normalized finite sum proves all multiplicities, including repeated directions. For each fixed source tuple, finite-cylinder convergence gives convergence of its moments, uniformly on $\widehat{K}$; near tuples form a finite set, and far tuples have the uniform tails in eq. (BB.30). Sending first the source cutoff and then the volume to infinity proves uniform convergence of finite derivatives through order three, and the sums are continuous by the same finite-near, uniform-tail argument. Passing finite coordinate fundamental-theorem-of-calculus identities to the limit proves that these are continuous derivatives, successively through order three, with agreeing mixed derivatives. At $j = 0$ the argument runs on right-sided coordinate segments and gives one-sided $C^3$ continuation. No infinite normalizer or null-event probability is differentiated.

### BB.7.2 Full local jets and interaction derivatives

For a free path, its conditional mean $\bar{z}$ given the other fine blocks is the product of $B$ hyperbolic tangents with fields $h + jw_e$, $|w_e| \leq 2$, on at most nine anchor cells. For $n \leq 3$, differentiating the

product in labeled directions gives

$$\|\partial^{\beta}\bar{z}\|_{\infty} \le g_n B^n \rho^{B-n}, \qquad g_n = 2^n n!, \quad |\beta| = n. \tag{BB.32}$$

The first three hyperbolic-tangent derivatives have bounds $1, 2, 6$, and each $j$ differentiation adds a factor at most two. There are at most $B^n$ assignments to product factors, and the product of their factorials is at most $n!$. Negative exponents when $B < n$ only enlarge the upper bound. Apply eqs. (BB.30) and (BB.31) to $\bar{z}$. With

$$D_n = g_n + \sum_{b'=0}^{n-1} \binom{n}{b'} g_{b'} S_{n-b'}(9), \qquad M_n(B) = D_n B^{2n+2} \rho^{B-n}, \quad M_0(B) = \rho^B, \tag{BB.33}$$

every order-$n$ derivative of a free-path mean is bounded by $M_n(B)$. These bounds tend to zero as $B$ increases.

For patterns agreeing through $Q_R$, the cell square of radius $R$, choose $a_R = \lfloor R/2 \rfloor - 1$ when $R \ge 2$. A near source tuple has joint support within $Q_{a_R+1}$; apply the group comparison on its remaining buffer and count $[2B^2(2a_R+1)^2]^n$ ordered tuples, and use eq. (BB.30) for each of the two far sums. Writing $\mathrm{c}'_n = 2^n \bar{k}_{n+1} \mathrm{c}_1 (1+4n)$, the resulting derivative modulus is

$$\begin{aligned} V_n(B,R) = \min\big\{ 2M_n(B), B^{2n+2} \big[ & \mathrm{c}'_n (2a_R+1)^{2n} (R-a_R) e^{-\tau(R-a_R)} \\ & + 2S_n(1)(a_R+2)^{2n+1} e^{-\tau(a_R+1)/n} \big] \big\}. \end{aligned} \tag{BB.34}$$

For $R = 0, 1$ take $V_n = 2M_n$, and for values use $V_0 = \min\{2\rho^B, 2\mathrm{c}_1 B^2 (R+1) e^{-\tau(R+1)}\}$. The same proof compares a finite whole-block conditional calculation against its infinite kernel: near sources are interior, and finite crossing sources and infinite far sources satisfy the two separate tail bounds. For $R \ge 2$, the bracket in eq. (BB.34) is at most $Z_n (R+1)^{2n+1} e^{-\tau R/(2n)}$, with $Z_n = \mathrm{c}'_n + 2e^{\tau/(2n)} S_n(1)$.

Every Bernoulli factor is $(1 + s\bar{z})/2$, with $1 + s\bar{z} \ge d$. Revealing both target parities sequentially retains all their dependence, and their positive four-state row is $C^3$. For one Bernoulli log factor the derivative amplitudes are

$$A_1 = \frac{M_1}{d}, \qquad A_2 = \frac{M_2}{d} + \frac{M_1^2}{d^2}, \qquad A_3 = \frac{M_3}{d} + \frac{3M_2 M_1}{d^2} + \frac{2M_1^3}{d^3}. \tag{BB.35}$$

The multiplicities follow by differentiating $\log(1 + s\bar{z})$ three times. Its derivative moduli are

$$\begin{aligned} T_1 &= V_1/d + M_1 V_0/d^2, \\ T_2 &= V_2/d + (M_2 V_0 + 2M_1 V_1)/d^2 + 2M_1^2 V_0/d^3, \\ T_3 &= V_3/d + (M_3 V_0 + 3M_2 V_1 + 3M_1 V_2)/d^2 \\ &\quad + (6M_2 M_1 V_0 + 6M_1^2 V_1)/d^3 + 6M_1^3 V_0/d^4. \end{aligned} \tag{BB.36}$$

Subtracting the derivative ratios and using $|r_1^{-k} - r_2^{-k}| \le k|r_1 - r_2|/d^{k+1}$ for $r_1, r_2 \ge d$ gives these formulas. One-cell log probabilities use $2A_n, 2T_n$, and reference-state log odds use $4A_n, 4T_n$. Each

modulus can also be capped by twice its amplitude. The same already revealed first parity is fixed when comparing the second factor, so the support patterns still agree in their common interior.

The complete interaction jets are summable, with explicit constants. Set $E_1 = D_1/d$, $E_2 = D_2/d + D_1^2/d^2$, and $E_3 = D_3/d + 3D_2D_1/d^2 + 2D_1^3/d^3$. Then $F_n(B) = 4A_n(B) \le 4E_nB^{4n}\rho^{B-n}$. Each $M_j$ is bounded over all $B$ by $U_j = D_j\rho^{1-j}(2j+2)!/(1-\rho)^{2j+3}$. Put $Z_0 = 2c_1$, and obtain $P_n$ from the right side of eq. (BB.36) by replacing every $M_j$ with $U_j$ and every $V_j$ with $Z_j$; these are explicit positive constants, and the powers of $d$ are unchanged. For $R \ge 2$, $T_n \le P_nB^{2n+2}(R+1)^{2n+1}e^{-\tau R/(2n)}$. The constant

$$H_n = 4P_n + 8E_n\rho^{-n}\frac{(2n-2)!}{(1-\rho)^{2n-1}}e^{\tau/(2n)} \tag{BB.37}$$

also covers shell radii one and two through the amplitude cap. Its order-$n$ canonical shell is therefore bounded by both $2F_n(B)$ and $H_nB^{2n+2}r^{2n+1}e^{-\tau(r-1)/(2n)}$. Choose one common $\lambda_f = \tau/48$. Geometric interpolation, $\operatorname{diam} A_{x,r} \le 2r$, and the translate count give

$$\|\partial^\alpha \mathscr{U}_B\|_{\lambda_f} \le F_n(B) + \sqrt{2F_n(B)H_n}B^{n+1}e^{\tau/(4n)}\sum_{r\ge1}(2r+1)^2r^{n+1}e^{-[\tau/(4n)-2\lambda_f]r}, \quad |\alpha| = n \le 3. \tag{BB.38}$$

The bracketed exponent $\tau/(4n) - 2\lambda_f$ is positive, the positive series is at most $9(n+3)!$ divided by $(1-e^{-[\tau/(4n)-2\lambda_f]})^{n+4}$, and its omitted tail has an explicit convergent bound. The right side decays as a constant times $B^{3n+1}\rho^{(B-n)/2} + B^{4n}\rho^{B-n}$. Values use the canonical interaction proof of appendix BB.3 in the same weaker $\lambda_f$ norm. Finite-shell maps are $C^3$, and their derivatives through order three converge uniformly in this Banach norm, so successive coordinate fundamental-theorem-of-calculus identities prove that the entire interaction is $C^3$. No shell or derivative is replaced by a finite truncation in the statement.

### BB.7.3 Moving expectations and marginal information

For any finite or infinite subset $S$ of the lexicographic past, let $p_S(s;y)$ be the four-state predictive row of the target state $s$ with exactly the past coordinates $y_S$ prescribed and the other paths, including the future, free. Let $l = \log p_S$, with subscripts denoting derivatives in the natural parameters, and use the indices $i,m,k,n$ for those derivatives. Define $\mathsf{I}_{S,im} = \sum_s p_S l_i l_m$, an observable of the prescribed past. Its first and second parameter derivatives, keeping $y$ fixed, are

$$\begin{aligned}
\mathsf{I}_{S,im,k} &= \sum_s p_S(l_k l_i l_m + l_{ik}l_m + l_i l_{mk}),\\
\mathsf{I}_{S,im,kn} &= \sum_s p_S\big[(l_{kn}+l_kl_n)l_il_m + l_k(l_{in}l_m + l_il_{mn}) + l_n(l_{ik}l_m + l_il_{mk})\\
&\quad + l_{ikn}l_m + l_{ik}l_{mn} + l_{in}l_{mk} + l_il_{mkn}\big].
\end{aligned} \tag{BB.39}$$

These formulas differentiate the probabilities as well as the scores. Put $L_i = 2A_i$ for $i = 1,2,3$,

$$E_B^F = \max\{L_1^2, L_1^3 + 2L_1L_2, L_1^4 + 5L_1^2L_2 + 2L_1L_3 + 2L_2^2\}, \qquad H_B^F = \max(1, L_1, L_2, L_3). \tag{BB.40}$$

Every matrix entry and its first two derivatives are bounded by $E_B^F$. For patterns agreeing in $Q_R$, every row probability changes by at most $V_0$, and each log jet by at most $2T_n$. Telescoping the at most ten expanded monomials in eq. (BB.39), with at most five factors per term and four states, gives the common modulus

$$\omega_B(R) = \min\{2E_B^F, 200(H_B^F)^4[V_0(B,R) + 2T_1(B,R) + 2T_2(B,R) + 2T_3(B,R)]\}. \tag{BB.41}$$

At fixed $B$ it is bounded by an explicit constant times $(R+1)^7 e^{-\tau R/6}$. These are differences of jets, not derivatives of an earlier scalar continuity bound.

Expectations now use the original unconstrained fine law, with coarse observables pulled back by the fixed path map. For a local $C^2$ observable on $Q_N$, put $a_N = (2N+1)^2$ and $W_B(N) = 1 + 2S_1(a_N)B^4 + S_2(a_N)B^6$. The moving response formula gives

$$\max_{|\alpha|\le 2} \sup_{(h,j)\in\widehat{K}} |\partial^\alpha \mathbb{E}_{h,j} f_N| \le W_B(N) \max_{|\alpha|\le 2} \sup_{(h,j),y} |\partial^\alpha f_N|. \tag{BB.42}$$

In particular, the second derivative includes $\mathbb{E}f_{im}$, both source covariance sums involving $f_i, f_m$, and the complete three-variable cumulant sum involving $f$. Formula eq. (BB.30) justifies each passage from finite volumes, and neither the observable nor the law is held fixed incorrectly. The function $W_B(N)$ is polynomial in $N$. For a shell series $f = \sum_N d_N$ with $\sum_N W_B(N)\|d_N\|_{C^2} < \infty$, finite partial sums and their expectation derivatives are uniformly Cauchy, and repeated fundamental-theorem-of-calculus identities give $C^2$ of its expectation with termwise differentiation.

Apply this to $f_N^S = \mathsf{I}_{S\cap Q_N}$. Its initial shell is bounded by $E_B^F$, and its later shell differences through order two by $\omega_B(N-1)$. Thus

$$Z_B^F = W_B(0)E_B^F + \sum_{N\ge 1} W_B(N)\omega_B(N-1) < \infty, \qquad \epsilon_B(R) = 2\sum_{N>R} W_B(N)\omega_B(N-1) \longrightarrow 0. \tag{BB.43}$$

If two past sets agree inside $Q_R$, their finite prefixes agree exactly through that radius, so their full-law Fisher-observable expectations differ in entrywise $C^2$ norm by at most $\epsilon_B(R)$, uniformly over the remote shape of the past. In particular the full-past expectation $\mathcal{I}_B = \mathbb{E}\mathsf{I}_{\mathrm{past}}$ is $C^2$.

The positive marginal on $\Lambda_L$ factors into its predictive rows in lexicographic order. Each score increment is centered conditionally on its earlier revealed cells, and two increments are orthogonal because the earlier one is measurable for the later past. The finite Fisher matrix is therefore the sum of expected row-Fisher matrices with their finite pasts, each a $C^2$ function by the preceding varying-expectation argument. Translation invariance removes the shifts. For $L \ge 2R$, all but at most $4RL$ cells have radius $R$ inside the square, and their finite pasts and full past agree on that radius; each remaining difference is bounded by $2Z_B^F$. Therefore

$$\|L^{-2}I_{B,\Lambda_L} - \mathcal{I}_B\|_{C^2,\mathrm{entry}} \le \epsilon_B(R) + 8Z_B^F R/L. \tag{BB.44}$$

Sending first $L$ to infinity at fixed $R$, then $R$ to infinity, proves the marginal $C^2$ limit, including all boundary-score derivatives. The canonical row with future fixed to vacuum is never substituted for the predictive row.

### BB.7.4 An exhibited positive chart and its curvature

At $j = 0$, with any fixed complete past, the blocks are independent product parity fibers, and the two target path parities are independent with mean $b^B$. For one fine edge of a target path with parity $Y$,

$$a(s) = \mathbb{E}[z_e \mid Y = s] - b = \frac{sb^{B-1}(1-b^2)}{1+sb^B}, \qquad v_B = \mathbb{E}a^2 = \frac{b^{2B-2}(1-b^2)^2}{1-b^{2B}} > 0. \tag{BB.45}$$

Multiplying the product law by its parity indicator and summing proves this formula; $\mathbb{E}a = 0$, and conditional-variance decomposition gives $v_B \le 1 - b^2 \le 1$. The field score is $\mathrm{s}_h = B(a_1 + a_2)$, and the plaquette score is $\mathrm{s}_j = b^2 a_1 a_2 + \alpha_1 a_1 + \alpha_2 a_2$, with $|\alpha_i| \le 2B$. For the mixed term, the plaquette at the common starting vertex is the unique one meeting both outgoing target paths; its other two edges are future paths at $B = 1$, or unused singleton edges at $B > 1$, and have mean $b$, so its centered contribution is $b^2 a_1 a_2 + b^3(a_1 + a_2)$. Every other incident plaquette meets only one target path and contributes a multiple of $a_i$ of magnitude at most one; there are $2B - 1$ such incidences. Nonincident plaquettes contribute zero by product-block independence. These are derivatives of the full family through its source formula and one-sided limit, not derivatives of a formula restricted to $j = 0$.

Independence and centering make $a_1 a_2$ orthogonal to both linear contrasts. The conditional Fisher entries are therefore $I_{hh} = 2B^2 v_B$, $I_{hj} = B v_B(\alpha_1 + \alpha_2)$, and $I_{jj} = v_B(\alpha_1^2 + \alpha_2^2) + b^4 v_B^2$. Their determinant is $B^2 v_B^2(\alpha_1 - \alpha_2)^2 + 2B^2 b^4 v_B^3$, and their trace is at most $11B^2 v_B$. Dividing the determinant by the trace bounds the smaller eigenvalue below by $2b^4 v_B^2/11$, uniformly in the past. On a compact positive-field interval with $b \in [b_-, b_+]$, this is at least

$$\eta_{0,B} = \frac{2}{11} b_-^{4B}(1 - b_+^2)^4 > 0. \tag{BB.46}$$

Use the bounds $A_1, A_2$ on the enlarged rectangle and put $Q_B = 16(A_1^3 + A_1 A_2) > 0$. The score norm is at most $2\sqrt{2}A_1$, and its $j$ derivative at most $2\sqrt{2}A_2$. Differentiating the four-state Fisher sum gives an operator-norm derivative at most $Q_B$, including the probability derivative term. One-sided $C^3$ and the fundamental theorem of calculus imply

$$\mathcal{I}_B(h, j) \succeq \frac{\eta_{0,B}}{2} I_2 \quad \text{for} \quad 0 < j < \delta_B := \min\{j_+/2, \eta_{0,B}/(2Q_B)\}. \tag{BB.47}$$

The same floor holds rowwise and hence after the full-law expectation. An enlarged field interval supplies an open neighborhood of each original compact interval. For a prescribed finite list of depths, take the finite minima of positive widths and floors. No positive minimum over every integer depth is asserted.

On a compact subchart of this region write $g_{mn} = \mathcal{I}_{B,mn}$, and use the indices $k, m, n, r, s$ for the natural parameters. The coefficients and curvature

$$\Gamma^r_{nk} = \frac{1}{2} g^{rs}(\partial_n g_{sk} + \partial_k g_{sn} - \partial_s g_{nk}), \qquad R^r{}_{kmn} = \partial_m \Gamma^r_{nk} - \partial_n \Gamma^r_{mk} + \Gamma^r_{ms}\Gamma^s_{nk} - \Gamma^r_{ns}\Gamma^s_{mk} \quad \text{(BB.48)}$$

define the Levi–Civita connection and its curvature: substitution shows torsion freedom and metric compatibility, and the cyclic combination of the compatibility equations recovers $\Gamma$. Since the metric is $C^2$ with a positive floor, its inverse is $C^2$, the connection is $C^1$, and the curvature and its Ricci and scalar contractions are continuous. By eq. (BB.44), the metrics $L^{-2} I_{B,\Lambda_L}$ converge to $\mathcal{I}_B$ in $C^2$ on a neighborhood of the subchart, where $\mathcal{I}_B$ is positive definite by eq. (BB.47). These are the hypotheses of proposition BC.9, which gives $C^2$ convergence of the inverses, $C^1$ convergence of the connections, and $C^0$ convergence of the curvatures, at this fixed $B$.

These are also local law charts. At a chart point choose a finite marginal with positive Fisher matrix. Its probability Jacobian has rank two because $I = (Dp)^\top \operatorname{diag}(1/p) Dp$, so the map $P$ from $(h, j)$ to two suitably chosen atom probabilities has a nonsingular Jacobian at the point. By the inverse function theorem [26, Thm. 9.24], $P$ has a $C^1$ local inverse, and the formula $D(P^{-1}) = (DP \circ P^{-1})^{-1}$ shows inductively that the inverse is $C^3$, because $P$ is. This exhibits a chart of the law without assuming a global quotient manifold. □

# Appendix BC

# Proofs for Thermodynamic Fisher Geometry

This appendix proves the statements of chapter B9 under hypothesis B9.1, with the three families of definition B9.2, and states the lattice-side label results that accompany the finite-mixture Fisher asymptotics of the main manuscript. Every volume limit is taken at a fixed word.

## BC.1 Native response and the finite-free boundary transfer

The response construction of appendices BB.2 and BB.7 uses finite all-touching specifications and marginals of the infinite law. The finite free law instead omits the plaquettes that leave the square. Lemma BC.1 shows that the estimates of those sections hold for the free law, and lemmas BC.2 to BC.4 prove what is new, the comparison of finite free and infinite quantities. The free law of a canonical coarse interaction is a fourth family and does not enter.

### BC.1.1 The free-boundary transfer

Throughout, $\mathfrak{D}$ is a finite whole-block inventory inside the fine square of a finite free law, with a prescribed-or-absent pattern of path parities and a fixed exterior completion, and its constrained law is the free law conditioned accordingly. For a target block $i$ and an exterior fine edge $e$, $n(i,e)$ counts the plaquettes meeting both; $\mathrm{cum}$ is the joint cumulant of bounded observables; $\mathsf{T}_{i,s}$ is the edge or plaquette source of the natural derivative direction $i \in \{h, j\}$ at the site $s$; $Q_R$ is the square of cells within Chebyshev distance $R$ of a target; $\nu_{\mathrm{g}}^{\mathfrak{D}}$ is the finite constrained law; and for a target cell and a past $S \subset \mathfrak{D}$ preceding it in a fixed lexicographic order, $p_S^{\mathfrak{D}}(\mathrm{g}, s; y)$ is the four-state row of the push conditional on $Y_S = y_S$, with $l = \log p_S^{\mathfrak{D}}$ and subscripts denoting natural derivatives.

**Lemma BC.1 (Free-boundary transfer).** The following hold with the constants eq. (B9.1), uniformly in the inventory, the pattern, the completion, and $\mathrm{g} \in \widehat{\mathcal{K}}$. Changing one incident plaquette moves the field of one coordinate from $\bar{h} + j$ to $\bar{h} - j$ for a baseline field $\bar{h}$, and the marginal of that coordinate changes in total variation by

$$\frac{\sinh(2j)}{\cosh(2\bar{h}) + \cosh(2j)} \leq \tanh j \leq \tanh j_+. \tag{BC.1}$$

For a target anchor set $A$, weights $w_i = e^{-\tau\,\mathrm{dist}_\infty(\mathrm{anchor}(i),A)}$, and the expected block Hamming discrepancies $D_i$ of a stationary coupling of two constrained laws with exteriors $\eta, \eta'$,

$$S \le \mathrm{q}S + \mathrm{q}\sum_{e\in\partial_f\mathfrak{D}} w_{\mathrm{block}(e)}\delta_e, \qquad S = \sum_{i\ \mathrm{free}} w_i D_i, \quad \delta_e = \mathbf{1}_{\{\eta_e\neq\eta'_e\}}, \tag{BC.2}$$

and for a vector $\mathbf{u}$ supported on $A$ and a vector $\mathbf{w}$ outside its radius-$R$ neighborhood,

$$\mathrm{TV}(\mathcal{L}(\mathbf{u},\mathbf{w}), \mathcal{L}(\mathbf{u})\otimes\mathcal{L}(\mathbf{w})) \le \min\{1, \mathrm{c}_1|A|B^2(R+1)e^{-\tau(R+1)}\}. \tag{BC.3}$$

The cumulants satisfy

$$\begin{aligned} \mathrm{cum}(F_0,\dots,F_r) &= \sum_{\pi}(-1)^{|\pi|-1}(|\pi|-1)!\prod_{D\in\pi}\mathbb{E}\prod_{i\in D}F_i, \\ (k_2,\bar{k}_2) = (2,6), \qquad (k_3,\bar{k}_3) &= (6,26), \qquad (k_4,\bar{k}_4) = (26,150), \end{aligned} \tag{BC.4}$$

where $k_m = \sum_\pi(|\pi|-1)!$ bounds an individual cumulant and $\bar{k}_m = 2\sum_\pi|\pi|!$ multiplies the total variation of a change of the joint law. For a bounded parameter-independent observable $f$ with anchor support $A$, $a_0 = |A| \ge 1$, and $1 \le r \le 3$, put

$$\begin{aligned} L_r(a_0) &= e^{4\tau}[k_{r+1} + \bar{k}_{r+1}\mathrm{c}_1(a_0+4r)], \\ S_r(a_0) &= \frac{(8a_0)^r L_r(a_0)(2r+1)!}{(1-e^{-\tau/r})^{2r+2}}, \qquad U_r(B,a_0) = S_r(a_0)B^{2r+2}, \quad U_0(B,a_0) = 1; \end{aligned} \tag{BC.5}$$

then, for ordered sources of the prescribed directions,

$$\begin{aligned} \sum_{s_1,\dots,s_r} |\,\mathrm{cum}(f,\mathsf{T}_{i_1,s_1},\dots,\mathsf{T}_{i_r,s_r})| &\le U_r(B,a_0)\|f\|_\infty, \\ \sum_{\max\mathrm{dist}(x_s,A)>n} |\,\mathrm{cum}(f,\mathsf{T}_{i_1,s_1},\dots,\mathsf{T}_{i_r,s_r})| &\le U_r(B,a_0)(n+2)^{2r+1}e^{-\tau(n+1)/r}\|f\|_\infty, \end{aligned} \tag{BC.6}$$

and a local $C^2$ observable $f_{\mathsf{g}}$ has, for $|\alpha| \le 2$,

$$\partial^\alpha\mathbb{E}_{\mathsf{g}}f_{\mathsf{g}} = \sum_{\beta\le\alpha}\binom{\alpha}{\beta}\sum_{s_1,\dots,s_{|\alpha-\beta|}}\mathrm{cum}_{\mathsf{g}}(\partial^\beta f_{\mathsf{g}}, \mathsf{T}_{i_1,s_1},\dots,\mathsf{T}_{i_{|\alpha-\beta|},s_{|\alpha-\beta|}}), \tag{BC.7}$$

the source directions being those of $\alpha-\beta$ and an empty source list meaning expectation. The rows are positive and $C^3$, with

$$p^{\mathfrak{D}}_S(\mathsf{g},s;y) \ge \frac{(1-\rho_*^B)^2}{4} \ge \frac{d_*^2}{4}; \tag{BC.8}$$

every order-$n$ derivative of a target parity mean is bounded by $M_n(B)$, where

$$\begin{aligned} D_n &= g_n + \sum_{k=0}^{n-1}\binom{n}{k}g_k S_{n-k}(9), \qquad g_n = 2^n n!, \\ M_0(B) = \rho_*^B, \qquad M_n(B) &= D_n B^{2n+2}\rho_*^{B-n}, \qquad 1 \le n \le 3; \end{aligned} \tag{BC.9}$$

the logarithmic derivatives of one Bernoulli factor $(1+sm)/2$ have amplitudes

$$A_1=\frac{M_1}{d_*},\qquad A_2=\frac{M_2}{d_*}+\frac{M_1^2}{d_*^2},\qquad A_3=\frac{M_3}{d_*}+\frac{3M_2M_1}{d_*^2}+\frac{2M_1^3}{d_*^3}, \tag{BC.10}$$

so that the order-$n$ derivatives of $l$ are bounded by $\widehat{A}_n=2A_n$; the row-Fisher observable $\mathsf{I}^{\mathfrak{D}}_{S,ik}=\sum_s p^{\mathfrak{D}}_S l_i l_k$ has, at fixed $y$,

$$\begin{aligned}\partial_m \mathsf{I}^{\mathfrak{D}}_{S,ik}&=\sum_s p^{\mathfrak{D}}_S(l_m l_i l_k+l_{im}l_k+l_i l_{km}),\\ \partial_n\partial_m \mathsf{I}^{\mathfrak{D}}_{S,ik}&=\sum_s p^{\mathfrak{D}}_S\big[(l_{mn}+l_m l_n)l_i l_k+l_m(l_{in}l_k+l_i l_{kn})+l_n(l_{im}l_k+l_i l_{km})\\ &\qquad +l_{imn}l_k+l_{im}l_{kn}+l_{in}l_{km}+l_i l_{kmn}\big],\end{aligned} \tag{BC.11}$$

with entries and first two derivatives bounded by

$$E^F_B=\max\{\widehat{A}_1^2,\widehat{A}_1^3+2\widehat{A}_1\widehat{A}_2,\widehat{A}_1^4+5\widehat{A}_1^2\widehat{A}_2+2\widehat{A}_1\widehat{A}_3+2\widehat{A}_2^2\},\qquad H^F_B=\max(1,\widehat{A}_1,\widehat{A}_2,\widehat{A}_3); \tag{BC.12}$$

if two target parity means have jet differences bounded by $v=(v_0,v_1,v_2,v_3)$, the one-factor log-jet differences are bounded by

$$\begin{aligned}\mathcal{T}_1(v)&=v_1/d_*+M_1v_0/d_*^2,\\ \mathcal{T}_2(v)&=v_2/d_*+(M_2v_0+2M_1v_1)/d_*^2+2M_1^2v_0/d_*^3,\\ \mathcal{T}_3(v)&=v_3/d_*+(M_3v_0+3M_2v_1+3M_1v_2)/d_*^2\\ &\qquad+(6M_2M_1v_0+6M_1^2v_1)/d_*^3+6M_1^3v_0/d_*^4,\end{aligned} \tag{BC.13}$$

and every entrywise row-Fisher derivative difference through degree two by

$$\mathcal{W}_B(v)=\min\{2E^F_B,200(H^F_B)^4[v_0+2\mathcal{T}_1(v)+2\mathcal{T}_2(v)+2\mathcal{T}_3(v)]\}; \tag{BC.14}$$

for two past patterns in the same $\mathfrak{D}$ agreeing through $Q_R$, with $a_R=\lfloor R/2\rfloor-1$ and $\mathrm{c}'_n=2^n\bar{k}_{n+1}\mathrm{c}_1(1+4n)$, the parity-jet differences obey

$$\begin{aligned}V_0(B,R)&=\min\{2\rho_*^B,2\mathrm{c}_1B^2(R+1)e^{-\tau(R+1)}\},\\ V_n(B,R)&=\min\big\{2M_n(B),B^{2n+2}\big[\mathrm{c}'_n(2a_R+1)^{2n}(R-a_R)e^{-\tau(R-a_R)}\\ &\qquad+2S_n(1)(a_R+2)^{2n+1}e^{-\tau(a_R+1)/n}\big]\big\},\qquad 1\le n\le 3,\end{aligned} \tag{BC.15}$$

with $V_n=2M_n$ for $R=0,1$, and consequently

$$\|\mathsf{I}^{\mathfrak{D}}_S-\mathsf{I}^{\mathfrak{D}}_{S'}\|_{C^2(\mathcal{K}),\infty,\mathrm{entry}}\le\omega_B(R):=\mathcal{W}_B(V_0,V_1,V_2,V_3), \tag{BC.16}$$

which at fixed $B$ is bounded by a constant times $(R+1)^7e^{-\tau R/6}$. Finally, the moving-expectation bound eq. (BB.42) and the shell bounds eq. (BB.43) hold for the finite free law and its finite-past rows, including targets at the free boundary, with the modulus eq. (BC.16). [ESTABLISHED]

*Proof.* Each assertion is the corresponding estimate of appendix BB.2 and sections BB.7.1 to BB.7.3 for all-touching laws, and we check that each step of those proofs uses the boundary of the finite law only through two facts. First, every sum over incident plaquettes or over sources is a sum of nonnegative terms over a subset of the all-touching list, so omitting the outer plaquettes only removes terms: $\sum_i n(i,e) \le 6$ replaces the column identity, the path fields keep $|a_e| \le 2$, and the source lists shrink. Second, when two conditionals of the same finite law are compared, any boundary inherited from the original inventory, including its omitted interactions, is identical on both sides and contributes no discrepancy.

A plaquette meets four distinct blocks and at most one edge of any retained path, so conditional on the other fine blocks a path row is a positive product sign law, possibly restricted to a parity fiber; the coupling of opposite parities at Hamming distance one in appendix BB.2 is a property of product laws. The tilt of a sign with baseline field $\bar{h}$ has total variation $(\tanh(\bar{h}+j) - \tanh(\bar{h}-j))/2$, which is eq. (BC.1) and the bound of eq. (BB.2), so the row costs and their telescoping over changed plaquettes are as there. The column identity enters eqs. (BB.5) and (BB.6) only as an upper bound, where $\sum_i n(i,e) \le 6$ suffices: each fine edge lies in at most two plaquettes, each meeting three other blocks. Positive finite heat baths are irreducible on the product support, and a stationary pair coupling exists by Cesàro averaging on the finite simplex. Incident anchors differ by at most one, which gives eq. (BC.2); a shell at anchor distance $R+1$ has at most $8|A|(R+1)$ cells and $2B^2$ fine edges per cell, which gives the joint-target bound. For group decoupling, condition on the free blocks outside the radius-$R$ neighborhood of $A$; by the second fact only the newly conditioned shell contributes, and integrating over the exterior gives eq. (BC.3), which passes to infinite constrained laws through finite-cylinder limits.

The cumulant formula and its constants are finite identities for bounded observables, obtained by expanding the finite logarithm of a moment-generating function, and a cumulant of two independent nonempty groups vanishes. The source-tail bounds follow, as in eq. (BB.30), from the minimum spanning tree of the labeled anchors and a root for $A$, from eq. (BC.3), and from counting at most $(8a_0)^r B^{2r}(M+1)^{2r}$ ordered tuples within anchor distance $M$; by the first fact the free source list is a subset of the counted one, and exterior-completed constant sources have zero higher cumulant. The moving response formula is finite differentiation of a normalized finite exponential family and holds for any finite source list.

Finite conditioning on past parities restricts the observed path fibers and keeps every plaquette of the free law. A free target path conditioned on the other fine blocks has independent signs with fields $h + ja_e$, $|a_e| \le 2$ by the first fact, so its parity mean has magnitude at most $\rho_*^B$ and support in at most nine anchors, and revealing the two target parities gives eq. (BC.8). The parity jets follow from eqs. (BC.6) and (BC.7) as in eq. (BB.33). The log amplitudes, the row-Fisher jets and their bounds, the log-jet difference map, and the row modulus map are algebraic identities and bounds for positive four-state rows with this floor, as in eqs. (BB.35), (BB.36) and (BB.39) to (BB.41). For the past modulus, split source tuples at $a_R$: near tuples occupy at most $1+4n$ anchors in $Q_{a_R+1}$, at most $[2B^2(2a_R+1)^2]^n$ of them, and group comparison on the remaining buffer gives the first term, the omitted interactions producing no near discrepancy by the second fact; each far sum obeys

eq. (BC.6). This is the argument of eq. (BB.34), including irregular intersections with the original inventory. The shell bounds of section BB.7.3 use only the moving response formula, the source tails, and the past modulus, and the shell series of a finite past terminates. No step uses any other property of the free boundary. □

**Lemma BC.2 (The finite-free local modulus).** Let a bounded parameter-independent observable $f$ have anchor support $A$ with $a_0 = |A| \geq 1$. Its finite-free expectations are $C^3$ on $\widehat{\mathcal{K}}$. If the finite inventory contains $A_R = \{x : \mathrm{dist}_\infty(x, A) \leq R\}$ and the finite and infinite constrained patterns agree there, then, with $\mu_{\mathrm{g}}$ the matching constrained infinite law,

$$\max_{|\alpha|=r} \sup_{\mathrm{g}\in\widehat{\mathcal{K}}} \left| \partial^\alpha \mathbb{E}_{\nu_{\mathrm{g}}^{\mathfrak{D}}} f - \partial^\alpha \mathbb{E}_{\mu_{\mathrm{g}}} f \right| \leq \Delta_r(B, a_0, R)\|f\|_\infty, \qquad 0 \leq r \leq 3, \tag{BC.17}$$

where, for $R \geq 4$,

$$\Delta_0(B, a_0, R) = \min\{2, 2\mathrm{c}_1 a_0 B^2 (R+1) e^{-\tau(R-1)}\}, \tag{BC.18}$$

and, with $n_R = \lfloor R/2 \rfloor - 2$,

$$\begin{aligned} \widetilde{\Delta}_r(B, a_0, R) &= \bar{k}_{r+1}[2B^2 a_0 (2n_R+1)^2]^r \mathrm{c}_1 (a_0 + 4r) B^2 (R - n_R + 1) e^{-\tau(R-n_R-2)} \\ &\quad + 2U_r(B, a_0)(n_R+2)^{2r+1} e^{-\tau(n_R+1)/r}, \\ \Delta_r(B, a_0, R) &= \min\{2U_r(B, a_0), \widetilde{\Delta}_r(B, a_0, R)\}, \qquad 1 \leq r \leq 3, \end{aligned} \tag{BC.19}$$

while $\Delta_r = 2U_r$ for $R < 4$, including $r = 0$. Each modulus tends to zero as $R \to \infty$ at fixed $B$ and $a_0$. [ESTABLISHED]

*Proof.* For values, condition both laws on a whole-block neighborhood one interaction layer inside $A_R$. Every plaquette touching that smaller region is present in both conditionals because the inventory contains $A_R$; this is the step that uses more than lemma BC.1. The exterior assignments may have different parameter-dependent laws, but at each $\mathrm{g}$ the comparison eq. (BC.3) is uniform over both, and integrating it proves eq. (BC.18). For derivatives, differentiate each finite normalized law with its own source list, which gives the ordered cumulant sums. A near source tuple and $f$ occupy at most $a_0 + 4r$ anchors inside $A_{n_R+1}$, whose neighborhood of radius $R - n_R - 2$ has a full extra interaction layer inside $A_R$, so their joint total variation is at most $\mathrm{c}_1(a_0 + 4r)B^2(R - n_R + 1)e^{-\tau(R-n_R-2)}$; there are at most $[2B^2 a_0(2n_R+1)^2]^r$ near tuples, and cumulant telescoping bounds their differences. The two far sums are bounded separately by eq. (BC.6). An omitted outer source cannot occur in the common near inventory, so it is covered by the far comparison. The infinite derivatives are those of section BB.7.1: exhausting the all-touching specifications, near tuples have uniformly convergent finite moment laws and far sums are uniformly small, so the finite derivatives through degree three are uniformly Cauchy and coordinate integration identifies continuous derivatives with agreeing mixed derivatives, also one-sidedly at $j = 0$. □

**Lemma BC.3 (Local moving observables).** For a local $C^2$ observable $f_{\mathrm{g}}$ with anchor support $A$ whose pointwise $C^2$ jets are bounded by $E_f$, the finite and infinite expectations of lemma BC.2

satisfy

$$\|\mathbb{E}_{\nu_{\mathfrak{D}}} f - \mathbb{E}_{\mu} f\|_{C^2(\mathcal{K})} \le E_f[\Delta_0(B, a_0, R) + 2\Delta_1(B, a_0, R) + \Delta_2(B, a_0, R)]. \tag{BC.20}$$

[ESTABLISHED]

*Proof.* By eq. (BC.7), each derivative of order at most two is a sum over $\beta \le \alpha$ of derivatives of order $|\alpha - \beta|$ of expectations of the fixed observables $\partial^\beta f$. Apply eq. (BC.17) to each; at degree two the total coefficients are $1, 2, 1$. □

**Lemma BC.4 (Finite and infinite predictive rows).** If a target has $Q_R \subset \mathfrak{D}$ after translation to the origin, its finite row-Fisher observable and the infinite row-Fisher observable $\mathsf{I}_{S'}$ with the same local past satisfy

$$\|\mathsf{I}_S^{\mathfrak{D}} - \mathsf{I}_{S'}\|_{C^2(\mathcal{K}),\infty,\mathrm{entry}} \le \Omega_B(R) := \mathcal{W}_B\big(\Delta_0(B,1,R), \Delta_1(B,1,R), \Delta_2(B,1,R), \Delta_3(B,1,R)\big) \longrightarrow 0, \tag{BC.21}$$

pointwise in the common past coordinates. [ESTABLISHED]

*Proof.* Apply eq. (BC.17) with $a_0 = 1$ separately to each target parity; when comparing the second factor the first parity is prescribed identically on both sides. Insert the resulting jet differences in eq. (BC.14). The bound controls row derivatives at fixed past coordinates and does not yet differentiate an expectation under a moving law. □

### BC.1.2 Fourth pressure derivatives and the native information tensor

*Proof of Theorem B9.3.* For a sign $z$, $1 + \tanh(h)z = e^{hz}/\cosh h$; applying this to the $2N$ edges and the $(L-1)^2$ plaquettes, with normalized counting in both partitions, gives the finite line of eq. (B9.6). The value limit of theorem B8.10 applies to the raw native interaction, whose first diameter moment is at most $3t_+/(1-t_+)$ by eq. (B8.12), so $\|\mathfrak{p}_L - \mathfrak{p}\|_{C^0(\widehat{\mathcal{K}})} \le 12t_+/[(1-t_+)L]$, and the conversion gives the value limit of $\mathsf{A}_L$. This supplies no derivative.

Put $f_h = z_{0,1} + z_{0,2}$ and $f_j = P_0$, with anchor supports of sizes one and three, and $m_i(\mathsf{g}) = \mathbb{E}_{\mu_{\mathsf{g}}} f_i$, which are $C^3$ by lemma BC.2. Finite differentiation gives

$$\partial_h \mathsf{A}_L = \frac{1}{N} \sum_{x \in \Lambda_L} \mathbb{E}_{\nu^{\mathrm{f}}_{L,\mathsf{g}}} f_h \circ \vartheta_x, \qquad \partial_j \mathsf{A}_L = \frac{1}{N} \sum_{x \in \{0,\dots,L-2\}^2} \mathbb{E}_{\nu^{\mathrm{f}}_{L,\mathsf{g}}} f_j \circ \vartheta_x, \tag{BC.22}$$

where $\vartheta_x$ translates the observable to $x$. For $0 \le r \le 3$ define

$$\mathcal{M}_r = \max\{2U_r(1,1), U_r(1,3)\}, \qquad \bar{\Delta}_r(R) = \max\{2\Delta_r(1,1,R), \Delta_r(1,3,R)\}. \tag{BC.23}$$

For $L \ge 2(R+2)$ at most $4(R+2)L$ cells lack a complete target support with its radius-$R$ neighborhood. The bulk differences of local derivatives are bounded by $\bar{\Delta}_r(R)$ and every boundary

derivative by $\mathcal{M}_r$, an omitted plaquette being an identically zero term, so

$$\max_{i\in\{h,j\},|\alpha|=r} \|\partial^\alpha \partial_i \mathsf{A}_L - \partial^\alpha m_i\|_{C^0(\mathcal{K})} \le \bar{\Delta}_r(R) + 8\mathcal{M}_r \frac{R+2}{L}, \qquad 0 \le r \le 3. \tag{BC.24}$$

Choosing first $R$ for the local modulus and then $L$ for the boundary fraction proves $C^3$ convergence of the gradients. On any coordinate segment of the enlarged rectangle,

$$\mathsf{A}_L(\mathsf{g} + ue_i) - \mathsf{A}_L(\mathsf{g}) = \int_0^u \partial_i \mathsf{A}_L(\mathsf{g} + ve_i)dv, \tag{BC.25}$$

and uniform value and gradient convergence pass through the integral, giving $\partial_i \mathsf{A} = m_i$. The same argument for the first, second, and third gradient derivatives proves $C^4$ regularity, convergence of every derivative through order four, and equality of mixed derivatives. Since $|(L-1)^2/N - 1| \le 2/L$ is parameter independent, the conversion transfers the result to the raw pressure. The fourth derivative uses only three local-response derivatives, because the first derivative is already a local source density, and no surface log-ratio estimate is differentiated. □

*Proof of Proposition B9.4.* The natural log score is $(\mathsf{T}_{h,L}, \mathsf{T}_{j,L}) - \mathbb{E}(\mathsf{T}_{h,L}, \mathsf{T}_{j,L})$, whose covariance is $N\nabla^2_{\mathsf{g}}\mathsf{A}_L$, as for every exponential family in its natural coordinates [1, Ch. 3]. Theorem B9.3 and its conversion give eq. (B9.7). The chain rule for the normalized log probability multiplies the source score by $D\mathsf{g}^\top$, which proves the finite pullback, and $C^2$ convergence follows from the $C^2$ limit of the source Hessian by the product and composition rules. In contrast,

$$\nabla^2_\theta(\mathsf{A}_L \circ \mathsf{g}) = D\mathsf{g}^\top(\nabla^2_{\mathsf{g}}\mathsf{A}_L)D\mathsf{g} + \sum_{i=1}^{2}(\partial_i \mathsf{A}_L)\nabla^2_\theta \mathsf{g}_i, \tag{BC.26}$$

and the last term is absent from the score covariance. □

**Proposition BC.5 (A native finite-energy information floor).** On an admitted source box put $H_* = h_+ + 2j_+$, $m_* = (1+e^{2H_*})^{-1}$, and $v_* = \operatorname{sech}^2 H_*$. For $L \ge 12$,

$$\frac{1}{N} I(\nu^{\mathrm{f}}_{L,\mathsf{g}}) \succeq \frac{v_* m_*^6}{8}\,\mathrm{Id}_2, \qquad \mathsf{H}(\mathsf{g}) \succeq \frac{v_* m_*^6}{8}\,\mathrm{Id}_2\,. \tag{BC.27}$$

Consequently the pullback in eq. (B9.8) has kernel $\ker D\mathsf{g}$ and rank $\operatorname{rank} D\mathsf{g} \le 2$. [ESTABLISHED]

*Proof.* Every edge conditioned on all other signs has field magnitude at most $H_*$, so either sign has conditional probability at least $m_*$ and the edge has conditional variance at least $v_*$. Choose horizontal edges based at $(1+3u, 1+3v)$ with $u, v \ge 0$ and both coordinates at most $L-2$; there are $\lfloor L/3 \rfloor^2 \ge N/16$ of them for $L \ge 12$, and no two share a plaquette, so conditional on the unselected signs they are independent. For $F = \psi_h \mathsf{T}_{h,L} + \psi_j \mathsf{T}_{j,L}$ a selected edge $e$ has coefficient $\psi_h + \psi_j R_e$, where $R_e$ is the sum of its two neighboring three-sign products, and

$$\operatorname{Var}(F) \ge \mathbb{E}\operatorname{Var}(F \mid \text{unselected signs}) \ge v_* \sum_{e \text{ selected}} \mathbb{E}(\psi_h + \psi_j R_e)^2. \tag{BC.28}$$

Setting the six neighboring signs of a selected edge positive forces $R_e = 2$; setting the horizontal edges immediately above and below negative and the four vertical neighbors positive forces $R_e = -2$. Each assignment has probability at least $m_*^6$, also under partial conditioning, because averaging the one-edge lower bound over unspecified signs preserves $m_*$. Therefore

$$\frac{\mathrm{Var}(F)}{N} \geq \frac{v_* m_*^6}{16}[(\psi_h + 2\psi_j)^2 + (\psi_h - 2\psi_j)^2] = \frac{v_* m_*^6}{8}(\psi_h^2 + 4\psi_j^2) \geq \frac{v_* m_*^6}{8}|\psi|^2, \tag{BC.29}$$

which is the finite bound, and the source-Hessian limit inherits it. A positive definite middle matrix in $D\mathsf{g}^\top \mathsf{H} D\mathsf{g}$ gives kernel $\ker D\mathsf{g}$. □

Only positivity of this floor is used below.

### BC.1.3 Exact finite-free pushes and the three-cutoff limit

*Proof of Theorem B9.6.* Order the cells of $\Lambda_L$ lexicographically and use the finite predictive rows of $P^{\mathrm{f}}_{B,L,\mathsf{g}}$. Their log scores add to the full score; each row score has conditional mean zero given its finite past, since differentiating its normalization gives zero, and an earlier row score is measurable for any later past, so all cross expectations vanish. With $S_{x,L} = \Lambda_L \cap \{y : y < x\}$,

$$I(P^{\mathrm{f}}_{B,L,\mathsf{g}}) = \sum_{x \in \Lambda_L} \mathbb{E}_{\nu^{\mathrm{f}}_{BL,\mathsf{g}}} \mathsf{I}^{\mathfrak{D}_L}_{S_{x,L}}(\mathsf{g}, Y), \tag{BC.30}$$

an identity of finite $C^2$ functions, so its first two derivatives are also exact.

Fix an inner past cutoff $n \geq 0$ and an outer radius $R \geq n + 4$, and consider a bulk target $x$ with $x + Q_R \subset \Lambda_L$, translated to the origin. Its finite past and the complete past have the same intersection with $Q_n$. Truncating their expected row observables to that common prefix costs at most $\epsilon_B(n)$ in entrywise $C^2$, where $W_B$, $Z_B^F$, and $\epsilon_B$ are the weight, bound, and tail of eqs. (BB.42) and (BB.43), valid for the finite free law by lemma BC.1. The finite and infinite truncated rows are functions of the same past coordinates, with pointwise $C^2$ difference at most $\Omega_B(R)$ by lemma BC.4, with $\mathfrak{D}_L$ the inventory of the fine square, so their difference under the moving finite law is at most $W_B(n)\Omega_B(R)$. It remains to compare expectations of the same infinite truncated row under $\nu^{\mathrm{f}}_{BL,\mathsf{g}}$ and $\mu_{\mathsf{g}}$; its jets are bounded by $E_B^F$, its support is $Q_n$, and this support has a radius-$(R-n)$ neighborhood in $\mathfrak{D}_L$, so lemma BC.3 bounds the remaining cost with $a_n = (2n+1)^2$. Together,

$$\begin{aligned}\mathcal{B}_B(n,R) = \epsilon_B(n) + W_B(n)\Omega_B(R) \\ + E_B^F[\Delta_0(B, a_n, R-n) + 2\Delta_1(B, a_n, R-n) + \Delta_2(B, a_n, R-n)].\end{aligned} \tag{BC.31}$$

At most $4RL$ cells lie outside the bulk for $L \geq 2R$, every finite and infinite expected-row jet is bounded by $Z_B^F$, also near the free outer boundary, and stationarity identifies the infinite target expectation at each bulk translation. Dividing eq. (BC.30) by $N$ gives

$$\left\| N^{-1} I(P^{\mathrm{f}}_{B,L,\mathsf{g}}) - \mathcal{I}_B(\mathsf{g}) \right\|_{C^2(\mathcal{K}),\mathrm{entry}} \leq \mathcal{B}_B(n,R) + 8 Z_B^F \frac{R}{L}. \tag{BC.32}$$

Given a tolerance, choose $n$ for the shell tail, then $R$ for the two local comparisons at that $n$, and finally $L$ for the boundary fraction; this order respects the growth of the constants in $a_n$, and $B$ is fixed throughout. For infinite-law marginals the same row identity with $\mu_{\mathsf{g}}$ as outer expectation gives $\epsilon_B(R) + 8Z_B^F R/L$, which is the proof of eq. (BB.44), so both limits equal $\mathcal{I}_B$. At $B = 1$ the channel is the identity, and eqs. (B9.7) and (B9.10) are limits of the same finite matrices, which proves eq. (B9.11). □

**Remark BC.6 (Area bound and native density inputs).** For every fixed $B$, $\mathcal{I}_B \preceq B^2\mathsf{H}$. If each $\mathsf{g}_\gamma$ is $C^4$ on a neighborhood of a compact $\mathcal{K}$ with image in a common admitted source box, the fine and coarse branch densities of a push along a word of total factor $B$ satisfy, in $C^2(\mathcal{K})$,

$$\frac{I(\nu^{\mathrm{f}}_{BL,\mathsf{g}_\gamma})}{B^2N} \longrightarrow D\mathsf{g}_\gamma^\top(\mathsf{H}\circ\mathsf{g}_\gamma)D\mathsf{g}_\gamma, \qquad \frac{I\big((C_B)_\#\nu^{\mathrm{f}}_{BL,\mathsf{g}_\gamma}\big)}{N} \longrightarrow D\mathsf{g}_\gamma^\top(\mathcal{I}_B\circ\mathsf{g}_\gamma)D\mathsf{g}_\gamma,$$

and the per-coarse-cell Fisher loss of the push converges to $D\mathsf{g}_\gamma^\top[(B^2\mathsf{H} - \mathcal{I}_B)\circ\mathsf{g}_\gamma]D\mathsf{g}_\gamma \succeq 0$. The coarse limit is a pulled-back push density, not the Hessian of a coarse pressure. [ESTABLISHED]

*Proof.* Proposition B7.16 with blocking factor $B$ applied to $\mathcal{I}_1$, and eq. (B9.11), give the area bound. The fine limit is proposition B9.4 on the square of side $BL$, the coarse limit is eq. (B9.16), and the congruence by $D\mathsf{g}_\gamma$ preserves the order of the area bound. □

### BC.1.4 Why scalar boundary control is insufficient

**Proposition BC.7 (A bounded boundary ratio with divergent information density).** Uniform control of log-density ratios and agreement of bulk cylinder limits do not imply convergence of volume-normalized Fisher tensors, even for positive smooth finite families on one carrier at each volume. At the native product control $j = 0$, the raw pressure has $\partial_h^2\mathfrak{p}(h,0) = 0$ while $\mathsf{H}_{hh}(h,0) = 2\,\mathrm{sech}^2 h > 0$, and the functions $L^{-1}\sin(L^2\theta)$ converge uniformly to zero with derivative $L$ at zero; hence neither a raw-pressure Hessian nor a differentiated value error supplies the Fisher density. [ESTABLISHED]

*Proof.* On the square carrier with $2N$ signs let $P_L$ be the fair product law, independent of a real parameter $\theta$, and let $Q_{L,\theta}$ change only one sign on the boundary of the centered square to the mean $a\sin(N\theta)$, with fixed $0 < a < 1/2$. All probabilities are positive and smooth on one open interval, and for every configuration

$$\left|\log\frac{Q_{L,\theta}}{P_L}\right| \le \max\{-\log(1-a), \log(1+a)\}. \tag{BC.33}$$

The varying sign leaves every fixed centered cylinder for large $L$, so both sequences have the same bulk cylinder limit. A sign of mean $m$ has Fisher information $(m')^2/(1-m^2)$, and independence of the other signs gives

$$I(Q_{L,\theta}) = \frac{a^2N^2\cos^2(N\theta)}{1 - a^2\sin^2(N\theta)}, \qquad I(Q_{L,0}) = a^2N^2, \qquad N^{-1}I(Q_{L,0}) = a^2N \longrightarrow \infty, \tag{BC.34}$$

whereas $I(P_L) = 0$. The witness violates the native local jet bounds, so it refutes the scalar boundary inference and not the native theorems. At $j = 0$ the raw partition is one for every $h$, and the variance of $\mathsf{T}_{h,L}$ is $2N \operatorname{sech}^2 h$. □

## BC.2 Proofs for retained labels, ties, and reciprocal-volume experiments

The lattice-free statements on finite labeled families, the labeled Fisher decomposition, the strict-phase, tie, and reciprocal-volume limits, and their counterexamples, are in section 10.8, with the families and norms of definition 10.23; this section supplies the native inputs, regular charts, residual scales, and comparison results that accompany them.

### BC.2.1 Native inputs and effective representatives

*Proof of Proposition B9.7.* The identity $1 + (\tanh h)z = e^{hz}/\cosh h$ on every edge and plaquette gives eq. (B9.12), and the normalized branch law is the exponential family with statistic $(\mathsf{T}_{h,L}, \mathsf{T}_{j,L})$, whose scalar factors cancel in its own normalization. Multiplying the centered natural score by $D\mathsf{g}_\gamma^\top$ gives eq. (B9.13) and, at each finite volume, the push identity eq. (B9.15). The chain and product rules through degree two use derivatives of $\mathsf{A}_L$ through degree four and of $\mathsf{g}_\gamma$ through degree three, and theorems B9.3 and B9.6 and proposition B9.4 give the limits, with $(L-1)^2/N \to 1$ for the raw pressure. No derivative of a $C^0$ pressure error is taken, and the word and $B$ are fixed before the volume limit. □

**Proposition BC.8 (Effective growth in both normalization modes).** Suppose the positive raw label weights have $C^3$ logarithms with

$$\|\log a_{\gamma,L}\|_{C^3(\mathcal{K})} \le C_{\mathcal{K}} N, \qquad N^{-1} \log a_{\gamma,L} \longrightarrow \alpha_\gamma \quad \text{in } C^3(\mathcal{K}), \tag{BC.35}$$

and the branches are those of proposition B9.7. Then

$$\lambda^{\mathrm{f}}_{\gamma,L} = \log a_{\gamma,L} + \log Z_{\gamma,L}, \qquad \mathsf{s}^{\mathrm{f}}_\gamma = \alpha_\gamma + \mathfrak{p}_\gamma, \tag{BC.36}$$

$$\lambda^{\mathrm{h}}_{\gamma,L} = \log a_{\gamma,L}, \qquad \mathsf{s}^{\mathrm{h}}_\gamma = \alpha_\gamma, \tag{BC.37}$$

are effective representatives of the latent experiment in the two modes, and each satisfies eq. (10.12) with bulk limit $\mathsf{s}_\gamma$; a fixed positive $C^3$ hierarchy $w_\gamma$ uses $\lambda_{\gamma,L} = \log w_\gamma$ and bulk limit zero. For the all-edge sensor of proposition B8.16 with mean parameter $\eta_\gamma(\theta)$, let

$$A^{\mathrm{sens}}_\gamma = \frac{1 + b_\gamma \eta_\gamma}{2}, \qquad b'_\gamma = \frac{b_\gamma + \eta_\gamma}{1 + b_\gamma \eta_\gamma}, \tag{BC.38}$$

assume these data are $C^4$ on a neighborhood of $\mathcal{K}$, and assume both the original sources and the shifted sources $(\operatorname{atanh} b'_\gamma, j_\gamma)$ lie in common compact admitted boxes. Then, up to a common log

mass, the posterior representatives

$$\lambda^{o,\mathrm{f}}_{\gamma,L} = \log a_{\gamma,L} + 2N \log A^{\mathrm{sens}}_{\gamma} + \log Z_{\gamma,L}(b'_{\gamma}, t_{\gamma}), \tag{BC.39}$$

$$\lambda^{o,\mathrm{h}}_{\gamma,L} = \log a_{\gamma,L} + 2N \log A^{\mathrm{sens}}_{\gamma} + \log Z_{\gamma,L}(b'_{\gamma}, t_{\gamma}) - \log Z_{\gamma,L}(b_{\gamma}, t_{\gamma}), \tag{BC.40}$$

satisfy eq. (10.12) with bulk limits $\alpha_\gamma + 2\log A^{\mathrm{sens}}_{\gamma} + \mathfrak{p}'_{\gamma}$ and $\alpha_\gamma + 2\log A^{\mathrm{sens}}_{\gamma} + \mathfrak{p}'_{\gamma} - \mathfrak{p}_{\gamma}$, where $\mathfrak{p}'_{\gamma}$ is the pressure at the shifted source, and the selected conditional branch is the shifted native free law. For a push along a fixed word, whose label weights are those of the fine volume $B^2N$, $\lambda_{\gamma,BL}/N \to B^2\mathsf{s}_{\gamma}$ in $C^3$ with the same strict winner. [ESTABLISHED]

*Proof.* The finite formulas eq. (10.11) give the two representatives, and proposition B9.7 gives the $C^3$ convergence and the bound $O(N)$ of $\log Z_{\gamma,L}$ after composition with the $C^4$ source maps. Adding a common $c_L(\theta)$ to every representative leaves the weights unchanged and adds $\nabla c_L$ to every score, so their covariance is unchanged with all its derivatives; common normalizers must therefore be removed before the growth condition is imposed, and the bound cannot in general be imposed on $\log W_\gamma$ at a tie: for $a_0 = 1$ and $a_1 = e^{N\theta}$ the second derivative of $\log(a_0 + a_1)$ at zero is $N^2/4$. For the sensor, eq. (B8.19) gives $Z^o_\gamma = (A^{\mathrm{sens}}_{\gamma})^{2N} Z_{\gamma,L}(b'_{\gamma}, t_{\gamma})$ by multiplication over exactly $2N$ signs, the hierarchical representative retains its original partition divisor, and theorem B9.3 at both sources gives the bulk limits with all sensor derivatives. Strict positivity makes the finite selected event well defined; the argument uses the exact source-shift identity and applies to no other likelihood. For a push, the label weights at coarse volume $N$ are those at fine volume $B^2N$, so $\lambda_{\gamma,BL}/N = B^2\lambda_{\gamma,BL}/(B^2N) \to B^2\mathsf{s}_\gamma$, and the winner and gaps scale by $B^2$. □

### BC.2.2 Regular charts

**Proposition BC.9 (Inverse, connection, and curvature on regular charts).** Let symmetric tensor fields $G_L \to G$ in $C^2$ on a neighborhood of a compact parameter chart, with $G$ positive definite there. Then $G_L^{-1} \to G^{-1}$ in $C^2$, the Levi-Civita coefficients converge in $C^1$, and the Riemann, Ricci, and scalar curvatures converge in $C^0$ on the chart. For a $C^1$ family of finite laws on an open $d$-dimensional parameter domain, with fixed positive support and positive definite Fisher tensor at a point, $d$ of its atom probabilities form a local chart there, and distinct nearby parameters give distinct laws. [ESTABLISHED]

*Proof.* Minimizing $z^\top G(\theta) z$ over the chart times the unit sphere gives a positive minimum $\epsilon$, and uniform convergence gives $G_L \succeq (\epsilon/2)\,\mathrm{Id}$ for large $L$; then $G_L^{-1} - G^{-1} = G_L^{-1}(G - G_L)G^{-1}$ proves uniform inverse convergence. Differentiating $G_L H_L = \mathrm{Id}$ with $H_L = G_L^{-1}$ gives

$$\partial_k H_L = -H_L(\partial_k G_L)H_L, \tag{BC.41}$$

$$\partial_{kl} H_L = H_L(\partial_l G_L)H_L(\partial_k G_L)H_L + H_L(\partial_k G_L)H_L(\partial_l G_L)H_L - H_L(\partial_{kl} G_L)H_L, \tag{BC.42}$$

and every factor converges uniformly. With

$$\Gamma^m_{kl}(G_L) = \frac{1}{2}(G_L)^{mn}(\partial_k(G_L)_{ln} + \partial_l(G_L)_{kn} - \partial_n(G_L)_{kl}), \tag{BC.43}$$

$$R^n{}_{mkl}(G_L) = \partial_k\Gamma^n_{lm} - \partial_l\Gamma^n_{km} + \Gamma^n_{kp}\Gamma^p_{lm} - \Gamma^n_{lp}\Gamma^p_{km}, \tag{BC.44}$$

the first line and its first derivatives converge, so the second line and its contractions converge continuously, using metric derivatives only through order two. For the chart, the Fisher tensor is $(Dp)^\top \operatorname{diag}(1/p)Dp$, so the probability Jacobian has full column rank $d$, so some $d$ atom probabilities have a nonsingular $d \times d$ Jacobian; the inverse function theorem [26, Thm. 9.24] makes that map one-to-one near the point with a $C^1$ inverse. □

**Remark BC.10 (Volume rescaling of curvature).** For a constant $c > 0$, replacing $G$ by $cG$ leaves the connection, the Riemann tensor of type $(1, 3)$, and the Ricci tensor unchanged, multiplies the fully covariant Riemann tensor by $c$, and divides the scalar curvature by $c$; in particular $N^{-1}I$ and $I$ have the same connection, and a vanishing sequence of metrics can have divergent scalar curvature. [ESTABLISHED]

*Proof.* Substitute $cG$ in eq. (BC.44): the factor $c$ cancels in $\Gamma$, lowering an index multiplies by $c$, and the scalar contraction uses $c^{-1}G^{-1}$. □

**Corollary BC.11 (Nonempty native regular charts).** On an admitted source box, proposition BC.9 applies to $N^{-1}I(\nu^{\mathrm{f}}_{L,\mathrm{g}}) \to \mathsf{H}$, and to a source pullback on a compact parameter chart on which $D\mathrm{g}$ has full column rank with singular values at least $\sigma_{\mathcal{K}} > 0$. For the pushes, write $\eta_B > 0$ and $\delta_B > 0$ for the floor and width of eqs. (BB.46) and (BB.47) on a compact positive field interval. For a prescribed finite list $\mathcal{B}$ of depths and a compact parameter chart whose source image lies in that interval with $0 < j < \min_{B\in\mathcal{B}} \delta_B$ and on which $D\mathrm{g}$ has full column rank with singular values at least $\sigma_{\mathcal{K}} > 0$, the pulled-back limits of eq. (B9.16) are bounded below by $(\sigma^2_{\mathcal{K}}/2) \min_{B\in\mathcal{B}} \eta_B$, and proposition BC.9 applies at every listed depth and to the strict-phase density of corollary B9.8 on such charts. [ESTABLISHED]

*Proof.* Proposition BC.5 bounds $\mathsf{H}$ below on the box, and $D\mathrm{g}^\top \mathsf{H} D\mathrm{g} \succeq cD\mathrm{g}^\top D\mathrm{g} \succeq c\sigma^2_{\mathcal{K}} \mathrm{Id}$ for any lower bound $c$. For the pushes, eq. (BB.47) gives $\mathcal{I}_B \succeq (\eta_B/2)\,\mathrm{Id}_2$ on the stated region by theorem B9.6, and the same congruence gives the common floor; the $C^2$ convergence is propositions B9.4 and B9.7, and the strict-phase limit is a winning branch density. □

The floor is not uniform in unbounded depth, and an arbitrary positive source image need not lie in these small-$j$ charts; a direction acting only on losing labels can disappear in the strict-phase limit, and constant rank alone gives no quotient.

### BC.2.3 Residual and rare-label scales

**Proposition BC.12 (Residual gradients and rare labels before score squaring).** At every finite volume the label contribution of a family of definition 10.23, with scores $\xi_{\gamma,L} = \nabla\lambda_{\gamma,L}$, is

$$\frac{1}{N}\operatorname{Cov}_{W_L}(\xi_{\gamma,L}) = \frac{1}{2N}\sum_{\gamma,\gamma'} W_{\gamma,L}W_{\gamma',L}(\xi_{\gamma,L}-\xi_{\gamma',L})(\xi_{\gamma,L}-\xi_{\gamma',L})^{\top}. \tag{BC.45}$$

Consequently, if the weighted branch-density term converges, convergence of the full density is equivalent to convergence of the right side, whose weights cannot be replaced by their limits before the score differences are squared. Write $\lambda_{\gamma,L} = N\mathsf{s}_\gamma + \varrho_{\gamma,L}$ with $\|\varrho_{\gamma,L}\|_{C^3(\mathcal{K})}/N \to 0$. In the setting of proposition 10.30, suppose every label of $S$ has positive limiting weight, fix a direction $z$ for which $z\cdot\mathsf{v}_\gamma$ is the same for all $\gamma\in S$, choose $\gamma_0\in S$, and assume for some $0\le\upsilon<1$ that

$$N^{-\upsilon}\partial_z(\varrho_{\gamma,L}-\varrho_{\gamma_0,L})(\theta_0)\longrightarrow d_\gamma \quad (\gamma\in S), \qquad d_{\gamma_0}=0. \tag{BC.46}$$

Then the directional label information divided by $N^{2\upsilon}$ tends to $\operatorname{Var}_w(d_\gamma)$; positive variance is below the extensive scale if $\upsilon<1/2$, gives a finite nonzero density if $\upsilon=1/2$, and makes the density diverge if $\upsilon>1/2$. If all slopes $\mathsf{v}_\gamma$ on $S$ agree, the relative residual gradients divided by $\sqrt{N}$ tend to vectors $d_\gamma$, and the branch densities converge at $\theta_0$ to $\mathcal{I}_\gamma(\theta_0)$, then

$$\frac{1}{N}I(P_L)(\theta_0)\longrightarrow\sum_{\gamma\in S}w_\gamma\mathcal{I}_\gamma(\theta_0)+\operatorname{Cov}_w(d_\gamma). \tag{BC.47}$$

These are limits at a fixed parameter, with no conclusion about derivatives, and a tied label of vanishing weight requires separate control. [ESTABLISHED]

*Proof.* Expanding the right side of eq. (BC.45), the two squared terms give $N^{-1}\sum_\gamma W_\gamma\xi_\gamma\xi_\gamma^\top$ and the two cross terms give $-N^{-1}(\sum_\gamma W_\gamma\xi_\gamma)(\sum_\gamma W_\gamma\xi_\gamma)^\top$, which proves the identity and, with the branch term of eq. (10.14), the equivalence. On $S$ equal bulk directional slopes cancel from directional score differences at $\theta_0$; dividing by $N^\upsilon$ and using eq. (BC.46), the limiting sum is $\frac{1}{2}\sum_{\gamma,\gamma'\in S}w_\gamma w_{\gamma'}(d_\gamma-d_{\gamma'})^2=\operatorname{Var}_w(d_\gamma)$. Pairs with a nonmaximizing label contribute at most $CN^{2-2\upsilon}e^{-cN}$, since all score differences are $O(N)$. Comparing $2\upsilon$ with one gives the thresholds, and the outer-product version with $\upsilon=1/2$ gives eq. (BC.47). □

The subextensive and rare-label families eqs. (10.26) and (10.27) of proposition 10.32 realize every threshold of this proposition.

### BC.2.4 Comparison kernels and moving centers

**Proposition BC.13 (Two parameter-independent comparison kernels).** Under the hypotheses of theorem 10.31, let $\Pi(\mathsf{w})$ be the finite-label experiment on the full catalog with masses $\bar{\mathsf{p}}_\gamma(\mathsf{w})$, including its structural zeros, and let $\mathcal{E}_L(\mathsf{w})=P_{L,\theta_0+\mathsf{w}/N}$. There are kernels in both directions between $\mathcal{E}_L(\mathsf{w})$ and $\Pi(\mathsf{w})$ that are independent of $\mathsf{w}$ and have uniformly vanishing total-variation

errors on every compact w-set. Replacing a conditional branch by its center law costs $O(N^{-1/2})$, and the label error has the rate of eq. (10.22). [ESTABLISHED]

*Proof.* Let $K_L$ extract the label and let $R_L$ sample the corresponding finite center branch:

$$K_L(\gamma' \mid \gamma, x) = \mathbf{1}_{\{\gamma'=\gamma\}}, \qquad R_L(\gamma', x \mid \gamma) = \mathbf{1}_{\{\gamma'=\gamma\}} p_{\gamma,L,\theta_0}(x). \tag{BC.48}$$

Both are normalized, preserve labels, and depend on $L$ but not on w; the center branch is defined for every label of positive finite mass. With $\varepsilon_L = \sup_{\mathsf{w}} \frac{1}{2} \sum_\gamma |W_{\gamma,L}(\theta_0 + \mathsf{w}/N) - \bar{\mathsf{p}}_\gamma(\mathsf{w})|$ over the chosen compact, the first comparison has error at most $\varepsilon_L \to 0$. For the reverse comparison each branch has $I(p_{\gamma,L,\theta}) \preceq CN\,\mathrm{Id}$ along the segments in question, by its bounded density, and on the fixed positive support

$$\left\| \frac{d}{d\sigma} \sqrt{p_{\gamma,L,\theta_0+\sigma\mathsf{w}/N}} \right\|_2^2 = \frac{1}{4N^2} \mathsf{w}^\top I(p_{\gamma,L,\theta_0+\sigma\mathsf{w}/N}) \mathsf{w} \le \frac{C|\mathsf{w}|^2}{4N}, \tag{BC.49}$$

for $0 \le \sigma \le 1$, so integrating gives square-root distance at most $C'|\mathsf{w}|/\sqrt{N}$ between the branch at $\theta_0 + \mathsf{w}/N$ and its center. For probability vectors, the Cauchy-Schwarz inequality gives

$$\|p - p'\|_{\mathrm{TV}} = \frac{1}{2} \sum_x |\sqrt{p(x)} - \sqrt{p'(x)}|(\sqrt{p(x)} + \sqrt{p'(x)}) \le \|\sqrt{p} - \sqrt{p'}\|_2, \tag{BC.50}$$

since $\|\sqrt{p} + \sqrt{p'}\|_2 \le 2$. Splitting $W_\gamma p_{\gamma,\theta} - \bar{\mathsf{p}}_\gamma p_{\gamma,\theta_0}$ into $W_\gamma(p_{\gamma,\theta} - p_{\gamma,\theta_0})$ and $(W_\gamma - \bar{\mathsf{p}}_\gamma) p_{\gamma,\theta_0}$ on the disjoint labeled carrier gives

$$\sup_{\mathsf{w}} \|\mathcal{E}_L(\mathsf{w}) - \Pi(\mathsf{w}) R_L\|_{\mathrm{TV}} \le C_{\mathcal{K}} N^{-1/2} + \varepsilon_L. \tag{BC.51}$$

In the terminology of Le Cam [19], the two kernels bound both deficiencies between the experiments restricted to a compact w-set by constant multiples of $\varepsilon_L$ and of the right side of eq. (BC.51), so their Le Cam distance tends to zero. □

**Proposition BC.14 (The Euclidean null-slope quotient).** Let

$$E = \mathrm{span}\{\mathsf{v}_\gamma - \mathsf{v}_{\gamma_0} : \gamma \in S\}, \qquad r = \dim E, \tag{BC.52}$$

and let $\mathbb{K}(\mathsf{w}) = \mathrm{Cov}_{\bar{\mathsf{p}}(\mathsf{w})}(\mathsf{v}_\gamma)$ be the window limit of theorem 10.31. For every w, $\ker \mathbb{K}(\mathsf{w}) = E^\perp$, and $\bar{\mathsf{p}}(\mathsf{w}) = \bar{\mathsf{p}}(\mathsf{w}')$ exactly when $\mathsf{w} - \mathsf{w}' \in E^\perp$. The limiting family descends to $\mathbb{R}^d/E^\perp$, identified with $E$, on which its probability map is injective and, if $r > 0$, its Fisher tensor is positive definite with a positive minimum on every compact chart. The restrictions of the finite window metrics to $E$ converge in $C^2$ with the inverse, connection, and curvature limits of proposition BC.9. [ESTABLISHED]

*Proof.* All limiting weights on $S$ are positive at every finite w, and the directional quadratic form of $\mathbb{K}$ is their variance of $z \cdot \mathsf{v}_\gamma$, which vanishes exactly when $z$ is orthogonal to all slope differences. The

probability ratios satisfy

$$\log \frac{\bar{\mathsf{p}}_\gamma(\mathsf{w})}{\bar{\mathsf{p}}_{\gamma_0}(\mathsf{w})} = \log \kappa_\gamma + \mathsf{w} \cdot (\mathsf{v}_\gamma - \mathsf{v}_{\gamma_0}), \tag{BC.53}$$

so equal laws are equivalent to orthogonality of $\mathsf{w} - \mathsf{w}'$ to every slope difference, and orthogonal projection onto $E$ gives the quotient. Choosing $r$ independent slope differences, their log ratios recover the coordinates on $E$, which proves injectivity and a smooth inverse on the image. In orthonormal coordinates on $E$ the restriction of $\mathbb{K}$ is positive definite with a positive minimum on each compact chart, and constant linear restriction preserves the $C^2$ convergence of eq. (10.24). When $r = 0$ the limiting experiment is constant and the quotient is a point. The finite metrics need not descend along $E^\perp$, and slower scales in the other directions are not excluded. □

**Proposition BC.15 (A nonempty native realization of the limiting experiment).** Every finite list of slopes $\mathsf{v}_\gamma \in \mathbb{R}^d$ and positive constants $\kappa_\gamma$ with $\kappa_{\gamma_0} = 1$ has a realization on identical native branches with strictly positive high-temperature plaquette coupling for which the window weights, the Fisher tensor, and both comparison kernels above are exact at every finite volume. In this realization the experiment that forgets the label and observes only the configuration has zero Fisher information, so the tie scale is information about the label. [ESTABLISHED]

*Proof.* Choose common $0 < b_0 < 1$ and $0 < t_0 < 1/12$, the free branch for every label on one common carrier, and $a_{\gamma,L}(\theta) = \kappa_\gamma e^{N\mathsf{v}_\gamma \cdot \theta}$. The weights are positive and smooth, their logarithms have $C^3$ norms $O(N)$ on compact sets and bulk limits $\mathsf{v}_\gamma \cdot \theta$, and the identical partitions cancel in all label ratios in both modes. At $\theta_0 = 0$ the center ratios are $\kappa_\gamma$ exactly, and at $\mathsf{w}/N$ the normalized weights are eq. (10.23). The branches do not depend on $\theta$, so their Fisher term is zero, the reconstruction kernel uses their unchanged law, and both comparison errors vanish; the configuration marginal $\sum_\gamma W_\gamma p = p$ does not depend on $\theta$ either. With two labels in one coordinate, $\mathsf{v}_0 = 0$, $\mathsf{v}_1 = 1$, and $\kappa_0 = \kappa_1 = 1$,

$$I_L^{(\mathsf{w})} = \mathbb{K}(\mathsf{w}) = \frac{1}{4\cosh^2(\mathsf{w}/2)} > 0, \tag{BC.54}$$

whose one-dimensional curvature is zero. □

**Theorem BC.16 (A transverse two-label moving center).** Assume the growth and branch $C^2$ hypotheses of theorem 10.31. Suppose exactly two labels $0, 1$ maximize at $\theta_0$, all others have a strict gap there, and for a fixed unit vector $n$, $a := \partial_n(\mathsf{s}_1 - \mathsf{s}_0)(\theta_0) > 0$ after reversing $n$ if necessary. On a sufficiently short segment through $\theta_0$ there is, for all large $L$, exactly one center $\theta_{c,L} = \theta_0 + t_L n$ with $\lambda_{1,L}(\theta_{c,L}) = \lambda_{0,L}(\theta_{c,L})$, and

$$|t_L| \le \frac{4|\varrho_{1,L}(\theta_0) - \varrho_{0,L}(\theta_0)|}{Na} \longrightarrow 0, \qquad \varrho_{\gamma,L} = \lambda_{\gamma,L} - N\mathsf{s}_\gamma. \tag{BC.55}$$

The experiment $P_{L,\theta_{c,L}+\mathsf{w}/N}$ has the $C^3$ weight and $C^2$ Fisher limits of theorem 10.31 with $S = \{0, 1\}$, $\kappa_0 = \kappa_1 = 1$, and slopes $\mathsf{v}_\gamma = \nabla \mathsf{s}_\gamma(\theta_0)$, and its comparison kernel samples $p_{\gamma,L,\theta_{c,L}}$, independently of $\mathsf{w}$. [ESTABLISHED]

*Proof.* Let $f = \mathsf{s}_1 - \mathsf{s}_0$ and choose $r_0 > 0$ so that the segment $\theta_0 + [-r_0, r_0]n$ lies in the parameter neighborhood, $\partial_n f \geq a/2$ on it, and labels $0, 1$ stay uniformly above every other bulk score there; then $f(\theta_0 \pm r_0 n)$ has sign $\pm$ with magnitude at least $ar_0/2$. For $F_L = \lambda_{1,L} - \lambda_{0,L}$, $C^1$ bulk convergence gives $\partial_n F_L \geq Na/4$ on the segment for large $L$, and uniform convergence gives opposite signs at its endpoints, so $F_L$ has exactly one zero. Since $F_L(\theta_0) = \varrho_{1,L}(\theta_0) - \varrho_{0,L}(\theta_0) = o(N)$, integrating the derivative bound gives eq. (BC.55). At the center the relative mass of the two labels is one and the others are exponentially small. For compact $\mathsf{w}$-sets, apply eq. (B.7) at the base point $\theta_{c,L}$; compact $C^3$ growth and $\theta_{c,L} \to \theta_0$ give the limits and bounds of eq. (B.8), and the covariance argument with the $O(N^{-1})$ branch bound of eq. (B.9) gives $C^2$ Fisher convergence. The kernels of eq. (BC.48) with the moving center branch and the square-root argument give the bound of eq. (BC.51). □

The bound permits $|t_L|$ much larger than $N^{-1/2}$, in which case sampling the branch at $\theta_0$ would cost up to $C\sqrt{N}|t_L|$ in total variation; no such replacement is used. The single transverse argument does not solve several relative-mass equations at once, so ties of more than two labels are not covered.

## BC.3 Superextensive scales and prescribed actions

In this section the channels are coarse channels of definition 10.1 that preserve the label, as in corollary 10.27, and a fixed word maps $\widetilde{N} = B^2 N$ fine cells to $N$ coarse cells, with $B$ and the word fixed before every volume limit. For one finite experiment and its image, $I_f$ and $I_c$ are the full Fisher tensors in the same parameter coordinates and $D = I_f - I_c$.

**Proposition BC.17 (Preservation of a proved superextensive scale).** Suppose that on a compact $\mathcal{K} \subset \Theta$ there is $C_{\mathcal{K}}$, independent of $\widetilde{N}$ and $\gamma$, with

$$I(p_{\gamma,\widetilde{N},\theta}) \preceq C_{\mathcal{K}} \widetilde{N}\, \mathrm{Id} \quad \text{for all } \theta \in \mathcal{K}. \tag{BC.56}$$

Then $0 \preceq D \preceq C_{\mathcal{K}} \widetilde{N}\, \mathrm{Id}$. If $a_{\widetilde{N}} > 0$ is a parameter-independent scale with $\widetilde{N}/a_{\widetilde{N}} \to 0$ and $I_f/a_{\widetilde{N}} \to \mathcal{H}$ pointwise or uniformly on $\mathcal{K}$, then $I_c/a_{\widetilde{N}} \to \mathcal{H}$ in the same sense; for a fixed $p > 1$, $I_f/\widetilde{N}^p \to \mathcal{H}$ implies

$$\frac{I_c}{N^p} \longrightarrow B^{2p}\mathcal{H}, \qquad \frac{D}{N^p} \longrightarrow 0. \tag{BC.57}$$

For a center $\theta_{c,L}$ independent of the local coordinate, compare the fine and coarse window coordinates

$$\mathsf{w} = \widetilde{N}(\theta - \theta_{c,L}), \qquad \mathsf{w}' = N(\theta - \theta_{c,L}), \qquad \mathsf{w} = B^2 \mathsf{w}'. \tag{BC.58}$$

In the common coordinate $\mathsf{w}$ the fine and coarse tensors $\widetilde{N}^{-2} I_f$ and $\widetilde{N}^{-2} I_c$ differ by at most $C_{\mathcal{K}}/\widetilde{N}$ uniformly on compact $\mathsf{w}$-sets, and if their common limit is $\mathcal{H}(\mathsf{w})$, the coarse tensor in its own coordinate satisfies

$$I_c^{(\mathsf{w}')}(\mathsf{w}') = B^4 I_c^{(\mathsf{w})}(B^2 \mathsf{w}') \longrightarrow B^4 \mathcal{H}(B^2 \mathsf{w}'). \tag{BC.59}$$

If moreover the branch densities $I(p_{\gamma,\widetilde{N}})/\widetilde{N}$ and $I(p_{\gamma,\widetilde{N}}C_\gamma)/\widetilde{N}$ have bounded $C^2$ norms on a compact neighborhood of the windows and the reparameterized label weights have bounded $C^2$ norms on compact w-sets, then the within-branch loss in the w chart,

$$\frac{1}{\widetilde{N}}\sum_\gamma W_{\gamma,\widetilde{N}}(\theta_{c,L}+\mathsf{w}/\widetilde{N})\left[\frac{I(p_{\gamma,\widetilde{N}})-I(p_{\gamma,\widetilde{N}}C_\gamma)}{\widetilde{N}}\right](\theta_{c,L}+\mathsf{w}/\widetilde{N}), \tag{BC.60}$$

is $O(\widetilde{N}^{-1})$ in $C^2$, so $C^2$ convergence of the fine window tensor transfers to the coarse one and eq. (BC.59) holds in $C^2$. [ESTABLISHED]

*Proof.* By eq. (10.20), $D=\sum_\gamma W_\gamma[I(p_\gamma)-I(p_\gamma C_\gamma)]\succeq 0$, and since each pushed branch tensor is a score Gram matrix, $D\preceq\sum_\gamma W_\gamma I(p_\gamma)\preceq C_{\mathcal{K}}\widetilde{N}\,\mathrm{Id}$. Hence $\|I_f/a_{\widetilde{N}}-I_c/a_{\widetilde{N}}\|\le C_{\mathcal{K}}\widetilde{N}/a_{\widetilde{N}}\to 0$ uniformly, and $\widetilde{N}^p=B^{2p}N^p$ gives the power case; no relation between scales at different volumes is implied for a general sequence. The differential of $\theta_{c,L}+\mathsf{w}/\widetilde{N}$ is $\widetilde{N}^{-1}\,\mathrm{Id}$, which gives the factor $\widetilde{N}^{-2}$ and the loss bound $C_{\mathcal{K}}/\widetilde{N}$; the differential of $\mathsf{w}=B^2\mathsf{w}'$ is $B^2\,\mathrm{Id}$, which gives the factor $B^4$ and the argument $B^2\mathsf{w}'$. The label law at the same w is unchanged by the channel at every finite volume. For the derivative version, every w-derivative of a composed branch density contributes a further $\widetilde{N}^{-1}$ and the weight derivatives are bounded, so two applications of the product and chain rules bound the $C^2$ norm of eq. (BC.60) by $O(\widetilde{N}^{-1})$. □

For native branches the branch bound and the $C^2$ inputs hold by remark BC.6. At a tie with convergent weights and $\widetilde{N}^{-1}\nabla\lambda_{\gamma,\widetilde{N}}\to\mathsf{v}_\gamma$, the score decomposition gives $I_f/\widetilde{N}^2\to\mathrm{Cov}_w(\mathsf{v}_\gamma)$, so the coarse tensor per squared coarse area tends to $B^4\,\mathrm{Cov}_w(\mathsf{v}_\gamma)$; in the window of theorem 10.31 or the moving center of theorem BC.16, taken in the fine area, the coarse limit is $B^4\mathcal{H}(B^2\mathsf{w}')$, equivalently coarse slopes $B^2\mathsf{v}_\gamma$. The label marginal is preserved at every finite volume, and the discarded state information cannot contribute at these larger scales; equality of bulk log-mass densities alone supplies neither the weight limit nor any information limit.

### BC.3.1 Prescribed geometry actions and relative label weights

**Definition BC.18 (Reweighted label ensemble).** Fix real functions $\mathscr{A}_{\gamma,L}(\theta)$ of the label alone and replace the positive raw label weights by $\widetilde{a}_{\gamma,L}=a_{\gamma,L}e^{-\mathscr{A}_{\gamma,L}}$; for a prescribed normalized hierarchy, reweight its label masses by the same factor and renormalize. Conditional branch laws and their physical partitions are unchanged. A factor depending on the branch state changes the branch and is not of this kind.

**Proposition BC.19 (Action growth and the exact relative multiplier).** Suppose $\log a_{\gamma,L}$ is $C^3$ on a neighborhood of each compact $\mathcal{K}\subset\Theta$ with

$$\|\log a_{\gamma,L}\|_{C^3(\mathcal{K})}\le C_{\mathcal{K}}N,\qquad N^{-1}\log a_{\gamma,L}\longrightarrow\alpha_\gamma\quad\text{in } C^3(\mathcal{K}), \tag{BC.61}$$

and the actions of definition BC.18 satisfy

$$\|\mathscr{A}_{\gamma,L}\|_{C^3(\mathcal{K})}\le C'_{\mathcal{K}}N,\qquad N^{-1}\mathscr{A}_{\gamma,L}\longrightarrow\mathscr{A}_\gamma^{\mathrm{bulk}}\quad\text{in } C^3(\mathcal{K}). \tag{BC.62}$$

Then the reweighted data have the same growth class, with bulk log weight $\alpha_\gamma - \mathscr{A}_\gamma^{\mathrm{bulk}}$, and in either normalization mode

$$\frac{\widetilde{W}_{\gamma,L}}{\widetilde{W}_{\gamma_0,L}} = \frac{W_{\gamma,L}}{W_{\gamma_0,L}} \exp\bigl[-\mathscr{A}_{\gamma,L} + \mathscr{A}_{\gamma_0,L}\bigr]. \tag{BC.63}$$

For two identical native branches with $0 < b < 1$, $0 < t < 1/12$, equal initial weights, and actions $\mathscr{A}_{0,L} = 0$, $\mathscr{A}_{1,L} = N^\upsilon\theta$ with $0 < \upsilon < 1$, the categorical Fisher information at $\theta = 0$ is $N^{2\upsilon}/4$, so a vanishing bulk action can leave the extensive density unchanged, add a finite correction, or dominate it as $\upsilon$ is below, equal to, or above $1/2$. [ESTABLISHED]

*Proof.* The new log raw weight $\log a_{\gamma,L} - \mathscr{A}_{\gamma,L}$ keeps the bound of order $N$ and has the stated bulk limit. The unnormalized label mass is $a_{\gamma,L}Z_{\gamma,L}$ in the global-factor ensemble and $a_{\gamma,L}$ in a hierarchy generated from raw weights; $Z_{\gamma,L}$ is unchanged, so either representative shifts by $-\mathscr{A}_{\gamma,L}$, and the ratio to $\gamma_0$ cancels the new common normalization, also for a reweighted prescribed hierarchy. In the example the common native law does not depend on $\theta$, the weight ratio is $e^{-N^\upsilon\theta}$, and eq. (10.25) with $f_L = -N^\upsilon\theta$ gives $q(0)(f_L')^2 = N^{2\upsilon}/4$. □

An extensive action $N\mathscr{A}_\gamma(\theta)$ changes the bulk log-mass density by $-\mathscr{A}_\gamma$, and a fixed order-one action has zero bulk rate. Real actions define positive finite weights even where they are negative; nonnegativity belongs to particular energy constructions, such as remark B8.4.

**Proposition BC.20 (Conditional limits and a bounded-action obstruction).** Let $\theta_{c,L}$ be centers at which the original positive relative weights converge to $k_\gamma \in (0,\infty)$. If

$$\mathscr{A}_{\gamma,L}(\theta_{c,L}) - \mathscr{A}_{\gamma_0,L}(\theta_{c,L}) \longrightarrow a_\gamma, \tag{BC.64}$$

then the new relative weights converge to $k_\gamma e^{-a_\gamma}$, and when these are all labels of the catalog the normalized weights converge to their normalization. Bounded actions, even with the bulk convergence of eq. (BC.62), do not imply eq. (BC.64) or a limiting ratio. [ESTABLISHED]

*Proof.* Evaluate eq. (BC.63) at the centers and use continuity of the exponential and of normalization on a finite positive vector; for fixed $C^3$ actions and centers tending to $\theta_0$, $a_\gamma = \mathscr{A}_\gamma(\theta_0) - \mathscr{A}_{\gamma_0}(\theta_0)$, and a common action cancels at every volume. For the obstruction take two identical native branches with $0 < b < 1$, $0 < t < 1/12$, unit initial weights, and

$$\mathscr{A}_{0,L} = 0, \qquad \mathscr{A}_{1,L} = 1 + (-1)^L. \tag{BC.65}$$

These actions are bounded by two, independent of $\theta$, and satisfy eq. (BC.62) with zero bulk actions, yet the ratio of label one to label zero is $e^{-2}$ for even $L$ and one for odd $L$, so the weight of label one alternates between $e^{-2}/(1+e^{-2})$ and $1/2$. The failure is the missing relative center limit, not a failure of normalization or of the native branch domain. □

# Appendix BD

# Notation, Open Problems, and Numerical Index

## BD.1 Notation

This supplement uses the symbols of the main manuscript (Table D.1) and of Supplement G (Table GB.1) with their meanings there. Table BD.1 lists the symbols that recur across the chapters of Parts B-I and B-II; a symbol used inside one chapter is declared at its opening, and the lattice part declares its symbols in Table B7.1. The word pullback denotes the connection-relative pullback of fiber tensors to the base of Definition B2.1; restriction of a tensor along an injective map of parameters, which Supplement G also calls a pullback, is a different construction.

**Table BD.1:** Notation of Supplement B.

| Symbol | Meaning | Introduced |
|---|---|---|
| **Base transport and multi-point designs** | | |
| $f$; $s$ | a base map; a section | chapter B0 |
| $u_i^b$, $u_i^m$, $h_i$ | belief and model frames of agent $i$ in two atlases; the relative frame, $u_i^m = u_i^b \cdot h_i$ | proposition B1.1 |
| $g_i^b$, $g_i^m$ | passive re-expressions of the two atlases, $u_i^{x\prime} = u_i^x \cdot (g_i^x)^{-1}$; the common frame has $g_i^b = g_i^m = g_i$ | proposition B1.1 |
| $\omega_b$, $\omega_m$; $H^b$, $H^m$; $H_e^x X$ | principal connections; their horizontal distributions; the horizontal lift of $X \in T_c\mathcal{C}$ at a point $e$ of $(\mathcal{E}_x)_c$ | definition B1.6 |
| $\gamma$; $\Omega_\gamma$, $\widetilde{\Omega}_\gamma$ | piecewise-smooth base curve; parallel transports of belief and model laws along it | definition B1.6 |
| $A_i^x = (u_i^x)^* \omega_x$ | local connection form in the frame $u_i^x$ | proposition B1.7 |
| $\mathcal{D}\Phi$, $\mathcal{D}\widetilde{\Phi}$ | covariant defects of the cross morphisms | definition B1.8 |
| $G_b \times G_m$ | structure group of the optional product-gauge extension | definition B1.16 |
| $D = \{c_d\}$, $d$; $I_D$, $D_i$, $V_d$ | finite design and its index; agent–design incidence set, design points of agent $i$, agents at the design point $c_d$ | definition B1.18 |
| $D_e = D_i \cap D_j$, $\Theta_{d,e}^x$, $g_{d,i}^x = g_i^x(c_d)$ | design points shared by the endpoints of $e : j \to i$; the link copy at $c_d$; the re-expression at an incidence | definition B1.25 |
| $\mathfrak{S}_i^x$ | measurable space of sections of channel $x$ for agent $i$ | hypothesis B1.19 |

*continued on the next page*

| Symbol | Meaning | Introduced |
|---|---|---|
| $f_\ell : \mathcal{C}_\ell \to \mathcal{C}_{\ell+1}$; $\pi_\ell : \mathscr{P}_\ell \to \mathcal{C}_\ell$; $\mathcal{P}_\ell$, $\kappa_\ell$ | base map between level bases; level principal bundles; principal scale map covering $f_\ell$ and its group homomorphism | proposition B1.31 |
| $D^\omega s$, $\mathrm{ver}^\omega$ | covariant first jet of a section; vertical projection of a connection | definition B2.1 |
| $h_s^\omega$, $c_s^\omega$ | connection-relative Fisher and Amari–Chentsov pullbacks of a section | definition B2.1 |
| $\Psi$, $\Psi_* = T^V\Psi$ | bundle morphism over a base map, and its vertical differential | section B2.7 |
| $A_\Psi = \mathcal{D}\Psi$; $\mathfrak{A}_\mathcal{P}$ | horizontal defect of $\Psi$; scale-connection defect form of a principal scale map | definitions B2.19 and B2.30 |
| $\Delta_F^\Psi$; $\delta_\Psi$, $\mathcal{X}_\Psi$, $\mathcal{Q}_\Psi$ | vertical Fisher defect of $\Psi$; its base pullback and the signed and positive anomaly tensors | definition B2.23 |
| $\mathcal{R}$ | residual of the base Fisher defect cocycle | theorem B2.35 |
| $\mathcal{Q}_\ell$, $\mathsf{G}_\ell$ | configuration manifold of sections over a base at scale $\ell$, and its configuration metric | definition B3.15 |
| **Infinite networks and Wilsonian asymptotics** | | |
| $\mathfrak{X}_\star$, $J_\ell$; $\widehat{\mathfrak{R}}_\ell$ | reference object and comparison isomorphisms into the scale-$\ell$ objects; the reference cocycle | definition B5.1 |
| $C_b$, $I_b$, $K_b = C_b I_b$; $\mathfrak{R}_b$ | structural coarse kernel, identification kernel, and rescaled kernel of a block ratio $b$; the measure-pair step | definition B5.4 |
| $v_{\ell,a}$, $\lambda_{\ell,a}$ | compatible cross-scale mode line and its coefficients | definition B5.7 |
| $b_k$; $\varsigma_{n\leftarrow\ell}$ | declared adjacent block ratios; cumulative log block scale | definition B5.9 |
| $\chi_{n\leftarrow\ell}$, $\upsilon_{n\leftarrow\ell,a}$ | finite-time growth rate and scalar mode exponent | definition B5.9 |
| $\mathfrak{R}_b^H$, $\mathfrak{b}_b^H$ | reference-dependent action map and discrete action beta | definition B5.23 |
| $\mathfrak{b}_\ell^{\mathrm{ex}}$, $\mathfrak{b}_\ell^{\mathrm{ret}}$ | exact and retained discrete interaction beta | definition B5.25 and proposition B5.26 |
| $\nabla^{\mathrm{scale}}$, $\mathfrak{b}^{\mathrm{scale}}$ | scale connection and continuous beta | definition B6.1 |
| $r_{\mathrm{ess}}$ | essential spectral radius, the spectral radius in the Calkin algebra | theorem B6.12 |

## BD.2 Open problems

Table BD.2 lists the open problems of this supplement, each with the statement at which it arises and a condition that would settle it. The first arises from a claim declined in Chapter B2 and is stated here. Open problems of the main manuscript that these chapters extend, such as the dynamic closure of the pointwise parent (Open Problem 12.40) and the slow-manifold reduction (Open Problem 8.14), are listed there.

**Open Problem BD.1 (A canonical nondegenerate pullback geometry).** The pullback tensors of a fixed section are passively gauge covariant for a chosen connection (Theorem B2.2) and change

by exact formulas when the connection changes (Proposition B2.4), and where the covariant first jet has constant rank they induce a positive-definite metric on a quotient vector bundle (Theorem B2.7). The collective rule of proposition B2.11 selects a minimizing connection class after common-target comparisons, law families, and weights are supplied; its isotropy component remains undetermined. The rule does not select those additional data from the model, and a combined belief–model tensor still needs channel weights (Definition B2.5). What is open is a pullback geometry determined by the data of the model. It is settled by a canonical or empirically specified connection together with channel weights under which the weighted pullback has constant rank, an integrable radical, a regular Hausdorff leaf space, and basic induced tensors, so that Theorem B2.7 and proposition B2.8 give a nondegenerate geometry on a quotient manifold; or by a proof that no connection in a stated class yields these properties.

**Table BD.2:** Open problems of Supplement B.

| Item | Locator | Settling condition |
|---|---|---|
| **Part B-I: base transport and multi-point designs** | | |
| Continuum law theory | open problem B1.22 | A refining family of designs, a projective system of normalized finite laws with a Kolmogorov-type extension, a topology and sigma-algebra on a space of sections with tightness, convergence of the evidence bound, connections, and observables, control of gauges and reference measures under refinement, and a normalizable continuum interaction energy. |
| Regular frame-coordinate quotient | open problem B1.30 | The generic stabilizers of the passive $G \times G$ coordinate action on the parameter space, and a proof of freeness and properness on declared strata on which its orbit space is a regular quotient manifold. |
| A canonical nondegenerate pullback geometry | open problem BD.1 | A canonical or empirically specified connection with declared channel weights under which the weighted pullback has constant rank, an integrable radical, a regular Hausdorff leaf space, and basic induced tensors; or a proof that no connection in a stated class yields them. |
| Collective interface preservation | open problem B2.15 | Structural conditions for coupled Gaussian quadratic-record inference and reducing linear block channels that preserve the comparisons and give a uniform positive aggregate-soldering singular-value bound through a positive horizon and nonzero coarse depth, with constants controlled along connected graphs of unbounded size. |
| Admission of declared recognition families to a configuration tier | open problem B3.14 | For a declared family, a manifold calculus, its statistical metric, a smooth quotient, and a metric-orthogonal splitting at every scale where a natural-gradient field or a Fisher duration is asserted; or a family for which one of them provably fails while the history theorems are still invoked. |

*continued on the next page*

| Item | Locator | Settling condition |
|---|---|---|
| Smooth structure of the projectable set | open problem B3.19 | A transversality or elliptic-regularity hypothesis on the base map and the recognition family under which the projectable set is proved to be a smooth submanifold carrying a smooth induced configuration map; or a family in which it provably is not. |
| Gluing of independently supplied local data | open problem B4.9 | A gluing theorem with hypotheses on the local laws, channels, and presentations of a cover alone, giving one joint extension compatible with every restriction and selected conditional version, or an obstruction computed from those data. |
| Graph-to-base identification | open problem B4.14 | A choice of vertex contexts, an embedding, and oriented edge curves for given graph data and a connection, with a proof that the independently specified represented links equal the corresponding base transports and hypotheses under which the realization is unique up to gauge; or graph data for which no such choice exists. |
| Connection recovery and record-class sensitivity | open problem B4.15 | A recovery theorem for a connection or its holonomy class from a declared class of population records; negatively, a theorem that every statistic in a declared admissible controlled class is insensitive to the holonomy class. |
| Quotients beyond the chosen metric splitting | open problem B4.19 | Constant-rank support data, compatible represented actions, and an admitted quotient or bundle map for which a canonical effective-support quotient, an intertwiner into a prescribed associated representation, and quotient geometry across rank changes are constructed; or an obstruction to them. |
| Informative base aggregation | open problem B4.21 | A construction of informative aggregation of laws, channels, and connections along a base map that collapses contexts, with the base prior, joint base-fiber model, and observation weighting it needs, and its descent and free-energy identities; or an obstruction for a declared base map. |
| **Part B-II: infinite networks and Wilsonian asymptotics** | | |
| Compatible rescaling kernels on agent networks | open problem B5.6 | A family of identification kernels acting on the gauge-retained network state of the main manuscript, with a proof of the compatibility law on a declared class of finite networks at $c_*$; or a proof that none exists for a stated class. |
| Multiplicative ergodic structure of scale cocycles | open problem B5.14 | A declared stationary family of blocking data over a measure-preserving base whose multiplier cocycle satisfies the hypotheses of a multiplicative ergodic theorem; or a proof that no such family exists for the exact interaction cocycle of the main manuscript. |
| Extensive relevance beyond the scalar Gaussian block | open problem B5.21 | For a correlated, anisotropic, or gauge-mixed block, a declared transported family of laws and tangent norms together with a diagonalization of its score operator. |
| Nonlinear attraction at the Gaussian block law | open problem B5.22 | A remainder estimate for the nonlinear block map in the score-tangent topology on a declared neighborhood of the Gaussian law. |

*continued on the next page*

| Item | Locator | Settling condition |
|---|---|---|
| Canonical continuous beta | open problem B6.4 | A canonical interpolation of a discrete blocking sequence with a proof of its uniqueness in a declared class; or a proof that the continuous beta is the same for every member of that class. |
| Two-index limits and universality | open problem B6.19 | A triangular family of finite agent networks at $c_*$ with projective consistency at fixed depth, uniform control of the coarse error, a diagonal scaling of depth with size, and a proof of convergence of a declared scheme-invariant observable; or an impossibility theorem for a stated class. |
| Primitive hierarchical identification | open problem B6.23 | For a stated family of graphs, a hierarchical identification under which sum-only aggregation yields one fixed primitive coupling matrix, with a Perron vector that factorizes for the bi-additive reading; or a proof that none exists. |
| **Part B-III: the lattice cluster** | | |
| Typing of the lattice sites | open problem B7.1 | A declared reading proved: either the lattice law realized as the link marginal of a normalized joint with belief and model channels on a countable agent set at $c_*$ satisfying the countable-index projective and specification hypotheses, or a base discretization with a connection whose transports are the raw links and whose holonomies are the plaquette characters, with the path channel a scale map over a declared base map. |
| Limits beyond the admitted high-temperature construction | open problem B7.17 | New parameter domains with the corresponding tightness, interaction, and scaling estimates; positive Fisher floors uniform in depth; joint volume and depth estimates for the labeled experiment; and an existence and regularity theorem for a normalized law whose curvature action is computed from itself. |
| Quenched evidence density for general records | open problem B8.17 | An every-exterior comparison and a pressure identity for site-dependent shifted fields of both signs in $(-1, 1)$, together with an ergodic theorem for the record-dependent evidence, giving almost-sure record-independent limits of the evidence density, the posterior label rates, and the specific free energy. |

## BD.3 Numerical index

Table BD.3 indexes the computations recorded for this supplement. The first three rows are the statements of this supplement whose status is NUMERICAL; their instances are deterministic, and the implementation that computed them has been retired, so the values were not recomputed. The remaining rows are finite exact comparisons for the lattice part, made in exact rational arithmetic without random sampling. They corroborate established statements, which rest on their proofs, and they supply no estimate as the lattice grows.

**Table BD.3:** Numerical index of Supplement B.

| Claim | Check | What is tested | Control | Seed |
|---|---|---|---|---|
| remark B5.5 | CHK-RG-RESCALING-COMPATIBILITY | Blocking by the full ratio against blocking in declared stages, for one declared rescaling family on nested cycles of six categorical agents. | A flat-link control, the 24 orderings of the edges, and gauge covariance of the composite step. | no randomness |
| remark B6.7 | CHK-RG-PASSIVE-BLOCKING | Iteration of the projected frozen-attention map at ratio two on a homogeneous six-cycle and ratio three on a homogeneous nine-cycle, and the pairwise block of its fixed structures. | The two ratios against each other, and the regeneration rule of the next row; no other control. | no randomness |
| remark B6.8 | CHK-RG-REGENERATED-ATTENTION | The declared row-regeneration rule on the same instances, and the pairwise block of its fixed structures. | The frozen-attention map from the same initial coupling, and a readout of equal cardinality that carries no gauge charge. | no randomness |
| theorem B9.3 and proposition B9.4 | CHK-RG-THERMO-NATIVE-JETS | Exhaustive summation over the 256 sign configurations of the side-two square at $(b,t)=(1/4,1/20)$, with one internal plaquette: every raw-factor and physical-partition derivative through total degree three. | An independent normalized-score Gram comparison of the Fisher matrix at derivative degree zero. | no randomness |
| proposition B9.4 | CHK-RG-THERMO-SINGULAR-FISHER | The side-one square at $(b,t)=(1/3,1/20)$: two signs, native partition one, a null plaquette coordinate, and the field Fisher derivatives. | None separate. | no randomness |
| proposition BC.8 | CHK-RG-THERMO-POSTERIOR-PUSH | Two side-one branches with global factor normalization and a binary sensor of mean $k_\gamma b$, $k_\gamma \in \{1/2, 1/3\}$, on one edge: posterior derivatives through total degree three against an independent two-coordinate Taylor-quotient calculation, and the induced pushed posterior and evidence derivatives. | The native partition is identically one, so a posterior with a nonconstant native partition divisor is not exercised. | no randomness |

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